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AGILEX™ 7 FPGA I-SERIES TRANSCEIVER-SOC DEVELOPMENT KIT (PRODUCTION-3 4 X F-TILE)

NOTES:

1. PROJECT DRAWING NUMBERS:

RAW PCB	100-0330690-D1
GERBER FILES	110-0330690-D1
PCB DESIGN FILES	120-0330690-D1
ASSEMBLY DRAWING	130-0330690-D1
FAB DRAWING	140-0330690-D1
SCHEMATIC DRAWING	150-0330690-D1
PCB FILM	160-0330690-D1
BILL OF MATERIALS	170-0330690-D1
SCHEMATIC DESIGN FILES	180-0330690-D1
FUNCTIONAL SPECIFICATION	210-0330690-D1
PCB LAYOUT GUIDELINES	220-0330690-D1
ASSEMBLY REWORK	320-0330690-D1

PAGE	DESCRIPTION	PAGE	DESCRIPTION
01	TITLE, NOTE, CONTENT	51	MAX10 PWR
02	BLOCK DIAGRAM	52	USB BLASTER
03	POWER TREE DIAGRAM	53	FT2232H
04	CLOCK TREE DIAGRAM	54	CLK SI5394
05	I2C TOPOLOGY	55	CLOCK - ZL30733 -1
06	JTAG TOPOLOGY	56	CLOCK - ZL30733 -2
07	FPGA BANK & CONFIG	57	CLK GEN 5391_A
08	FPGA SDM	58	CLK GEN 5391_B
09	FPGA BANK 2D	59	CLK GEN 5332
10	FPGA BANK 2C	60	CLK VCXO
11	FPGA BANK 2F	61	CLK MUX -1
12	FPGA BANK 2E	62	CLK MUX -2
13	FPGA BANK 3D	63	CLK MUX -3
14	FPGA BANK 3C	64	CLK MUX -4
15	FPGA BANK 3B	65	CLK MUX -5
16	FPGA BANK 3A	66	RESERVED
17	FPGA HPS	67	FPGA SW LEDS
18	FPGA F-TILE 12C	68	SYS SW LEDS
19	FPGA F-TILE 12A	69	LVL SHIFT-1
20	FPGA F-TILE 13A	70	LVL SHFT-2
21	FPGA F-TILE 13C	71	RESERVED
22	FPGA PWR1	72	TEMP SENSORS
23	FPGA PWR2	73	POWER SENSE
24	FPGA GND1	74	PWR INPUT
25	FPGA GND2	75	PWR BIAS
26	FPGA DECOUPLING-1	76	PWR 3.3V/1.8V_STB
27	FPGA DECOUPLING-2	77	PWR P3V3_BUS
28	FPGA DECOUPLING-3	78	PWR P1V2_BUS
29	DDR4 HPS COMP-1	79	PWR VCCL -1
30	DDR4 HPS COMP-2	80	PWR VCCL -2
31	DDR4 HPS COMP-3	81	PWR VCCL -3
32	DDR4 DIMM-2A	82	PWR P0V8_G1
33	DDR4 DIMM-2B	83	PWR VCC_HSSI_U1X_G1
34	DIMM PWR FLT	84	PWR VCC_HSSI_U2X_G1
35	USB3.X/DP MUX	85	PWR VCCERT1_13A/13C
36	SDI BNC	86	PWR VCCERT2_U20/U22
37	PCIE MCIO	87	PWR VCCERT_UX
38	QSFP-DD CONN 0	88	PWR P1V5_G2/P2V5_G2
39	QSFP-DD CONN 1	89	PWR P1V8_G2
40	QSFP-DD-800 CONN	90	PWR P1V2_G3
41	FMC+_A -1	91	PWR P1V2_IO
42	FMC+_A -2	92	PWR P1V8_G3
43	FMC+_B-1	93	PWR VCCH_SDM/P1V0_G2
44	FMC+_B-2	94	PWR LOAD SWITCHES
45	USB2.0	95	PWR DDR4 VTT
46	SMA	96	PD CONTROLLER
47	PWR MAX10 -1	97	BLANK PAGE
48	PWR MAX10 -2		
49	SYS MAX10 -1		
50	SYS MAX10 -2		

REVISION HISTORY

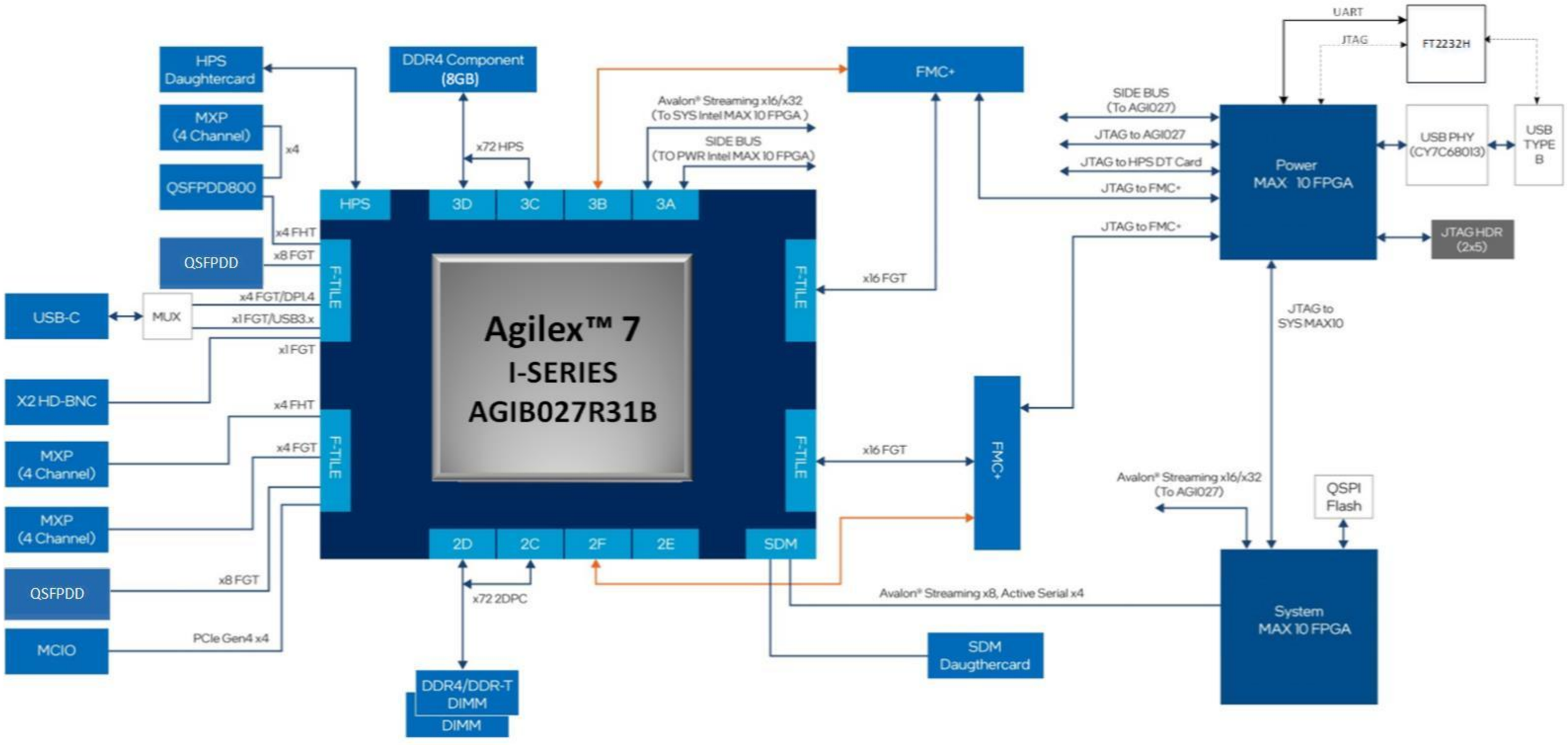
REV	DATE	PAGES	DESCRIPTION
1P0	07-NOV-2022	ALL	REPLACED ALL THE ENPIRION VR, UPDATED CLOCKS WITH ALTERNATE PART NUMBER MADE I2C CONNECTION FROM FPGA TO FMC CONNECTOR. REMOVED PG SHORTS AT U79, UPDATED FP FOR CORE SUPPLY INDUCTORS,CORRECTED NETNAME AT P1V8_G3 SUPPLY RAIL
2P0	16-NOV-2022	ALL	REPLACED U94,U96 WITH ADP1762ACPZ-R7
3P0	06-03-2025	ALL	ADDED FT2232H TO SUPPORT ON-BOARD BLASTER III CHANGE RCOMP RESISTOR TO 0.1% RATED RESISTOR REPLACED LSF01080 BY TXB0304 UPDATED FPGA OPN TO 99D3CA AND REMOVE EMPTY ADDED IOPLL CLOCK TO ZL30733 & MODIFIED CLOCK TREE ACCORDINGLY UPDATED TA & PBA IPN UPDATED U123 CLOCK OUT TERMINATIONS UPDATED BLOCK DIAGRAMS UPDATED J47 & J28 LANES CONNECTION IN ORDER UPDATED CLOCK TREE TO PROVIDE IOPLL CLOCK INPUT TO ZL30733 REPLACED EOL COMPONENTS WITH ACTIVE COMPONENTS UPDATED DESIGN TO MATCH BOM CHANGES
4P0	04-09-2025	ALL	UPDATE TO ALTERA SCHEMATIC TEMPLATES

DEVKIT _OPN	DK-SI-AGI027FD
MMID	99CMK6
DEVICE_OPN	AGIB027R31B1E1VB
PBA	N76441-101
PB	N73826-002
PCB REV	D
PCB P/N#	100-0330690-D1

Mon May 18 23:37:53 2026

DEPARTMENT	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	SIZE	CODE	DOCUMENT NUMBER	REV
		C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET	1 OF 97

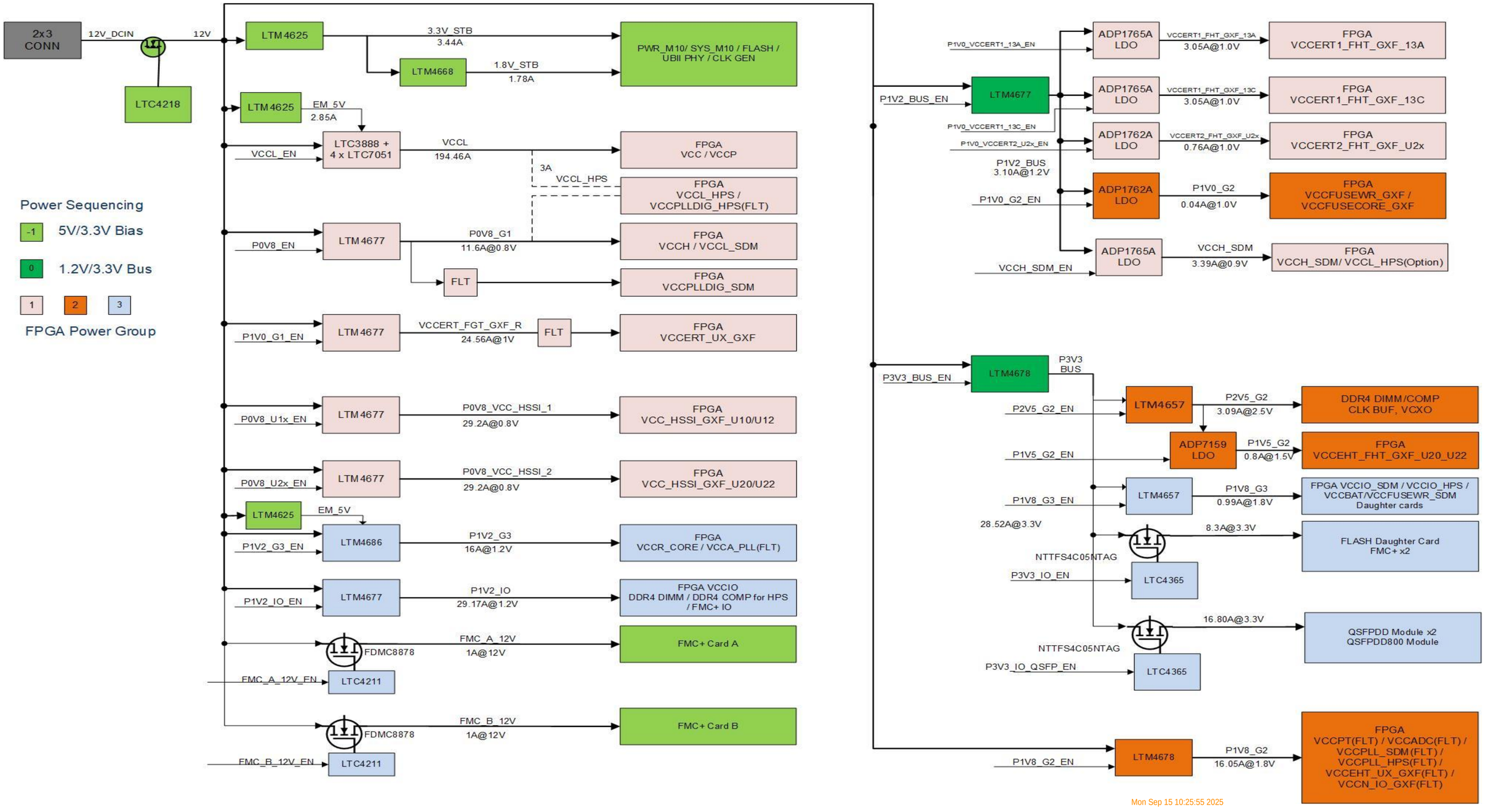
BLOCK DIAGRAM



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	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
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POWER TREE



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CLOCK TREE

NOTE:

One of QSFP_REC_CLK_x config can be:
REC_CLK = 39.0625M (25G/100G w/o FEC)
REC_CLK = 39.0625M(25G/100G with FEC)
REC_CLK = 26.0417M (10G)
REC_CLK can be programmed to any one of above frequencies.

XX_CLK → PTP Clocks
XX_CLK → SyncE Clocks
XX_CLK → Local Clocks

TCXO/XO 114.285M

AGIB027R31B
REFCLK8 / REFCLK9 (13C) → MUX → QSFP_REC_CLK_1/2
REFCLK8 / REFCLK8 (12C / 12A) → MUX → FMC_RCVD_CLK
IOPLL [2E]
FPGA_1PPS_OUT
QSFP_RCVD_CLK

ZL30733 (1588 PTP / SyncE)
REFIN0 ← QSFP_REC_CLK_1/2
REFIN2 ← FMC_RCVD_CLK
REFIN4p ← FPGA_1PPS_OUT
REFIN1 ← QSFP_RCVD_CLK
REFIN3p ← 1PPS_SMA_IN (SW)
REFIN3n ← 10MHz_SMA_IN (SW)
REFIN4n ← OCXO 20MHz OX-6011 (SiT5711 footprint option)
I2C/SPI ← AGILEX FPGA / AGIB027R31B
CTRL/STAT
NVM
OUT0 → ToD_CLK_100M (LVPECL-2.5V)
OUT1p → 1PPS_CLK (LVCMOS-2.5V) → LSF0108 → LVCMOS-1.2V
OUT1n → 1PPS_SMA_OUT
OUT2p → 10MHz_SMA_OUT
OUT2n → SPARE_CLK (from PTP DPLL) Jitter=150fs (max for 156.25M) → LSF0108 → 100M (SI5332-OUT7) LVCMOS-1.2V
OUT3 → ETH_REFCLK_156.25M (LVPECL-2.5V)
OUT4 → CPRI_REFCLK_153.6M (LVPECL-2.5V)
OUT5 → CPRI_REFCLK_184.32M (LVPECL-2.5V)
OUT6 → FMCA_REFCLK_122.88M
OUT7 → FMCA_REFCLK7_156.25M
OUT9 → FMCB_Si5391_REFIN
OUT8 → FMCA_Si5391_REFIN

AGIB027R31B
ToD (IOPLL) ← ToD_CLK_100M
QSFP F-Tile (13C) UX
REFCLK0 ← 100M (SI5332-OUT7)
REFCLK6
REFCLK4 [ETH] ← ETH_REFCLK_156.25M
REFCLK5 [CPRI] ← CPRI_REFCLK_153.6M
REFCLK7 [CPRI] ← CPRI_REFCLK_184.32M
REFCLK1 [SDI12G] ← VCXO 148.5M/148.35M (Si566)
REFCLK2 ← 135M
BARAK
REFCLK0 ← XTAL Si5394J OUT3 / OUT1 / OUT0
REFCLK1 ← 156.25M (13A BRK REFCLK0)
FMCA F-Tile (12C)
REFCLK0 ← Si53307 ← IOPLL_3A ← 100M FMCB RCLK1
REFCLK1 ← Si5332E OUT4 / FMCA_P0 ← FMCB RCLK7
REFCLK7 ← 156.25M ← IOPLL_3A ← FMCA_P1
REFCLK2 ← 156.25M ← IOPLL_3A
REFCLK3 ← 148.5M ← FMCA_P2
REFCLK4 ← 153.6M ← FMCA_P3
REFCLK5 ← 184.32M
REFCLK6 ← 368.63M
REFCLK8
REFCLK9 → Recovered_CLK
FMCB F-Tile (12A)
REFCLK0 ← 100M ← Si53307 MUX
REFCLK1 ← FMCB RCLK1
REFCLK7 ← 156.25M FMCB RCLK7
REFCLK2 ← 156.25M ← IOPLL_3C ← FMCB-P0
REFCLK3 ← 148.5M ← IOPLL_3C ← FMCB-P1
REFCLK4 ← 153.6M ← FMCB-P2
REFCLK5 ← 184.32M ← FMCB-P3
REFCLK6 ← 368.63M ← IOPLL ← Si53307 MUX
REFCLK8
REFCLK9 → Recovered_CLK
SDM OSC_CLK ← 125M ← OUT3 Si5391 FMC-A
Bank 3D ← 166.66M
EMIF-DDR4
Bank 2C ← 166.66M
IO PLL Bank 2D ← 100M
PCIe/MXP F-Tile (13A) UX
REFCLK0 ← 100M ← MUX ← FMCA F-Tile RCLK0
REFCLK3 ← 153.6M
REFCLK2 ← 184.32M
REFCLK4 ← 156.25M
BARAK
REFCLK0 ← OUT0 Si5394J
REFCLK1

Si5332E
OUT7 ← 100M ← 13C REFCLK0
OUT0 → 166.66M
OUT1 → 166.66M
OUT6 → 100M
OUT4 → 100M
OUT5 → 100M
OUT2
OUT3 → 100M
XTAL (Built-in)
NVM
I2C ↔ MAX10
CTRL/STAT
CLKIN2 ← EP_REFIN ← SPDT MUX ← REFCLK ← MCIO
RP_REFOUT
XTAL 100M

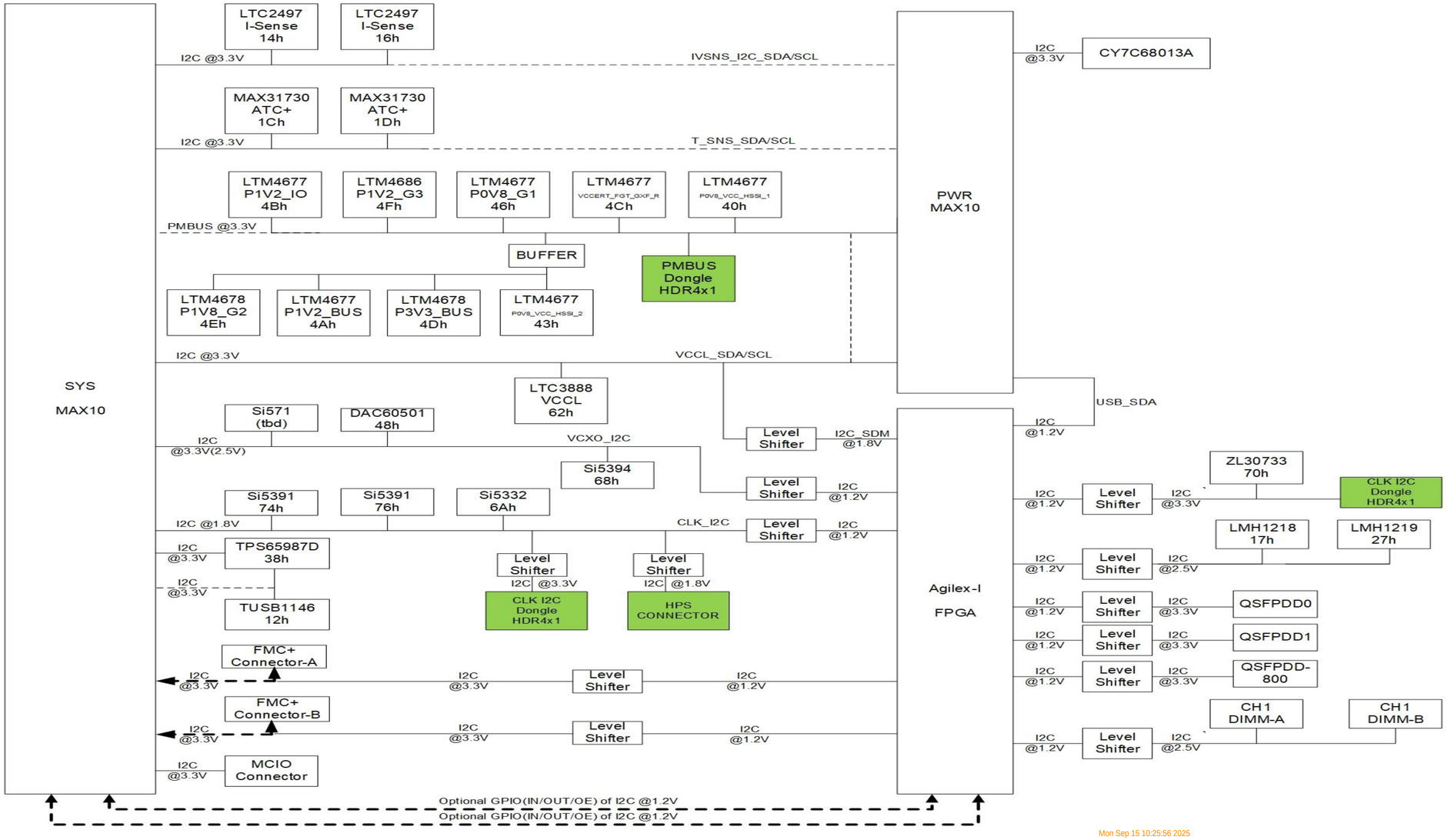
FMCA [FMC-A] Si5391
XA/XB ← XTAL 48.00M
IN0 ← 156.25M
OUT2 → 156.25M → IOPLL_REFCLK 13C
OUT3 → 125M → SDM_125M
OUT1 → 156.25M
OUT0 → 156.25M
OUT6 → 148.5M
OUT9 → 153.6M
OUT5 → 184.32M
OUT7 → 368.63M
OUT0A → 156.25M → IN1 FMCB Si5391
I2C ↔ MAX10
CTRL/STAT
NVM

FMCB [FMC-B] Si5391
XA/XB ← XTAL 48.00M
IN0 ← 156.25M ← OUT9 ZL3079x
IN1 ← 156.25M ← OUT0A FMCA Si5391
I2C ↔ MAX10
CTRL/STAT
NVM
OUT9A ← 153.6M
OUT4 ← 184.32M
OUT1 ← 156.25M
OUT2 ← 156.25M ← IOPLL_REFCLK 13A
OUT0 → 156.25M
OUT6 → 148.5M
OUT9 → 153.6M
OUT5 → 184.32M
OUT7 → 368.63M

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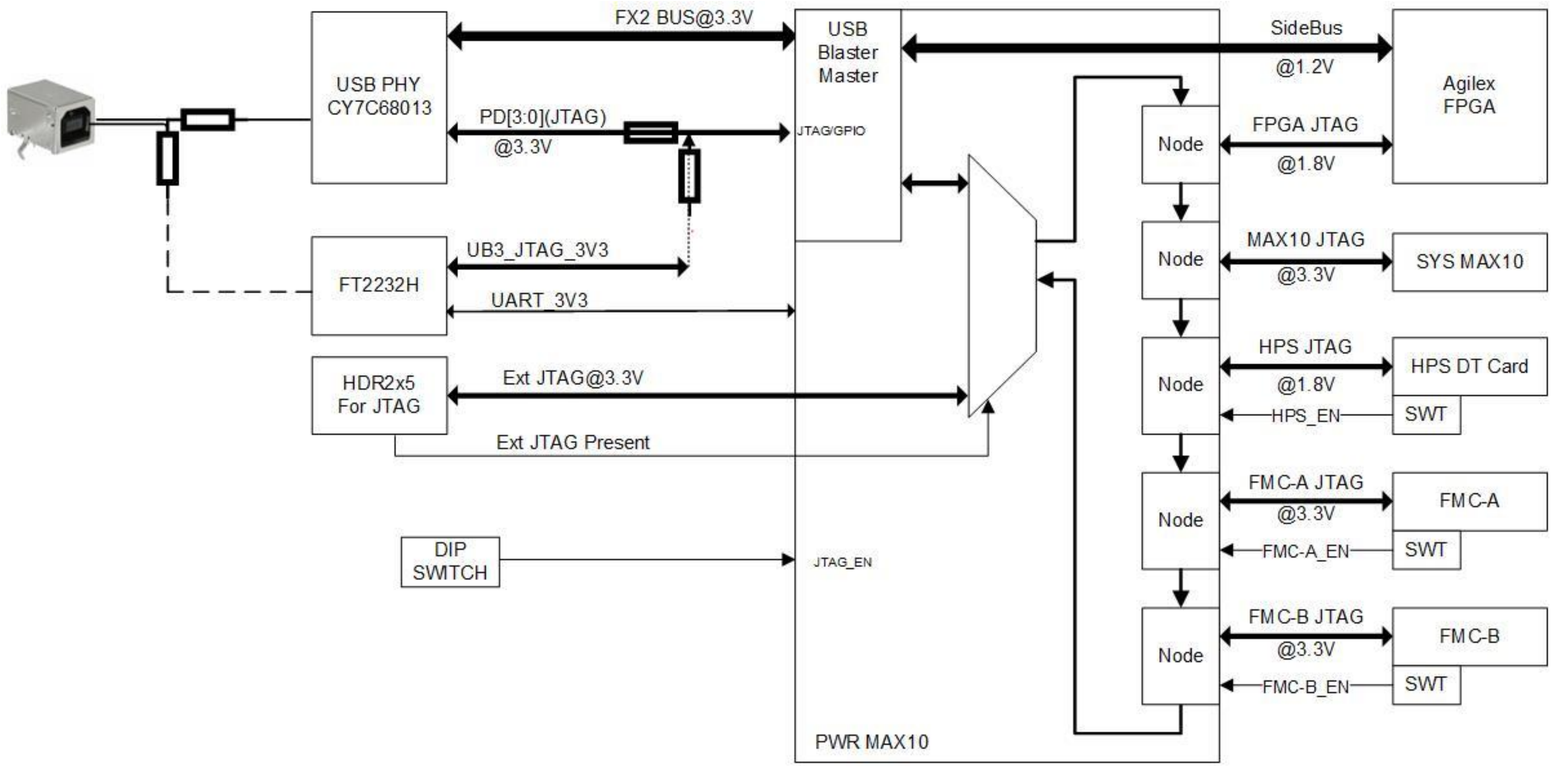
I2C TREE



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	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
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JTAG TOPOLOGY



Note:
1. HPS JTAG and FMC JTAG can be added into or removed from the JTAG chain by the setting of DIP switches
2. The JTAG of Agilex FPGA can't be added to the chain before fully powered up

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SDM IO MAPPING

CONFIGURATION MODE		MSEL[2:0]
PASSIVE	AVSTX32	000
	AVSTX16	101
	AVSTX8	110
	JTAG	111
	CVP	001
ACTIVE	AS - FAST	001
	AS - NORMAL	011

SDM PINS	MSEL FUNCTIONS	CONFIGURATION SOURCE FUNCTION			
		AVSTX8	AS	AVSTX16	AVSTX32
SDM_IO0		PWR_SCL	PWR_SCL	PWR_SCL	PWR_SCL
SDM_IO1		DATA2	DATA1		
SDM_IO2		DATA0	CLK		
SDM_IO3		DATA3	DATA2		
SDM_IO4		DATA1	DATA0		
SDM_IO5	MSEL0	HPS_CRSTN	NCSO0		
SDM_IO6		DATA4	DATA3		
SDM_IO7	MSEL1		NCSO2		
SDM_IO8		READY	NCSO3		
SDM_IO9	MSEL2		NCSO1		
SDM_IO10		DATA7			
SDM_IO11		VALID	HPS_CRSTN	HPS_CRSTN	HPS_CRSTN
SDM_IO12		PWR_SDA		PWR_SDA	PWR_SDA
SDM_IO13		DATA5			
SDM_IO14		CLK			
SDM_IO15		DATA6	RESETN		
SDM_IO16		CFG_DONE	CFG_DONE	CFG_DONE	CFG_DONE

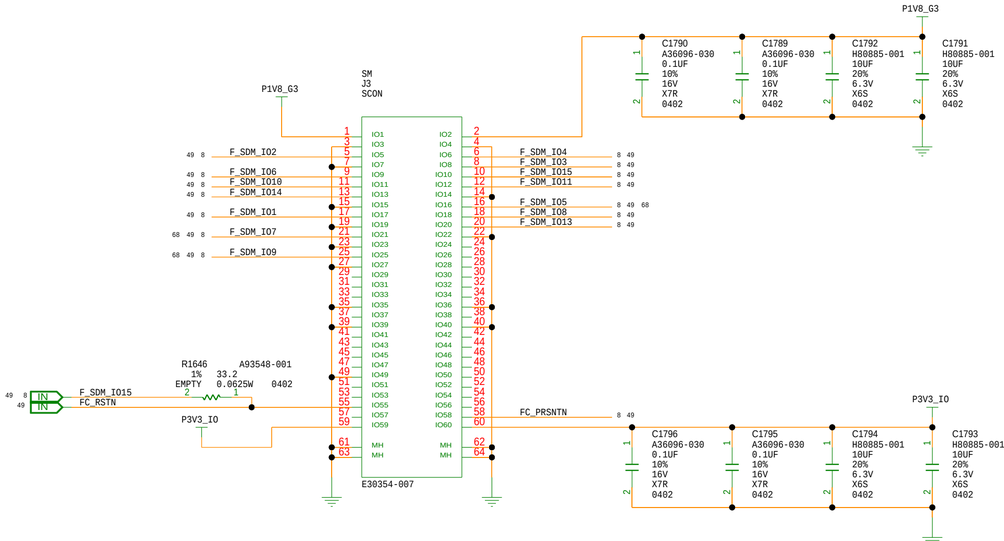
SYSTEM MAX IS THE SOURCE OF HPS_CRSTN FUNCTION

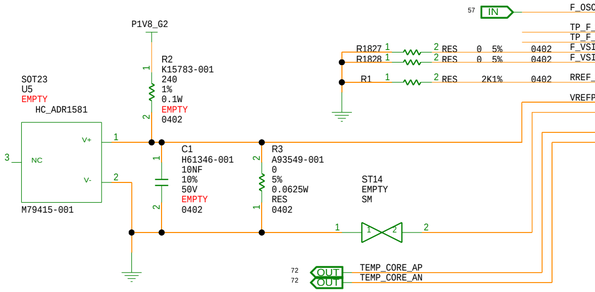
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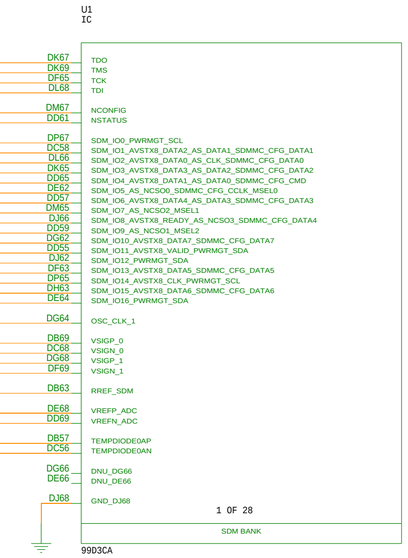
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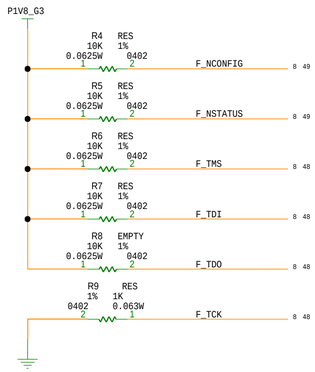
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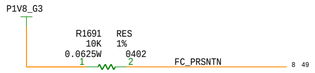
FPGA SDM











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		C+	34649	150-0330690-D1	D1
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CAD NOTE:
BIT-SWAPPINGS ALLOWED WITHIN A BYTE.
BYTE-SWAPPING IS ALLOWED WITH PRIOR APPROVAL
FROM DESIGN ENGINEER

FPGA BANK 2D

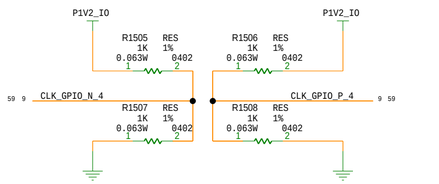
U1
IC

33	32	BI	DDR4_DIMM_2_DQ52	DA20	IO_DIFF_RX_2D1N_DQ0	IO_CDR_DIFF_RX_2D13N_DQ4	CL24	QSFPDD0_MODSEL_L	OUT 38
33	32	BI	DDR4_DIMM_2_DQ55	DB19	IO_DIFF_RX_2D1P_DQ0	IO_CDR_DIFF_RX_2D13P_DQ4	CM23	QSFPDD0_RESET_L	OUT 38
33	32	BI	DDR4_DIMM_2_DQ54	DE20	IO_DIFF_TX_2D1N_DQ0	IO_DIFF_TX_2D13N_DQ4	CR24	QSFPDD0_MODPRS_L	IN 38
33	32	BI	DDR4_DIMM_2_DQ53	DD19	IO_DIFF_TX_2D1P_DQ0	IO_DIFF_TX_2D13P_DQ4	CP23	QSFPDD0_LPMODE	OUT 38
33	32	BI	DDR4_DIMM_2_DQSN15	DA22	IO_DIFF_RX_2D2N_DQ0	IO_DIFF_TX_2D14N_DQ4	CL26	QSFPDD0_INT_L	IN 38
33	32	BI	DDR4_DIMM_2_DQSP15	DB21	IO_DIFF_RX_2D2P_DQ0	IO_DIFF_TX_2D14P_DQ4	CM25	QSFPDD0_EPPS	OUT 38
33	32	BI	DDR4_DIMM_2_DQSN6	DE22	IO_DIFF_TX_2D2N_DQSN0	IO_DIFF_TX_2D14N_DQSN4	CR26	QSFPDD800_MODSEL_L	OUT 40
33	32	BI	DDR4_DIMM_2_DQSP6	DD21	IO_DIFF_TX_2D2P_DQS0	IO_DIFF_TX_2D14P_DQS4	CP25	QSFPDD800_RESET_L	OUT 40
33	32	BI	DDR4_DIMM_2_DQ49	DA24	IO_CDR_DIFF_RX_2D3N_DQ0	IO_CDR_DIFF_RX_2D15N_DQ4	CL28	QSFPDD800_MODPRS_L	IN 40
33	32	BI	DDR4_DIMM_2_DQ51	DB23	IO_CDR_DIFF_RX_2D3P_DQ0	IO_CDR_DIFF_RX_2D15P_DQ4	CM27	QSFPDD800_LPMODE	OUT 40
33	32	BI	DDR4_DIMM_2_DQ48	DE24	IO_DIFF_TX_2D3N_DQ0	IO_DIFF_TX_2D15N_DQ4	CR28	QSFPDD800_INT_L	IN 40
33	32	BI	DDR4_DIMM_2_DQ50	DD23	IO_DIFF_TX_2D3P_DQ0	IO_DIFF_TX_2D15P_DQ4	CP27	QSFPDD800_EPPS	OUT 38 40
33	32	BI	DDR4_DIMM_2_DQ60	DG20	IO_DIFF_RX_2D4N_DQ1	IO_CDR_DIFF_RX_2D16N_DQ5	CU24	I2C_QSFPDD800_SCL	OUT 40
33	32	BI	DDR4_DIMM_2_DQ61	DF19	IO_DIFF_RX_2D4P_DQ1	IO_CDR_DIFF_RX_2D16P_DQ5	CT23	I2C_QSFPDD800_SDA	BI 40
33	32	BI	DDR4_DIMM_2_DQ62	DJ20	IO_DIFF_TX_2D4N_DQ1	IO_DIFF_TX_2D16N_DQ5	CW24		
33	32	BI	DDR4_DIMM_2_DQ63	DK19	IO_DIFF_TX_2D4P_DQ1	IO_DIFF_TX_2D16P_DQ5	CY23	DDR4_DIMM_2_SCL_1V2	BI 34
33	32	BI	DDR4_DIMM_2_DQSN16	DF21	IO_DIFF_RX_2D5N_DQ1	IO_DIFF_RX_2D17N_DQ5	CU26	DDR4_DIMM_2_SDA_1V2	BI 34
33	32	BI	DDR4_DIMM_2_DQSP16	DG22	IO_DIFF_RX_2D5P_DQ1	IO_DIFF_RX_2D17P_DQ5	CT25		
33	32	BI	DDR4_DIMM_2_DQSN7	DJ22	IO_PLL_2D_T_CLKOUT1N_DIFF_TX_2D5N_DQSN1	IO_PLL_2D_B_CLKOUT1N_DIFF_TX_2D17N_DQSN5	CW26		
33	32	BI	DDR4_DIMM_2_DQSP7	DK21	IO_PLL_2D_T_CLKOUT1P_PLL_2D_T_CLKOUT1_PLL_2D_T_FB1_DIFF_TX_2D5P_DQS1	IO_PLL_2D_B_CLKOUT1P_PLL_2D_B_CLKOUT1_PLL_2D_B_FB1_DIFF_TX_2D17P_DQS5	CY25		
33	32	BI	DDR4_DIMM_2_DQ59	DF23	IO_CDR_DIFF_RX_2D6N_DQ1	IO_CDR_DIFF_RX_2D18N_DQ5	CU28	DDR4_DIMM_2_EVENTn_1V2	IN 34
33	32	BI	DDR4_DIMM_2_DQ58	DG24	IO_RZQ_T_2D_CDR_DIFF_RX_2D6P_DQ1	IO_RZQ_B_2D_CDR_DIFF_RX_2D18P_DQ5	CT27	DDR4_DIMM_2_SAVEn_1V2	OUT 34
33	32	BI	DDR4_DIMM_2_DQ57	DK23	IO_CLK_T_2D_1N_DIFF_TX_2D6N_DQ1	IO_CLK_B_2D_1N_DIFF_TX_2D18N_DQ5	CW28		
33	32	BI	DDR4_DIMM_2_DQ56	DJ24	IO_CLK_T_2D_1P_DIFF_TX_2D6P_DQ1	IO_CLK_B_2D_1P_DIFF_TX_2D18P_DQ5	CY27		
33	32	BI	DDR4_DIMM_2_DQ36	DL18	IO_CLK_T_2D_0N_DIFF_RX_2D7N_DQ2	IO_CLK_B_2D_0N_CDR_DIFF_RX_2D19N_DQ6	CL30	CLK_GPIO_N_4	IN 9 59
33	32	BI	DDR4_DIMM_2_DQ39	DM17	IO_CLK_T_2D_0P_DIFF_RX_2D7P_DQ2	IO_CLK_B_2D_0P_CDR_DIFF_RX_2D19P_DQ6	CM29	CLK_GPIO_P_4	IN 9 59
33	32	BI	DDR4_DIMM_2_DQ38	DR18	IO_DIFF_TX_2D7N_DQ2	IO_DIFF_TX_2D19N_DQ6	CR30		
33	32	BI	DDR4_DIMM_2_DQ37	DP17	IO_DIFF_TX_2D7P_DQ2	IO_DIFF_TX_2D19P_DQ6	CP29	QSFPDD1_MODSEL_L	OUT 39
33	32	BI	DDR4_DIMM_2_DQSN13	DL20	IO_PLL_2D_T_CLKOUT0N_DIFF_RX_2D8N_DQ2	IO_PLL_2D_B_CLKOUT0N_DIFF_RX_2D20N_DQ6	CL32	QSFPDD1_RESET_L	OUT 39
33	32	BI	DDR4_DIMM_2_DQSP13	DM19	IO_PLL_2D_T_CLKOUT0P_PLL_2D_T_CLKOUT0_PLL_2D_T_FB0_DIFF_RX_2D8P_DQ2	IO_PLL_2D_B_CLKOUT0P_PLL_2D_B_CLKOUT0_PLL_2D_B_FB0_DIFF_RX_2D20P_DQ6	CM31	QSFPDD1_MODPRS_L	IN 39
33	32	BI	DDR4_DIMM_2_DQSN4	DP19	IO_DIFF_TX_2D8N_DQSN2	IO_DIFF_TX_2D20N_DQSN6	CR32	QSFPDD1_LPMODE	OUT 39
33	32	BI	DDR4_DIMM_2_DQSP4	DR20	IO_DIFF_TX_2D8P_DQS2	IO_DIFF_TX_2D20P_DQS6	CP31	QSFPDD1_INT_L	IN 39
33	32	BI	DDR4_DIMM_2_DQ33	DL22	IO_CDR_DIFF_RX_2D9N_DQ2	IO_CDR_DIFF_RX_2D21N_DQ6	CL34	QSFPDD1_EPPS	OUT 39
33	32	BI	DDR4_DIMM_2_DQ35	DM21	IO_CDR_DIFF_RX_2D9P_DQ2	IO_CDR_DIFF_RX_2D21P_DQ6	CM33	I2C_QSFPDD1_SCL	OUT 39
33	32	BI	DDR4_DIMM_2_DQ34	DP21	IO_DIFF_TX_2D9N_DQ2	IO_DIFF_TX_2D21N_DQ6	CR34	I2C_QSFPDD1_SDA	BI 39
33	32	BI	DDR4_DIMM_2_DQ32	DR22	IO_DIFF_TX_2D9P_DQ2	IO_DIFF_TX_2D21P_DQ6	CP33		
33	32	BI	DDR4_DIMM_2_DQ45	DM23	IO_DIFF_RX_2D10N_DQ3	IO_CDR_DIFF_RX_2D22N_DQ7	DA26	DDR4_DIMM_2_DQ68	BI 32 33
33	32	BI	DDR4_DIMM_2_DQ47	DL24	IO_DIFF_RX_2D10P_DQ3	IO_CDR_DIFF_RX_2D22P_DQ7	DB25	DDR4_DIMM_2_DQ69	BI 32 33
33	32	BI	DDR4_DIMM_2_DQ44	DR24	IO_DIFF_TX_2D10N_DQ3	IO_DIFF_TX_2D22N_DQ7	DD25	DDR4_DIMM_2_DQ70	BI 32 33
33	32	BI	DDR4_DIMM_2_DQ46	DP23	IO_DIFF_TX_2D10P_DQ3	IO_DIFF_TX_2D22P_DQ7	DE26	DDR4_DIMM_2_DQ71	BI 32 33
33	32	BI	DDR4_DIMM_2_DQSN14	DL26	IO_DIFF_RX_2D11N_DQ3	IO_DIFF_RX_2D23N_DQ7	DA28	DDR4_DIMM_2_DQSN17	BI 32 33
33	32	BI	DDR4_DIMM_2_DQSP14	DM25	IO_DIFF_RX_2D11P_DQ3	IO_DIFF_RX_2D23P_DQ7	DB27	DDR4_DIMM_2_DQSP17	BI 32 33
33	32	BI	DDR4_DIMM_2_DQSN5	DR26	IO_DIFF_TX_2D11N_DQSN3	IO_DIFF_TX_2D23N_DQSN7	DE28	DDR4_DIMM_2_DQSN8	BI 32 33
33	32	BI	DDR4_DIMM_2_DQSP5	DP25	IO_DIFF_TX_2D11P_DQS3	IO_DIFF_TX_2D23P_DQS7	DD27	DDR4_DIMM_2_DQSP8	BI 32 33
33	32	BI	DDR4_DIMM_2_DQ41	DM27	IO_CDR_DIFF_RX_2D12N_DQ3	IO_CDR_DIFF_RX_2D24N_DQ7	DA30	DDR4_DIMM_2_DQ65	BI 32 33
33	32	BI	DDR4_DIMM_2_DQ42	DL28	IO_CDR_DIFF_RX_2D12P_DQ3	IO_CDR_DIFF_RX_2D24P_DQ7	DB29	DDR4_DIMM_2_DQ64	BI 32 33
33	32	BI	DDR4_DIMM_2_DQ43	DR28	IO_DIFF_TX_2D12N_DQ3	IO_DIFF_TX_2D24N_DQ7	DE30	DDR4_DIMM_2_DQ66	BI 32 33
33	32	BI	DDR4_DIMM_2_DQ40	DP27	IO_DIFF_TX_2D12P_DQ3	IO_DIFF_TX_2D24P_DQ7	DD29	DDR4_DIMM_2_DQ67	BI 32 33

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IO BANK 2D

99D3CA



CAD NOTE:
PLACE THE TERMINATION RESISTORS CLOSE TO THE FPGA

Mon Sep 15 10:27:08 2025

DEPARTMENT		SIZE	CODE	DOCUMENT NUMBER	REV
	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 9 OF 97	

ALTERA CONFIDENTIAL

FPGA BANK 2C

U1
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FPGA BANK 2F

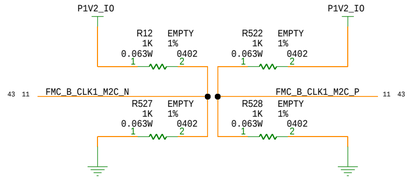
U1
IC

43	BI	FMC_B_LA_N12	DG38	IO_DIFF_RX_2F1N_DQ16	IO_CDR_DIFF_RX_2F13N_DQ20	DA38	FMC_B_LA_N6
43	BI	FMC_B_LA_P12	DF37	IO_DIFF_RX_2F1P_DQ16	IO_CDR_DIFF_RX_2F13P_DQ20	DB37	FMC_B_LA_P6
43	BI	FMC_B_LA_N7	DJ38	IO_DIFF_TX_2F1N_DQ16	IO_DIFF_TX_2F13N_DQ20	DE38	FMC_B_LA_N2
43	BI	FMC_B_LA_P7	DK37	IO_DIFF_TX_2F1P_DQ16	IO_DIFF_TX_2F13P_DQ20	DD37	FMC_B_LA_P2
43	BI	FMC_B_LA_N11	DG40	IO_DIFF_RX_2F2N_DQ16	IO_DIFF_RX_2F14N_DQ20	DA40	FMC_B_LA_N15
43	BI	FMC_B_LA_P11	DF39	IO_DIFF_RX_2F2P_DQ16	IO_DIFF_RX_2F14P_DQ20	DB39	FMC_B_LA_P15
43	BI	FMC_B_LA_N8	DJ40	IO_DIFF_TX_2F2N_DQSN16	IO_DIFF_TX_2F14N_DQSN20	DE40	FMC_B_LA_N4
43	BI	FMC_B_LA_P8	DK39	IO_DIFF_TX_2F2P_DQS16	IO_DIFF_TX_2F14P_DQS20	DD39	FMC_B_LA_P4
43	BI	FMC_B_LA_N5	DG42	IO_CDR_DIFF_RX_2F3N_DQ16	IO_CDR_DIFF_RX_2F15N_DQ20	DA42	FMC_B_LA_N20
43	BI	FMC_B_LA_P5	DF41	IO_CDR_DIFF_RX_2F3P_DQ16	IO_CDR_DIFF_RX_2F15P_DQ20	DB41	FMC_B_LA_P20
43	BI	FMC_B_LA_N22	DJ42	IO_DIFF_TX_2F3N_DQ16	IO_DIFF_TX_2F15N_DQ20	DE42	FMC_B_LA_N21
43	BI	FMC_B_LA_P22	DK41	IO_DIFF_TX_2F3P_DQ16	IO_DIFF_TX_2F15P_DQ20	DD41	FMC_B_LA_P21
43	BI	FMC_B_LA_N23	DL42	IO_DIFF_RX_2F4N_DQ17	IO_CDR_DIFF_RX_2F16N_DQ21	DA44	FMC_B_LA_N32
43	BI	FMC_B_LA_P23	DM41	IO_DIFF_RX_2F4P_DQ17	IO_CDR_DIFF_RX_2F16P_DQ21	DB43	FMC_B_LA_P32
43	BI	FMC_B_LA_N13	DR42	IO_DIFF_TX_2F4N_DQ17	IO_DIFF_TX_2F16N_DQ21	DE44	FMC_B_LA_N26
43	BI	FMC_B_LA_P13	DP41	IO_DIFF_TX_2F4P_DQ17	IO_DIFF_TX_2F16P_DQ21	DD43	FMC_B_LA_P26
43	BI	FMC_B_LA_N10	DL44	IO_DIFF_RX_2F5N_DQ17	IO_DIFF_RX_2F17N_DQ21	DA46	FMC_B_LA_N24
43	BI	FMC_B_LA_P10	DM43	IO_DIFF_RX_2F5P_DQ17	IO_DIFF_RX_2F17P_DQ21	DB45	FMC_B_LA_P24
43	BI	FMC_B_LA_N9	DR44	IO_PLL_2F_T_CLKOUT1N_DIFF_TX_2F5N_DQSN17	IO_PLL_2F_B_CLKOUT1N_DIFF_TX_2F17N_DQSN21	DE46	FMC_B_LA_N0
43	BI	FMC_B_LA_P9	DP43	IO_PLL_2F_T_CLKOUT1P_PLL_2F_T_CLKOUT1_PLL_2F_T_FB1_DIFF_TX_2F5P_DQS17	IO_PLL_2F_B_CLKOUT1P_PLL_2F_B_CLKOUT1_PLL_2F_B_FB1_DIFF_TX_2F17P_DQS21	DD45	FMC_B_LA_P0
43	BI	FMC_B_LA_N14	DL46	IO_CDR_DIFF_RX_2F6N_DQ17	IO_CDR_DIFF_RX_2F18N_DQ21	DA48	FMC_B_LA_N3
43	BI	FMC_B_LA_P14	DM45	IO_RZQ_T_2F_CDR_DIFF_RX_2F6P_DQ17	IO_RZQ_B_2F_CDR_DIFF_RX_2F18P_DQ21	DB47	FMC_B_LA_P3
43 11	IN	FMC_B_CLK2_M2C_N	DR46	IO_CLK_T_2F_1N_DIFF_TX_2F6N_DQ17	IO_CLK_B_2F_1N_DIFF_TX_2F18N_DQ21	DE48	FMC_B_CLK1_M2C_N
43 11	IN	FMC_B_CLK2_M2C_P	DP45	IO_CLK_T_2F_1P_DIFF_TX_2F6P_DQ17	IO_CLK_B_2F_1P_DIFF_TX_2F18P_DQ21	DD47	FMC_B_CLK1_M2C_P
43 11	IN	FMC_B_CLK0_M2C_N	DG44	IO_CLK_T_2F_0N_DIFF_RX_2F7N_DQ18	IO_CLK_B_2F_0N_CDR_DIFF_RX_2F19N_DQ22	CT41	FMC_B_REFCLK_M2C_N
43 11	IN	FMC_B_CLK0_M2C_P	DF43	IO_CLK_T_2F_0P_DIFF_RX_2F7P_DQ18	IO_CLK_B_2F_0P_CDR_DIFF_RX_2F19P_DQ22	CU42	FMC_B_REFCLK_M2C_P
43 11	IN	FMC_B_SYNC_M2C_N	DK43	IO_DIFF_TX_2F7N_DQ18	IO_DIFF_TX_2F19N_DQ22	CY41	FMC_B_LA_N33
43 11	IN	FMC_B_SYNC_M2C_P	DJ44	IO_DIFF_TX_2F7P_DQ18	IO_DIFF_TX_2F19P_DQ22	CW42	FMC_B_LA_P33
43	OUT	FMC_B_REFCLK_C2M_N	DG46	IO_PLL_2F_T_CLKOUT0N_DIFF_RX_2F8N_DQ18	IO_PLL_2F_B_CLKOUT0N_DIFF_RX_2F20N_DQ22	CU44	FMC_B_LA_N19
43	OUT	FMC_B_REFCLK_C2M_P	DF45	IO_PLL_2F_T_CLKOUT0P_PLL_2F_T_CLKOUT0_PLL_2F_T_FB0_DIFF_RX_2F8P_DQ18	IO_PLL_2F_B_CLKOUT0P_PLL_2F_B_CLKOUT0_PLL_2F_B_FB0_DIFF_RX_2F20P_DQ22	CT43	FMC_B_LA_P19
43	OUT	FMC_B_SYNC_C2M_N	DJ46	IO_DIFF_TX_2F8N_DQSN18	IO_DIFF_TX_2F20N_DQSN22	CW44	FMC_B_LA_N28
43	OUT	FMC_B_SYNC_C2M_P	DK45	IO_DIFF_TX_2F8P_DQS18	IO_DIFF_TX_2F20P_DQS22	CY43	FMC_B_LA_P28
43	OUT	FMC_B_LA_N25	DG48	IO_CDR_DIFF_RX_2F9N_DQ18	IO_CDR_DIFF_RX_2F21N_DQ22	CU46	FMC_B_LA_N1
43	BI	FMC_B_LA_P25	DF47	IO_CDR_DIFF_RX_2F9P_DQ18	IO_CDR_DIFF_RX_2F21P_DQ22	CT45	FMC_B_LA_P1
	BI		DJ48	IO_DIFF_TX_2F9N_DQ18	IO_DIFF_TX_2F12N_DQ22	CY45	FMC_B_LA_N29
	BI		DK47	IO_DIFF_TX_2F9P_DQ18	IO_DIFF_TX_2F12P_DQ22	CW46	FMC_B_LA_P29
	BI		DL48	IO_DIFF_RX_2F10N_DQ19	IO_CDR_DIFF_RX_2F22N_DQ23	CM41	FMC_B_LA_N18
	BI		DM47	IO_DIFF_RX_2F10P_DQ19	IO_CDR_DIFF_RX_2F22P_DQ23	CL42	FMC_B_LA_P18
	BI		DR48	IO_DIFF_TX_2F10N_DQ19	IO_DIFF_TX_2F22N_DQ23	CR42	FMC_B_LA_N17
	BI		DP47	IO_DIFF_TX_2F10P_DQ19	IO_DIFF_TX_2F22P_DQ23	CP41	FMC_B_LA_P17
	BI		DL50	IO_DIFF_RX_2F11N_DQ19	IO_DIFF_RX_2F23N_DQ23	CL44	FMC_B_LA_N16
	BI		DM49	IO_DIFF_RX_2F11P_DQ19	IO_DIFF_RX_2F23P_DQ23	CM43	FMC_B_LA_P16
	BI		DR50	IO_DIFF_TX_2F11N_DQSN19	IO_DIFF_TX_2F23N_DQSN23	CR44	FMC_B_LA_N27
	BI		DP49	IO_DIFF_TX_2F11P_DQS19	IO_DIFF_TX_2F23P_DQS23	CP43	FMC_B_LA_P27
	BI		DL52	IO_CDR_DIFF_RX_2F12N_DQ19	IO_CDR_DIFF_RX_2F24N_DQ23	CM45	FMC_B_LA_N30
	BI		DM51	IO_CDR_DIFF_RX_2F12P_DQ19	IO_CDR_DIFF_RX_2F24P_DQ23	CL46	FMC_B_LA_P30
	BI		DR52	IO_DIFF_TX_2F12N_DQ19	IO_DIFF_TX_2F24N_DQ23	CP45	FMC_B_LA_N31
	BI		DP51	IO_DIFF_TX_2F12P_DQ19	IO_DIFF_TX_2F24P_DQ23	CR46	FMC_B_LA_P31

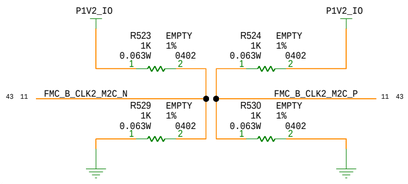
5 OF 28

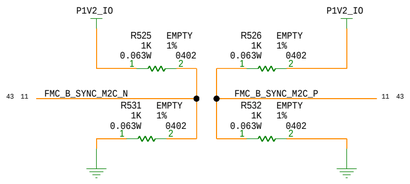
IO BANK 2F

99D3CA



CAD NOTE:
PLACE THE TERMINATION RESISTORS CLOSE TO THE FPGA





CAD NOTE:
PLACE THE TERMINATION RESISTORS CLOSE TO THE FPGA

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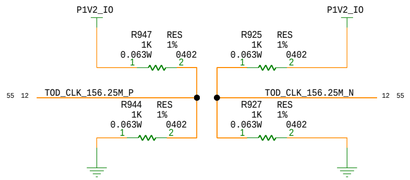
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	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 11 OF 97	

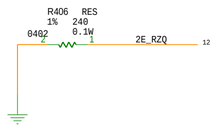
ALTERA CONFIDENTIAL

FPGA BANK 2E

U1
IC







CAD NOTE:
PLACE THE TERMINATION RESISTORS CLOSE TO THE FPGA

Mon Sep 15 10:27:08 2025

DEPARTMENT		SIZE	CODE	DOCUMENT NUMBER	REV
	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 12 OF 97	

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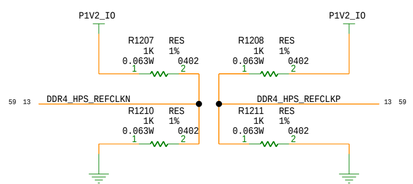
FPGA BANK 3D

CAD NOTE:
BIT-SWAPPINGS ALLOWED WITHIN A BYTE.
BYTE-SWAPPING IS ALLOWED WITH PRIOR APPROVAL
FROM DESIGN ENGINEER (EXCEPT ECC)

U1
IC



99D3CA



CAD NOTE:
PLACE THE TERMINATION RESISTORS CLOSE TO THE FPGA
PLACE TE PAD ON THE TRACE

Mon Sep 15 10:27:08 2025

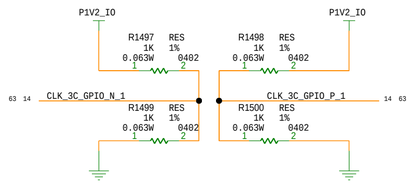
DEPARTMENT		SIZE	CODE	DOCUMENT NUMBER	REV
	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
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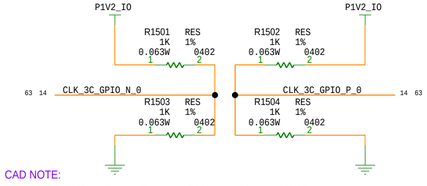
FPGA BANK 3C

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CAD NOTE:
PLACE THE TERMINATION RESISTORS CLOSE TO THE FPGA



Mon Sep 15 10:27:09 2025

DEPARTMENT		SIZE	CODE	DOCUMENT NUMBER	REV
	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
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FPGA BANK 3B

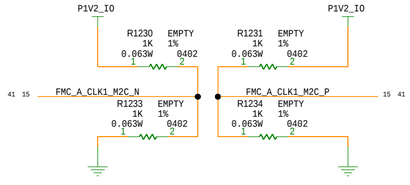
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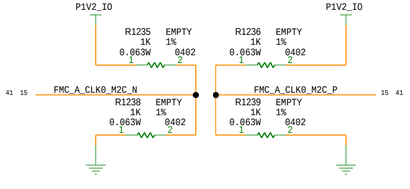
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41	BI	FMC_A_LA_P13	E60	IO_DIFF_RX_3B1P_DQ80	IO_CDR_DIFF_RX_3B13P_DQ84	AE46	FMC_A_LA_P8
41	BI	FMC_A_LA_N20	B61	IO_DIFF_TX_3B1N_DQ80	IO_DIFF_TX_3B13N_DQ84	AB47	FMC_A_LA_N2
41	BI	FMC_A_LA_P20	A60	IO_DIFF_TX_3B1P_DQ80	IO_DIFF_TX_3B13P_DQ84	AA46	FMC_A_LA_P2
41	BI	FMC_A_LA_N23	D59	IO_DIFF_RX_3B2N_DQ80	IO_DIFF_RX_3B14N_DQ84	AD45	FMC_A_LA_N12
41	BI	FMC_A_LA_P23	E58	IO_DIFF_RX_3B2P_DQ80	IO_DIFF_RX_3B14P_DQ84	AE44	FMC_A_LA_P12
41	BI	FMC_A_LA_N22	B59	IO_DIFF_TX_3B2N_DQSN80	IO_DIFF_TX_3B14N_DQSN84	AA44	FMC_A_LA_N4
41	BI	FMC_A_LA_P22	A58	IO_DIFF_TX_3B2P_DQS80	IO_DIFF_TX_3B14P_DQS84	AB45	FMC_A_LA_P4
41	BI	FMC_A_LA_N26	D57	IO_CDR_DIFF_RX_3B3N_DQ80	IO_CDR_DIFF_RX_3B15N_DQ84	AD43	FMC_A_LA_N33
41	BI	FMC_A_LA_P26	E56	IO_CDR_DIFF_RX_3B3P_DQ80	IO_CDR_DIFF_RX_3B15P_DQ84	AE42	FMC_A_LA_P33
41	BI	FMC_A_LA_N25	B57	IO_DIFF_TX_3B3N_DQ80	IO_DIFF_TX_3B15N_DQ84	AB43	FMC_A_LA_N6
41	BI	FMC_A_LA_P25	A56	IO_DIFF_TX_3B3P_DQ80	IO_DIFF_TX_3B15P_DQ84	AA42	FMC_A_LA_P6
41	BI	FMC_A_LA_N14	K57	IO_DIFF_RX_3B4N_DQ81	IO_CDR_DIFF_RX_3B16N_DQ85	Y47	FMC_A_LA_N7
41	BI	FMC_A_LA_P14	J56	IO_DIFF_RX_3B4P_DQ81	IO_CDR_DIFF_RX_3B16P_DQ85	W46	FMC_A_LA_P7
41	BI	FMC_A_LA_N29	F57	IO_DIFF_TX_3B4N_DQ81	IO_DIFF_TX_3B16N_DQ85	T47	FMC_A_LA_N11
41	BI	FMC_A_LA_P29	G56	IO_DIFF_TX_3B4P_DQ81	IO_DIFF_TX_3B16P_DQ85	U46	FMC_A_LA_P11
41	BI	FMC_A_LA_N9	K55	IO_DIFF_RX_3B5N_DQ81	IO_DIFF_RX_3B17N_DQ85	Y45	FMC_A_LA_N15
41	BI	FMC_A_LA_P9	J54	IO_DIFF_RX_3B5P_DQ81	IO_DIFF_RX_3B17P_DQ85	W44	FMC_A_LA_P15
41	BI	FMC_A_LA_N31	F55	IO_PLL_3B_T_CLKOUT1N_DIFF_TX_3B5N_DQSN81	IO_PLL_3B_B_CLKOUT1N_DIFF_TX_3B17N_DQSN85	T45	FMC_A_LA_N1
41	BI	FMC_A_LA_P31	G54	IO_PLL_3B_T_CLKOUT1P_PLL_3B_T_CLKOUT1_PLL_3B_T_FB1_DIFF_TX_3B5P_DQS81	IO_PLL_3B_B_CLKOUT1P_PLL_3B_B_CLKOUT1_PLL_3B_B_FB1_DIFF_TX_3B17P_DQS85	U44	FMC_A_LA_P1
41	BI	FMC_A_LA_N10	K53	IO_CDR_DIFF_RX_3B6N_DQ81	IO_CDR_DIFF_RX_3B18N_DQ85	Y43	FMC_A_LA_N16
41	BI	FMC_A_LA_P10	J52	IO_RZQ_T_3B_CDR_DIFF_RX_3B6P_DQ81	IO_RZQ_B_3B_CDR_DIFF_RX_3B18P_DQ85	W42	FMC_A_LA_P16
41 15	IN	FMC_A_CLK2_M2C_N	F53	IO_CLK_T_3B_1N_DIFF_TX_3B6N_DQ81	IO_CLK_B_3B_1N_DIFF_TX_3B18N_DQ85	T43	FMC_A_CLK1_M2C_N
41 15	IN	FMC_A_CLK2_M2C_P	G52	IO_CLK_T_3B_1P_DIFF_TX_3B6P_DQ81	IO_CLK_B_3B_1P_DIFF_TX_3B18P_DQ85	U42	FMC_A_CLK1_M2C_P
41 15	IN	FMC_A_CLK0_M2C_N	D55	IO_CLK_T_3B_0N_DIFF_RX_3B7N_DQ82	IO_CLK_B_3B_0N_CDR_DIFF_RX_3B19N_DQ86	P51	FMC_A_REFCLK_M2C_N
41 15	IN	FMC_A_CLK0_M2C_P	E54	IO_CLK_T_3B_0P_DIFF_RX_3B7P_DQ82	IO_CLK_B_3B_0P_CDR_DIFF_RX_3B19P_DQ86	R50	FMC_A_REFCLK_M2C_P
41 15	IN	FMC_A_SYNC_M2C_N	B55	IO_DIFF_TX_3B7N_DQ82	IO_DIFF_TX_3B19N_DQ86	M51	FMC_A_LA_N5
41 15	IN	FMC_A_SYNC_M2C_P	A54	IO_DIFF_TX_3B7P_DQ82	IO_DIFF_TX_3B19P_DQ86	L50	FMC_A_LA_P5
41	OUT	FMC_A_REFCLK_C2M_N	D53	IO_PLL_3B_T_CLKOUTON_DIFF_RX_3B8N_DQ82	IO_PLL_3B_B_CLKOUTON_DIFF_RX_3B20N_DQ86	P49	FMC_A_LA_N3
41	OUT	FMC_A_REFCLK_C2M_P	E52	IO_PLL_3B_T_CLKOUT0P_PLL_3B_T_CLKOUT0_PLL_3B_T_FB0_DIFF_RX_3B8P_DQ82	IO_PLL_3B_B_CLKOUT0P_PLL_3B_B_CLKOUT0_PLL_3B_B_FB0_DIFF_RX_3B20P_DQ86	R48	FMC_A_LA_P3
41	OUT	FMC_A_SYNC_C2M_N	B53	IO_DIFF_TX_3B8N_DQSN82	IO_DIFF_TX_3B20N_DQSN86	M49	FMC_A_LA_N24
41	OUT	FMC_A_SYNC_C2M_P	A52	IO_DIFF_TX_3B8P_DQS82	IO_DIFF_TX_3B20P_DQS86	L48	FMC_A_LA_P24
			D51	IO_CDR_DIFF_RX_3B9N_DQ82	IO_CDR_DIFF_RX_3B21N_DQ86	P47	FMC_A_LA_N8
			E50	IO_CDR_DIFF_RX_3B9P_DQ82	IO_CDR_DIFF_RX_3B21P_DQ86	R46	FMC_A_LA_P8
41	BI	FMC_A_LA_N30	B51	IO_DIFF_TX_3B9N_DQ82	IO_DIFF_TX_3B21N_DQ86	M47	FMC_A_LA_N27
41	BI	FMC_A_LA_P30	A50	IO_DIFF_TX_3B9P_DQ82	IO_DIFF_TX_3B21P_DQ86	L46	FMC_A_LA_P27
					IO_CDR_DIFF_RX_3B22N_DQ87	K51	FMC_A_LA_N17
					IO_CDR_DIFF_RX_3B22P_DQ87	J50	FMC_A_LA_P17
					IO_DIFF_TX_3B22N_DQ87	F51	FMC_A_LA_N18
					IO_DIFF_TX_3B22P_DQ87	G50	FMC_A_LA_P18
					IO_DIFF_RX_3B23N_DQ87	K49	FMC_A_LA_N19
					IO_DIFF_RX_3B23P_DQ87	J48	FMC_A_LA_P19
					IO_DIFF_TX_3B23N_DQSN87	F49	FMC_A_LA_N28
					IO_DIFF_TX_3B23P_DQS87	G48	FMC_A_LA_P28
					IO_CDR_DIFF_RX_3B24N_DQ87	K47	FMC_A_LA_N21
					IO_CDR_DIFF_RX_3B24P_DQ87	J46	FMC_A_LA_P21
					IO_DIFF_TX_3B24N_DQ87	F47	FMC_A_LA_N32
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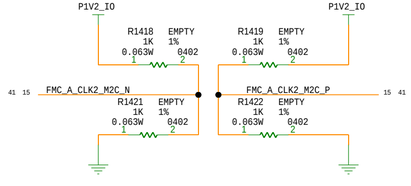
7 OF 28

IO BANK 3B

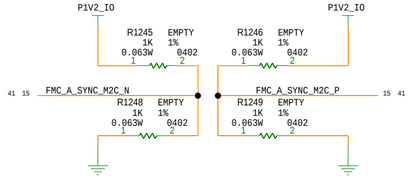
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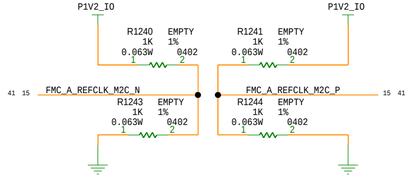






CAD NOTE:
PLACE THE TERMINATION RESISTORS CLOSE TO THE FPGA





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DEPARTMENT	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	SIZE: C+	CODE: 34649	DOCUMENT NUMBER: 150-0330690-D1
		SCALE: DO NOT SCALE DRAWING		SHEET: 15 OF 97

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FPGA BANK 3A

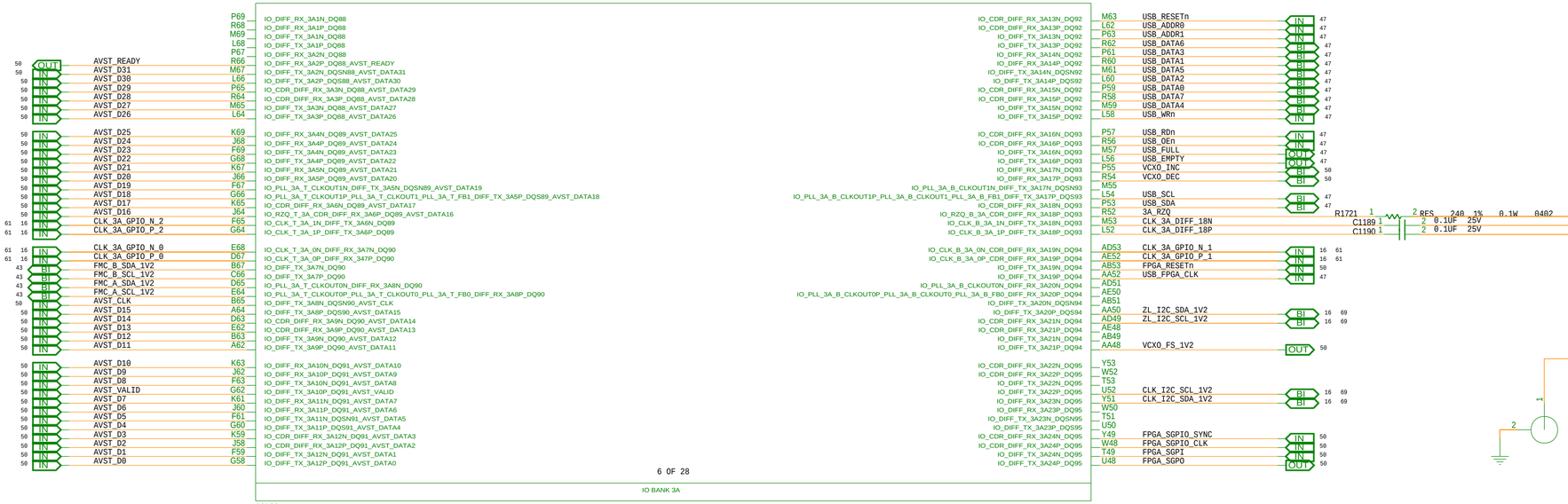
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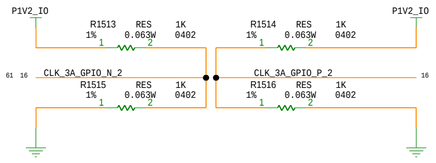
			P69	IO_DIFF_RX_3A1N_DQ88			
			R68	IO_DIFF_RX_3A1P_DQ88			
			M69	IO_DIFF_TX_3A1N_DQ88			
			L68	IO_DIFF_TX_3A1P_DQ88			
			P67	IO_DIFF_RX_3A2N_DQ88			
50	OUT	AVST_READY	R66	IO_DIFF_RX_3A2P_DQ88_AVST_READY			
50	IN	AVST_D31	M67	IO_DIFF_TX_3A2N_DQSN88_AVST_DATA31			
50	IN	AVST_D30	L66	IO_DIFF_TX_3A2P_DQS88_AVST_DATA30			
50	IN	AVST_D29	P65	IO_CDR_DIFF_RX_3A3N_DQ88_AVST_DATA29			
50	IN	AVST_D28	R64	IO_CDR_DIFF_RX_3A3P_DQ88_AVST_DATA28			
50	IN	AVST_D27	M65	IO_DIFF_TX_3A3N_DQ88_AVST_DATA27			
50	IN	AVST_D26	L64	IO_DIFF_TX_3A3P_DQ88_AVST_DATA26			
50	IN	AVST_D25	K69	IO_DIFF_RX_3A4N_DQ89_AVST_DATA25			
50	IN	AVST_D24	J68	IO_DIFF_RX_3A4P_DQ89_AVST_DATA24			
50	IN	AVST_D23	F69	IO_DIFF_TX_3A4N_DQ89_AVST_DATA23			
50	IN	AVST_D22	G68	IO_DIFF_TX_3A4P_DQ89_AVST_DATA22			
50	IN	AVST_D21	K67	IO_DIFF_RX_3A5N_DQ89_AVST_DATA21			
50	IN	AVST_D20	J66	IO_DIFF_RX_3A5P_DQ89_AVST_DATA20			
50	IN	AVST_D19	F67	IO_PLL_3A_T_CLKOUT1N_DIFF_TX_3A5N_DQSN89_AVST_DATA19			
50	IN	AVST_D18	G66	IO_PLL_3A_T_CLKOUT1P_PLL_3A_T_CLKOUT1_PLL_3A_T_FB1_DIFF_TX_3A5P_DQS89_AVST_DATA18			
50	IN	AVST_D17	K65	IO_CDR_DIFF_RX_3A6N_DQ89_AVST_DATA17			
50	IN	AVST_D16	J64	IO_RZQ_T_3A_CDR_DIFF_RX_3A6P_DQ89_AVST_DATA16			
61 16	IN	CLK_3A_GPIO_N_2	F65	IO_CLK_T_3A_1N_DIFF_TX_3A6N_DQ89			
61 16	IN	CLK_3A_GPIO_P_2	G64	IO_CLK_T_3A_1P_DIFF_TX_3A6P_DQ89			
61 16	IN	CLK_3A_GPIO_N_0	E68	IO_CLK_T_3A_0N_DIFF_RX_3A7N_DQ90			
61 16	IN	CLK_3A_GPIO_P_0	D67	IO_CLK_T_3A_0P_DIFF_RX_3A7P_DQ90			
43	BI	FMC_B_SDA_1V2	B67	IO_DIFF_TX_3A7N_DQ90			
43	BI	FMC_B_SCL_1V2	C66	IO_DIFF_TX_3A7P_DQ90			
43	BI	FMC_A_SDA_1V2	D65	IO_PLL_3A_T_CLKOUT0N_DIFF_RX_3A8N_DQ90			
43	BI	FMC_A_SCL_1V2	E64	IO_PLL_3A_T_CLKOUT0P_PLL_3A_T_CLKOUT0_PLL_3A_T_FB0_DIFF_RX_3A8P_DQ90			
50	BI	AVST_CLK	B65	IO_DIFF_TX_3A8N_DQSN90_AVST_CLK			
50	IN	AVST_D15	A64	IO_DIFF_TX_3A8P_DQS90_AVST_DATA15			
50	IN	AVST_D14	D63	IO_CDR_DIFF_RX_3A9N_DQ90_AVST_DATA14			
50	IN	AVST_D13	E62	IO_CDR_DIFF_RX_3A9P_DQ90_AVST_DATA13			
50	IN	AVST_D12	B63	IO_DIFF_TX_3A9N_DQ90_AVST_DATA12			
50	IN	AVST_D11	A62	IO_DIFF_TX_3A9P_DQ90_AVST_DATA11			
50	IN	AVST_D10	K63	IO_DIFF_RX_3A10N_DQ91_AVST_DATA10			
50	IN	AVST_D9	J62	IO_DIFF_RX_3A10P_DQ91_AVST_DATA9			
50	IN	AVST_D8	F63	IO_DIFF_TX_3A10N_DQ91_AVST_DATA8			
50	IN	AVST_VALID	G62	IO_DIFF_TX_3A10P_DQ91_AVST_VALID			
50	IN	AVST_D7	K61	IO_DIFF_RX_3A11N_DQ91_AVST_DATA7			
50	IN	AVST_D6	J60	IO_DIFF_RX_3A11P_DQ91_AVST_DATA6			
50	IN	AVST_D5	F61	IO_DIFF_TX_3A11N_DQSN91_AVST_DATA5			
50	IN	AVST_D4	G60	IO_DIFF_TX_3A11P_DQS91_AVST_DATA4			
50	IN	AVST_D3	K59	IO_CDR_DIFF_RX_3A12N_DQ91_AVST_DATA3			
50	IN	AVST_D2	J58	IO_CDR_DIFF_RX_3A12P_DQ91_AVST_DATA2			
50	IN	AVST_D1	F59	IO_DIFF_TX_3A12N_DQ91_AVST_DATA1			
50	IN	AVST_D0	G58	IO_DIFF_TX_3A12P_DQ91_AVST_DATA0			
					M63	USB_RESETn	
				IO_CDR_DIFF_RX_3A13N_DQ92	L62	USB_ADDR0	
				IO_CDR_DIFF_RX_3A13P_DQ92	P63	USB_ADDR1	
				IO_DIFF_TX_3A13N_DQ92	R62	USB_DATA6	
				IO_DIFF_TX_3A13P_DQ92	P61	USB_DATA3	
				IO_DIFF_RX_3A14N_DQ92	R60	USB_DATA1	
				IO_DIFF_RX_3A14P_DQ92	M61	USB_DATA5	
				IO_DIFF_TX_3A14N_DQSN92	L60	USB_DATA2	
				IO_DIFF_TX_3A14P_DQS92	P59	USB_DATA0	
				IO_CDR_DIFF_RX_3A15N_DQ92	R58	USB_DATA7	
				IO_CDR_DIFF_RX_3A15P_DQ92	M59	USB_DATA4	
				IO_DIFF_TX_3A15N_DQ92	L58	USB_WRn	
				IO_DIFF_TX_3A15P_DQ92			
				IO_CDR_DIFF_RX_3A16N_DQ93	P57	USB_RDn	
				IO_CDR_DIFF_RX_3A16P_DQ93	R56	USB_OEn	
				IO_DIFF_TX_3A16N_DQ93	M57	USB_FULL	
				IO_DIFF_TX_3A16P_DQ93	L56	USB_EMPTY	
				IO_DIFF_RX_3A17N_DQ93	P55	VCXO_INC	
				IO_DIFF_RX_3A17P_DQ93	R54	VCXO_DEC	
				IO_PLL_3A_B_CLKOUT1N_DIFF_TX_3A17N_DQSN93	M55		
				IO_PLL_3A_B_CLKOUT1P_PLL_3A_B_FB1_DIFF_TX_3A17P_DQS93	L54	USB_SCL	
				IO_CDR_DIFF_RX_3A18N_DQ93	P53	USB_SDA	
				IO_RZQ_B_3A_CDR_DIFF_RX_3A18P_DQ93	R52	3A_RZQ	
				IO_CLK_B_3A_1N_DIFF_TX_3A18N_DQ93	M53	CLK_3A_DIFF_18N	
				IO_CLK_B_3A_1P_DIFF_TX_3A18P_DQ93	L52	CLK_3A_DIFF_18P	
				IO_CLK_B_3A_0N_CDR_DIFF_RX_3A19N_DQ94	AD53	CLK_3A_GPIO_N_1	
				IO_CLK_B_3A_0P_CDR_DIFF_RX_3A19P_DQ94	AE52	CLK_3A_GPIO_P_1	
				IO_DIFF_TX_3A19N_DQ94	AB53	FPGA_RESETn	
				IO_DIFF_TX_3A19P_DQ94	AA52	USB_FPGA_CLK	
				IO_PLL_3A_B_CLKOUT0N_DIFF_RX_3A20N_DQ94	AD51		
				IO_PLL_3A_B_CLKOUT0P_PLL_3A_B_CLKOUT0_PLL_3A_B_FB0_DIFF_RX_3A20P_DQ94	AE50		
				IO_DIFF_TX_3A20N_DQSN94	AB51		
				IO_DIFF_TX_3A20P_DQS94	AA50	ZL_I2C_SDA_1V2	
				IO_CDR_DIFF_RX_3A21N_DQ94	AD49	ZL_I2C_SCL_1V2	
				IO_CDR_DIFF_RX_3A21P_DQ94	AE48		
				IO_DIFF_TX_3A21N_DQ94	AB49		
				IO_DIFF_TX_3A21P_DQ94	AA48	VCXO_FS_1V2	
				IO_CDR_DIFF_RX_3A22N_DQ95	Y53		
				IO_CDR_DIFF_RX_3A22P_DQ95	W52		
				IO_DIFF_TX_3A22N_DQ95	T53		
				IO_DIFF_TX_3A22P_DQ95	U52	CLK_I2C_SCL_1V2	
				IO_DIFF_RX_3A23N_DQ95	Y51	CLK_I2C_SDA_1V2	
				IO_DIFF_RX_3A23P_DQ95	W50		
				IO_DIFF_TX_3A23N_DQSN95	T51		
				IO_DIFF_TX_3A23P_DQS95	U50		
				IO_CDR_DIFF_RX_3A24N_DQ95	Y49	FPGA_SGPIO_SYNC	
				IO_CDR_DIFF_RX_3A24P_DQ95	W48	FPGA_SGPIO_CLK	
				IO_DIFF_TX_3A24N_DQ95	T49	FPGA_SGPI	
				IO_DIFF_TX_3A24P_DQ95	U48	FPGA_SGPO	

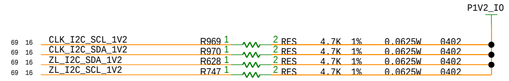
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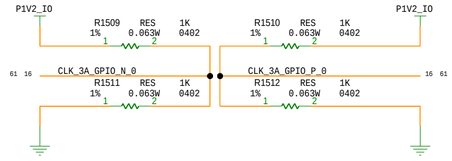
IO BANK 3A

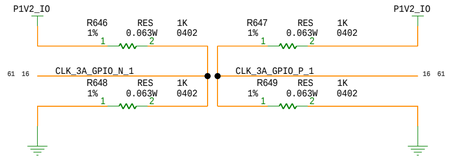
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CAD NOTE:
PLACE THE TERMINATION RESISTORS CLOSE TO THE FPGA

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DEPARTMENT		SIZE	CODE	DOCUMENT NUMBER	REV
	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 16 OF 97	

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FPGA BANK HPS

HPS DAUGHTER CARD CONNECTOR



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	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 17 OF 97	

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CAD NOTE:
PLACE THESE 100-OHM RES RIGHT BELOW FPGA BALLS
ON THE BREAKOUT VIAS.

R1819 A36092-043 100 1% 0.0625W EMPTY 0402 — CLK_A_12C_FGT_P_0 (18 61) / CLK_A_12C_FGT_N_0 (18 61)
R1820 A36092-043 100 1% 0.0625W EMPTY 0402 — CLK_A_12C_FGT_P_1 (18 61) / CLK_A_12C_FGT_N_1 (18 61)
R1821 A36092-043 100 1% 0.0625W EMPTY 0402 — CLK_A_12C_FGT_P_2 (18 61) / CLK_A_12C_FGT_N_2 (18 61)
R1822 A36092-043 100 1% 0.0625W EMPTY 0402 — CLK_A_12C_FGT_P_3 (18 61) / CLK_A_12C_FGT_N_3 (18 61)
R1823 A36092-043 100 1% 0.0625W EMPTY 0402 — CLK_A_12C_FGT_P_4 (18 62) / CLK_A_12C_FGT_N_4 (18 62)
R1824 A36092-043 100 1% 0.0625W EMPTY 0402 — CLK_A_12C_FGT_P_5 (18 62) / CLK_A_12C_FGT_N_5 (18 62)
R1825 A36092-043 100 1% 0.0625W EMPTY 0402 — CLK_A_12C_FGT_P_6 (18 62) / CLK_A_12C_FGT_N_6 (18 62)
R1826 A36092-043 100 1% 0.0625W EMPTY 0402 — CLK_A_12C_FGT_P_7 (18 62) / CLK_A_12C_FGT_N_7 (18 62)

CAD NOTE: PLEASE FOLLOW THE ARC ROUTING FOR THESE HIGH SPEED NETS.
FMC_A_RX*

U1
IC

FPGA BANK 12C

FMC + : A

42 IN	FMC_A_RX_N_0	BH65	FGTL12C_RX_Q0_CH0N	FGTL12C_TX_Q0_CH0N	BH59 FMC_A_TX_N_0 OUT 42
42 IN	FMC_A_RX_P_0	BJ66	FGTL12C_RX_Q0_CH0P	FGTL12C_TX_Q0_CH0P	BJ60 FMC_A_TX_P_0 OUT 42
42 IN	FMC_A_RX_N_1	BG68	FGTL12C_RX_Q0_CH1N	FGTL12C_TX_Q0_CH1N	BG62 FMC_A_TX_N_1 OUT 42
42 IN	FMC_A_RX_P_1	BF69	FGTL12C_RX_Q0_CH1P	FGTL12C_TX_Q0_CH1P	BF63 FMC_A_TX_P_1 OUT 42
42 IN	FMC_A_RX_N_2	BD65	FGTL12C_RX_Q0_CH2N	FGTL12C_TX_Q0_CH2N	BD59 FMC_A_TX_N_2 OUT 42
42 IN	FMC_A_RX_P_2	BE66	FGTL12C_RX_Q0_CH2P	FGTL12C_TX_Q0_CH2P	BE60 FMC_A_TX_P_2 OUT 42
42 IN	FMC_A_RX_N_3	BC68	FGTL12C_RX_Q0_CH3N	FGTL12C_TX_Q0_CH3N	BC62 FMC_A_TX_N_3 OUT 42
42 IN	FMC_A_RX_P_3	BB69	FGTL12C_RX_Q0_CH3P	FGTL12C_TX_Q0_CH3P	BB63 FMC_A_TX_P_3 OUT 42
61 18 IN	CLK_A_12C_FGT_N_0	BE56	REFCLK_FGTL12C_Q0_RX_CH0N		
61 18 IN	CLK_A_12C_FGT_P_0	BF57	REFCLK_FGTL12C_Q0_RX_CH0P		
61 18 IN	CLK_A_12C_FGT_N_1	BA56	REFCLK_FGTL12C_Q0_RX_CH1N		
61 18 IN	CLK_A_12C_FGT_P_1	AY57	REFCLK_FGTL12C_Q0_RX_CH1P		
42 IN	FMC_A_RX_N_4	AY65	FGTL12C_RX_Q1_CH0N	FGTL12C_TX_Q1_CH0N	AY59 FMC_A_TX_N_4 OUT 42
42 IN	FMC_A_RX_P_4	BA66	FGTL12C_RX_Q1_CH0P	FGTL12C_TX_Q1_CH0P	BA60 FMC_A_TX_P_4 OUT 42
42 IN	FMC_A_RX_N_5	AW68	FGTL12C_RX_Q1_CH1N	FGTL12C_TX_Q1_CH1N	AW62 FMC_A_TX_N_5 OUT 42
42 IN	FMC_A_RX_P_5	AV69	FGTL12C_RX_Q1_CH1P	FGTL12C_TX_Q1_CH1P	AV63 FMC_A_TX_P_5 OUT 42
42 IN	FMC_A_RX_N_6	AT65	FGTL12C_RX_Q1_CH2N	FGTL12C_TX_Q1_CH2N	AT59 FMC_A_TX_N_6 OUT 42
42 IN	FMC_A_RX_P_6	AU66	FGTL12C_RX_Q1_CH2P	FGTL12C_TX_Q1_CH2P	AU60 FMC_A_TX_P_6 OUT 42
42 IN	FMC_A_RX_N_7	AR68	FGTL12C_RX_Q1_CH3N	FGTL12C_TX_Q1_CH3N	AR62 FMC_A_TX_N_7 OUT 42
42 IN	FMC_A_RX_P_7	AP69	FGTL12C_RX_Q1_CH3P	FGTL12C_TX_Q1_CH3P	AP63 FMC_A_TX_P_7 OUT 42
61 18 IN	CLK_A_12C_FGT_N_2	BD57	REFCLK_FGTL12C_Q1_RX_CH2N		
61 18 IN	CLK_A_12C_FGT_P_2	BB57	REFCLK_FGTL12C_Q1_RX_CH2P		
61 18 IN	CLK_A_12C_FGT_N_3	AT57	REFCLK_FGTL12C_Q1_RX_CH3N		
61 18 IN	CLK_A_12C_FGT_P_3	AR56	REFCLK_FGTL12C_Q1_RX_CH3P		
42 IN	FMC_A_RX_N_8	AM65	FGTL12C_RX_Q2_CH0N	FGTL12C_TX_Q2_CH0N	AM59 FMC_A_TX_N_8 OUT 42
42 IN	FMC_A_RX_P_8	AN66	FGTL12C_RX_Q2_CH0P	FGTL12C_TX_Q2_CH0P	AN60 FMC_A_TX_P_8 OUT 42
42 IN	FMC_A_RX_N_9	AL68	FGTL12C_RX_Q2_CH1N	FGTL12C_TX_Q2_CH1N	AL62 FMC_A_TX_N_9 OUT 42
42 IN	FMC_A_RX_P_9	AK69	FGTL12C_RX_Q2_CH1P	FGTL12C_TX_Q2_CH1P	AK63 FMC_A_TX_P_9 OUT 42
42 IN	FMC_A_RX_N_10	AH65	FGTL12C_RX_Q2_CH2N	FGTL12C_TX_Q2_CH2N	AH59 FMC_A_TX_N_10 OUT 42
42 IN	FMC_A_RX_P_10	AJ66	FGTL12C_RX_Q2_CH2P	FGTL12C_TX_Q2_CH2P	AJ60 FMC_A_TX_P_10 OUT 42
42 IN	FMC_A_RX_N_11	AG68	FGTL12C_RX_Q2_CH3N	FGTL12C_TX_Q2_CH3N	AG62 FMC_A_TX_N_11 OUT 42
42 IN	FMC_A_RX_P_11	AF69	FGTL12C_RX_Q2_CH3P	FGTL12C_TX_Q2_CH3P	AF63 FMC_A_TX_P_11 OUT 42
62 18 IN	CLK_A_12C_FGT_N_4	AP57	REFCLK_FGTL12C_Q2_RX_CH4N		
62 18 IN	CLK_A_12C_FGT_P_4	AM57	REFCLK_FGTL12C_Q2_RX_CH4P		
62 18 IN	CLK_A_12C_FGT_N_5	AA56	REFCLK_FGTL12C_Q2_RX_CH5N		
62 18 IN	CLK_A_12C_FGT_P_5	Y57	REFCLK_FGTL12C_Q2_RX_CH5P		
64 OUT	RCVD_CLKO_12C_CH8_N1	AB57	REFCLK_FGTL12C_Q2_CH8N		
64 OUT	RCVD_CLKO_12C_CH8_P1	AD57	REFCLK_FGTL12C_Q2_CH8P		
42 IN	FMC_A_RX_N_12	AD65	FGTL12C_RX_Q3_CH0N	FGTL12C_TX_Q3_CH0N	AD59 FMC_A_TX_N_12 OUT 42
42 IN	FMC_A_RX_P_12	AE66	FGTL12C_RX_Q3_CH0P	FGTL12C_TX_Q3_CH0P	AE60 FMC_A_TX_P_12 OUT 42
42 IN	FMC_A_RX_N_13	AC68	FGTL12C_RX_Q3_CH1N	FGTL12C_TX_Q3_CH1N	AC62 FMC_A_TX_N_13 OUT 42
42 IN	FMC_A_RX_P_13	AB69	FGTL12C_RX_Q3_CH1P	FGTL12C_TX_Q3_CH1P	AB63 FMC_A_TX_P_13 OUT 42
42 IN	FMC_A_RX_N_14	Y65	FGTL12C_RX_Q3_CH2N	FGTL12C_TX_Q3_CH2N	Y59 FMC_A_TX_N_14 OUT 42
42 IN	FMC_A_RX_P_14	AA66	FGTL12C_RX_Q3_CH2P	FGTL12C_TX_Q3_CH2P	AA60 FMC_A_TX_P_14 OUT 42
42 IN	FMC_A_RX_N_15	W68	FGTL12C_RX_Q3_CH3N	FGTL12C_TX_Q3_CH3N	W62 FMC_A_TX_N_15 OUT 42
42 IN	FMC_A_RX_P_15	V69	FGTL12C_RX_Q3_CH3P	FGTL12C_TX_Q3_CH3P	V63 FMC_A_TX_P_15 OUT 42
62 18 IN	CLK_A_12C_FGT_N_6	AE57	REFCLK_FGTL12C_Q3_RX_CH6N		
62 18 IN	CLK_A_12C_FGT_P_6	AE56	REFCLK_FGTL12C_Q3_RX_CH6P		
62 18 IN	CLK_A_12C_FGT_N_7	W56	REFCLK_FGTL12C_Q3_RX_CH7N		
62 18 IN	CLK_A_12C_FGT_P_7	V55	REFCLK_FGTL12C_Q3_RX_CH7P		
RCVD_CLKO_12C_CH9_N1		AK57	REFCLK_FGTL12C_Q3_CH9N		
RCVD_CLKO_12C_CH9_P1		AL56	REFCLK_FGTL12C_Q3_CH9P		
49 18 IN	FM_GXF_12C_PERST_N	T59	I_PIN_PERST_N_12C_GXF	RCOMP_N_Q2_CH1_FGT_12C_GXF	AJ56 RCOMP_12C_N
				RCOMP_P_Q2_CH1_FGT_12C_GXF	AH57 RCOMP_12C_P
49 IN	CLK_GXF_IO_PLL	AE54	IO_PLL_REFCLK_12C_GXF	ENB_GXF_FHT12C	AL54
	TP_12C_GXF_I_JTAG_TDI_DFX	U60	DNU_U60		
	TP_12C_GXF_IO_JTAG_TDO_DFX	AK53	DNU_AK53		
	TP_12C_GXF_APROBE_0_ATB	AH53	DNU_AH53		
	TP_12C_GXF_APROBE_1_ATB	AG52	DNU_AG52		
	TP_12C_GXF_APROBE_Q0_CH3_FGT	AW56	APROBE_GXF_FGT12C_Q0_CH3		
	TP_12C_GXF_APROBE_Q2_CH3_FGT	AV57	APROBE_GXF_FGT12C_Q2_CH3		
	TP_12C_GXF_APROBE_Q3_CH3_FGT	AH55	APROBE_GXF_FGT12C_Q3_CH3		
	TP_12C_GXF_APROBE2_Q3_CH3_FGT	AG56	APROBE2_GXF_FGT12C_Q3_CH3		
72 OUT	TEMPDIODE_12C_GXF_N	AN52	TEMPDIODE3N		
72 OUT	TEMPDIODE_12C_GXF_P	AL52	TEMPDIODE3P		
		AJ52	GND_AJ52		

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F - TILE BANK 12C

99D3CA

CAD NOTE: PLEASE FOLLOW THE ARC ROUTING FOR THESE HIGH SPEED NETS.
FMC_A_TX*

CAD NOTE:
BREAK-OUT EVERY TP* NETS TO AN
EXPOSED PAD ON THE BOTTOM SIDE.

J12
K10975-001
CON
TH
1 — SMA_12C_CH9_N1
5 4 3 2

J13
K10975-001
CON
TH
1 — SMA_12C_CH9_P1
5 4 3 2

C233 1 — 2 0.1UF 16V
C234 1 — 2 0.1UF 16V

49 18 IN — FM_GXF_12C_PERST_N — R1783 1 — 2 RES 0 0.0625W 5% 0402 — FM_GXF_12C_PERST_N_R

R16 RES 499 0.1% 0.0625W 0402 — RCOMP_12C_N / RCOMP_12C_P

VCCN_IO_GXF

R1003
A93548-034
10K
1%
0.0625W
RES
0402

49 18 IN — FM_GXF_12C_PERST_N

C696
H61346-001
10NF
10%
50V
EMPTY
0402

CAD Note:
Place RCOMP res right below FPGA balls
on the breakout vias.

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DEPARTMENT		SIZE	CODE	DOCUMENT NUMBER	REV
	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 18 OF 97	

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CAD NOTE:
PLACE THESE 100-OHM RES RIGHT BELOW FPGA BALLS
ON THE BREAKOUT VIAS.

R1811 A36092-043 100 1% 0.0625W EMPTY 0402 — CLK_B_12A_FGT_P_0 (19 63) / CLK_B_12A_FGT_N_0 (19 63)
R1812 A36092-043 100 1% 0.0625W EMPTY 0402 — CLK_B_12A_FGT_P_1 (19 61) / CLK_B_12A_FGT_N_1 (19 61)
R1813 A36092-043 100 1% 0.0625W EMPTY 0402 — CLK_B_12A_FGT_P_2 (19 63) / CLK_B_12A_FGT_N_2 (19 63)
R1814 A36092-043 100 1% 0.0625W EMPTY 0402 — CLK_B_12A_FGT_P_3 (19 63) / CLK_B_12A_FGT_N_3 (19 63)
R1815 A36092-043 100 1% 0.0625W EMPTY 0402 — CLK_B_12A_FGT_P_4 (19 64) / CLK_B_12A_FGT_N_4 (19 64)
R1816 A36092-043 100 1% 0.0625W EMPTY 0402 — CLK_B_12A_FGT_P_5 (19 64) / CLK_B_12A_FGT_N_5 (19 64)
R1817 A36092-043 100 1% 0.0625W EMPTY 0402 — CLK_B_12A_FGT_P_6 (19 64) / CLK_B_12A_FGT_N_6 (19 64)
R1818 A36092-043 100 1% 0.0625W EMPTY 0402 — CLK_B_12A_FGT_P_7 (19 62) / CLK_B_12A_FGT_N_7 (19 62)

CAD NOTE: PLEASE FOLLOW THE ARC ROUTING FOR THESE HIGH SPEED NETS

FMC_B_RX*

FPGA BANK 12A

U1
IC

FMC + : B

44	IN	FMC_B_RX_N_0	CY65	FGTL12A_RX_Q0_CH0N	FGTL12A_TX_Q0_CH0N	CY59	FMC_B_TX_N_0
44	IN	FMC_B_RX_P_0	DA66	FGTL12A_RX_Q0_CH0P	FGTL12A_TX_Q0_CH0P	DA60	FMC_B_TX_P_0
44	IN	FMC_B_RX_N_1	CW68	FGTL12A_RX_Q0_CH1N	FGTL12A_TX_Q0_CH1N	CW62	FMC_B_TX_N_1
44	IN	FMC_B_RX_P_1	CV69	FGTL12A_RX_Q0_CH1P	FGTL12A_TX_Q0_CH1P	CV63	FMC_B_TX_P_1
44	IN	FMC_B_RX_N_2	CT65	FGTL12A_RX_Q0_CH2N	FGTL12A_TX_Q0_CH2N	CT59	FMC_B_TX_N_2
44	IN	FMC_B_RX_P_2	CU66	FGTL12A_RX_Q0_CH2P	FGTL12A_TX_Q0_CH2P	CU60	FMC_B_TX_P_2
44	IN	FMC_B_RX_N_3	CR68	FGTL12A_RX_Q0_CH3N	FGTL12A_TX_Q0_CH3N	CR62	FMC_B_TX_N_3
44	IN	FMC_B_RX_P_3	CP69	FGTL12A_RX_Q0_CH3P	FGTL12A_TX_Q0_CH3P	CP63	FMC_B_TX_P_3
63 19	IN	CLK_B_12A_FGT_N_0	CW56	REFCLK_FGTL12A_Q0_RX_CH0N			
63 19	IN	CLK_B_12A_FGT_P_0	CV57	REFCLK_FGTL12A_Q0_RX_CH0P			
61 19	IN	CLK_B_12A_FGT_N_1	CR56	REFCLK_FGTL12A_Q0_RX_CH1N			
61 19	IN	CLK_B_12A_FGT_P_1	CU56	REFCLK_FGTL12A_Q0_RX_CH1P			
44	IN	FMC_B_RX_N_4	CM65	FGTL12A_RX_Q1_CH0N	FGTL12A_TX_Q1_CH0N	CM59	FMC_B_TX_N_4
44	IN	FMC_B_RX_P_4	CN66	FGTL12A_RX_Q1_CH0P	FGTL12A_TX_Q1_CH0P	CN60	FMC_B_TX_P_4
44	IN	FMC_B_RX_N_5	CL68	FGTL12A_RX_Q1_CH1N	FGTL12A_TX_Q1_CH1N	CL62	FMC_B_TX_N_5
44	IN	FMC_B_RX_P_5	CK69	FGTL12A_RX_Q1_CH1P	FGTL12A_TX_Q1_CH1P	CK63	FMC_B_TX_P_5
44	IN	FMC_B_RX_N_6	CH65	FGTL12A_RX_Q1_CH2N	FGTL12A_TX_Q1_CH2N	CH59	FMC_B_TX_N_6
44	IN	FMC_B_RX_P_6	CJ66	FGTL12A_RX_Q1_CH2P	FGTL12A_TX_Q1_CH2P	CJ60	FMC_B_TX_P_6
44	IN	FMC_B_RX_N_7	CG68	FGTL12A_RX_Q1_CH3N	FGTL12A_TX_Q1_CH3N	CG62	FMC_B_TX_N_7
44	IN	FMC_B_RX_P_7	CF69	FGTL12A_RX_Q1_CH3P	FGTL12A_TX_Q1_CH3P	CF63	FMC_B_TX_P_7
63 19	IN	CLK_B_12A_FGT_N_2	CP57	REFCLK_FGTL12A_Q1_RX_CH2N			
63 19	IN	CLK_B_12A_FGT_P_2	CN56	REFCLK_FGTL12A_Q1_RX_CH2P			
63 19	IN	CLK_B_12A_FGT_N_3	CG56	REFCLK_FGTL12A_Q1_RX_CH3N			
63 19	IN	CLK_B_12A_FGT_P_3	CE56	REFCLK_FGTL12A_Q1_RX_CH3P			
44	IN	FMC_B_RX_N_8	CD65	FGTL12A_RX_Q2_CH0N	FGTL12A_TX_Q2_CH0N	CD59	FMC_B_TX_N_8
44	IN	FMC_B_RX_P_8	CE66	FGTL12A_RX_Q2_CH0P	FGTL12A_TX_Q2_CH0P	CE60	FMC_B_TX_P_8
44	IN	FMC_B_RX_N_9	CC68	FGTL12A_RX_Q2_CH1N	FGTL12A_TX_Q2_CH1N	CC62	FMC_B_TX_N_9
44	IN	FMC_B_RX_P_9	CB69	FGTL12A_RX_Q2_CH1P	FGTL12A_TX_Q2_CH1P	CB63	FMC_B_TX_P_9
44	IN	FMC_B_RX_N_10	BY65	FGTL12A_RX_Q2_CH2N	FGTL12A_TX_Q2_CH2N	BY59	FMC_B_TX_N_10
44	IN	FMC_B_RX_P_10	CA66	FGTL12A_RX_Q2_CH2P	FGTL12A_TX_Q2_CH2P	CA60	FMC_B_TX_P_10
44	IN	FMC_B_RX_N_11	BW68	FGTL12A_RX_Q2_CH3N	FGTL12A_TX_Q2_CH3N	BW62	FMC_B_TX_N_11
44	IN	FMC_B_RX_P_11	BV69	FGTL12A_RX_Q2_CH3P	FGTL12A_TX_Q2_CH3P	BV63	FMC_B_TX_P_11
64 19	IN	CLK_B_12A_FGT_N_4	CJ56	REFCLK_FGTL12A_Q2_RX_CH4N			
64 19	IN	CLK_B_12A_FGT_P_4	CH57	REFCLK_FGTL12A_Q2_RX_CH4P			
64 19	IN	CLK_B_12A_FGT_N_5	CB57	REFCLK_FGTL12A_Q2_RX_CH5N			
64 19	IN	CLK_B_12A_FGT_P_5	BY57	REFCLK_FGTL12A_Q2_RX_CH5P			
64	OUT	RCVD_CLKO_12A_CH8_N1	CD57	REFCLK_FGTL12A_Q2_CH8N			
64	OUT	RCVD_CLKO_12A_CH8_P1	CC56	REFCLK_FGTL12A_Q2_CH8P			
44	IN	FMC_B_RX_N_12	BT65	FGTL12A_RX_Q3_CH0N	FGTL12A_TX_Q3_CH0N	BT59	FMC_B_TX_N_12
44	IN	FMC_B_RX_P_12	BU66	FGTL12A_RX_Q3_CH0P	FGTL12A_TX_Q3_CH0P	BU60	FMC_B_TX_P_12
44	IN	FMC_B_RX_N_13	BR68	FGTL12A_RX_Q3_CH1N	FGTL12A_TX_Q3_CH1N	BR62	FMC_B_TX_N_13
44	IN	FMC_B_RX_P_13	BP69	FGTL12A_RX_Q3_CH1P	FGTL12A_TX_Q3_CH1P	BP63	FMC_B_TX_P_13
44	IN	FMC_B_RX_N_14	BM65	FGTL12A_RX_Q3_CH2N	FGTL12A_TX_Q3_CH2N	BM59	FMC_B_TX_N_14
44	IN	FMC_B_RX_P_14	BN66	FGTL12A_RX_Q3_CH2P	FGTL12A_TX_Q3_CH2P	BN60	FMC_B_TX_P_14
44	IN	FMC_B_RX_N_15	BL68	FGTL12A_RX_Q3_CH3N	FGTL12A_TX_Q3_CH3N	BL62	FMC_B_TX_N_15
44	IN	FMC_B_RX_P_15	BK69	FGTL12A_RX_Q3_CH3P	FGTL12A_TX_Q3_CH3P	BK63	FMC_B_TX_P_15
64 19	IN	CLK_B_12A_FGT_N_6	BR56	REFCLK_FGTL12A_Q3_RX_CH6N			
64 19	IN	CLK_B_12A_FGT_P_6	BP57	REFCLK_FGTL12A_Q3_RX_CH6P			
62 19	IN	CLK_B_12A_FGT_N_7	BN56	REFCLK_FGTL12A_Q3_RX_CH7N			
19	IN	CLK_B_12A_FGT_P_7	BL56	REFCLK_FGTL12A_Q3_RX_CH7P			
62		C_RCVD_CLKO_12A_CH9_N1	BJ56	REFCLK_FGTL12A_Q3_CH9N			
		C_RCVD_CLKO_12A_CH9_P1	BK57	REFCLK_FGTL12A_Q3_CH9P			
19 49	IN	FM_GXF_12A_PERST_N	CG52	I_PIN_PERST_N_12A_GXF	RCOMP_N_Q2_CH1_FGT_12A_GXF	BW56	RCOMP_12A_N
					RCOMP_P_Q2_CH1_FGT_12A_GXF	BV57	RCOMP_12A_P
		FM_GXF_12A_IO_PLL_REFCLK_12A	CW54	IO_PLL_REFCLK_12A_GXF			
			CT53	DNU_CT53	ENB_GXF_FHT12A	CF53	
			CE54	DNU_CE54			
		TP_F_GXF_DNU1	CP53	DNU_CP53			
		TP_F_GXF_DNU2	CR54	DNU_CR54			
		TP_F_GXF_DNU3	CL56	APROBE_GXF_FGT12A_Q0_CH3			
		TP_F_GXF_DNU4	CK57	APROBE_GXF_FGT12A_Q2_CH3			
		TP_F_GXF_DNU5	BT57	APROBE_GXF_FGT12A_Q3_CH3			
		TP_F_GXF_DNU6	BU56	APROBE2_GXF_FGT12A_Q3_CH3			
72	OUT	TEMPDIODE_12A_GXF_DN	BK53	TEMPDIODE1N			
72	OUT	TEMPDIODE_12A_GXF_DP	BJ52	TEMPDIODE1P			
			CG54	GND_CG54			

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F - TILE BANK 12A

99D3CA

CAD NOTE: PLEASE FOLLOW THE ARC ROUTING FOR THESE HIGH SPEED NETS

FMC_B_TX*

CAD Note:
Place RCOMP res right below FPGA balls
on the breakout vias

R39 RES 499 0.1% 0.0625W 0402 — BW56 RCOMP_12A_N / BV57 RCOMP_12A_P

J15
K10975-001
CON
TH

J16
K10975-001
CON
TH

RCVD_CLKO_12A_CH9_N1 — C235 0.1UF 16V — C_RCVD_CLKO_12A_CH9_N1
RCVD_CLKO_12A_CH9_P1 — C236 0.1UF 16V — C_RCVD_CLKO_12A_CH9_P1

FM_GXF_12A_PERST_N — R1784 RES 0402 0 0.0625W 5% — FM_GXF_12A_PERST_N_R
TP87 — R1232 RES 0402 0 0.0625W 5% — FM_GXF_12A_IO_PLL_REFCLK

VCCN_IO_GXF
R1788 A93548-034 10K 1% 0.0625W RES 0402
C3052 H61346-001 10NF 10% 50V EMPTY 0402
49 19 IN FM_GXF_12A_PERST_N

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DEPARTMENT		SIZE	CODE	DOCUMENT NUMBER	REV
	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 19 OF 97	

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FPGA BANK 13A

CAD NOTE: PLEASE FOLLOW THE ARC ROUTING FOR THESE HIGH SPEED NETS

U1
IC

FPGA BANK 13A

20	IN	FHT_MXPM_RX_N4	DD9	FHTR13A_RX_CH0N	FHTR13A_TX_CH0N	DG6	FHT_MXPM_TX_N4
20	IN	FHT_MXPM_RX_P4	DC10	FHTR13A_RX_CH0P	FHTR13A_TX_CH0P	DF7	FHT_MXPM_TX_P4
20	IN	FHT_MXPM_RX_N5	DE12	FHTR13A_RX_CH1N	FHTR13A_TX_CH1N	DH9	FHT_MXPM_TX_N5
20	IN	FHT_MXPM_RX_P5	DD13	FHTR13A_RX_CH1P	FHTR13A_TX_CH1P	DG10	FHT_MXPM_TX_P5
20	IN	FHT_MXPM_RX_N6	DH13	FHTR13A_RX_CH2N	FHTR13A_TX_CH2N	DJ10	FHT_MXPM_TX_N6
20	IN	FHT_MXPM_RX_P6	DJ12	FHTR13A_RX_CH2P	FHTR13A_TX_CH2P	DM9	FHT_MXPM_TX_P6
20	IN	FHT_MXPM_RX_N7	DJ16	FHTR13A_RX_CH3N	FHTR13A_TX_CH3N	DM13	FHT_MXPM_TX_N7
20	IN	FHT_MXPM_RX_P7	DK15	FHTR13A_RX_CH3P	FHTR13A_TX_CH3P	DN12	FHT_MXPM_TX_P7

R260 RES 1.5K 0.1% 0.063W 0402
CAD Note: Place RCOMP res right below FPGA balls on the breakout vias.

54 20	IN	CLK_FHT_13A_N_0	DB15	REFCLK_FHTR13A_CH0N	RCOMP_N_FHT_13A_GXF	CU18	GXF_13A_FHT_RCOMP_N_0
20	IN	CLK_FHT_13A_P_0	DA16	REFCLK_FHTR13A_CH0P	RCOMP_P_FHT_13A_GXF	CR18	GXF_13A_FHT_RCOMP_P_0
54	IN	CLK_13A_FHT_N1	DE16	REFCLK_FHTR13A_CH1N			
		CLK_13A_FHT_P1	DF15	REFCLK_FHTR13A_CH1P			

C698 1 2 0.1UF 25V
C699 1 2 0.1UF 25V

J38
J85530-001
SCON
SM

J45
J85530-001
SCON
SM

37	IN	PCIE_RX_N_0	BL2	FGTR13A_RX_Q0_CH0N	FGTR13A_TX_Q0_CH0N	BK11	PCIE_TX_N_0
37	IN	PCIE_RX_P_0	BM1	FGTR13A_RX_Q0_CH0P	FGTR13A_TX_Q0_CH0P	BJ10	PCIE_TX_P_0
37	IN	PCIE_RX_N_1	BP5	FGTR13A_RX_Q0_CH1N	FGTR13A_TX_Q0_CH1N	BL8	PCIE_TX_N_1
37	IN	PCIE_RX_P_1	BN4	FGTR13A_RX_Q0_CH1P	FGTR13A_TX_Q0_CH1P	BM7	PCIE_TX_P_1
37	IN	PCIE_RX_N_2	BR2	FGTR13A_RX_Q0_CH2N	FGTR13A_TX_Q0_CH2N	BP11	PCIE_TX_N_2
37	IN	PCIE_RX_P_2	BT1	FGTR13A_RX_Q0_CH2P	FGTR13A_TX_Q0_CH2P	BN10	PCIE_TX_P_2
37	IN	PCIE_RX_N_3	BV5	FGTR13A_RX_Q0_CH3N	FGTR13A_TX_Q0_CH3N	BR8	PCIE_TX_N_3
37	IN	PCIE_RX_P_3	BU4	FGTR13A_RX_Q0_CH3P	FGTR13A_TX_Q0_CH3P	BT7	PCIE_TX_P_3
63 20	IN	CLK_13A_FGT_REFCLK0_N	BU14	REFCLK_FGTR13A_Q0_RX_CH0N			
63 20	IN	CLK_13A_FGT_REFCLK0_P	BR14	REFCLK_FGTR13A_Q0_RX_CH0P			
			BJ14	REFCLK_FGTR13A_Q0_RX_CH1N			
			BK13	REFCLK_FGTR13A_Q0_RX_CH1P			
20	IN	FGT_MXPM_RX_N_0	BW2	FGTR13A_RX_Q1_CH0N	FGTR13A_TX_Q1_CH0N	BV11	FGT_MXPM_TX_N_0
20	IN	FGT_MXPM_RX_P_0	BY1	FGTR13A_RX_Q1_CH0P	FGTR13A_TX_Q1_CH0P	BU10	FGT_MXPM_TX_P_0
20	IN	FGT_MXPM_RX_N_1	CB5	FGTR13A_RX_Q1_CH1N	FGTR13A_TX_Q1_CH1N	BW8	FGT_MXPM_TX_N_1
20	IN	FGT_MXPM_RX_P_1	CA4	FGTR13A_RX_Q1_CH1P	FGTR13A_TX_Q1_CH1P	BY7	FGT_MXPM_TX_P_1
20	IN	FGT_MXPM_RX_N_2	CC2	FGTR13A_RX_Q1_CH2N	FGTR13A_TX_Q1_CH2N	CB11	FGT_MXPM_TX_N_2
20	IN	FGT_MXPM_RX_P_2	CD1	FGTR13A_RX_Q1_CH2P	FGTR13A_TX_Q1_CH2P	CA10	FGT_MXPM_TX_P_2
20	IN	FGT_MXPM_RX_N_3	CF5	FGTR13A_RX_Q1_CH3N	FGTR13A_TX_Q1_CH3N	CC8	FGT_MXPM_TX_N_3
20	IN	FGT_MXPM_RX_P_3	CE4	FGTR13A_RX_Q1_CH3P	FGTR13A_TX_Q1_CH3P	CD7	FGT_MXPM_TX_P_3
58 20	IN	CLK_13A_FGT_REFCLK2_N	BP13	REFCLK_FGTR13A_Q1_RX_CH2N			
58 20	IN	CLK_13A_FGT_REFCLK2_P	BN14	REFCLK_FGTR13A_Q1_RX_CH2P			
58 20	IN	CLK_13A_FGT_REFCLK3_N	CC14	REFCLK_FGTR13A_Q1_RX_CH3N			
58 20	IN	CLK_13A_FGT_REFCLK3_P	CE14	REFCLK_FGTR13A_Q1_RX_CH3P			
39	IN	QSFPDD1_RX_N0	CG2	FGTR13A_RX_Q2_CH0N	FGTR13A_TX_Q2_CH0N	CF11	QSFPDD1_TX_N0
39	IN	QSFPDD1_RX_P0	CH1	FGTR13A_RX_Q2_CH0P	FGTR13A_TX_Q2_CH0P	CE10	QSFPDD1_TX_P0
39	IN	QSFPDD1_RX_N1	CK5	FGTR13A_RX_Q2_CH1N	FGTR13A_TX_Q2_CH1N	CG8	QSFPDD1_TX_N1
39	IN	QSFPDD1_RX_P1	CJ4	FGTR13A_RX_Q2_CH1P	FGTR13A_TX_Q2_CH1P	CH7	QSFPDD1_TX_P1
39	IN	QSFPDD1_RX_N2	CL2	FGTR13A_RX_Q2_CH2N	FGTR13A_TX_Q2_CH2N	CK11	QSFPDD1_TX_N2
39	IN	QSFPDD1_RX_P2	CM1	FGTR13A_RX_Q2_CH2P	FGTR13A_TX_Q2_CH2P	CJ10	QSFPDD1_TX_P2
39	IN	QSFPDD1_RX_N3	CP5	FGTR13A_RX_Q2_CH3N	FGTR13A_TX_Q2_CH3N	CL8	QSFPDD1_TX_N3
39	IN	QSFPDD1_RX_P3	CN4	FGTR13A_RX_Q2_CH3P	FGTR13A_TX_Q2_CH3P	CM7	QSFPDD1_TX_P3
58 20	IN	CLK_13A_FGT_REFCLK4_N	CL16	REFCLK_FGTR13A_Q2_RX_CH4N			
58 20	IN	CLK_13A_FGT_REFCLK4_P	CM17	REFCLK_FGTR13A_Q2_RX_CH4P			
			BW14	REFCLK_FGTR13A_Q2_RX_CH5N			
			BV13	REFCLK_FGTR13A_Q2_RX_CH5P			
			CU14	REFCLK_FGTR13A_Q2_CH6N			
			CR14	REFCLK_FGTR13A_Q2_CH6P			
39	IN	QSFPDD1_RX_N4	CR2	FGTR13A_RX_Q3_CH0N	FGTR13A_TX_Q3_CH0N	CP11	QSFPDD1_TX_N4
39	IN	QSFPDD1_RX_P4	CT1	FGTR13A_RX_Q3_CH0P	FGTR13A_TX_Q3_CH0P	CN10	QSFPDD1_TX_P4
39	IN	QSFPDD1_RX_N5	CV5	FGTR13A_RX_Q3_CH1N	FGTR13A_TX_Q3_CH1N	CR8	QSFPDD1_TX_N5
39	IN	QSFPDD1_RX_P5	CU4	FGTR13A_RX_Q3_CH1P	FGTR13A_TX_Q3_CH1P	CT7	QSFPDD1_TX_P5
39	IN	QSFPDD1_RX_N6	CW2	FGTR13A_RX_Q3_CH2N	FGTR13A_TX_Q3_CH2N	CV11	QSFPDD1_TX_N6
39	IN	QSFPDD1_RX_P6	CY1	FGTR13A_RX_Q3_CH2P	FGTR13A_TX_Q3_CH2P	CU10	QSFPDD1_TX_P6
39	IN	QSFPDD1_RX_N7	DC2	FGTR13A_RX_Q3_CH3N	FGTR13A_TX_Q3_CH3N	CW8	QSFPDD1_TX_N7
39	IN	QSFPDD1_RX_P7	DD1	FGTR13A_RX_Q3_CH3P	FGTR13A_TX_Q3_CH3P	CY7	QSFPDD1_TX_P7
			CF13	REFCLK_FGTR13A_Q3_RX_CH6N			
			CG14	REFCLK_FGTR13A_Q3_RX_CH6P			
			CN14	REFCLK_FGTR13A_Q3_RX_CH7N			
			CP13	REFCLK_FGTR13A_Q3_RX_CH7P			
			CV13	REFCLK_FGTR13A_Q3_CH9N			
			CW14	REFCLK_FGTR13A_Q3_CH9P			

CAD Note: Place RCOMP res right below FPGA balls on the breakout vias.

R1781 RES 0402 5% 0 0.0625W
R57 RES 0402 5% 0 0.0625W

69	IN	FM_GXF_13A_PERST_N	FM_GXF_13A_PERST_N_R	CR20	I_PIN_PERST_N_13A_GXF	CA14	GXF_13A_FGT_RCOMP_N_0
58	IN	CLK_13A_IO_PLL	CLK_13A_IO_PLL_R	CM21	IO_PLL_REFCLK_13A_GXF	CB13	GXF_13A_FGT_RCOMP_P_0
		TP_13A_GXF_IO_JTAG_TDO_DFX		CN22	DNU_CN22	CH23	U20_GXF_IO_DFX_0
		TP_13A_GXF_I_JTAG_TMS_DFX		CW22	DNU_CW22		
		TP_13A_GXF_APROBE_0_ATB		CU22	DNU_CU22		
		TP_13A_GXF_APROBE_1_ATB		CT21	DNU_CT21		
		TP_13A_GXF_APROBE_Q0_CH3_FGT		BL14	APROBE_GXF_FGT13A_Q0_CH3		
		TP_13A_GXF_APROBE_Q2_CH3_FGT		BM13	APROBE_GXF_FGT13A_Q2_CH3		
		TP_13A_GXF_APROBE_Q3_CH3_FGT		CK13	APROBE_GXF_FGT13A_Q3_CH3		
		TP_13A_GXF_APROBE2_Q3_CH3_FGT		CL14	APROBE2_GXF_FGT13A_Q3_CH3		
		TP_13A_GXF_APROBE1_COM_FHT		CV19	APROBE1_GXF_FHT13A		
		TP_13A_GXF_APROBE2_COM_FHT		CW18	APROBE2_GXF_FHT13A	ENB_GXF_FHT13A	
72	OUT	TEMPDIODE_13A_GXF_DN		CL22	TEMPDIODE4N		
72	OUT	TEMPDIODE_13A_GXF_DP		CK21	TEMPDIODE4P		
				CP21	GND_CP21		

R17 RES 499 0.1% 0.0625W 0402

13 OF 28

F - TILE BANK 13A

99D3CA

VCCN_IO_GXF

R59
A93549-001
0
5%
0.0625W
RES
0402

R60
A93548-034
10K
1%
0.0625W
EMPTY
0402

CAD NOTE:
BREAK-OUT EVERY TP* NETS TO AN
EXPOSED PAD ON THE BOTTOM SIDE.

P1V2_IO

R1797 EMPTY 0402 1% 1K 0.063W
R1798 EMPTY 0402 1% 1K 0.063W
R1799 EMPTY 0402 1% 1K 0.063W
R1800 EMPTY 0402 1% 1K 0.063W

54 20 CLK_FHT_13A_P_0
CLK_FHT_13A_N_0 20 54

SM
J2
SCON

20 OUT	FHT_MXPM_RX_N7	1	IO1
20 OUT	FHT_MXPM_RX_P7	2	IO2
20 OUT	FHT_MXPM_RX_N6	3	IO3
20 OUT	FHT_MXPM_RX_P6	4	IO4
20 OUT	FHT_MXPM_RX_N5	5	IO5
20 OUT	FHT_MXPM_RX_P5	6	IO6
20 OUT	FHT_MXPM_RX_N4	7	IO7
20 OUT	FHT_MXPM_RX_P4	8	IO8
20 IN	FHT_MXPM_TX_N6	9	IO9
20 IN	FHT_MXPM_TX_P6	10	IO10
20 IN	FHT_MXPM_TX_N7	11	IO11
20 IN	FHT_MXPM_TX_P7	12	IO12
20 IN	FHT_MXPM_TX_N4	13	IO13
20 IN	FHT_MXPM_TX_P4	14	IO14
20 IN	FHT_MXPM_TX_N5	15	IO15
20 IN	FHT_MXPM_TX_P5	16	IO16
		MH	MH 17
		MH	MH 18
		MH	MH 19

M34670-001

SM
J28
SCON

20 OUT	FGT_MXPM_RX_N_3	1	IO1
20 OUT	FGT_MXPM_RX_P_3	2	IO2
20 OUT	FGT_MXPM_RX_N_2	3	IO3
20 OUT	FGT_MXPM_RX_P_2	4	IO4
20 OUT	FGT_MXPM_RX_N_0	5	IO5
20 OUT	FGT_MXPM_RX_P_0	6	IO6
20 OUT	FGT_MXPM_RX_N_1	7	IO7
20 OUT	FGT_MXPM_RX_P_1	8	IO8
20 IN	FGT_MXPM_TX_N_2	9	IO9
20 IN	FGT_MXPM_TX_P_2	10	IO10
20 IN	FGT_MXPM_TX_N_3	11	IO11
20 IN	FGT_MXPM_TX_P_3	12	IO12
20 IN	FGT_MXPM_TX_P_1	13	IO13
20 IN	FGT_MXPM_TX_N_1	14	IO14
		MH	MH 17
20 IN	FGT_MXPM_TX_P_0	15	IO15 MH 18
20 IN	FGT_MXPM_TX_N_0	16	IO16 MH 19

M34670-001

CAD NOTE:
PLACE THESE 100-OHM RES RIGHT BELOW FPGA BALLS
ON THE BREAKOUT VIAS.

R1808 A36092-043 100 1% 0.0625W EMPTY 0402
CLK_FHT_13A_P_0 20 54
CLK_FHT_13A_N_0 20 54

R1809 A36092-043 100 1% 0.0625W EMPTY 0402
CLK_13A_FGT_REFCLK2_P 20 58
CLK_13A_FGT_REFCLK2_N 20 58

R1810 A36092-043 100 1% 0.0625W EMPTY 0402
CLK_13A_FGT_REFCLK3_P 20 58
CLK_13A_FGT_REFCLK3_N 20 58

R998 A36092-043 100 1% 0.0625W EMPTY 0402
CLK_13A_FGT_REFCLK4_P 20 58
CLK_13A_FGT_REFCLK4_N 20 58

R999 A36092-043 100 1% 0.0625W EMPTY 0402
CLK_13A_FGT_REFCLK0_P 20 63
CLK_13A_FGT_REFCLK0_N 20 63

Mon Sep 15 10:27:10 2025

DEPARTMENT	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	SIZE: C+	CODE: 34649	DOCUMENT NUMBER: 150-0330690-D1 REV: D1
		SCALE: DO NOT SCALE DRAWING		SHEET 20 OF 97

ALTERA CONFIDENTIAL

CAD NOTE:
PLACE THESE 100-OHM RES RIGHT BELOW FPGA BALLS
ON THE BREAKOUT VIAS.

R268 A36092-043 100 1% 0.0625W EMPTY 0402 — CLK_FHT_13C_N_0 — CLK_FHT_13C_P_0
R277 A36092-043 100 1% 0.0625W EMPTY 0402 — CLK_13C_FHT_N1 — CLK_13C_FHT_P1
R311 A36092-043 100 1% 0.0625W EMPTY 0402 — ETH_REFCLK_156.25M_N — ETH_REFCLK_156.25M_P
R312 A36092-043 100 1% 0.0625W EMPTY 0402 — CPRI_REFCLK_153.6M_N — CPRI_REFCLK_153.6M_P
R313 A36092-043 100 1% 0.0625W EMPTY 0402 — CPRI_REFCLK_184.32M_N — CPRI_REFCLK_184.32M_P
R314 A36092-043 100 1% 0.0625W EMPTY 0402 — CLK_13C_FGT_N3 — CLK_13C_FGT_P3
R315 A36092-043 100 1% 0.0625W EMPTY 0402 — CLK_VCXO_MUX_2N — CLK_VCXO_MUX_2P
R316 A36092-043 100 1% 0.0625W EMPTY 0402 — SI5332_OUT7_N — SI5332_OUT7_P
R317 A36092-043 100 1% 0.0625W EMPTY 0402 — CLK_FGT_13C_REFCLK_2N — CLK_FGT_13C_REFCLK_2P
R1278 A36092-043 100 1% 0.0625W EMPTY 0402 — R_QSFP_RCVD_CLK1_N — R_QSFP_RCVD_CLK1_P
R1279 A36092-043 100 1% 0.0625W EMPTY 0402 — R_QSFP_RCVD_CLK2_N — R_QSFP_RCVD_CLK2_P

FPGA BANK 13C

U1
IC

CAD NOTE: PLEASE FOLLOW THE ARC ROUTING FOR THESE HIGH SPEED NETS.

QSFPDD800_RX_N0 AR6 / QSFPDD800_RX_P0 AT5 / QSFPDD800_RX_N1 AW6 / QSFPDD800_RX_P1 AY5 / QSFPDD800_RX_N2 BD5 / QSFPDD800_RX_P2 BC6 / QSFPDD800_RX_N3 BH5 / QSFPDD800_RX_P3 BG6
CLK_FHT_13C_N_0 BC10 / CLK_FHT_13C_P_0 BB11
C471 0.1UF 25V CLK_13C_FHT_N1 — C477 0.1UF 25V CLK_13C_FHT_P1 — BE8 / BF9
CLK_SMP_13C_A_N / CLK_SMP_13C_A_P
J23 J85530-001 SCON SM — J24 J85530-001 SCON SM
FGT_13C_RX0_DN B11 / FGT_13C_RX0_DP C10 / E8 / D7 / C4 / B3
SDI_RX_N J8 / SDI_RX_P H7
SI5332_OUT7_N AD17 / SI5332_OUT7_P AF17 / CLK_VCXO_MUX_2N AA16 / CLK_VCXO_MUX_2P W16
F5 / G4 / K5 / L4 / J2 / H1 / P5 / R4
CLK_FGT_13C_REFCLK_2N N14 / CLK_FGT_13C_REFCLK_2P P13
C521 0.1UF 25V CLK_13C_FGT_N3 — C523 0.1UF 25V CLK_13C_FGT_P3 — AM11 / AL12
CLK_SMP_13C_B_N / CLK_SMP_13C_B_P
J25 J85530-001 SCON SM — J26 J85530-001 SCON SM
QSFPDD0_RX_N0 N2 / QSFPDD0_RX_P0 M1 / QSFPDD0_RX_N1 V5 / QSFPDD0_RX_P1 W4 / QSFPDD0_RX_N2 U2 / QSFPDD0_RX_P2 T1 / QSFPDD0_RX_N3 AB5 / QSFPDD0_RX_P3 AC4
ETH_REFCLK_156.25M_N U14 / ETH_REFCLK_156.25M_P R14 / CPRI_REFCLK_153.6M_N AU12 / CPRI_REFCLK_153.6M_P AV11
QSFP_RCVD_CLK1_N — C261 0.1UF 16V — R_QSFP_RCVD_CLK1_N AB13
QSFP_RCVD_CLK1_P — C263 0.1UF 16V — R_QSFP_RCVD_CLK1_P AA14
QSFPDD0_RX_N4 AA2 / QSFPDD0_RX_P4 Y1 / QSFPDD0_RX_N5 AF5 / QSFPDD0_RX_P5 AG4 / QSFPDD0_RX_N6 AE2 / QSFPDD0_RX_P6 AD1 / QSFPDD0_RX_N7 AJ2 / QSFPDD0_RX_P7 AH1
ZL_SPARE_CLK — R934 EMPTY 0 5% 0.0625W 0402 — ZL_SPARE_CLK_R AE14
CPRI_REFCLK_184.32M_N AC14 / CPRI_REFCLK_184.32M_P AG14
QSFP_RCVD_CLK2_N — C264 0.1UF 16V — R_QSFP_RCVD_CLK2_N AF13
QSFP_RCVD_CLK2_P — C265 0.1UF 16V — R_QSFP_RCVD_CLK2_P AR10 / AT9
FM_GXF_13C_SPARE_N — R1782 RES 0.5% 0.0625W 0402 — FM_GXF_13C_SPARE_N_R Y21
CLK_13C_IO_PLL — R263 RES 0.5% 0.0625W 0402 — CLK_13C_IO_PLL_R W18
VCCN_IO_GXF — R278 RES 10K 1% 0.0625W 0402
TP_13C_GXF_I_JTAG_TCK_DFX CL54 / TP_13C_GXF_I_JTAG_TDI_DFX AA20 / TP_13C_GXF_IO_JTAG_TDO_DFX Y19 / TP_13C_GXF_I_JTAG_TMS_DFX CU54 / U22_GXF_I_JTAG_TRST_N_DFX CN54
TP_13C_GXF_APROBE_0_ATB U20 / TP_13C_GXF_APROBE_1_ATB W20 / TP_13C_GXF_APROBE_Q0_CH3_FGT AE16 / TP_13C_GXF_APROBE_Q2_CH3_FGT AG16 / TP_13C_GXF_APROBE_Q3_CH3_FGT AP13 / TP_13C_GXF_APROBE2_Q3_CH3_FGT AR14 / TP_13C_GXF_APROBE1_COM_FHT BE14 / TP_13C_GXF_APROBE2_COM_FHT BC14
TEMPDIODE_13C_GXF_DN BG22 / TEMPDIODE_13C_GXF_DP BJ22
R20

14 OF 28
F - TILE BANK 13C
99D3CA

FHTR13C_RX_CH0N / CH0P / CH1N / CH1P / CH2N / CH2P / CH3N / CH3P
REFCLK_FHTR13C_CH0N / CH0P / CH1N / CH1P
FGTR13C_RX_Q0_CH0N / CH0P / CH1N / CH1P / CH2N / CH2P / CH3N / CH3P
REFCLK_FGTR13C_Q0_RX_CH0N / CH0P / CH1N / CH1P
FGTR13C_RX_Q1_CH0N / CH0P / CH1N / CH1P / CH2N / CH2P / CH3N / CH3P
REFCLK_FGTR13C_Q1_RX_CH2N / CH2P / CH3N / CH3P
FGTR13C_RX_Q2_CH0N / CH0P / CH1N / CH1P / CH2N / CH2P / CH3N / CH3P
REFCLK_FGTR13C_Q2_RX_CH4N / CH4P / CH5N / CH5P / CH8N / CH8P
FGTR13C_RX_Q3_CH0N / CH0P / CH1N / CH1P / CH2N / CH2P / CH3N / CH3P
REFCLK_FGTR13C_Q3_RX_CH6N / CH6P / CH7N / CH7P / CH9N / CH9P
I_PIN_PERST_N_13C_GXF
IO_PLL_REFCLK_13C_GXF
DNU_CL54 / DNU_AA20 / DNU_Y19 / DNU_CU54 / DNU_CN54
DNU_U20 / DNU_W20
APROBE_GXF_FGT13C_Q0_CH3 / APROBE_GXF_FGT13C_Q2_CH3 / APROBE_GXF_FGT13C_Q3_CH3 / APROBE2_GXF_FGT13C_Q3_CH3 / APROBE1_GXF_FHT13C / APROBE2_GXF_FHT13C
TEMPDIODE6N / TEMPDIODE6P
GND_R20

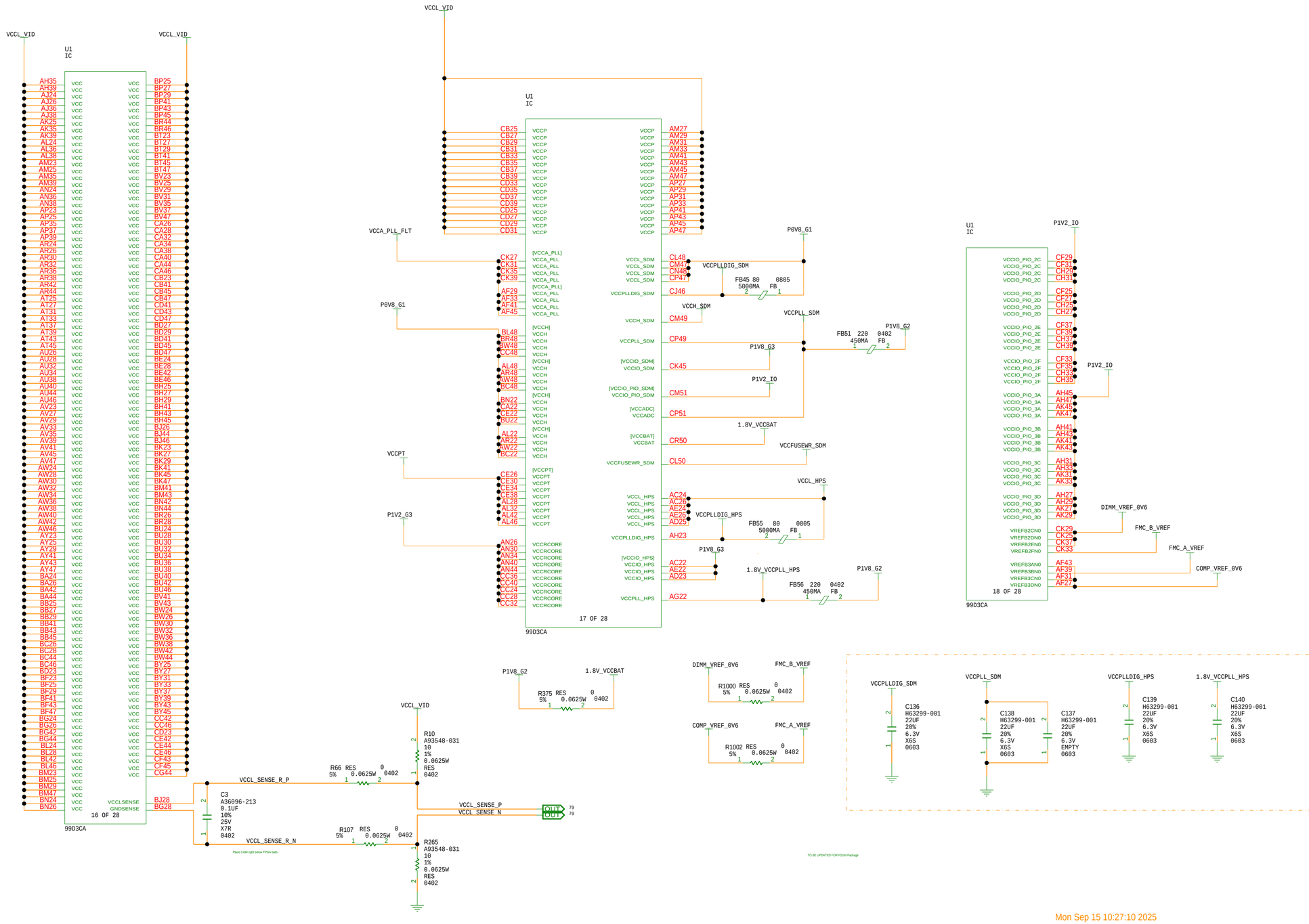
FHTR13C_TX_CH0N / CH0P / CH1N / CH1P / CH2N / CH2P / CH3N / CH3P
AP3 QSFPDD800_TX_N0 / AN2 QSFPDD800_TX_P0 / AV3 QSFPDD800_TX_N1 / AU2 QSFPDD800_TX_P1 / BA2 QSFPDD800_TX_N2 / BB3 QSFPDD800_TX_P2 / BE2 QSFPDD800_TX_N3 / BF3 QSFPDD800_TX_P3
RCOMP_N_FHT_13C_GXF — BF13 GXF_13C_FHT_RCOMP_N_0
RCOMP_P_FHT_13C_GXF — BE12 GXF_13C_FHT_RCOMP_P_0 — R323 RES 1.5K 0.1% 0.063W 0402
CAD Note: Place RCOMP res right below FPGA balls on the breakout vias.
FGTR13C_TX_Q0_CH0N / CH0P / CH1N / CH1P / CH2N / CH2P / CH3N / CH3P
B17 C_FGT_13C_TX0_DN — C558 220NF 6.3V FGT_DP_13C_TX0_DN
C16 C_FGT_13C_TX0_DP — C576 220NF 6.3V FGT_DP_13C_TX0_DP
F17 / G16 / E14 / D13
J14 SDI_TX_N / H13 SDI_TX_P
CAD Note: Transceiver Pairs may be swapped to optimize layout (Except LVDS CH0).
FGTR13C_TX_Q1_CH0N / CH0P / CH1N / CH1P / CH2N / CH2P / CH3N / CH3P
F11 C_FGT_DP_13C_TX0_DN — C915 220NF 6.3V FGT_DP_13C_TX0_DN
G10 C_FGT_DP_13C_TX0_DP — C919 220NF 6.3V FGT_DP_13C_TX0_DP
K11 C_FGT_DP_13C_TX1_DN — C933 220NF 6.3V FGT_DP_13C_TX1_DN
L10 C_FGT_DP_13C_TX1_DP — C934 220NF 6.3V FGT_DP_13C_TX1_DP
N8 C_FGT_DP_13C_TX2_DN — C274 220NF 6.3V FGT_DP_13C_TX2_DN
M7 C_FGT_DP_13C_TX2_DP — C275 220NF 6.3V FGT_DP_13C_TX2_DP
P11 C_FGT_DP_13C_TX3_DN — C935 220NF 6.3V FGT_DP_13C_TX3_DN
R10 C_FGT_DP_13C_TX3_DP — C937 220NF 6.3V FGT_DP_13C_TX3_DP
FGTR13C_TX_Q2_CH0N / CH0P / CH1N / CH1P / CH2N / CH2P / CH3N / CH3P
U8 QSFPDD0_TX_N0 / T7 QSFPDD0_TX_P0 / V11 QSFPDD0_TX_N1 / W10 QSFPDD0_TX_P1 / AA8 QSFPDD0_TX_N2 / Y7 QSFPDD0_TX_P2 / AB11 QSFPDD0_TX_N3 / AC10 QSFPDD0_TX_P3
FGTR13C_TX_Q3_CH0N / CH0P / CH1N / CH1P / CH2N / CH2P / CH3N / CH3P
AE8 QSFPDD0_TX_N4 / AD7 QSFPDD0_TX_P4 / AF11 QSFPDD0_TX_N5 / AG10 QSFPDD0_TX_P5 / AJ8 QSFPDD0_TX_N6 / AH7 QSFPDD0_TX_P6 / AN8 QSFPDD0_TX_N7 / AM7 QSFPDD0_TX_P7
CAD Note: Place RCOMP res right below FPGA balls on the breakout vias.
RCOMP_N_Q2_CH1_FGT_13C_GXF — AJ14 GXF_13C_FGT_RCOMP_N_0
RCOMP_P_Q2_CH1_FGT_13C_GXF — AH13 GXF_13C_FGT_RCOMP_P_0 — R325 RES 499 0.1% 0.0625W 0402
ENB_GXF_FHT13C / GND_CN52 / DNU_CH53 / DNU_CJ52 / DNU_CM53 / DNU_CJ54
AB19 U22_GXF_IO_DFX_0 / CN52 U22_GXF_IO_DFX_1 / CH53 TP_13C_GXF_IO_DFX_2 / CJ52 TP_13C_GXF_IO_DFX_3 / CM53 TP_13C_GXF_IO_DFX_4 / CJ54 TP_13C_GXF_IO_DFX_6
R329 EMPTY
VCCN_IO_GXF — R326 A93549-001 0 5% 0.0625W RES 0402 — R330 A93548-034 10K 1% 0.0625W EMPTY 0402
VCCN_IO_GXF — R1787 A93548-034 10K 1% 0.0625W RES 0402 — FM_GXF_13C_SPARE_N — C3051 H61346-001 10NF 10% 50V EMPTY 0402

P1V2_IO
R946 EMPTY 1K 0402 0.063W — R948 EMPTY 1K 0402 0.063W — ETH_REFCLK_156.25M_P — R955 EMPTY 1K 0402 0.063W
R948 EMPTY 1K 0402 0.063W — ETH_REFCLK_156.25M_N — R956 EMPTY 1K 0402 0.063W
R957 EMPTY 1K 0402 0.063W — R958 EMPTY 1K 0402 0.063W — CPRI_REFCLK_153.6M_P — R959 EMPTY 1K 0402 0.063W
R958 EMPTY 1K 0402 0.063W — CPRI_REFCLK_153.6M_N — R960 EMPTY 1K 0402 0.063W
R961 EMPTY 1K 0402 0.063W — CPRI_REFCLK_184.32M_P — R963 EMPTY 1K 0402 0.063W
R962 EMPTY 1K 0402 0.063W — CPRI_REFCLK_184.32M_N — R964 EMPTY 1K 0402 0.063W
R1801 EMPTY 1K 0402 0.063W — CLK_FHT_13C_P_0 — R1803 EMPTY 1K 0402 0.063W
R1802 EMPTY 1K 0402 0.063W — CLK_FHT_13C_N_0 — R1804 EMPTY 1K 0402 0.063W

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DEPARTMENT	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	SIZE C+	CODE 34649	DOCUMENT NUMBER 150-0330690-D1	REV D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 21 OF 97	

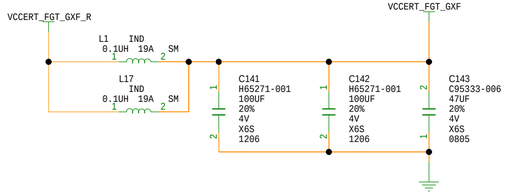
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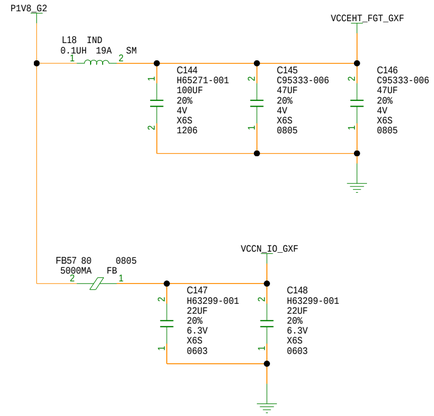


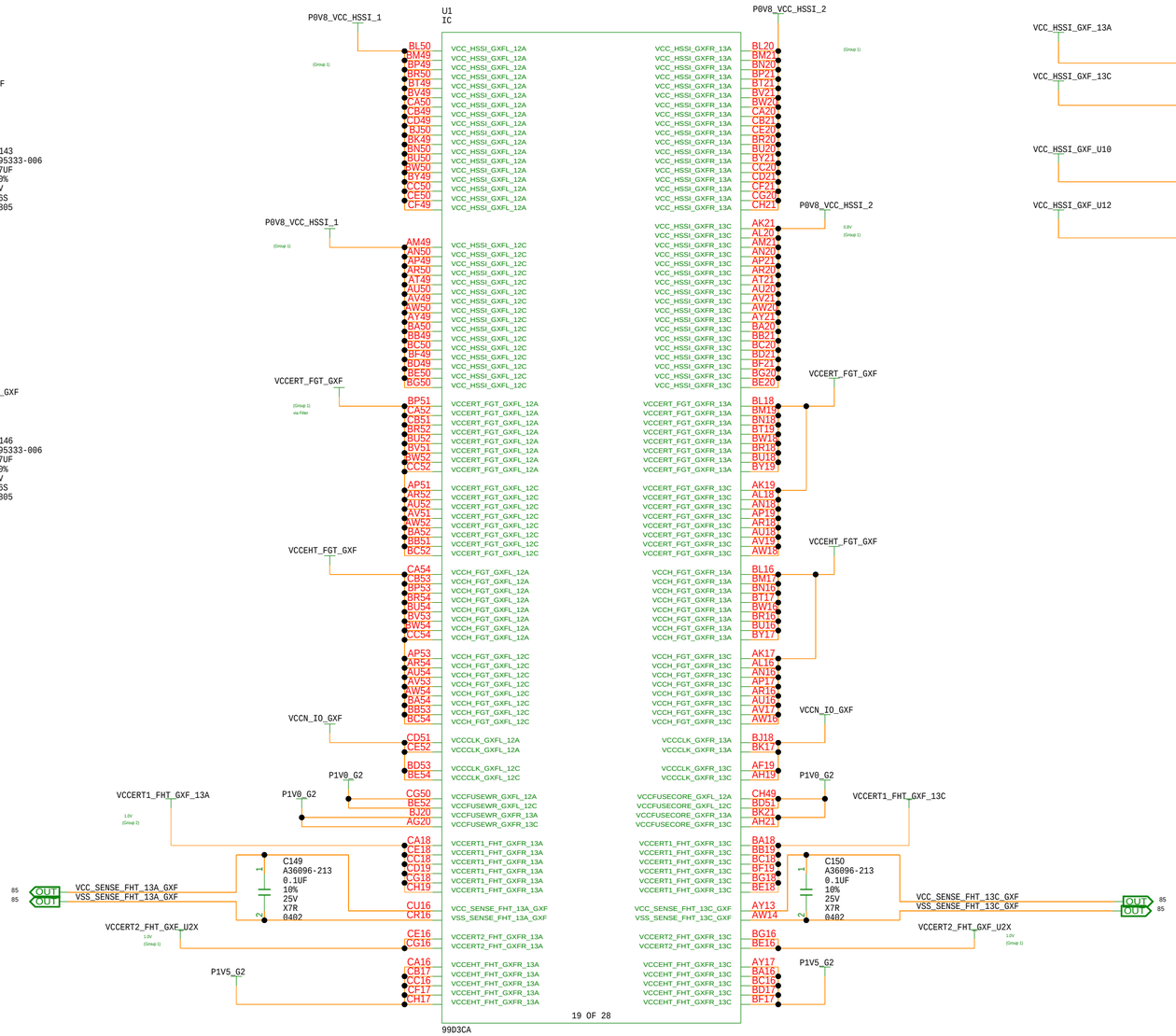
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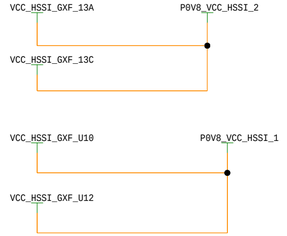
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	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 22 OF 97	

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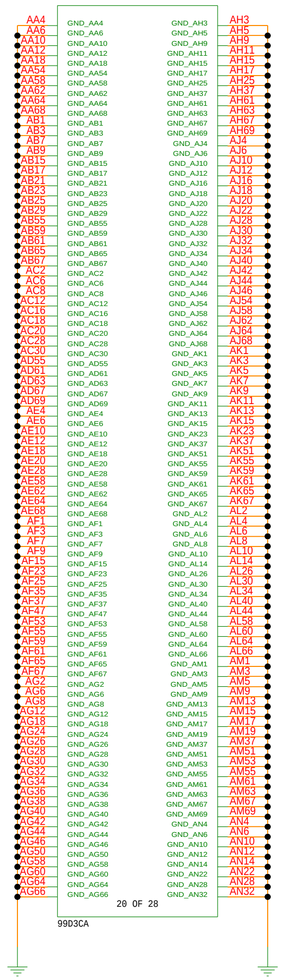
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DEPARTMENT	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	SIZE	CODE	DOCUMENT NUMBER	REV
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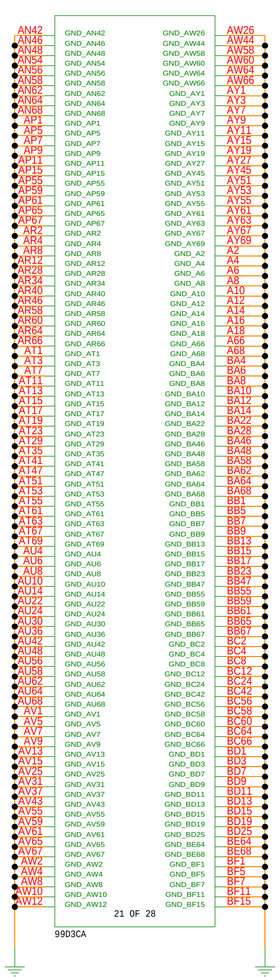
ALTERA CONFIDENTIAL

FPGA GND- I

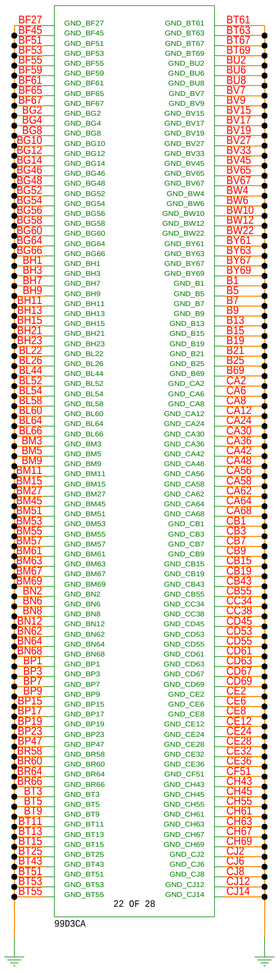
U1
IC



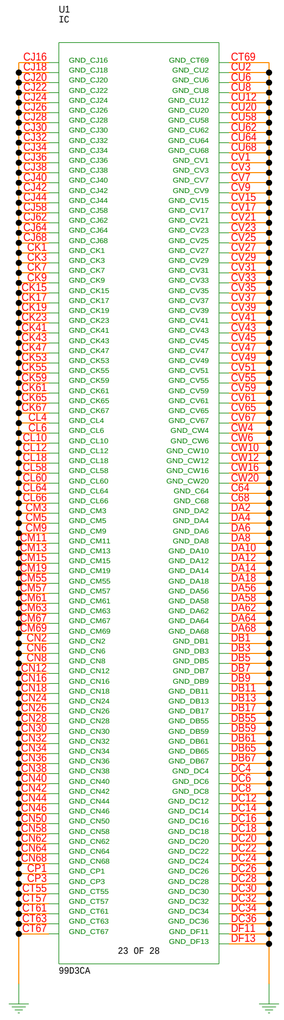
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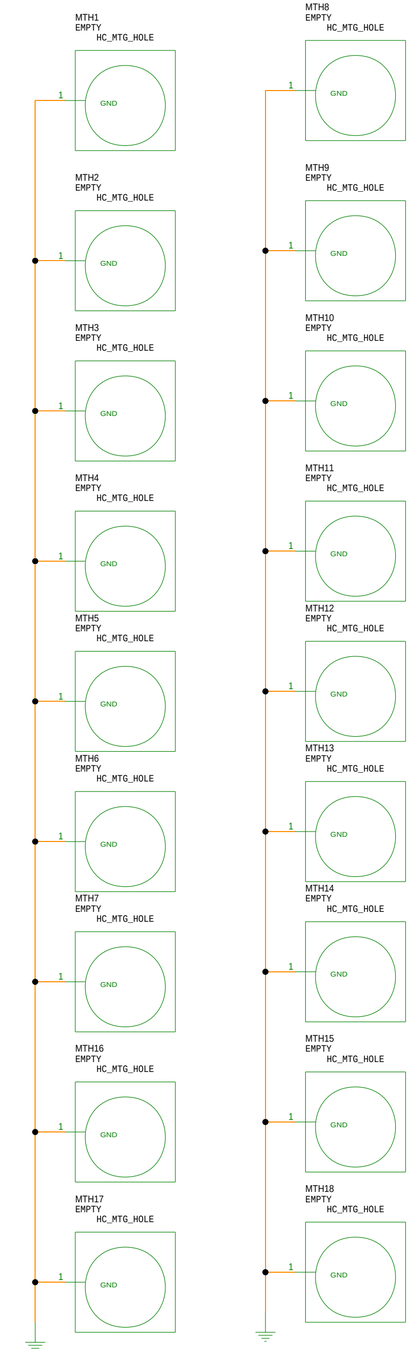


U1
IC



U1
IC





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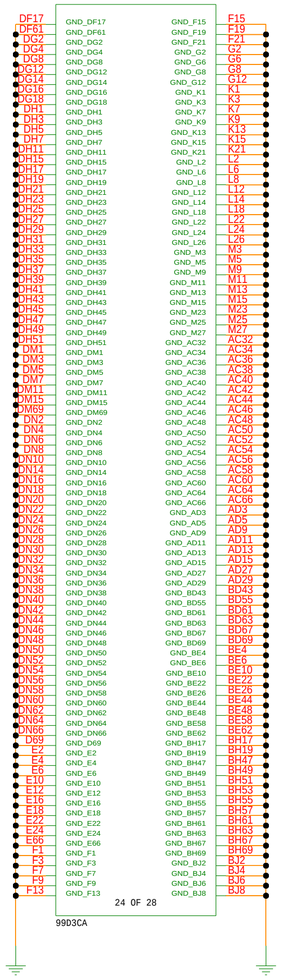
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		SCALE:	DO NOT SCALE DRAWING	SHEET: 24 OF 97

REV: D1

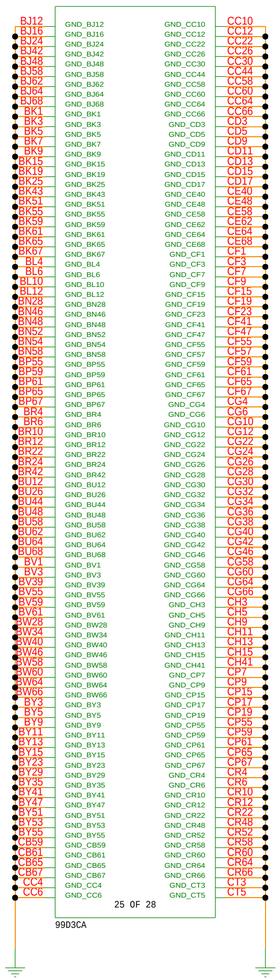
ALTERA CONFIDENTIAL

FPGA GND- II

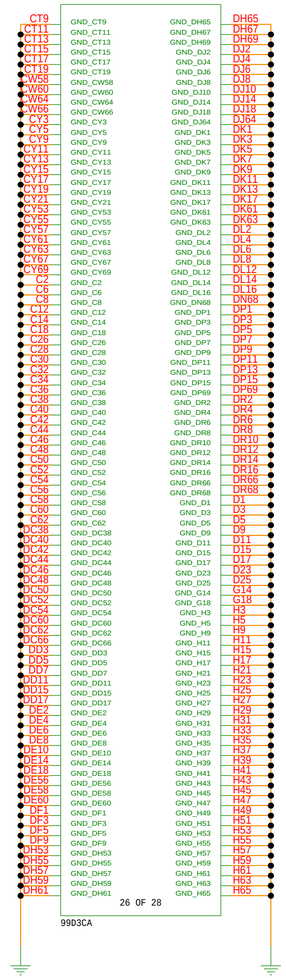
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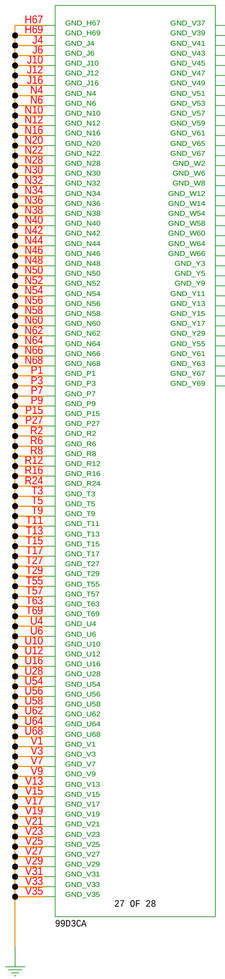
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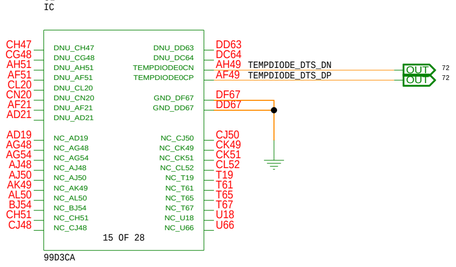
U1
IC



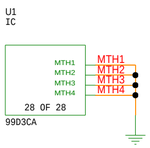
U1
IC



U1
IC



U1
IC



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DEPARTMENT	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	SIZE: C+	CODE: 34649	DOCUMENT NUMBER: 150-0330690-D1 — REV: D1
		SCALE:	DO NOT SCALE DRAWING	SHEET: 25 OF 97

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VCCL_VID

FPGA DECOUPLING - I

C2247 H94455-003 100UF 20% 2.5V X6S 0805
C2244 H94455-003 100UF 20% 2.5V X6S 0805
C2245 H94455-003 100UF 20% 2.5V X6S 0805
C2246 H94455-003 100UF 20% 2.5V X6S 0805
C2249 H94455-003 100UF 20% 2.5V X6S 0805
C2250 H94455-003 100UF 20% 2.5V X6S 0805
C2370 H94455-003 100UF 20% 2.5V X6S 0805
C2353 H94455-003 100UF 20% 2.5V X6S 0805
C2251 H94455-003 100UF 20% 2.5V X6S 0805
C2252 H94455-003 100UF 20% 2.5V X6S 0805

BOTTOM FPGA CAVITY

C2267 H80885-001 10UF 20% 6.3V X6S 0402
C2268 H80885-001 10UF 20% 6.3V X6S 0402
C2269 H80885-001 10UF 20% 6.3V X6S 0402
C2270 H80885-001 10UF 20% 6.3V X6S 0402
C2271 H80885-001 10UF 20% 6.3V X6S 0402
C2272 H80885-001 10UF 20% 6.3V X6S 0402
C2273 H80885-001 10UF 20% 6.3V X6S 0402
C2274 H80885-001 10UF 20% 6.3V X6S 0402
C2275 H80885-001 10UF 20% 6.3V X6S 0402
C2276 H80885-001 10UF 20% 6.3V X6S 0402
C2277 H80885-001 10UF 20% 6.3V X6S 0402
C2278 H80885-001 10UF 20% 6.3V X6S 0402
C2279 H80885-001 10UF 20% 6.3V X6S 0402
C2280 H80885-001 10UF 20% 6.3V X6S 0402
C2281 H80885-001 10UF 20% 6.3V X6S 0402
C2282 H80885-001 10UF 20% 6.3V X6S 0402
C2283 H80885-001 10UF 20% 6.3V X6S 0402
C2284 H80885-001 10UF 20% 6.3V X6S 0402
C2285 H80885-001 10UF 20% 6.3V X6S 0402
C2286 H80885-001 10UF 20% 6.3V X6S 0402

C2287 H80885-001 10UF 20% 6.3V X6S 0402
C2288 H80885-001 10UF 20% 6.3V X6S 0402
C2289 H80885-001 10UF 20% 6.3V X6S 0402
C2290 H80885-001 10UF 20% 6.3V X6S 0402
C2291 H80885-001 10UF 20% 6.3V X6S 0402
C2292 H80885-001 10UF 20% 6.3V X6S 0402
C2293 H80885-001 10UF 20% 6.3V X6S 0402
C2294 H80885-001 10UF 20% 6.3V X6S 0402
C2295 H80885-001 10UF 20% 6.3V X6S 0402
C2296 H80885-001 10UF 20% 6.3V X6S 0402
C2297 H80885-001 10UF 20% 6.3V X6S 0402
C2298 H80885-001 10UF 20% 6.3V X6S 0402
C2299 H80885-001 10UF 20% 6.3V X6S 0402
C2300 H80885-001 10UF 20% 6.3V X6S 0402
C2301 H80885-001 10UF 20% 6.3V X6S 0402
C2302 H80885-001 10UF 20% 6.3V X6S 0402
C2303 H80885-001 10UF 20% 6.3V X6S 0402
C2304 H80885-001 10UF 20% 6.3V X6S 0402
C2305 H80885-001 10UF 20% 6.3V X6S 0402
C2306 H80885-001 10UF 20% 6.3V X6S 0402

BOTTOM FPGA

C2307 H94455-003 100UF 20% 2.5V X6S 0805
C2308 H94455-003 100UF 20% 2.5V X6S 0805
C2309 H94455-003 100UF 20% 2.5V X6S 0805
C2310 H94455-003 100UF 20% 2.5V X6S 0805
C2311 H94455-003 100UF 20% 2.5V X6S 0805
C2337 H94455-003 100UF 20% 2.5V X6S 0805
C2243 H94455-003 100UF 20% 2.5V X6S 0805
C2248 H94455-003 100UF 20% 2.5V X6S 0805
C2354 C95333-006 47UF 20% 4V X6S 0805
C2355 C95333-006 47UF 20% 4V X6S 0805
C2352 C95333-006 47UF 20% 4V X6S 0805
C2371 C95333-006 47UF 20% 4V X6S 0805

PERIPHERAL SURROUNDING FPGA

P0V8_G1

16 VCCH BALLS

C2392 H63299-001 22UF 20% 6.3V X6S 0603
C2393 H63299-001 22UF 20% 6.3V X6S 0603
C2394 H63299-001 22UF 20% 6.3V X6S 0603
C2395 H63299-001 22UF 20% 6.3V X6S 0603
C2396 H63299-001 22UF 20% 6.3V X6S 0603
C2397 H63299-001 22UF 20% 6.3V X6S 0603
C2412 C95333-006 47UF 20% 4V X6S 0805
C2413 C95333-006 47UF 20% 4V X6S 0805

PERIPHERAL SURROUNDING FPGA
VCCH

C2427 C95333-006 47UF 20% 4V X6S 0805
C2426 C95333-006 47UF 20% 4V X6S 0805
C2414 C95333-006 47UF 20% 4V X6S 0805
C2415 C95333-006 47UF 20% 4V X6S 0805
C2416 C95333-006 47UF 20% 4V X6S 0805
C2417 C95333-006 47UF 20% 4V X6S 0805

PERIPHERAL SURROUNDING FPGA

P0V8_G1

C2411 H80885-001 10UF 20% 6.3V X6S 0402
C2420 H80885-001 10UF 20% 6.3V X6S 0402
C2419 H80885-001 10UF 20% 6.3V X6S 0402
C2425 H81594-001 1UF 10% 25V EMPTY 0402
C2423 H80885-001 10UF 20% 6.3V X6S 0402
C2424 H80885-001 10UF 20% 6.3V X6S 0402

VCCH
BOTTOM FPGA

P0V8_G1

C2418 H80885-001 10UF 20% 6.3V EMPTY 0402
C2421 H80885-001 10UF 20% 6.3V EMPTY 0402
C2422 H80885-001 10UF 20% 6.3V EMPTY 0402

VCCL_SDM
PERIPHERAL SURROUNDING FPGA

C2428 H81594-001 1UF 10% 25V EMPTY 0402
C2429 H81594-001 1UF 10% 25V X6S 0402

VCCL_SDM

BOTTOM FPGA

VCCPLLDIG_SDM

C2225 H81594-001 1UF 10% 25V X6S 0402
C2226 H81594-001 1UF 10% 25V EMPTY 0402

BOTTOM FPGA

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	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 26 OF 97	

ALTERA CONFIDENTIAL

FPGA DECOUPLING - II

P1V2_IO

C2467 H80885-001 10UF 20% 6.3V X6S 0402
C2456 K19409-001 4.7UF 20% 6.3V X6S 0402
C610 K19409-001 4.7UF 20% 6.3V X6S 0402
C1028 K19409-001 4.7UF 20% 6.3V X6S 0402
C2472 H80885-001 10UF 20% 6.3V X6S 0402
PER CHANNEL BOTTOM FPGA

C2470 K19409-001 4.7UF 20% 6.3V X6S 0402
C2457 K19409-001 4.7UF 20% 6.3V X6S 0402
C1040 K19409-001 4.7UF 20% 6.3V X6S 0402
C2473 H80885-001 10UF 20% 6.3V X6S 0402
PER CHANNEL BOTTOM FPGA

C2474 K19409-001 4.7UF 20% 6.3V X6S 0402
C1039 K19409-001 4.7UF 20% 6.3V X6S 0402
C2458 K19409-001 4.7UF 20% 6.3V X6S 0402
C2475 H80885-001 10UF 20% 6.3V X6S 0402
PER CHANNEL BOTTOM FPGA

C981 K19409-001 4.7UF 20% 6.3V X6S 0402
C2469 K19409-001 4.7UF 20% 6.3V X6S 0402
C2465 K19409-001 4.7UF 20% 6.3V X6S 0402
PER CHANNEL BOTTOM FPGA

P1V2_IO

C1041 H80885-001 10UF 20% 6.3V X6S 0402
C2462 K19409-001 4.7UF 20% 6.3V X6S 0402
C1043 K19409-001 4.7UF 20% 6.3V X6S 0402
C2466 K19409-001 4.7UF 20% 6.3V X6S 0402
C2461 H80885-001 10UF 20% 6.3V X6S 0402
PER CHANNEL BOTTOM FPGA

C2471 K19409-001 4.7UF 20% 6.3V X6S 0402
C1044 K19409-001 4.7UF 20% 6.3V X6S 0402
C2468 K19409-001 4.7UF 20% 6.3V X6S 0402
C2463 H80885-001 10UF 20% 6.3V X6S 0402
PER CHANNEL BOTTOM FPGA

C1042 K19409-001 4.7UF 20% 6.3V X6S 0402
C2464 K19409-001 4.7UF 20% 6.3V X6S 0402
C1027 K19409-001 4.7UF 20% 6.3V X6S 0402
C611 H80885-001 10UF 20% 6.3V X6S 0402
PER CHANNEL BOTTOM FPGA

C2454 K19409-001 4.7UF 20% 6.3V X6S 0402
C2476 K19409-001 4.7UF 20% 6.3V X6S 0402
C854 K19409-001 4.7UF 20% 6.3V X6S 0402
PER CHANNEL BOTTOM FPGA

DIMM_VREF_0V6

C1030 A36096-030 0.1UF 10% 16V X7R 0402
C1029 A36096-030 0.1UF 10% 16V X7R 0402
C1031 A36096-030 0.1UF 10% 16V X7R 0402
C1032 A36096-030 0.1UF 10% 16V X7R 0402

COMP_VREF_0V6

C1033 A36096-030 0.1UF 10% 16V X7R 0402
C1034 A36096-030 0.1UF 10% 16V X7R 0402
C1035 A36096-030 0.1UF 10% 16V X7R 0402
C1036 A36096-030 0.1UF 10% 16V X7R 0402

P1V8_G2

L13 IND 19A SM 0.1UH

VCCPI

C3049 K33438-001 220UF 20% 4V EMPTY 1206
C2158 C95333-006 47UF 20% 4V X6S 0805
C3050 H65271-001 100UF 20% 4V X6S 1206
C2160 C95333-006 47UF 20% 4V X6S 0805
C2161 H63299-001 22UF 20% 6.3V X6S 0603
PERIPHERY

C2165 H80885-001 10UF 20% 6.3V X6S 0402
C2164 K19409-001 4.7UF 20% 6.3V X6S 0402
C2163 K19409-001 4.7UF 20% 6.3V X6S 0402
C2162 K19409-001 4.7UF 20% 6.3V X6S 0402
C2166 H81594-001 1UF 10% 25V X6S 0402
C2167 H81594-001 1UF 10% 25V X6S 0402
BOTTOM FPGA

P1V2_IO

C2665 H81594-001 1UF 10% 25V X6S 0402
C2664 H81594-001 1UF 10% 25V EMPTY 0402
VCCIO_PIO_SDM BOTTOM FPGA

1.8V_VCCPLL_HPS

C2145 H81594-001 1UF 10% 25V X6S 0402
BOTTOM FPGA

1.8V_VCCBAT

C824 H81594-001 1UF 10% 25V X6S 0402
BOTTOM FPGA

VCCPLLDIG_HPS

C2142 H81594-001 1UF 10% 25V X6S 0402
C2141 H81594-001 1UF 10% 25V EMPTY 0402
BOTTOM FPGA

P1V2_G3

10 VCCRCORE BALLS

C2168 H80885-001 10UF 20% 6.3V X6S 0402
C2169 K19409-001 4.7UF 20% 6.3V X6S 0402
C2170 K19409-001 4.7UF 20% 6.3V X6S 0402
C2171 K19409-001 4.7UF 20% 6.3V X6S 0402
VCCRCOREBOTTOM FPGA

P1V8_G3

C2154 H80885-001 10UF 20% 6.3V X6S 0402
C2155 H81594-001 1UF 10% 25V EMPTY 0402
C2153 H81594-001 1UF 10% 25V X6S 0402
VCCIO_HPS BOTTOM FPGA

C2152 H80885-001 10UF 20% 6.3V X6S 0402
C2156 H81594-001 1UF 10% 25V X6S 0402
C2157 H81594-001 1UF 10% 25V EMPTY 0402
VCCIO_SDM BOTTOM FPGA

VCCFUSEWR_SDM

C2227 K19409-001 4.7UF 20% 6.3V X6S 0402
C2228 H81594-001 1UF 10% 25V X6S 0402
BOTTOM FPGA

P1V2_G3

L14 IND 19A SM 0.1UH

VCCA_PLL_FLT

C2204 H63299-001 22UF 20% 6.3V X6S 0603
C1037 H65271-001 100UF 20% 4V X6S 1206
C1026 C95333-006 47UF 20% 4V X6S 0805
C2205 H80885-001 10UF 20% 6.3V X6S 0402
C2206 H80885-001 10UF 20% 6.3V X6S 0402
C2207 H80885-001 10UF 20% 6.3V X6S 0402
C2208 H63299-001 22UF 20% 6.3V X6S 0603
C2209 H63299-001 22UF 20% 6.3V X6S 0603
VCCA_PLL BOTTOM FPGA

C2210 H81594-001 1UF 10% 25V X6S 0402
C2211 H81594-001 1UF 10% 25V X6S 0402
C2212 H81594-001 1UF 10% 25V X6S 0402
EXTRA PER PDG

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		SCALE:	DO NOT SCALE DRAWING	SHEET	27 OF 97

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FPGA DECOUPLING - III

VCCL_VID — R1649 0% RES 0.75W 1206 — VCCL_HPS
P0V8_G1 — R1647 0% EMPTY 0.75W 1206
VCCH_SDM — R431 0% EMPTY 0.75W 1206

C2130 K19409-001 4.7UF 20% 6.3V X6S 0402
C2139 K19409-001 4.7UF 20% 6.3V X6S 0402
C2138 K19409-001 4.7UF 20% 6.3V X6S 0402
C2137 K19409-001 4.7UF 20% 6.3V EMPTY 0402
C2140 K19409-001 4.7UF 20% 6.3V EMPTY 0402
BOTTOM FPGA

C2143 H94455-003 100UF 20% 2.5V EMPTY 0805
C2144 H63299-001 22UF 20% 6.3V X6S 0603
PERIPHERAL SURROUNDING FPGA

VCCH_SDM
C2214 H81594-001 1UF 10% 25V X6S 0402
C2215 H81594-001 1UF 10% 25V EMPTY 0402
BOTTOM FPGA

P1V0_G2
C2444 H80885-001 10UF 20% 6.3V X6S 0402
C2445 H80885-001 10UF 20% 6.3V X6S 0402
C2446 H80885-001 10UF 20% 6.3V X6S 0402
C374 H80885-001 10UF 20% 6.3V X6S 0402
VCCFUSEWR_FGT_U*X
BOTTOM FPGA

P1V0_G2
C379 H80885-001 10UF 20% 6.3V X6S 0402
C384 H80885-001 10UF 20% 6.3V X6S 0402
C385 H80885-001 10UF 20% 6.3V X6S 0402
C386 H80885-001 10UF 20% 6.3V X6S 0402
VCCFUSECORE_FGT_U*X
BOTTOM FPGA

VCCPLL_SDM
VCCADC_SDM
C135 H81594-001 1UF 10% 25V X6S 0402
C134 H81594-001 1UF 10% 25V X6S 0402
BOTTOM FPGA

P0V8_VCC_HSSI_1
C711 C95333-006 47UF 20% 4V X6S 0805
C712 C95333-006 47UF 20% 4V X6S 0805
C713 H80885-001 10UF 20% 6.3V X6S 0402
C714 H80885-001 10UF 20% 6.3V X6S 0402
BOTTOM FPGA

P0V8_VCC_HSSI_2
C715 C95333-006 47UF 20% 4V X6S 0805
C716 C95333-006 47UF 20% 4V X6S 0805
C717 H80885-001 10UF 20% 6.3V X6S 0402
C718 H80885-001 10UF 20% 6.3V X6S 0402
BOTTOM FPGA

P0V8_VCC_HSSI_1
C719 C95333-006 47UF 20% 4V X6S 0805
C720 C95333-006 47UF 20% 4V X6S 0805
C721 H80885-001 10UF 20% 6.3V X6S 0402
C722 H80885-001 10UF 20% 6.3V X6S 0402
BOTTOM FPGA

P0V8_VCC_HSSI_2
C723 C95333-006 47UF 20% 4V X6S 0805
C724 C95333-006 47UF 20% 4V X6S 0805
C725 H80885-001 10UF 20% 6.3V X6S 0402
C726 H80885-001 10UF 20% 6.3V X6S 0402
BOTTOM FPGA

VCCERT_FGT_GXF
C2229 C95333-006 47UF 20% 4V X6S 0805
C2230 C95333-006 47UF 20% 4V X6S 0805
C2231 C95333-006 47UF 20% 4V X6S 0805
C2232 C95333-006 47UF 20% 4V X6S 0805
C2233 H80885-001 10UF 20% 6.3V X6S 0402
C2234 H80885-001 10UF 20% 6.3V X6S 0402
BOTTOM FPGA

VCCERT_FGT_GXF
C727 C95333-006 47UF 20% 4V X6S 0805
C728 C95333-006 47UF 20% 4V X6S 0805
C729 C95333-006 47UF 20% 4V X6S 0805
C730 C95333-006 47UF 20% 4V X6S 0805
C731 H80885-001 10UF 20% 6.3V X6S 0402
C732 H80885-001 10UF 20% 6.3V X6S 0402
PERIPHERAL SURROUNDING FPGA
BOTTOM FPGA

VCCERT_FGT_GXF
C733 C95333-006 47UF 20% 4V X6S 0805
C734 C95333-006 47UF 20% 4V X6S 0805
C735 C95333-006 47UF 20% 4V X6S 0805
C736 C95333-006 47UF 20% 4V X6S 0805
C737 H80885-001 10UF 20% 6.3V X6S 0402
C738 H80885-001 10UF 20% 6.3V X6S 0402
PERIPHERAL SURROUNDING FPGA
BOTTOM FPGA

VCCERT_FGT_GXF
C739 C95333-006 47UF 20% 4V X6S 0805
C740 C95333-006 47UF 20% 4V X6S 0805
C741 C95333-006 47UF 20% 4V X6S 0805
C742 C95333-006 47UF 20% 4V X6S 0805
C743 H80885-001 10UF 20% 6.3V X6S 0402
C744 H80885-001 10UF 20% 6.3V X6S 0402
PERIPHERAL SURROUNDING FPGA
BOTTOM FPGA

VCCERT2_FHT_GXF_U2X
C2237 H80885-001 10UF 20% 6.3V X6S 0402
BOTTOM FPGA

VCCERT2_FHT_GXF_U2X
C2442 H80885-001 10UF 20% 6.3V X6S 0402
BOTTOM FPGA

VCCERT1_FHT_GXF_13A
C2235 H80885-001 10UF 20% 6.3V X6S 0402
C2236 H80885-001 10UF 20% 6.3V X6S 0402
BOTTOM FPGA

VCCERT1_FHT_GXF_13C
C2434 H80885-001 10UF 20% 6.3V X6S 0402
C2435 H80885-001 10UF 20% 6.3V X6S 0402
BOTTOM FPGA

P1V5_G2
C579 H80885-001 10UF 20% 6.3V X6S 0402
C580 H80885-001 10UF 20% 6.3V X6S 0402
VCCEHT_FHT_GXF_U22
BOTTOM FPGA

P1V5_G2
C2241 H80885-001 10UF 20% 6.3V X6S 0402
C2242 H80885-001 10UF 20% 6.3V X6S 0402
VCCEHT_FHT_GXF_U20
BOTTOM FPGA

VCCEHT_FGT_GXF U10
C612 H80885-001 10UF 20% 6.3V X6S 0402
C613 H80885-001 10UF 20% 6.3V X6S 0402

VCCEHT_FGT_GXF U20
C745 H80885-001 10UF 20% 6.3V X6S 0402
C746 H80885-001 10UF 20% 6.3V X6S 0402

VCCEHT_FGT_GXF 12C
C747 H80885-001 10UF 20% 6.3V X6S 0402
C748 H80885-001 10UF 20% 6.3V X6S 0402

VCCEHT_FGT_GXF U22
C749 H80885-001 10UF 20% 6.3V X6S 0402
C750 H80885-001 10UF 20% 6.3V X6S 0402

VCCN_IO_GXF U10
C751 H80885-001 10UF 20% 6.3V X6S 0402
C752 H80885-001 10UF 20% 6.3V X6S 0402
BOTTOM FPGA

VCCN_IO_GXF U20
C753 H80885-001 10UF 20% 6.3V X6S 0402
C754 H80885-001 10UF 20% 6.3V X6S 0402

VCCN_IO_GXF 12C
C755 H80885-001 10UF 20% 6.3V X6S 0402
C756 H80885-001 10UF 20% 6.3V X6S 0402

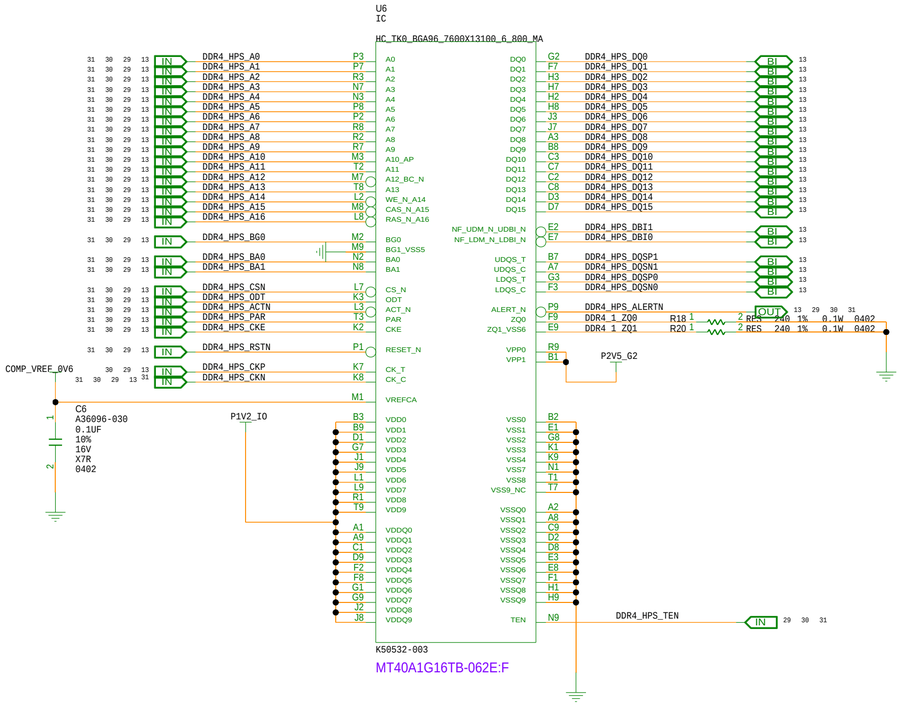
VCCN_IO_GXF U22
C757 H80885-001 10UF 20% 6.3V X6S 0402
C758 H80885-001 10UF 20% 6.3V X6S 0402

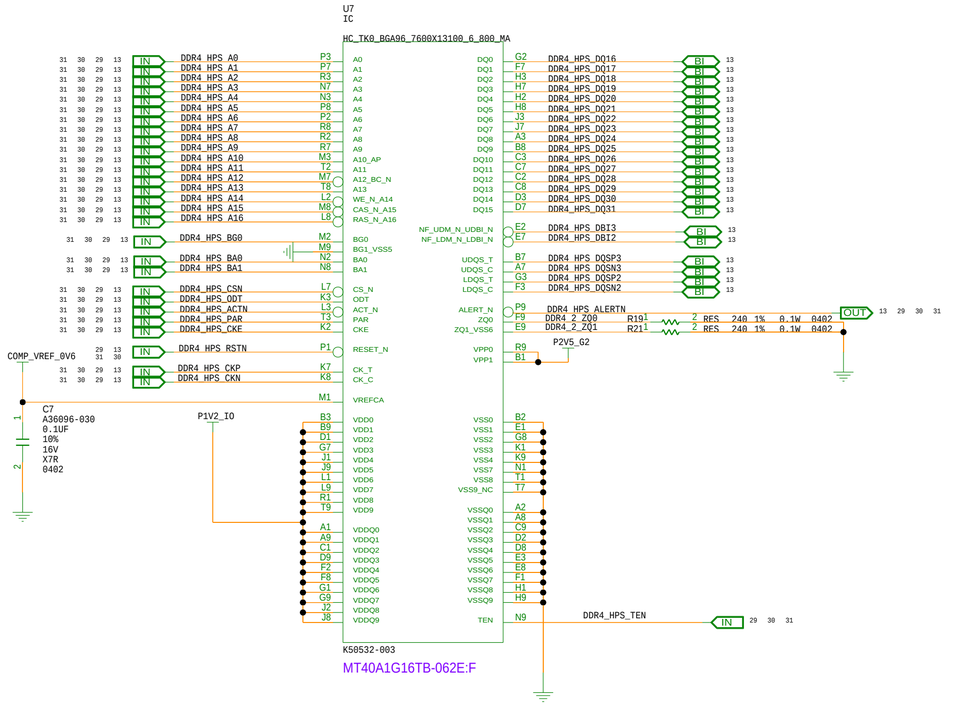
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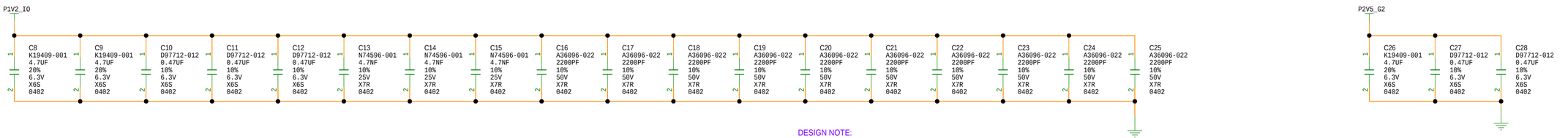
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	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 28 OF 97	

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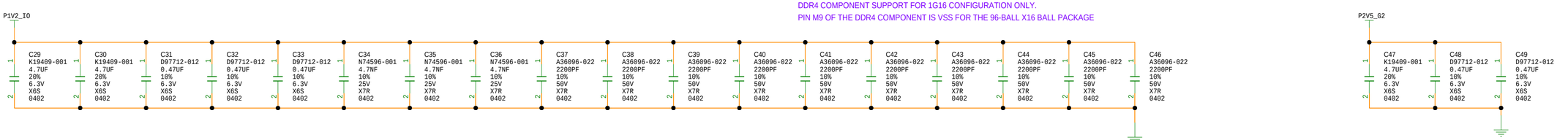
DDR4 COMPONENT-I







DESIGN NOTE:
DDR4 COMPONENT SUPPORT FOR 1G16 CONFIGURATION ONLY.
PIN M9 OF THE DDR4 COMPONENT IS VSS FOR THE 96-BALL X16 BALL PACKAGE

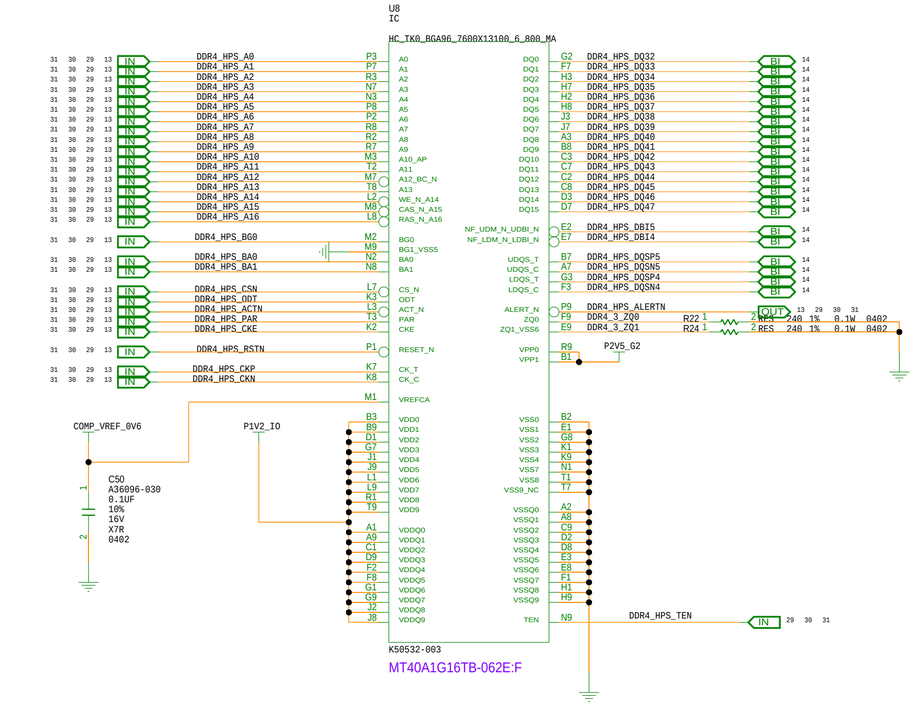


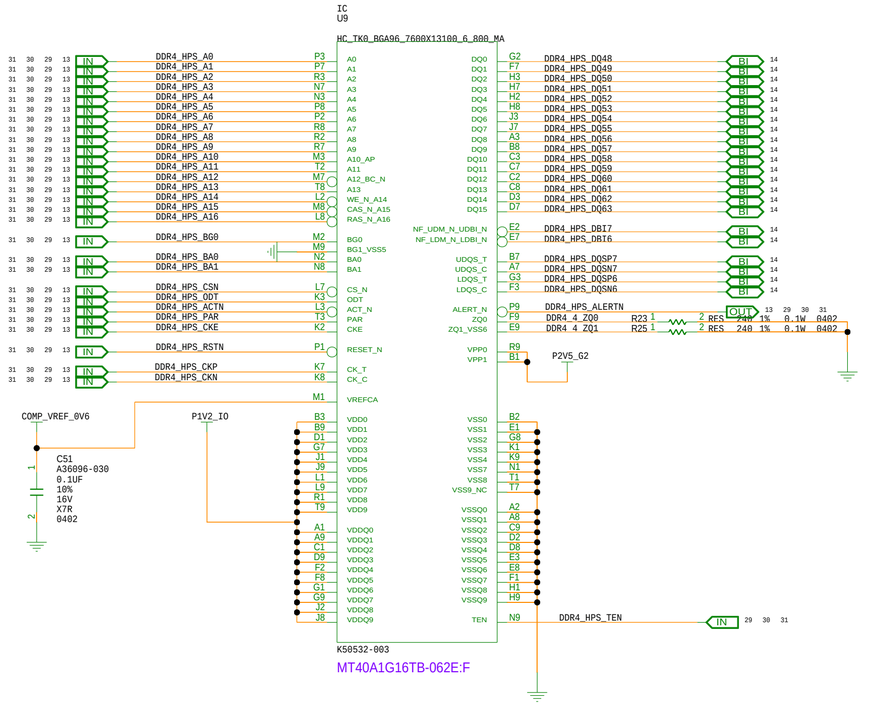
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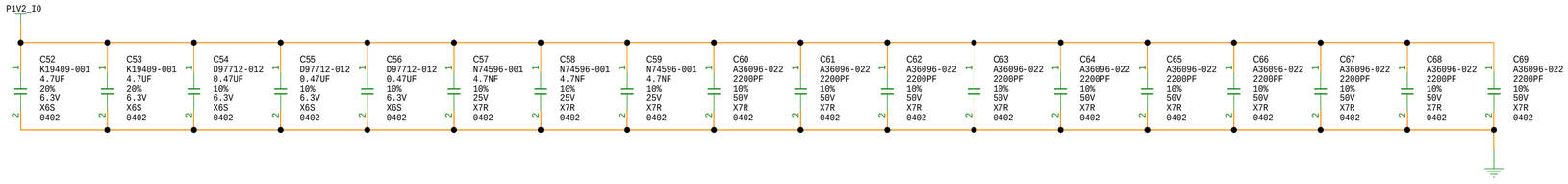
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	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 29 OF 97	

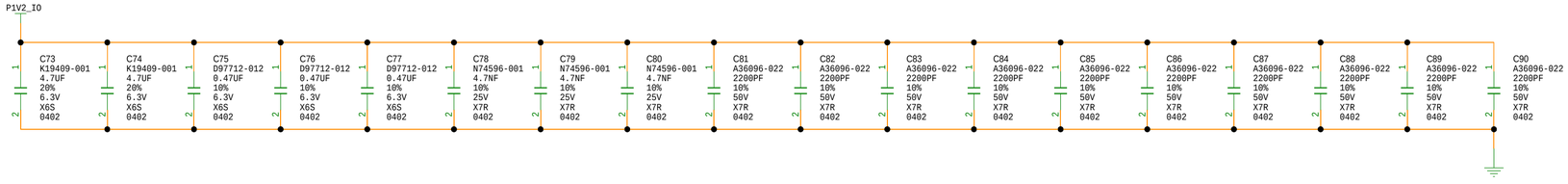
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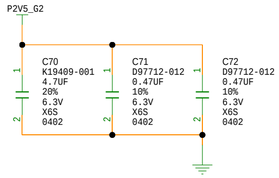
DDR4 COMPONENT-II

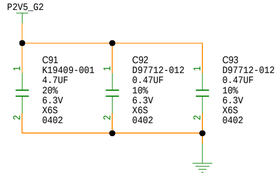












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		SCALE:	DO NOT SCALE DRAWING	SHEET 30 OF 97	

DDR4 COMPONENT-III

U10
IC
HC_TK0_BGA96_7600X13100_6_800_MA
HC_DDR4_96P

DDR4_HPS_A0 P3 A0
DDR4_HPS_A1 P7 A1
DDR4_HPS_A2 R3 A2
DDR4_HPS_A3 N7 A3
DDR4_HPS_A4 N3 A4
DDR4_HPS_A5 P8 A5
DDR4_HPS_A6 P2 A6
DDR4_HPS_A7 R8 A7
DDR4_HPS_A8 R2 A8
DDR4_HPS_A9 R7 A9
DDR4_HPS_A10 M3 A10_AP
DDR4_HPS_A11 T2 A11
DDR4_HPS_A12 M7 A12_BC_N
DDR4_HPS_A13 T8 A13
DDR4_HPS_A14 L2 WE_N_A14
DDR4_HPS_A15 M8 CAS_N_A15
DDR4_HPS_A16 L8 RAS_N_A16
DDR4_HPS_BG0 M2 BG0
M9 BG1_VSS5
DDR4_HPS_BA0 N2 BA0
DDR4_HPS_BA1 N8 BA1
DDR4_HPS_CSN L7 CS_N
DDR4_HPS_ODT K3 ODT
DDR4_HPS_ACTN L3 ACT_N
DDR4_HPS_PAR T3 PAR
DDR4_HPS_CKE K2 CKE
DDR4_HPS_RSTN P1 RESET_N
DDR4_HPS_CKP K7 CK_T
DDR4_HPS_CKN K8 CK_C
M1 VREFCA

B3 VDD0
B9 VDD1
D1 VDD2
G7 VDD3
J1 VDD4
J9 VDD5
L1 VDD6
L9 VDD7
R1 VDD8
T9 VDD9
A1 VDDQ0
A9 VDDQ1
C1 VDDQ2
D9 VDDQ3
F2 VDDQ4
F8 VDDQ5
G1 VDDQ6
G9 VDDQ7
J2 VDDQ8
J8 VDDQ9

B2 VSS0
E1 VSS1
G8 VSS2
K1 VSS3
K9 VSS4
N1 VSS7
T1 VSS8
T7 VSS9_NC
A2 VSSQ0
A8 VSSQ1
C9 VSSQ2
D2 VSSQ3
D8 VSSQ4
E3 VSSQ5
E8 VSSQ6
F1 VSSQ7
H1 VSSQ8
H9 VSSQ9
N9 TEN

G2 DDR4_HPS_DQ64 DQ0
F7 DDR4_HPS_DQ65 DQ1
H3 DDR4_HPS_DQ66 DQ2
H7 DDR4_HPS_DQ67 DQ3
H2 DDR4_HPS_DQ68 DQ4
H8 DDR4_HPS_DQ69 DQ5
J3 DDR4_HPS_DQ70 DQ6
J7 DDR4_HPS_DQ71 DQ7
A3 DQ8
B8 DQ9
C3 DQ10
C7 DQ11
C2 DQ12
C8 DQ13
D3 DQ14
D7 DQ15
E2 NF_UDM_N_UDBI_N
E7 DDR4_HPS_DBI8 NF_LDM_N_LDBI_N
B7 UDQS_T
A7 UDQS_C
G3 DDR4_HPS_DQSP8 LDQS_T
F3 DDR4_HPS_DQSN8 LDQS_C
P9 DDR4_HPS_ALERTN ALERT_N
F9 DDR4_5_ZQ0 ZQ0
E9 DDR4_5_ZQ1 ZQ1_VSS6
R9 VPP0
B1 VPP1

K50532-003
MT40A1G16TB-062E:F

R26 R27 R28 R29 R30 R31 R32 R33 R34 R35 R36 RES 1K 1% 0.063W 0402
R37 R38 RES 240 1% 0.1W 0402
P1V2_IO
P2V5_G2
DDR4_HPS_TEN
R445 A36092-043 100 1% 0.0625W RES 0402

COMP_VREF_0V6
C94 A36096-030 0.1UF 10% 16V X7R 0402

COMP_VTT_0V6
C2682 C2683 C2684 C2685 D97712-023 1UF 10% 16V X6S 0402
C1497 C1498 C1499 H63299-001 22UF 20% 6.3V X6S 0603
C1500 H63299-001 22UF 20% 6.3V X6S 0603

COMP_VTT_0V6
DDR4_HPS_A0 DDR4_HPS_A1 DDR4_HPS_A2 DDR4_HPS_A3 R1009 R1010 R1011 R1012 RES 36.1% 0.05W 0201
DDR4_HPS_A4 DDR4_HPS_A5 DDR4_HPS_A6 DDR4_HPS_A7 R1013 R1014 R1015 R1016 RES 36.1% 0.05W 0201
DDR4_HPS_A8 DDR4_HPS_A9 DDR4_HPS_A10 DDR4_HPS_A11 R1017 R1018 R1019 R1020 RES 36.1% 0.05W 0201
COMP_VTT_0V6
DDR4_HPS_A12 DDR4_HPS_A13 DDR4_HPS_A14 DDR4_HPS_A15 R1021 R1022 R1023 R1024 RES 36.1% 0.05W 0201
DDR4_HPS_A16 DDR4_HPS_BG0 R1025 R1026 RES 36.1% 0.05W 0201
DDR4_HPS_BA0 R1028 RES 36.1% 0.05W 0201
DDR4_HPS_BA1 DDR4_HPS_ODT DDR4_HPS_ACTN DDR4_HPS_PAR R1029 R1030 R1031 R1032 RES 36.1% 0.05W 0201
DDR4_HPS_CSN R1033 RES 36.1% 0.05W 0201
DDR4_HPS_CKE R1034 EMPTY 36.1% 0.05W 0201
R41 RES 1K 1% 0.063W 0402

P1V2_IO
DDR4_HPS_CKP R42 RES 36.1% 0.05W 0201 C95 10NF 50V
DDR4_HPS_CKN R43 RES 36.1% 0.05W 0201
P1V2_IO
DDR4_HPS_ALERTN R44 RES 49.9 1% 0.0625W 0402
DDR4_HPS_RSTN R45 EMPTY 1K 1% 0.063W 0402
R46 RES 1K 1% 0.063W 0402

P1V2_IO
C97 C98 C99 C100 C101 K19409-001 4.7UF 20% 6.3V X6S 0402
C102 C103 C104 N74596-001 4.7NF 10% 25V X7R 0402
C105 C106 C107 C108 C109 C110 C111 C112 C113 A36096-022 2200PF 10% 50V X7R 0402
C114 A36096-022 2200PF 10% 50V X7R 0402

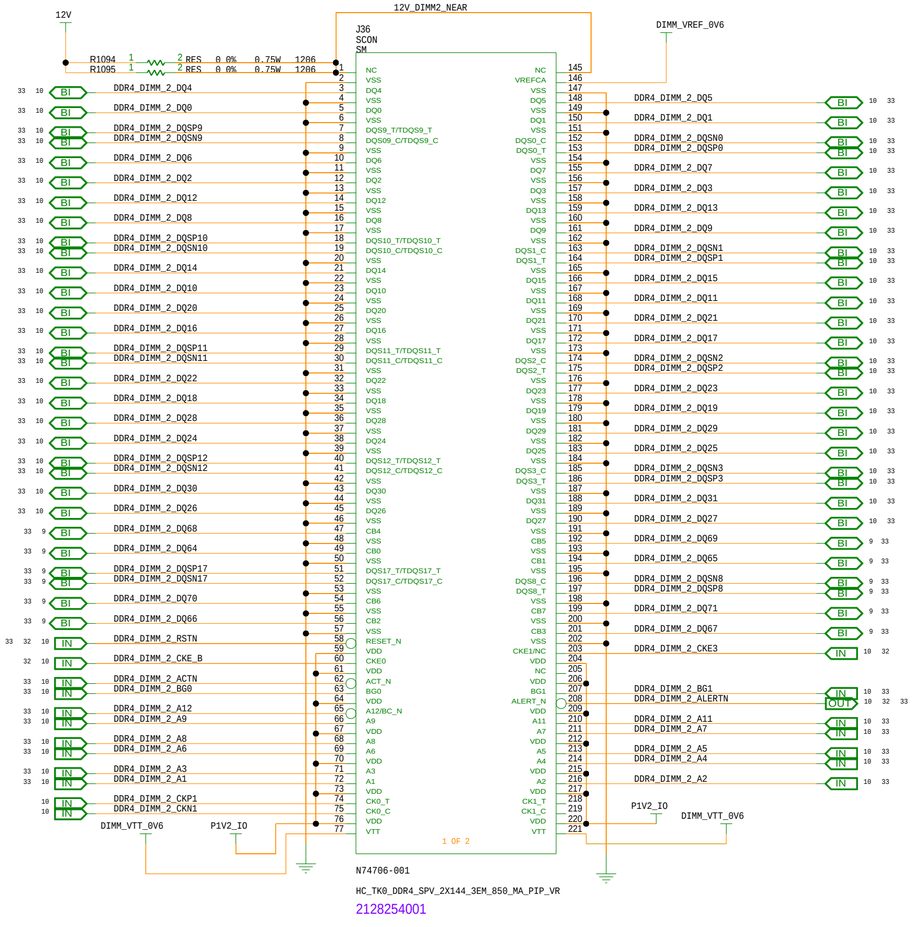
P2V5_G2
C115 K19409-001 4.7UF 20% 6.3V X6S 0402
C116 D97712-012 0.47UF 10% 6.3V X6S 0402
C117 D97712-012 0.47UF 10% 6.3V X6S 0402

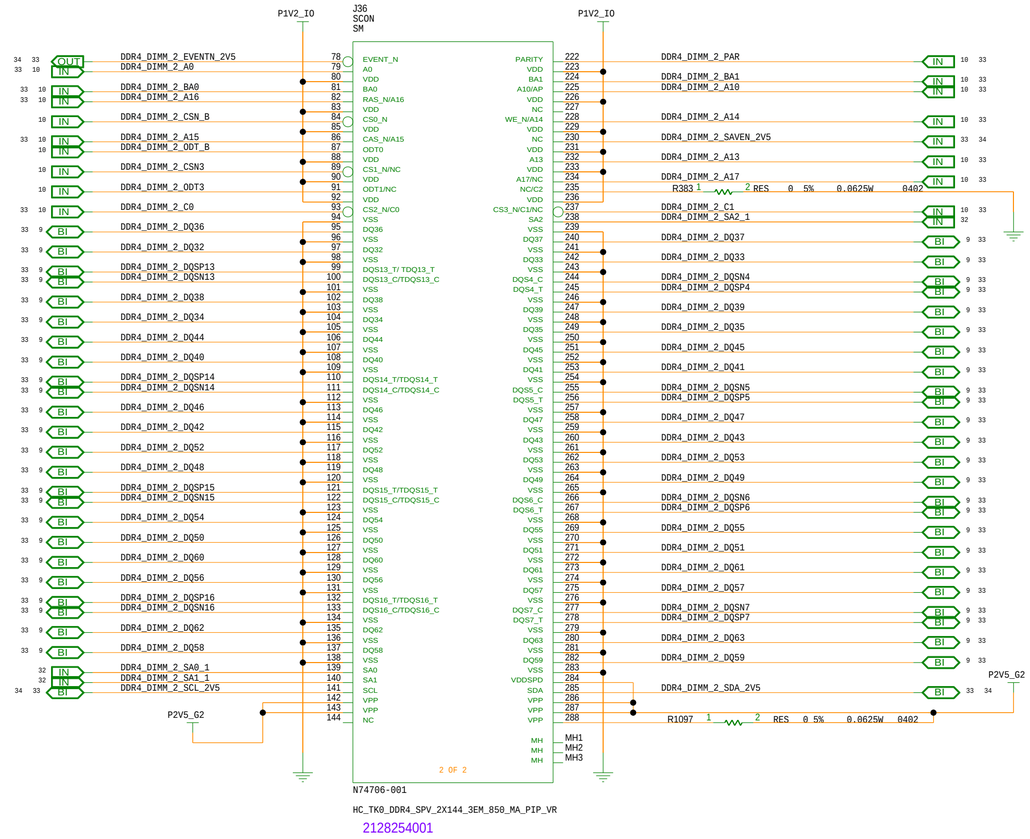
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		C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET	31 OF 97

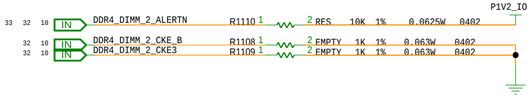
ALTERA CONFIDENTIAL

2DPC DIMM-I











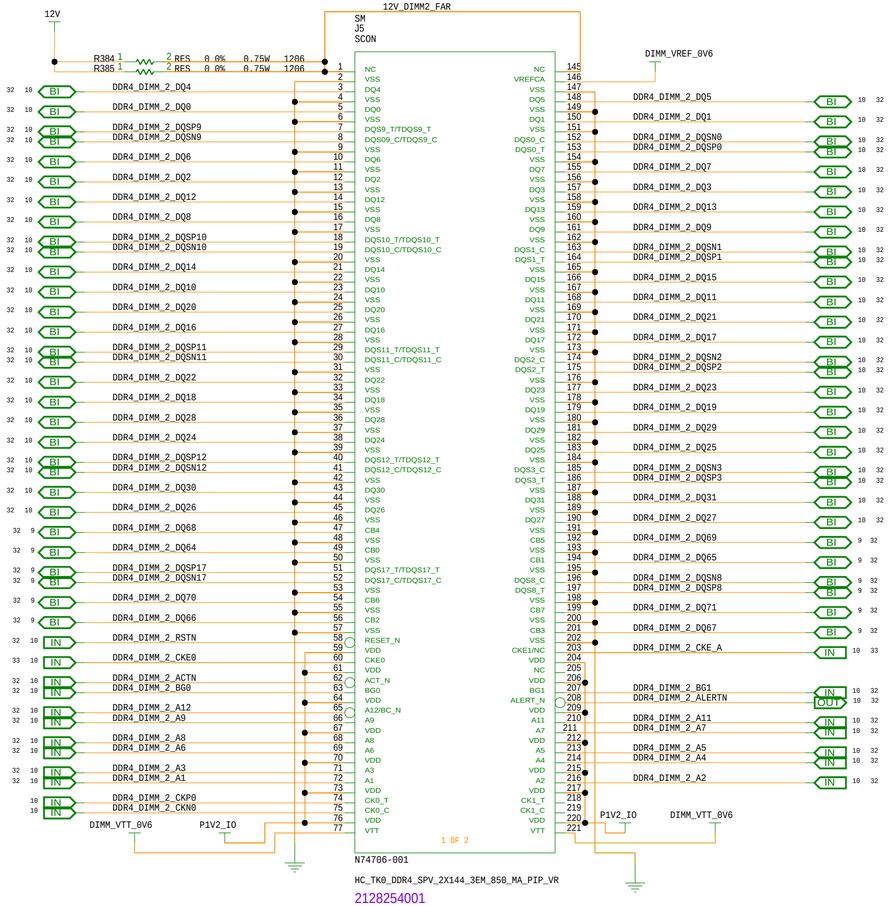
SIMULATION BASED ON
MTA18ASF2G72PZ-3G2E2 16GB SRX4 DDR4-3200
MTA36ASF4G72PZ-2G6E1 32GB DRX4 DDR4-2666

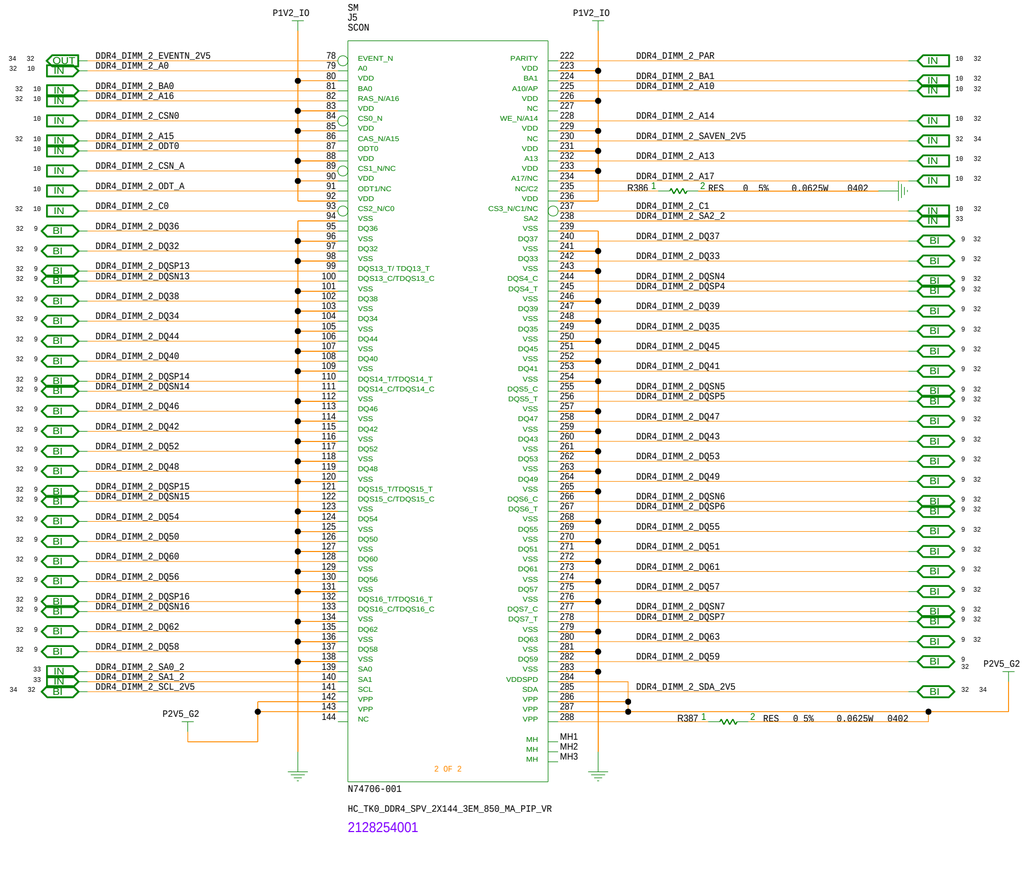
Mon Sep 15 10:27:12 2025

DEPARTMENT		SIZE	CODE	DOCUMENT NUMBER	REV
	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 32 OF 97	

ALTERA CONFIDENTIAL

2DPC DIMM-II









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DEPARTMENT		SIZE	CODE	DOCUMENT NUMBER	REV
	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 33 OF 97	

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2DPC DIMM-FILTER AND LVL SHIFTER

P1V2_IO

C1797 C95333-006 47UF 20% 4V X6S 0805
C1798 C95333-006 47UF 20% 4V X6S 0805
C1799 C95333-006 47UF 20% 4V X6S 0805
C1800 C95333-006 47UF 20% 4V X6S 0805
C1801 C95333-006 47UF 20% 4V X6S 0805
C1802 C95333-006 47UF 20% 4V X6S 0805
C1803 C95333-006 47UF 20% 4V X6S 0805
C1804 C95333-006 47UF 20% 4V X6S 0805
C1805 D97712-023 1UF 10% 16V X6S 0402
C1806 D97712-023 1UF 10% 16V X6S 0402
C1807 D97712-023 1UF 10% 16V X6S 0402
C1808 D97712-023 1UF 10% 16V X6S 0402
C1809 D97712-023 1UF 10% 16V X6S 0402
C1810 D97712-023 1UF 10% 16V X6S 0402
C1811 D97712-023 1UF 10% 16V X6S 0402
C1812 D97712-023 1UF 10% 16V X6S 0402

DIMM_VTT_0V6

C1813 C95333-006 47UF 20% 4V X6S 0805
C1814 D97712-023 1UF 10% 16V X6S 0402
C1815 D97712-023 1UF 10% 16V X6S 0402

P2V5_G2

C1816 H65935-001 47UF 20% 10V X6S 1206
C1817 D97712-023 1UF 10% 16V X6S 0402
C1818 A36096-030 0.1UF 10% 16V X7R 0402
C1819 D97712-023 1UF 10% 16V X6S 0402

DIMM_VREF_0V6

C1820 D97712-023 1UF 10% 16V X6S 0402
C1821 A36096-030 0.1UF 10% 16V X7R 0402

12V

C2701 H65936-001 22UF 10% 25V X6S 1206
C2702 A36096-213 0.1UF 10% 25V X7R 0402

P1V2_IO

C1847 C95333-006 47UF 20% 4V X6S 0805
C1848 C95333-006 47UF 20% 4V X6S 0805
C1849 C95333-006 47UF 20% 4V X6S 0805
C1850 C95333-006 47UF 20% 4V X6S 0805
C1851 C95333-006 47UF 20% 4V X6S 0805
C1852 C95333-006 47UF 20% 4V X6S 0805
C1853 C95333-006 47UF 20% 4V X6S 0805
C1854 C95333-006 47UF 20% 4V X6S 0805
C1855 D97712-023 1UF 10% 16V X6S 0402
C1856 D97712-023 1UF 10% 16V X6S 0402
C1857 D97712-023 1UF 10% 16V X6S 0402
C1858 D97712-023 1UF 10% 16V X6S 0402
C1859 D97712-023 1UF 10% 16V X6S 0402
C1860 D97712-023 1UF 10% 16V X6S 0402
C1861 D97712-023 1UF 10% 16V X6S 0402
C1862 D97712-023 1UF 10% 16V X6S 0402

DIMM_VTT_0V6

C1863 C95333-006 47UF 20% 4V X6S 0805
C1864 D97712-023 1UF 10% 16V X6S 0402
C1865 D97712-023 1UF 10% 16V X6S 0402

P2V5_G2

C1866 H65935-001 47UF 20% 10V X6S 1206
C1867 D97712-023 1UF 10% 16V X6S 0402
C1868 A36096-030 0.1UF 10% 16V X7R 0402
C1869 D97712-023 1UF 10% 16V X6S 0402

DIMM_VREF_0V6

C1870 D97712-023 1UF 10% 16V X6S 0402
C1871 A36096-030 0.1UF 10% 16V X7R 0402

12V

C2705 H65936-001 22UF 10% 25V X6S 1206
C2706 A36096-213 0.1UF 10% 25V X7R 0402

PLACED CLOSE TO J5/J36

U92
IC
HC_MAX3378EEBC
K13865-001
MAX3378EEBC+ 6

P2V5_G2
P1V2_IO

BI	DDR4_DIMM_2_SCL_2V5	B1 / C1	A1 / DDR4_DIMM_2_SCL_1V2
BI	DDR4_DIMM_2_SDA_2V5	C2	A2 / DDR4_DIMM_2_SDA_1V2
IN	DDR4_DIMM_2_EVENTN_2V5	C3	A3 / DDR4_DIMM_2_EVENTN_1V2
OUT	DDR4_DIMM_2_SAVEN_2V5	C4	A4 / DDR4_DIMM_2_SAVEN_1V2

VCC, IO_VCC1, IO_VCC2, IO_VCC3, IO_VCC4, THREE_STATE_N, VL, IO_VL1, IO_VL2, IO_VL3, IO_VL4, GND

P1V2_IO
R1645 RES 4.7K 1% 0.0625W 0402
B3
B4

50 IN DIMM_IO_EN

P2V5_G2

BI	DDR4_DIMM_2_SCL_2V5	R1098	RES 4.7K 1% 0.0625W 0402
BI	DDR4_DIMM_2_SDA_2V5	R1099	RES 4.7K 1% 0.0625W 0402
IN	DDR4_DIMM_2_EVENTN_2V5	R1100	RES 4.7K 1% 0.0625W 0402
OUT	DDR4_DIMM_2_SAVEN_2V5	R1101	RES 4.7K 1% 0.0625W 0402

P1V2_IO

BI	DDR4_DIMM_2_SCL_1V2	R1102	RES 4.7K 1% 0.0625W 0402
BI	DDR4_DIMM_2_SDA_1V2	R1103	RES 4.7K 1% 0.0625W 0402
OUT	DDR4_DIMM_2_EVENTN_1V2	R1104	RES 4.7K 1% 0.0625W 0402
IN	DDR4_DIMM_2_SAVEN_1V2	R1105	RES 4.7K 1% 0.0625W 0402

P2V5_G2
C1874 A36096-030 0.1UF 10% 16V X7R 0402

P1V2_IO
C1875 A36096-030 0.1UF 10% 16V X7R 0402

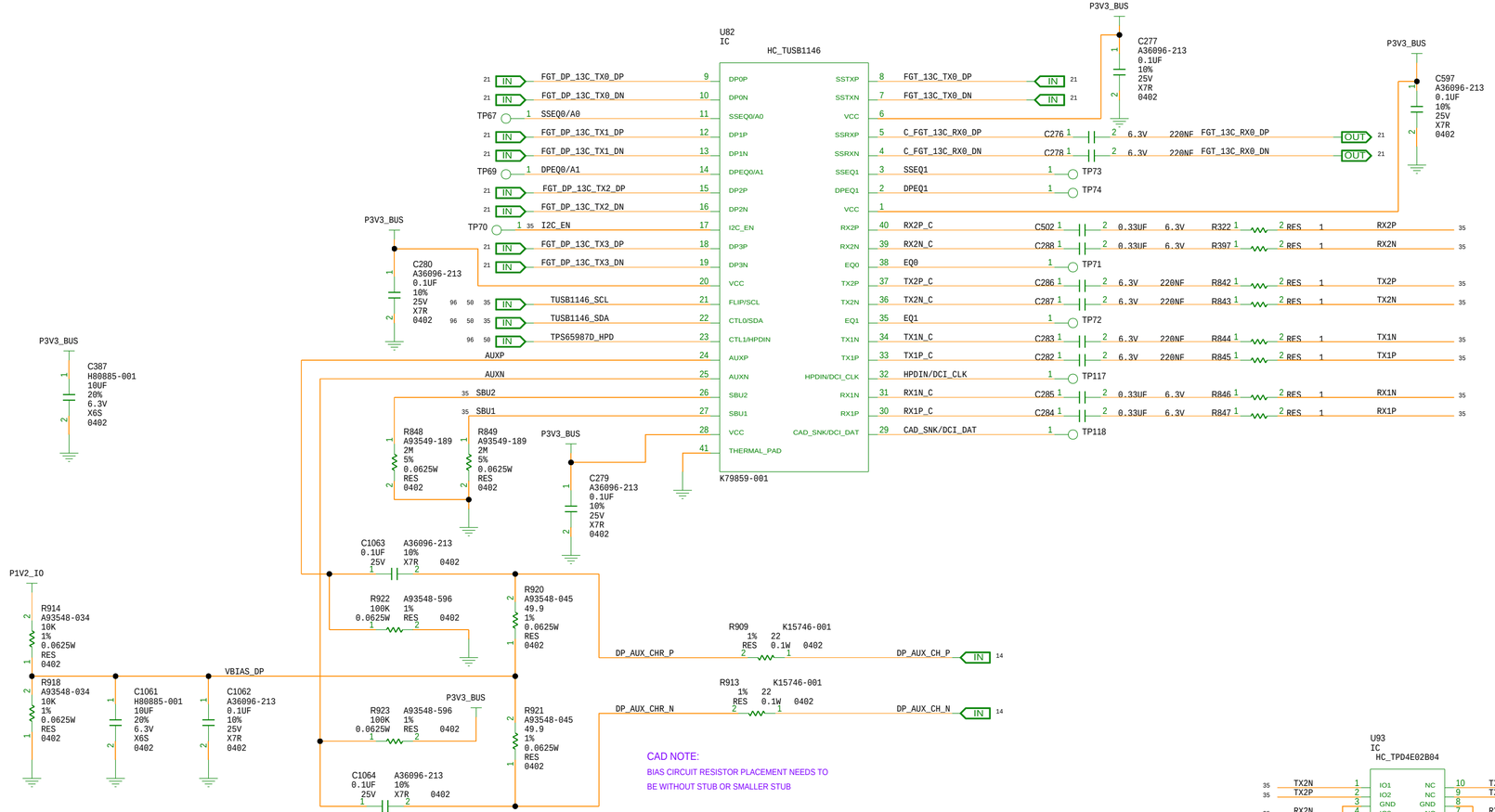
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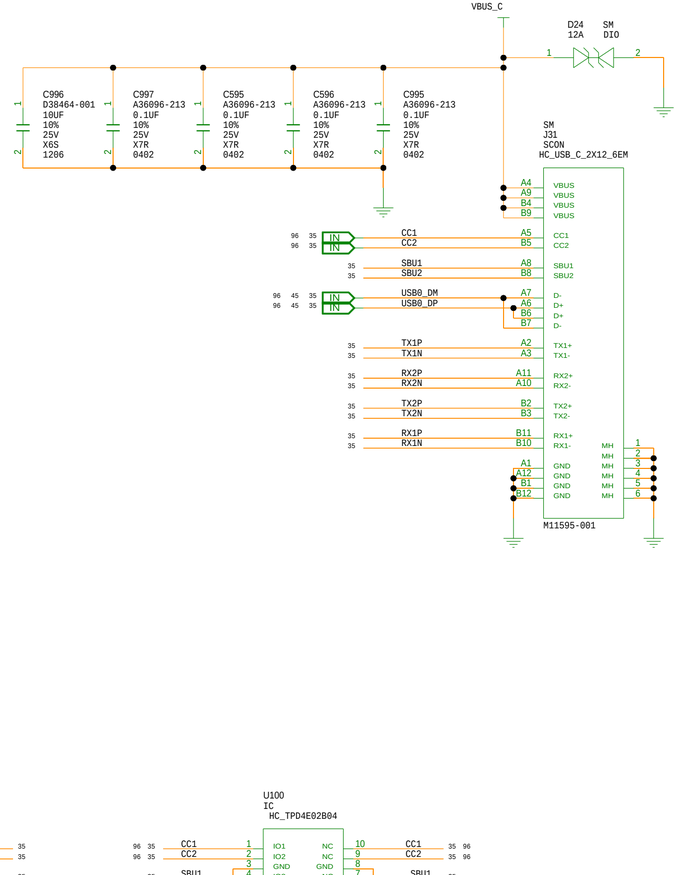
DEPARTMENT	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	SIZE: C+	CODE: 34649	DOCUMENT NUMBER: 150-0330690-D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 34 OF 97

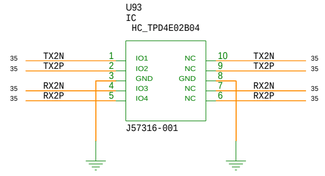
REV: D1

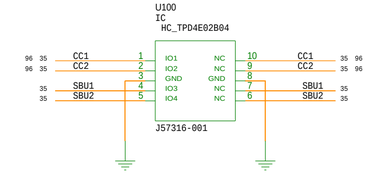
ALTERA CONFIDENTIAL

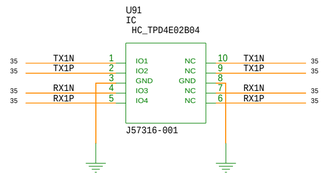
USB3.X/DP1.4 ALTMODE MUX

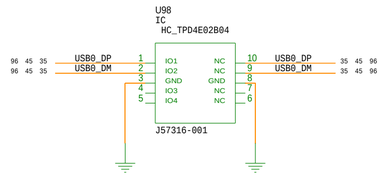










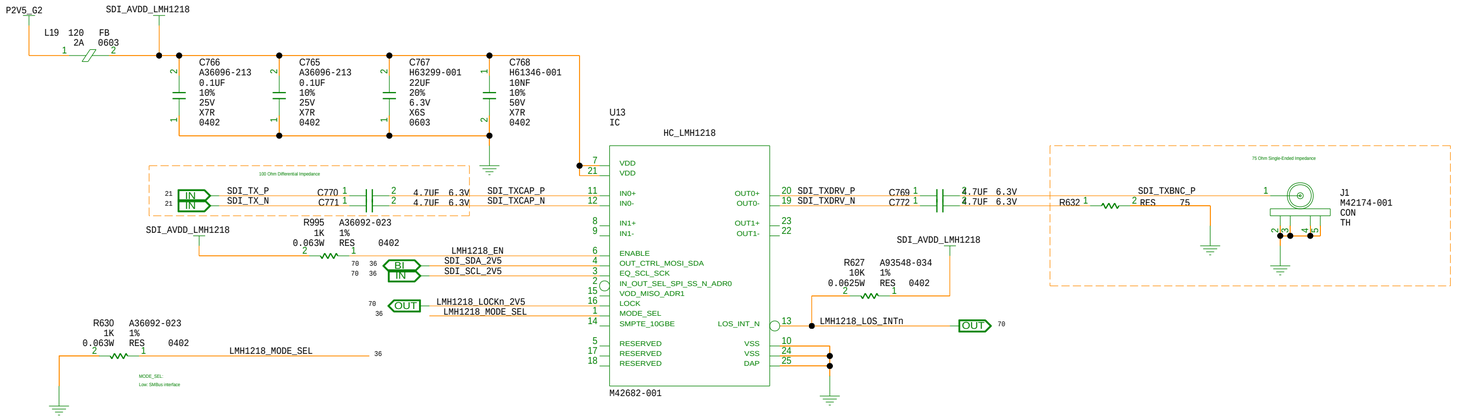


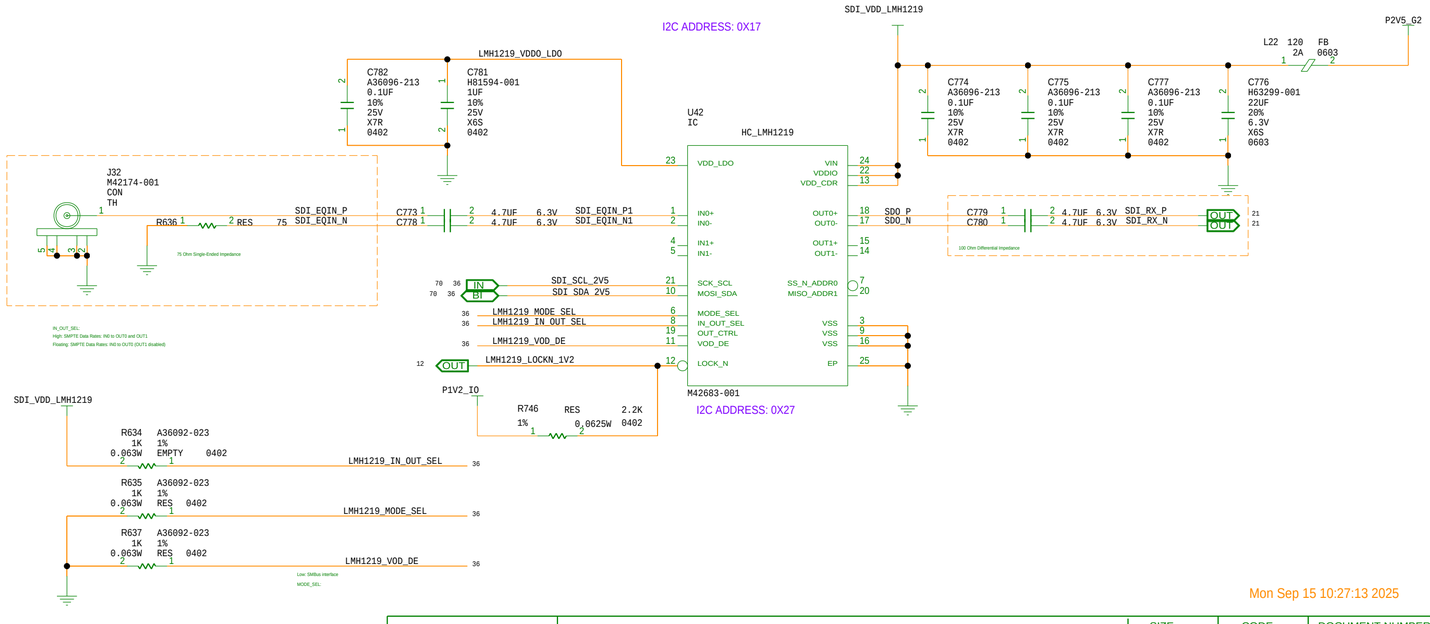
Mon Sep 15 10:27:13 2025

DEPARTMENT		SIZE	CODE	DOCUMENT NUMBER	REV
	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 35 OF 97	

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SDI CABLE DRIVER, EQUALIZER, BNC



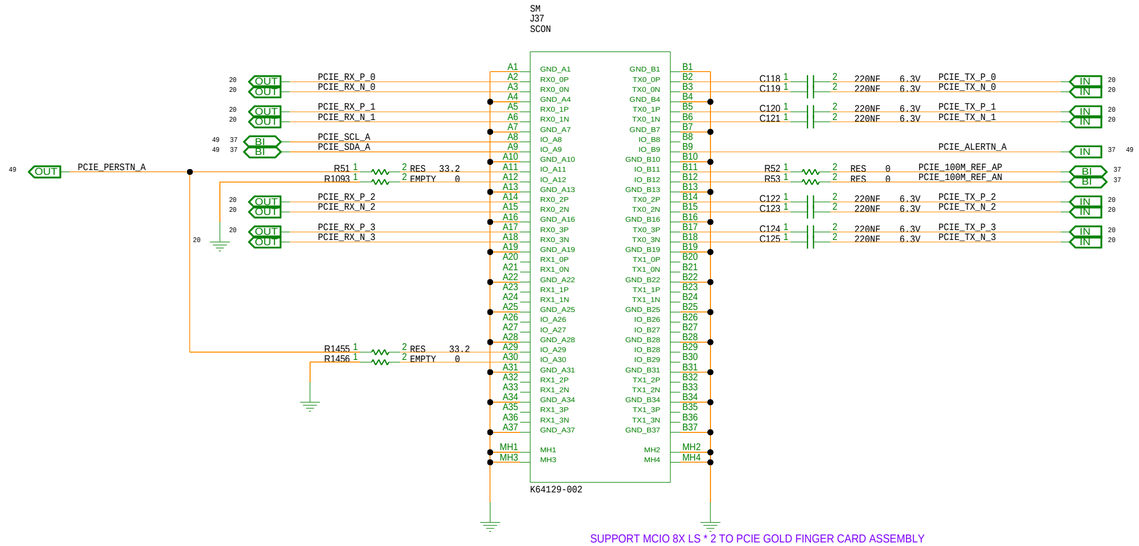


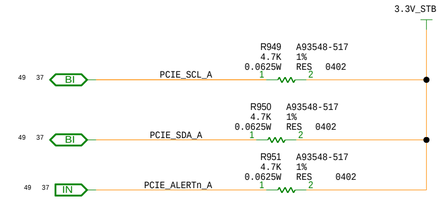
Mon Sep 15 10:27:13 2025

DEPARTMENT		SIZE	CODE	DOCUMENT NUMBER	REV
	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 36 OF 97	

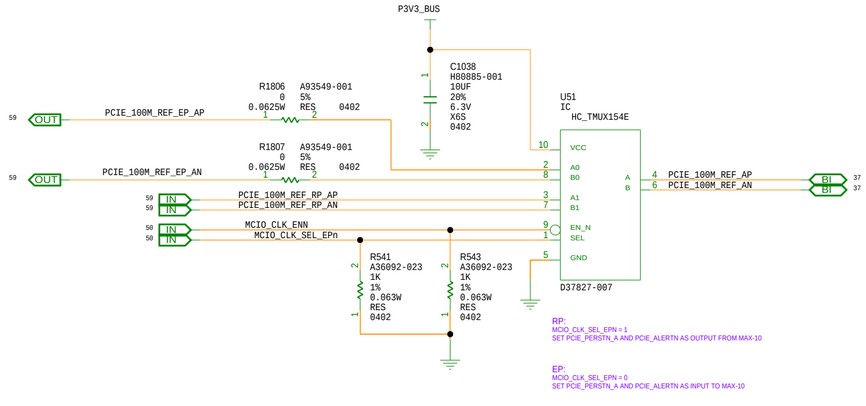
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MCIO





SUPPORT MCIO 8X LS * 2 TO PCIE GOLD FINGER CARD ASSEMBLY



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DEPARTMENT	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	SIZE	CODE	DOCUMENT NUMBER	REV
		C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 37 OF 97	

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QSFP-DD CONN #0

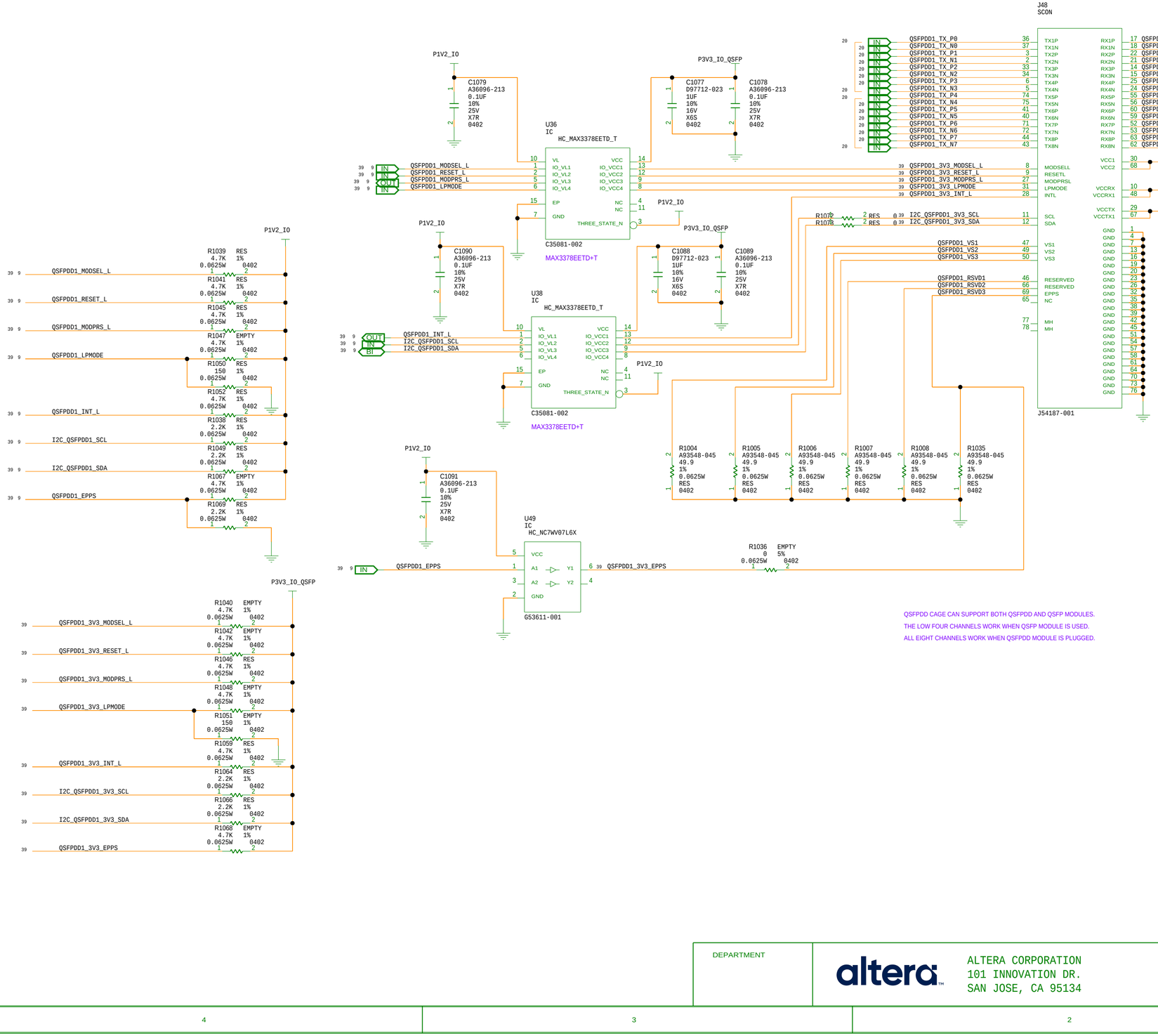
P1V2_IO
C537 A36096-213 0.1UF 10% 25V X7R 0402
P3V3_IO_QSFP
C535 D97712-023 1UF 10% 16V X6S 0402
C536 A36096-213 0.1UF 10% 25V X7R 0402
U39 IC HC_MAX3378EETD_T
C35081-002
MAX3378EETD+T
QSFPDD0_MODSEL_L
QSFPDD0_RESET_L
QSFPDD0_MODPRS_L
QSFPDD0_LPMODE
P1V2_IO
C548 A36096-213 0.1UF 10% 25V X7R 0402
P3V3_IO_QSFP
C546 D97712-023 1UF 10% 16V X6S 0402
C547 A36096-213 0.1UF 10% 25V X7R 0402
U40 IC HC_MAX3378EETD_T
C35081-002
MAX3378EETD+T
QSFPDD0_INT_L
I2C_QSFPDD0_SCL
I2C_QSFPDD0_SDA
P1V2_IO
C549 A36096-213 0.1UF 10% 25V X7R 0402
U41 IC HC_NC7WV07L6X
G53611-001
QSFPDD0_EPPS
QSFPDD800_EPPS
QSFPDD0_3V3_EPPS
QSFPDD800_3V3_EPPS
R343 EMPTY 0 5% 0.0625W 0402
SM J27 SCON
J54187-001
QSFPDD0_TX_P0 … QSFPDD0_TX_N7
QSFPDD0_RX_P0 … QSFPDD0_RX_N7
QSFPDD0_VCC
QSFPDD0_VCCR
QSFPDD0_VCCT
QSFPDD0_3V3_MODSEL_L
QSFPDD0_3V3_RESET_L
QSFPDD0_3V3_MODPRS_L
QSFPDD0_3V3_LPMODE
QSFPDD0_3V3_INT_L
I2C_QSFPDD0_3V3_SCL
I2C_QSFPDD0_3V3_SDA
QSFPDD0_VS1
QSFPDD0_VS2
QSFPDD0_VS3
QSFPDD0_RSVD1
QSFPDD0_RSVD2
QSFPDD0_RSVD3
R337 R338 R339 R340 R341 R342 A93548-045 49.9 1% 0.0625W RES 0402
QSFPDD CAGE CAN SUPPORT BOTH QSFPDD AND QSFP MODULES.
THE LOW FOUR CHANNELS WORK WHEN QSFP MODULE IS USED.
ALL EIGHT CHANNELS WORK WHEN QSFPDD MODULE IS PLUGGED.
CAD NOTE:
PLACE THE TP AT EASILY ACCESSIBLE SPOT ON THE PCB.
AVOID UNDER HEATSINK AND ADD-ON CARD SHADOW AREA
NOTE 1: BYPASS CAPACITORS SHOULD BE PLACED AS CLOSE TO THE ASSOCIATED 20-PIN CONNECTOR AS POSSIBLE.
NOTE 2: ZQSFP 100-OHM TERMINATION IS IMPLEMENTED VIA THE FPGA ON-CHIP TERMINATION.
NOTE 3: DC BLOCKING CAPACITORS ARE IN THE MODULE FOR RX AND TX.
NOTE 4: 1UH INDUCTORS SHOULD HAVE A DC RESISTANCE OF LESS THAN 0.1-OHM.
P3V3_IO_QSFP
C529 A36096-213 0.1UF 10% 25V X7R 0402
C530 H63299-001 22UF 20% 6.3V X6S 0603
L10 1UH 2.5A IND
C531 A36096-213 0.1UF 10% 25V X7R 0402
C532 E15431-002 10UF 20% 6.3V X6S 0603
C533 H63299-001 22UF 20% 6.3V X6S 0603
C534 H63299-001 22UF 20% 6.3V X6S 0603
QSFPDD0_VCCR
L11 1UH 2.5A IND
C538 A36096-213 0.1UF 10% 25V X7R 0402
C539 E15431-002 10UF 20% 6.3V X6S 0603
C540 H63299-001 22UF 20% 6.3V X6S 0603
C541 H63299-001 22UF 20% 6.3V X6S 0603
QSFPDD0_VCC
L16 1UH 2.5A IND
C542 A36096-213 0.1UF 10% 25V X7R 0402
C543 E15431-002 10UF 20% 6.3V X6S 0603
C544 H63299-001 22UF 20% 6.3V X6S 0603
C545 H63299-001 22UF 20% 6.3V X6S 0603
PLACE CLOSE TO QSFPDD0 CONNECTOR
TH XJ2 CON
HC_HS_10EM
M79404-001
R344 10K 1% 0.0625W 0402
C550 4700PF X7R 10% 50V 0402
GND_QSFPDD0_CAGE
P3V3_IO_QSFP
R346 R348 R350 R358 R360 R362 R364 R368 R371 EMPTY 4.7K 1% 0.0625W 0402
R346 QSFPDD0_3V3_MODSEL_L
R348 QSFPDD0_3V3_RESET_L
R350 RES 4.7K 1% 0.0625W 0402 QSFPDD0_3V3_MODPRS_L
R358 QSFPDD0_3V3_LPMODE
R360 EMPTY 150 1% 0.0625W 0402
R362 RES 4.7K 1% 0.0625W 0402 QSFPDD0_3V3_INT_L
R364 RES 2.2K 1% 0.0625W 0402 I2C_QSFPDD0_3V3_SCL
R368 RES 2.2K 1% 0.0625W 0402 I2C_QSFPDD0_3V3_SDA
R371 EMPTY 4.7K 1% 0.0625W 0402 QSFPDD0_3V3_EPPS
P3V3_BUS
EM_5V
C1066 H81594-001 1UF 10% 25V X6S 0402
Q16 M11185-002 CSD17575Q3
R885 RES 4.7K 1% 0.0625W 0402
C1067 H61346-001 10NF 10% 50V X7R 0402
U68 IC HC_LTC4365
K13848-001
P3V3_IO_QSFP_EN
R886 RES 0
R889 A36092-023 1K 1% 0.063W RES 0402
3.3V_STB
R888 A93548-517 4.7K 1% 0.0625W RES 0402
P3V3_IO_QSFP_POK
TP131 EMPTY SM
C990 K13620-001 100UF 20% 10V XSR 1206

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	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 38 OF 97	

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QSFP-DD CONN #1



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	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 39 OF 97	

ALTERA CONFIDENTIAL

QSFP-DD-800 CONN

CAD NOTE: PLEASE FOLLOW THE ARC ROUTING FOR THESE HIGH SPEED NETS.

DESIGN NOTE: CHANNELS MAY BE SWAPPED
WITHIN EACH BK BLOCK TO OPTIMIZE ROUTINGS.
SWAP BOTH TX & RX TOGETHER.

NOTE 4: 1UH INDUCTORS SHOULD HAVE A DC RESISTANCE OF LESS THAN 0.1-OHM.
NOTE 2: ZQSFP 100-OHM TERMINATION IS IMPLEMENTED VIA THE FPGA ON-CHIP TERMINATION.
NOTE 1: BYPASS CAPACITORS SHOULD BE PLACED AS CLOSE TO THE ASSOCIATED 20-PIN CONNECTOR AS POSSIBLE.
NOTE 3: DC BLOCKING CAPACITORS ARE IN THE MODULE FOR RX AND TX.

PLACE CLOSE TO QSFPDD800 CONNECTOR

1x1 Cage

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DEPARTMENT	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	SIZE	CODE	DOCUMENT NUMBER	REV
		C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 40 OF 97	

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FMC+ CONNECTOR A (NORTH)

J7
SCON
HC_B2B_14X40
1 OF 7
K12425-001

FMC_A_LA_P9	R1423	RES 0 5% 0.0625W 0402	G6	LA00_P_CC	LA16_P	G18	FMC_A_LA_P16_LT
FMC_A_LA_N0	R1424	RES 0 5% 0.0625W 0402	G7	LA00_N_CC	LA16_N	G19	FMC_A_LA_N16_LT
FMC_A_LA_P1	R1435	RES 0 5% 0.0625W 0402	D8	LA01_P_CC	LA17_P_CC	D20	FMC_A_LA_P17
FMC_A_LA_N1	R1436	RES 0 5% 0.0625W 0402	D9	LA01_N_CC	LA17_N_CC	D21	FMC_A_LA_N17
FMC_A_LA_P2	R1769	RES 0 5% 0.0625W 0402	H7	LA02_P	LA18_P_CC	C22	FMC_A_LA_P18
FMC_A_LA_N2	R1770	RES 0 5% 0.0625W 0402	H8	LA02_N	LA18_N_CC	C23	FMC_A_LA_N18
FMC_A_LA_P3	R1447	RES 0 5% 0.0625W 0402	G9	LA03_P	LA19_P	H22	FMC_A_LA_P19
FMC_A_LA_N3	R1448	RES 0 5% 0.0625W 0402	G10	LA03_N	LA19_N	H23	FMC_A_LA_N19
FMC_A_LA_P4	R1767	RES 0 5% 0.0625W 0402	H10	LA04_P	LA20_P	G21	FMC_A_LA_P20_LT
FMC_A_LA_N4	R1768	RES 0 5% 0.0625W 0402	H11	LA04_N	LA20_N	G22	FMC_A_LA_N20_LT
FMC_A_LA_P5			D11	LA05_P	LA21_P	H25	FMC_A_LA_P21
FMC_A_LA_N5			D12	LA05_N	LA21_N	H26	FMC_A_LA_N21
FMC_A_LA_P6_LT			C10	LA06_P	LA22_P	G24	FMC_A_LA_P22
FMC_A_LA_N6_LT			C11	LA06_N	LA22_N	G25	FMC_A_LA_N22
FMC_A_LA_P7			H13	LA07_P	LA23_P	D23	FMC_A_LA_P23
FMC_A_LA_N7			H14	LA07_N	LA23_N	D24	FMC_A_LA_N23
FMC_A_LA_P8			G12	LA08_P	LA24_P	H28	FMC_A_LA_P24
FMC_A_LA_N8			G13	LA08_N	LA24_N	H29	FMC_A_LA_N24
FMC_A_LA_P9			D14	LA09_P	LA25_P	G27	FMC_A_LA_P25
FMC_A_LA_N9			D15	LA09_N	LA25_N	G28	FMC_A_LA_N25
FMC_A_LA_P10_LT			C14	LA10_P	LA26_P	D26	FMC_A_LA_P26
FMC_A_LA_N10_LT			C15	LA10_N	LA26_N	D27	FMC_A_LA_N26
FMC_A_LA_P11			H16	LA11_P	LA27_P	C26	FMC_A_LA_P27
FMC_A_LA_N11			H17	LA11_N	LA27_N	C27	FMC_A_LA_N27
FMC_A_LA_P12			G15	LA12_P	LA28_P	H31	FMC_A_LA_P28
FMC_A_LA_N12			G16	LA12_N	LA28_N	H32	FMC_A_LA_N28
FMC_A_LA_P13			D17	LA13_P	LA29_P	G30	FMC_A_LA_P29
FMC_A_LA_N13			D18	LA13_N	LA29_N	G31	FMC_A_LA_N29
FMC_A_LA_P14_LT			C18	LA14_P	LA30_P	H34	FMC_A_LA_P30
FMC_A_LA_N14_LT			C19	LA14_N	LA30_N	H35	FMC_A_LA_N30
FMC_A_LA_P15			H19	LA15_P	LA31_P	G33	FMC_A_LA_P31
FMC_A_LA_N15			H20	LA15_N	LA31_N	G34	FMC_A_LA_N31
					LA32_P	H37	FMC_A_LA_P32_CON
					LA32_N	H38	FMC_A_LA_N32_CON
					LA33_P	G36	FMC_A_LA_P33_CON
					LA33_N	G37	FMC_A_LA_N33_CON

P3V3_IO
FMC_A_12V
FMC_A_VADJ
FMC_A_VREF
FMC_A_PG_M2C
P3V3_IO

FMC_A_PG_C2M	D1	PG_C2M	PG_M2C
FMC_A_PRSNT_N	H2	PRSNT_M2C_L	RES2
FMC_A_SDA_3V3_CON	C31	HSPC_PRSNT_M2C_L	RES3
FMC_A_SCL_3V3_CON	C30	SDA	TRST_L
FMC_A_CLK2_M2C_P	K4	SCL	TMS
FMC_A_CLK2_M2C_N	K5	CLK2_BIDIR_P	TDO
	J2	CLK2_BIDIR_N	TDI
	J3	CLK3_BIDIR_P	TCK
	B1	CLK3_BIDIR_N	CLK0_M2C_P
	C34	CLK_DIR	CLK0_M2C_N
	D35	GA0	CLK1_M2C_P
	L20	GA1	CLK1_M2C_N
	L21	REFCLK_C2M_P	RES0
	L24	REFCLK_C2M_N	RES1
	L25	REFCLK_M2C_P	SYNC_C2M_P
		REFCLK_M2C_N	SYNC_C2M_N
			SYNC_M2C_P
			SYNC_M2C_N

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FMC_A_REFCLK_C2M_P	R1732	RES 0	R_FMC_A_REFCLK_C2M_P	FMC_A_REFCLK_C2M_P
FMC_A_REFCLK_C2M_N	R1733	RES 0	R_FMC_A_REFCLK_C2M_N	FMC_A_REFCLK_C2M_N
FMC_A_REFCLK_M2C_P	C2494	0.22UF 16V	C_FMC_A_REFCLK_M2C_P	FMC_A_REFCLK_M2C_P
FMC_A_REFCLK_M2C_N	C2495	0.22UF 16V	C_FMC_A_REFCLK_M2C_N	FMC_A_REFCLK_M2C_N

R68 EMPTY 1K
C2680 C2681 0.22UF 16V C_FMC_A_CLK2_M2C_P C_FMC_A_CLK2_M2C_N
R69 R70 RES 10K 1% 0.0625W 0402
FMC_A_CLK_DIR

FMC_A_TMS	R71	RES 10K 1%	0.0625W 0402
FMC_A_TDI	R72	RES 10K 1%	0.0625W 0402
FMC_A_TDO	R73	RES 10K 1%	0.0625W 0402
FMC_A_PRSNT_N	R74	RES 10K 1%	0.0625W 0402
FMC_A_PG_M2C	R75	RES 10K 1%	0.0625W 0402
FMC_A_SDA_3V3_CON	R1212	RES 4.7K 1%	0.0625W 0402
FMC_A_SCL_3V3_CON	R1213	RES 4.7K 1%	0.0625W 0402

R1416 R1417 RES 10K 1% 0.0625W 0402 FMC_A_CLK_DIR

FMC_A_LA_P33_CON	R1250	RES 0 5%	0.0625W 0402	FMC_A_LA_P33
FMC_A_LA_N33_CON	R1251	RES 0 5%	0.0625W 0402	FMC_A_LA_N33
	R1252	EMPTY 5% 0	0.0625W 0402	FMC_A_PCIE_PERSTN_3V3
	R1253	EMPTY 5% 0	0.0625W 0402	FMC_A_PCIE_WAKEN_3V3
FMC_A_LA_P32_CON	R1254	RES 0 5%	0.0625W 0402	FMC_A_LA_P32
FMC_A_LA_N32_CON	R1255	RES 0 5%	0.0625W 0402	FMC_A_LA_N32

R1256 EMPTY 100 1% 0.0625W 0402

FMC_A_TCK
FMC_A_PG_C2M

R76 A36092-023 1K 1% 0.063W RES 0402
R77 A36092-023 1K 1% 0.063W RES 0402

P1V2_IO
P1V8_G2
FMC_A_VADJ
R1122 RES 0 0% 0.75W 1206
R1123 EMPTY 0 0% 0.75W 1206

5: SUPPORT ONLY FOR SPECIFIC 1.8V IO STANDARD FMC+ CARDS.

COMP_VTT_0V6

FMC_A_REFCLK_C2M_N	R1738	EMPTY 49.9 0402	R1739	FMC_A_REFCLK_C2M_P
FMC_A_SYNC_C2M_N	R1736	EMPTY 49.9 0402	R1737	FMC_A_SYNC_C2M_P
FMC_A_LA_N6	R1777	EMPTY 49.9 0402	R1778	FMC_A_LA_P6
FMC_A_LA_N0	R1425	EMPTY 49.9 0402	R1426	FMC_A_LA_P0
FMC_A_LA_N17	R1429	EMPTY 49.9 0402	R1430	FMC_A_LA_P17
FMC_A_LA_N18	R1431	EMPTY 49.9 0402	R1432	FMC_A_LA_P18
FMC_A_LA_N1	R1433	EMPTY 49.9 0402	R1434	FMC_A_LA_P1
FMC_A_LA_N27	R1443	EMPTY 49.9 0402	R1444	FMC_A_LA_P27
FMC_A_LA_N3	R1445	EMPTY 49.9 0402	R1446	FMC_A_LA_P3
FMC_A_LA_N2	R1771	EMPTY 49.9 0402	R1772	FMC_A_LA_P2
FMC_A_LA_N4	R1773	EMPTY 49.9 0402	R1774	FMC_A_LA_P4

C2686 D97712-023 1UF 10% 16V X6S 0402
C2687 D97712-023 1UF 10% 16V X6S 0402
C2688 D97712-023 1UF 10% 16V X6S 0402
C2689 D97712-023 1UF 10% 16V X6S 0402
C1510 H63299-001 22UF 20% 6.3V X6S 0603
C1511 H63299-001 22UF 20% 6.3V X6S 0603
C1512 H63299-001 22UF 20% 6.3V X6S 0603
C1513 H63299-001 22UF 20% 6.3V X6S 0603

P1V2_IO
C133 A36096-213 0.1UF 10% 25V X7R 0402
U74 IC
HC_14-TDFN_EP
HC_MAX3378EETD_T
C35081-002
MAX3378EETD+T

FMC_A_LA_P10	R1124	EMPTY 5% 0 0.0625W 0402	FMC_A_LA_P10_LT
FMC_A_LA_N10	R1125	EMPTY 5% 0 0.0625W 0402	FMC_A_LA_N10_LT
FMC_A_LA_P14	R1126	EMPTY 5% 0 0.0625W 0402	FMC_A_LA_P14_LT
FMC_A_LA_N14	R1127	EMPTY 5% 0 0.0625W 0402	FMC_A_LA_N14_LT
	R1128	EMPTY 0 0.0625W 0402	
	R1129	EMPTY 0 0.0625W 0402	
	R1130	EMPTY 0 0.0625W 0402	
	R1131	EMPTY 0 0.0625W 0402	

FMC_A_VADJ
C131 D97712-022 1UF 10% 16V X6S 0402
C132 A36096-213 0.1UF 10% 25V X7R 0402
P1V2_IO

R1132 R1135 R1136 R1138 RES 0 5% 0.0625W 0402

CAD NOTE: PLEASE PLACE THESE THREE RESISTOR FOR EACH SIGNAL AS TRIPAD.

P1V2_IO
C1094 A36096-213 0.1UF 10% 25V X7R 0402
U89 IC
HC_MAX3378EETD_T
C35081-002
MAX3378EETD+T

FMC_A_LA_P6	R1161	EMPTY 5% 0 0.0625W 0402	FMC_A_LA_P6_LT
FMC_A_LA_N6	R1165	EMPTY 5% 0 0.0625W 0402	FMC_A_LA_N6_LT
FMC_A_CLK1_M2C_P	R1169	EMPTY 5% 0 0.0625W 0402	FMC_A_CLK1_M2C_P_LT
FMC_A_CLK1_M2C_N	R1178	EMPTY 5% 0 0.0625W 0402	FMC_A_CLK1_M2C_N_LT
	R1164	EMPTY 0 0402	
	R1167	EMPTY 0 0402	
	R1177	EMPTY 0 0402	
	R1179	EMPTY 0 0402	

C1012 D97712-023 1UF 10% 16V X6S 0402
C1093 A36096-213 0.1UF 10% 25V X7R 0402
P1V2_IO

R1775 R1776 RES 0.5% 0.0625W 0402
C2490 C2491 X7R 0.22UF 10% 16V 0402

CAD NOTE: PLEASE PLACE THESE THREE RESISTOR FOR EACH SIGNAL AS TRIPAD.

C1011 A36096-213 0.1UF 10% 25V X7R 0402
U80 IC
HC_14-TDFN_EP
HC_MAX3378EETD_T
C35081-002
MAX3378EETD+T

FMC_A_LA_P16	R1141	EMPTY 5% 0 0.0625W 0402	FMC_A_LA_P16_LT
FMC_A_LA_N16	R1144	EMPTY 5% 0 0.0625W 0402	FMC_A_LA_N16_LT
FMC_A_LA_P20	R1146	EMPTY 5% 0 0.0625W 0402	FMC_A_LA_P20_LT
FMC_A_LA_N20	R1150	EMPTY 5% 0 0.0625W 0402	FMC_A_LA_N20_LT
	R1142	EMPTY 5% 0 0.0625W 0402	
	R1145	EMPTY 5% 0 0.0625W 0402	
	R1147	EMPTY 5% 0 0.0625W 0402	
	R1151	EMPTY 5% 0 0.0625W 0402	

C431 D97712-023 1UF 10% 16V X6S 0402
C526 A36096-213 0.1UF 10% 25V X7R 0402
FMC_A_VADJ
P1V2_IO

R1154 R1155 R1156 R1158 RES 0.5% 0.0625W 0402

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DEPARTMENT	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	SIZE: C+ / CODE: 34649	DOCUMENT NUMBER: 150-0330690-D1 / REV: D1
		SCALE:	DO NOT SCALE DRAWING / SHEET: 41 OF 97

ALTERA CONFIDENTIAL

FMC+ CONNECTOR A (NORTH)

J7
SCON
HC_B2B_14X40
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J7
SCON
HC_B2B_14X40
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J7
SCON
HC_14X40_ASP_VITA57P4
HC_B2B_14X40
2 OF 7
K12425-001

J7
SCON
HC_14X40_ASP_VITA57P4
HC_B2B_14X40
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J7
SCON
HC_B2B_14X40
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K12425-001

FMC_A_CLK_M2C_P3 R1221 0 R_FMC_A_CLK_M2C_P3
FMC_A_CLK_M2C_N3 R1222 0 R_FMC_A_CLK_M2C_N3
FMC_A_CLK_M2C_P2 R1219 0 R_FMC_A_CLK_M2C_P2
FMC_A_CLK_M2C_N2 R1220 0 R_FMC_A_CLK_M2C_N2
FMC_A_CLK_M2C_P3 R1223 0 R_FMC_A_CLK_M2C_P3_1
FMC_A_CLK_M2C_N3 R1224 0 R_FMC_A_CLK_M2C_N3_1

REWORK RESISTORS TO ACCESS GBTCLK4 AND GBTCLK5
OVERLAP THE RESISTORS FOR FMC_A_CLK_M2C_P3/N3

FMC_A_CLK_M2C_P2 R1217 0
FMC_A_CLK_M2C_N2 R1218 0

FMC_A_VADJ
FMC_A_VREF
P3V3_IO
FMC_A_12V

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DEPARTMENT	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	SIZE	CODE	DOCUMENT NUMBER	REV
		C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 42 OF 97	

ALTERA CONFIDENTIAL

J9
SCON
HC_14X40_ASP_VITA57P4
HC_B2B_14X40
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K12425-001

LA00_P_CC	G6	FMC_B_LA_P0	R402	RES 0 5% 0.0625W 0402							FMC_B_LA_P16
LA00_N_CC	G7	FMC_B_LA_N0	R403	RES 0 5% 0.0625W 0402							FMC_B_LA_N16
LA01_P_CC	D8	FMC_B_LA_P1	R405	RES 0 5% 0.0625W 0402							FMC_B_LA_P17
LA01_N_CC	D9	FMC_B_LA_N1	R417	RES 0 5% 0.0625W 0402							FMC_B_LA_N17
LA02_P	H7	FMC_B_LA_P2	R420	RES 0 5% 0402							FMC_B_LA_P18
LA02_N	H8	FMC_B_LA_N2	R423	RES 0 5% 0.0625W 0402							FMC_B_LA_N18
LA03_P	G9	FMC_B_LA_P3	R446	RES 0 5% 0.0625W 0402							FMC_B_LA_P19
LA03_N	G10	FMC_B_LA_N3	R447	RES 0 5% 0.0625W 0402							FMC_B_LA_N19
LA04_P	H10	FMC_B_LA_P4	R448	RES 0 5% 0.0625W 0402	G21	R1192	RES 0 5% 0.0625W 0402				FMC_B_LA_P20
LA04_N	H11	FMC_B_LA_N4	R449	RES 0 5% 0.0625W 0402	G22	R1194	RES 0 5% 0.0625W 0402				FMC_B_LA_N20
LA05_P	D11	FMC_B_LA_P5			H25	R1203	RES 0 5% 0.0625W 0402				FMC_B_LA_P21
LA05_N	D12	FMC_B_LA_N5			H26	R1204	RES 0 5% 0.0625W 0402				FMC_B_LA_N21
LA06_P	C10	FMC_B_LA_P6	R450	RES 0 5% 0.0625W 0402	G24	R1197	RES 0 5% 0.0625W 0402				FMC_B_LA_P22
LA06_N	C11	FMC_B_LA_N6	R451	RES 0 5% 0.0625W 0402	G25	R1198	RES 0 5% 0.0625W 0402				FMC_B_LA_N22
LA07_P	H13	FMC_B_LA_P7			D23	R1190	RES 0 5% 0.0625W 0402				FMC_B_LA_P23
LA07_N	H14	FMC_B_LA_N7			D24	R1191	RES 0 5% 0.0625W 0402				FMC_B_LA_N23
LA08_P	G12	FMC_B_LA_P8			H28	R1209	RES 0 5% 0.0625W 0402				FMC_B_LA_P24
LA08_N	G13	FMC_B_LA_N8			H29	R1214	RES 0 5% 0.0625W 0402				FMC_B_LA_N24
LA09_P	D14	FMC_B_LA_P9			G27	R1199	RES 0 5% 0.0625W 0402				FMC_B_LA_P25
LA09_N	D15	FMC_B_LA_N9			G28	R1200	RES 0 5% 0.0625W 0402				FMC_B_LA_N25
LA10_P	C14	FMC_B_LA_P10			D26						FMC_B_LA_P26
LA10_N	C15	FMC_B_LA_N10			D27						FMC_B_LA_N26
LA11_P	H16	FMC_B_LA_P11			C26	R452	RES 0 5% 0.0625W 0402				FMC_B_LA_P27
LA11_N	H17	FMC_B_LA_N11			C27	R456	RES 0 5% 0.0625W 0402				FMC_B_LA_N27
LA12_P	G15	FMC_B_LA_P12			H31						FMC_B_LA_P28
LA12_N	G16	FMC_B_LA_N12			H32						FMC_B_LA_N28
LA13_P	D17	FMC_B_LA_P13			G30	R1226	RES 0 5% 0.0625W 0402				FMC_B_LA_P29
LA13_N	D18	FMC_B_LA_N13			G31	R1227	RES 0 5% 0.0625W 0402				FMC_B_LA_N29
LA14_P	C18	FMC_B_LA_P14			H34	R1187	RES 0 5% 0.0625W 0402				FMC_B_LA_P30
LA14_N	C19	FMC_B_LA_N14			H35	R1833	RES 0 5% 0.0625W 0402				FMC_B_LA_N30
LA15_P	H19	FMC_B_LA_P15			G33						FMC_B_LA_P31
LA15_N	H20	FMC_B_LA_N15			G34						FMC_B_LA_N31
					H37						FMC_B_LA_P32_CON
					H38						FMC_B_LA_N32_CON
					G36						FMC_B_LA_P33_CON
					G37						FMC_B_LA_N33_CON

FMC_B_VADJ
P1V2_IO
P1V8_G2
R1180 RES 0 0% 0.75W 1206
R1181 EMPTY 0 0% 0.75W 1206
0 OHM OPTION PROVIDED ONLY TO ISOLATE VADJ FROM THE FMC CONNECTOR

P3V3_IO
FMC_B_12V
J9
SCON
HC_14X40_ASP_VITA57P4
HC_B2B_14X40
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3P3VAUX	D32		
3P3V	D40	VADJ	E39
3P3V	C39	VADJ	F40
3P3V	D36	VADJ	G39
3P3V	D38	VADJ	H40
3P3V	Z40	VIO_B_M2C	K40
12P0V	C35	VIO_B_M2C	J39
12P0V	C37	VREF_B_M2C	K1
12P0V	L36	VREF_A_M2C	H1
12P0V	L37		
12P0V	L40		
PG_C2M	D1	PG_M2C	F1
PRSNT_M2C_L	H2	RES2	L32
HSPC_PRSNT_M2C_L	Z1	RES3	L33
SDA	C31	TRST_L	D34
SCL	C30	TMS	D33
CLK2_BIDIR_P	K4	TDO	D31
CLK2_BIDIR_N	K5	TDI	D30
CLK3_BIDIR_P	J2	TCK	D29
CLK3_BIDIR_N	J3	CLK0_M2C_P	H4
CLK_DIR	B1	CLK0_M2C_N	H5
GA0	C34	CLK1_M2C_P	G2
GA1	D35	CLK1_M2C_N	G3
REFCLK_C2M_P	L20	RES0	B40
REFCLK_C2M_N	L21	RES1	L1
REFCLK_M2C_P	L24	SYNC_C2M_P	L16
REFCLK_M2C_N	L25	SYNC_C2M_N	L17
		SYNC_M2C_P	L28
		SYNC_M2C_N	L29

R1126 2 EMPTY 0 0402
R457 2 RES 10K 1% 0.0625W 0402
FMC_B_PG_M2C
FMC_B_TMS
FMC_B_TDO
FMC_B_TDI
FMC_B_TCK
C_FMC_B_CLK0_M2C_P
C_FMC_B_CLK0_M2C_N
C_FMC_B_CLK1_M2C_P
C_FMC_B_CLK1_M2C_N
R_FMC_B_SYNC_C2M_P
R_FMC_B_SYNC_C2M_N
R_FMC_B_SYNC_M2C_P
R_FMC_B_SYNC_M2C_N

FMC_B_PG_C2M
FMC_B_PRSNT_N
R458 2 1 EMPTY 1K
FMC_B_SDA_3V3_CON
FMC_B_SCL_3V3_CON
FMC_B_CLK2_M2C_P
FMC_B_CLK2_M2C_N
C151 C152 0.22UF 16V C_FMC_B_CLK2_M2C_P C_FMC_B_CLK2_M2C_N
FMC_B_CLK_DIR
R459 R460 10K 1% 0.0625W 0402
FMC_B_REFCLK_C2M_P
FMC_B_REFCLK_C2M_N
FMC_B_REFCLK_M2C_P
FMC_B_REFCLK_M2C_N
R468 R473 0 0402 R_FMC_B_REFCLK_C2M_P R_FMC_B_REFCLK_C2M_N
C157 C446 0.22UF 16V C_FMC_B_REFCLK_M2C_P C_FMC_B_REFCLK_M2C_N

FMC_B_LA_P33_CON
FMC_B_LA_N33_CON
R482 R483 0 0.0625W 0402 FMC_B_LA_P33 FMC_B_LA_N33
R484 R485 EMPTY 0 0402 FMC_B_PCIE_PERSTN_3V3 FMC_B_PCIE_WAKEN_3V3
FMC_B_LA_P32_CON
FMC_B_LA_N32_CON
R486 R487 0 0.0625W 0402 FMC_B_LA_P32 FMC_B_LA_N32
R488 EMPTY 100 1% 0.0625W 0402

P3V3_IO
R520 R521 RES 10K 1% 0.0625W 0402 EMPTY 10K 1% 0.0625W 0402
FMC_B_CLK_DIR

FMC_A_SDA_3V3_CON
FMC_A_SCL_3V3_CON
R1182 R1183 EMPTY 0 5% 0.0625W 0402 FMC_A_SDA_3V3 FMC_A_SCL_3V3
R673 R674 RES 0 0402 FMC_A_SDA_3V3_IO FMC_A_SCL_3V3_IO
NOTE : PLEASE CONNECT THESE AS TRIPAD.

FMC_B_SDA_3V3_CON
FMC_B_SCL_3V3_CON
R1184 R1185 EMPTY 0 5% 0.0625W 0402 FMC_B_SDA_3V3 FMC_B_SCL_3V3
R677 R681 RES 0 0402 FMC_B_SDA_3V3_IO FMC_B_SCL_3V3_IO
NOTE : PLEASE CONNECT THESE AS TRIPAD.

COMP_VTT_0V6

FMC_B_REFCLK_C2M_N	R489	EMPTY	49.9	0402	R490	EMPTY 49.9 0402	FMC_B_REFCLK_C2M_P
FMC_B_SYNC_C2M_N	R491	EMPTY	49.9	0402	R492	EMPTY 49.9 0402	FMC_B_SYNC_C2M_P
FMC_B_LA_N6	R493	EMPTY	49.9	0402	R494	EMPTY 49.9 0402	FMC_B_LA_P6
FMC_B_LA_N0	R495	EMPTY	49.9	0402	R496	EMPTY 49.9 0402	FMC_B_LA_P0
FMC_B_LA_N17	R497	EMPTY	49.9	0402	R498	EMPTY 49.9 0402	FMC_B_LA_P17
FMC_B_LA_N18	R499	EMPTY	49.9	0402	R500	EMPTY 49.9 0402	FMC_B_LA_P18
FMC_B_LA_N1	R501	EMPTY	49.9	0402	R502	EMPTY 49.9 0402	FMC_B_LA_P1
FMC_B_LA_N27	R503	EMPTY	49.9	0402	R504	EMPTY 49.9 0402	FMC_B_LA_P27
FMC_B_LA_N3	R505	EMPTY	49.9	0402	R506	EMPTY 49.9 0402	FMC_B_LA_P3
FMC_B_LA_N2	R507	EMPTY	49.9	0402	R508	EMPTY 49.9 0402	FMC_B_LA_P2
FMC_B_LA_N4	R509	EMPTY	49.9	0402	R510	EMPTY 49.9 0402	FMC_B_LA_P4

C452	D97712-023	1UF	10%	16V X6S 0402
C551	D97712-023	1UF	10%	16V X6S 0402
C552	D97712-023	1UF	10%	16V X6S 0402
C553	D97712-023	1UF	10%	16V X6S 0402
C554	H63299-001	22UF	20%	6.3V X6S 0603
C555	H63299-001	22UF	20%	6.3V X6S 0603
C556	H63299-001	22UF	20%	6.3V X6S 0603
C557	H63299-001	22UF	20%	6.3V X6S 0603

R518 A36092-023 1K 1% 0.063W RES 0402 FMC_B_TCK
R519 A36092-023 1K 1% 0.063W RES 0402 FMC_B_PG_C2M

P1V2_IO
C503 A36096-213 0.1UF 10% 25V X7R 0402
P3V3_IO
C504 D97712-022 1UF 10% 16V X6S 0402
C505 A36096-213 0.1UF 10% 25V X7R 0402

U62
IC
HC_MAX3378EETD_T
C35081-002
MAX3378EETD+T

VL	10	VCC	14
IO_VL1	1	IO_VCC1	13
IO_VL2	2	IO_VCC2	12
IO_VL3	5	IO_VCC3	9
IO_VL4	6	IO_VCC4	8
EP	15	NC	4
GND	7	NC	11
		THREE_STATE_N	3

FMC_A_SDA_1V2
FMC_A_SCL_1V2
FMC_B_SDA_1V2
FMC_B_SCL_1V2
FMC_A_SDA_3V3_IO
FMC_A_SCL_3V3_IO
FMC_B_SDA_3V3_IO
FMC_B_SCL_3V3_IO

P1V2_IO
R1302 R1371 R1562 R1563 4.7K 1% 0.0625W 0402
FMC_B_SDA_1V2
FMC_B_SCL_1V2
FMC_A_SDA_1V2
FMC_A_SCL_1V2

P3V3_IO
FMC_B_TMS
FMC_B_TDI
FMC_B_TDO
FMC_B_PRSNT_N
FMC_B_PG_M2C
FMC_B_SDA_3V3_CON
FMC_B_SCL_3V3_CON
R511 R512 R513 R514 R515 R516 R517 10K 1% 0.0625W 0402
R516 4.7K 1% 0.0625W 0402
R517 4.7K 1% 0.0625W 0402

FMC+ CONNECTOR B (EAST)

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FMC+ CONNECTOR B (EAST)

SM
J9
SCON
HC_B2B_14X40

6 OF 7
K12425-001

SM
J9
SCON
HC_B2B_14X40

7 OF 7
K12425-001

SM
J9
SCON
HC_B2B_14X40

4 OF 7
K12425-001

REWORK RESISTORS TO ACCESS GBTCLK4 AND GBTCLK5
OVERLAP THE RESISTORS FOR FMC_B_CLK_M2C_P3/N3

SM
J9
SCON
HC_B2B_14X40

3 OF 7
K12425-001

SM
J9
SCON
HC_B2B_14X40

2 OF 7
K12425-001

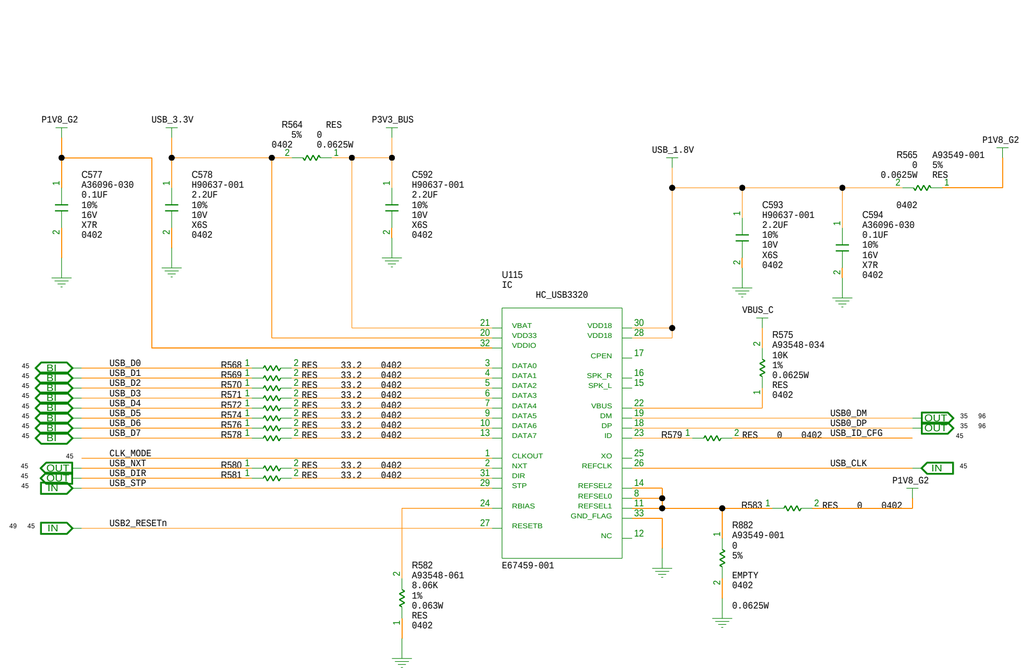
FMC_B_VADJ
FMC_B_VREF
P3V3_IO
FMC_B_12V

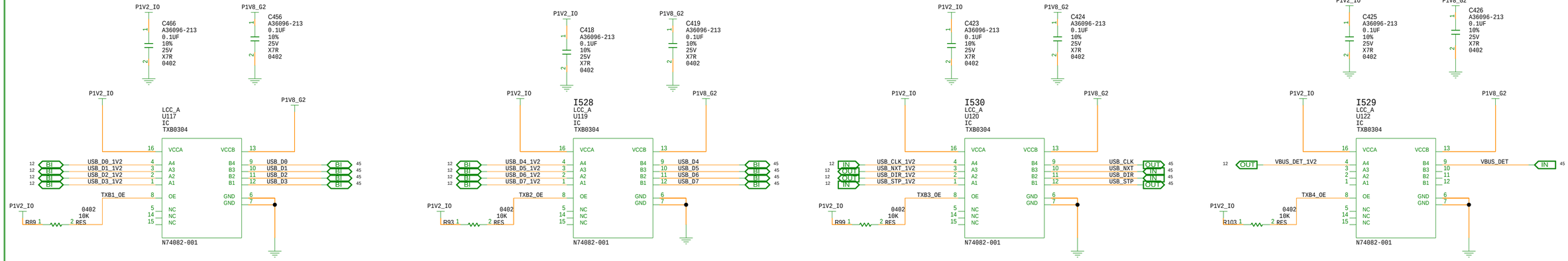
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	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
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USB2.0





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		C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET	45 OF 97

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1PPS INOUT

Pin 6 of TLV3501A =0 Output ON
Pin 6 of TLV3501A =1 Output High-Z

J14 CON TH — SMA_1PPS_INOUT — D27 A53075-002 PGB1010603MR 0A DIO 0603

R135 A93548-045 49.9 1% 0.0625W RES 0402

R136 A93548-048 249 1% 0.0625W RES 0402

VIN MIN 0-0.5V
VIN MAX 0-5.5V

TO BE CHECKED FOR SLOWER RAMP RATE.

GDZ3V3LP3
VRMAX VZENER (V) =
X3-DFN0603-2

C251 D70542-023 15PF 5% 25V C0G 0201

D28 GDZ3V3LP3-7 SMA DIO SM — COMPE_VREF_0P257V

TP96 EMPTY SM

TP94 EMPTY SM

TP97 EMPTY SM

Q3 H31491-001 CSD17381F4 FET SM — SEL_1PPS_INOUT_R

R139 1K 0.063W RES 1% 0402

R140 150K 0.0625W RES 0.1% 0402

R144 20K 0.063W RES 0.1% 0402

R145 1.69K 0.0625W A93548-582 1% RES 0402

R147 1K 0.063W A36092-023 1% RES 0402 — OD_SEL_1PPS_INOUT

U63 H11519-001 HC_TLV3501 IC SOT23 — COMPE_VREF_1P65V

C247 25V X6S 1UF 10% 0402

C248 25V X6S 1UF 10% 0402

P3V3_BUS_SMA

U72 H11519-001 HC_TLV3501 IC SOT23 — 1PPS_SMA_IN

R134 A93548-517 4.7K 1% 0.0625W RES 0402

R142 A36092-023 1K 1% 0.063W RES 0402

TP92 EMPTY SM

TP93 EMPTY SM

Q2 H31491-001 CSD17381F4 FET SM

C252 H61346-001 10NF 10% 50V X7R 0402

EM_5V_SMA

C253 25V X6S 1UF 10% 0402

U76 H11519-001 HC_TLV3501 IC SOT23 — COMPE_VREF_1P65V

SHDN_10MHZ_IN — Q12 SM FET CSD25481F4

TERM_ON_IN_P R150 A36092-043 100 1% 0.0625W RES 0402

C255 16V X7R 0.1UF 10% 0402

SMA_10MHZ_INOUT

R740 A36092-043 100 1% 0.0625W RES 0402 TERM_ON_IN_N

J17 CON TH — SMA_1PPS_10MHZ_INOUT — D29 A53075-002 PGB1010603MR 0A DIO 0603

R776 A93548-517 4.7K 1% 0.0625W RES 0402 — SHDN_10MIN_RN

TP99 EMPTY SM

Q14 H31491-001 CSD17381F4 FET SM

Q15 H31491-001 CSD17381F4 FET SM

TP102 EMPTY SM

SHDN_10M_IN_R

R830 1K 0.063W A36092-023 1% RES 0402

C254 25V X6S 1UF 10% 0402

U78 H11519-001 HC_TLV3501 IC SOT23 — 10MHZ_SMA_OUT — COMPE_VREF_1P65V

TP98

R741 A36092-023 1K 1% 0.063W RES 0402

C256 25V X6S 1UF 10% 0402

R744 150K 0.0625W RES 0.1% 0402

U83 H11519-001 HC_TLV3501 IC SOT23 — 10MHZ_SMA_IN

TP100 EMPTY SM

TP101 EMPTY SM

R821 20K 0.063W RES 0.1% 0402

R828 20K 0.063W RES 0.1% 0402

1.65V

R829 A36092-023 1K 1% 0.063W RES 0402 — SHDN_10MHZ_IN

PIN 6 OF TLV3501A =0 OUTPUT ON
PIN 6 OF TLV3501A =1 OUTPUT HIGH-Z

0-3.3V

SHDN_10MHZ_IN =0 INPUT ENABLED
SHDN_10MHZ_IN =1 INPUT DISABLED

R138 C48657-447 51 1% 0.05W RES 0201

R143 C48657-447 51 1% 0.05W RES 0201

U75 IC — M44051-001 (VH, VDD, OA, OB, IN_A, IN_B, GND, PD)

C249 H81594-001 1UF 10% 25V X6S 0402

C250 H81594-001 1UF 10% 25V X6S 0402

1PPS_SMA_IN — 1PPS_SMA_OUT — PD_SMA_OUT — TP95

R141 A93548-517 4.7K 1% 0.0625W RES 0402

R901 A93548-517 4.7K 1% 0.0625W RES 0402

OD_SEL_1PPS_INOUT

Q1 H31491-001 CSD17381F4 FET SM — SEL_1PPS_INOUT

R137 A93548-517 4.7K 1% 0.0625W RES 0402

SEL_1PPS_INOUT=0 Input (---->) Default
SEL_1PPS_INOUT=1 Output (<----)

C1048 H81594-001 1UF 10% 25V X6S 0402

C1047 H81594-001 1UF 10% 25V X6S 0402

450MA FB30 220 FB 0402

C1070 H81594-001 1UF 10% 25V X6S 0402

C1069 H81594-001 1UF 10% 25V X6S 0402

R148 A36092-023 1K 1% 0.063W RES 0402

OD_SHDN_1PPS_TO_10MHZ

Q11 H31491-001 CSD17381F4 FET SM — SHDN_1PPS_TO_10MHZ

R146 A93548-517 4.7K 1% 0.0625W RES 0402

SHDN_1PPS_TO_10MHZ= 0 DEFAULT, NO CLOCK
SHDN_1PPS_TO_10MHZ= 1 DRIVING 1PPS CLOCK

10MHZ INOUT
OR
1PPS OUT

R739 A36092-023 1K 1% 0.063W RES 0402 — OD_SHDN_10MHZ_OUT

Q13 H31491-001 CSD17381F4 FET SM — SHDN_10MHZ_OUT

R149 A93548-517 4.7K 1% 0.0625W RES 0402

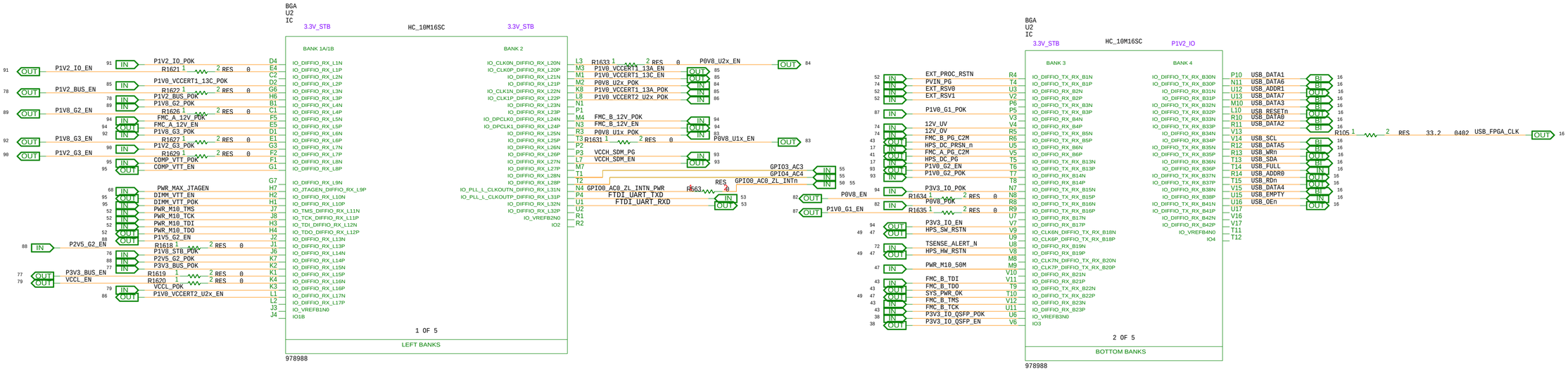
SHDN_10MHZ_OUT= 0 DEFAULT, NO CLOCK
SHDN_10MHZ_OUT= 1 DRIVING 10MHZ CLOCK

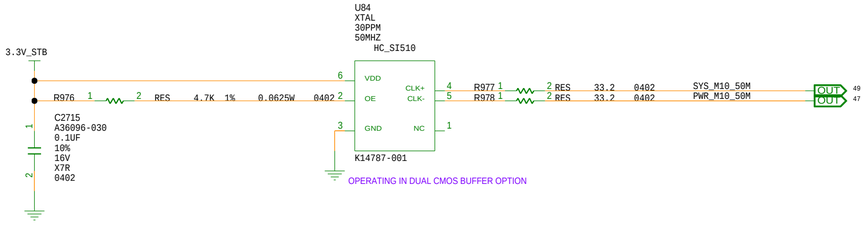
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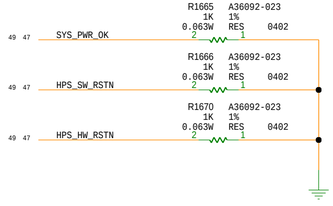
DEPARTMENT	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	SIZE	CODE	DOCUMENT NUMBER	REV
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PWR MAX-10- I





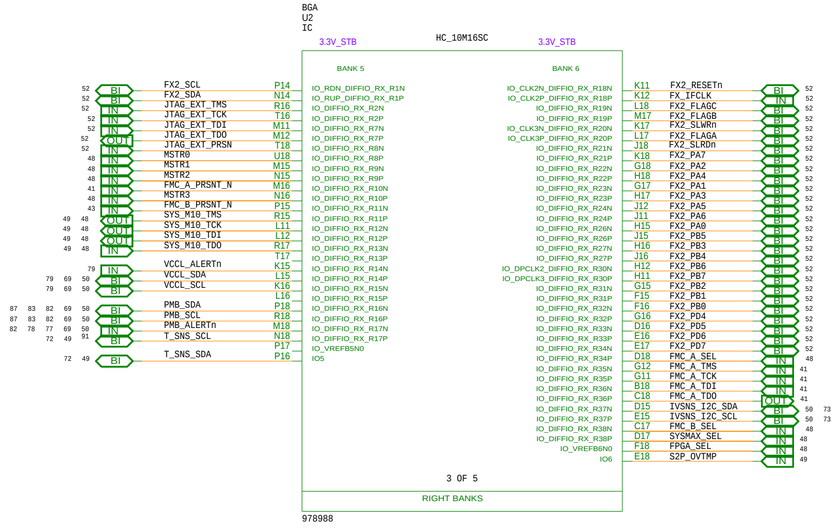


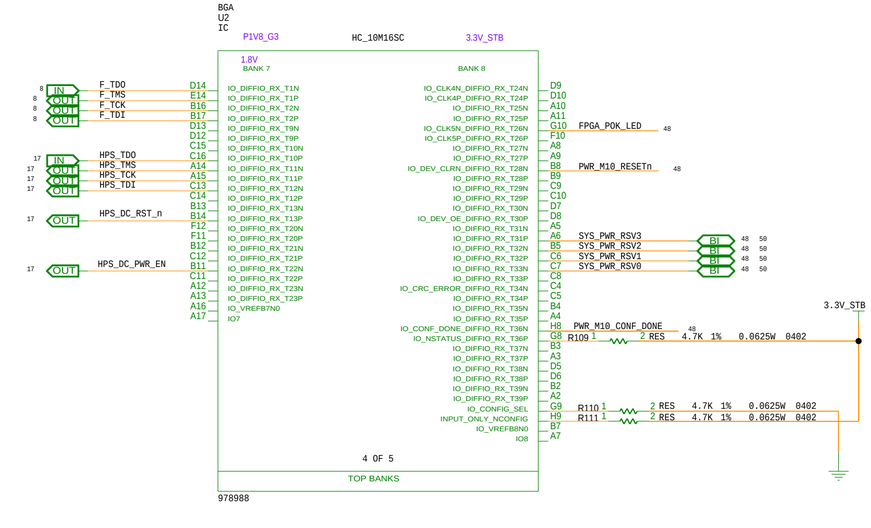
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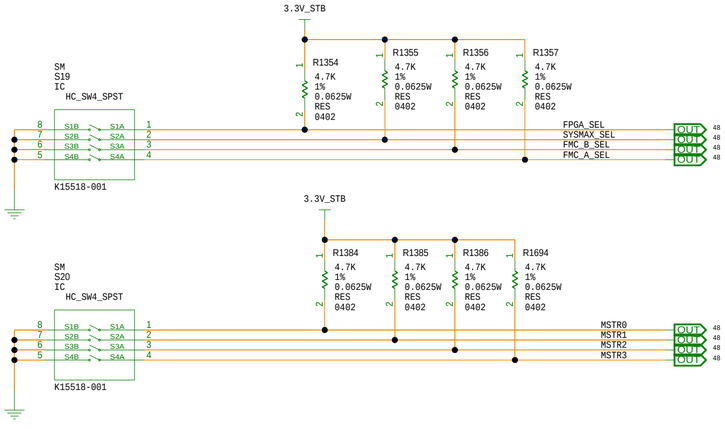
DEPARTMENT		SIZE	CODE	DOCUMENT NUMBER	REV
	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 47 OF 97	

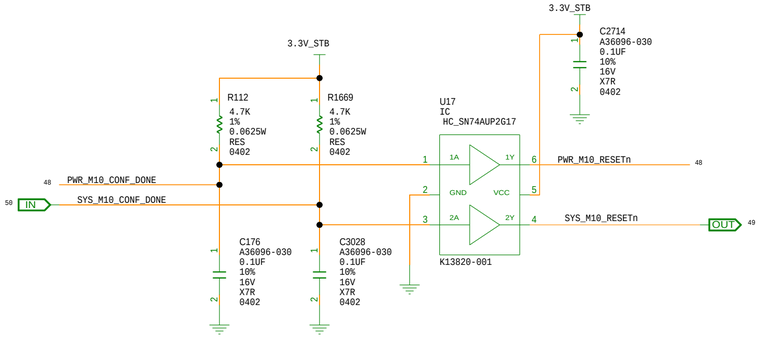
ALTERA CONFIDENTIAL

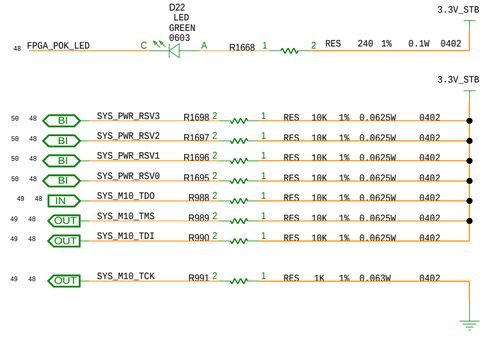
PWR MAX-10- II











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		SCALE:	DO NOT SCALE DRAWING	SHEET 48 OF 97	

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SYS MAX-10- I

BGA
U3
IC

HC_10M16SC

3.3V_STB

BANK 1A/1B

68	IN	FPGA_MSEL0	D4	IO_DIFFIO_RX_L1N
68	IN	FPGA_MSEL1	E4	IO_DIFFIO_RX_L1P
68	IN	FPGA_MSEL2	C2	IO_DIFFIO_RX_L2N
48	OUT	S2P_OVTMP	D2	IO_DIFFIO_RX_L2P
47	IN	HPS_SW_RSTN	G6	IO_DIFFIO_RX_L3N
72 48	BI	T_SNS_SCL	H6	IO_DIFFIO_RX_L3P
72 48	BI	T_SNS_SDA	B1	IO_DIFFIO_RX_L4N
69 60 54	BI	VCXO_SDA	C1	IO_DIFFIO_RX_L4P
69 60 54	BI	VCXO_SCL	F5	IO_DIFFIO_RX_L5N
37	OUT	PCIE_PERSTn_A	E5	IO_DIFFIO_RX_L5P
37	BI	PCIE_SCL_A	D1	IO_DIFFIO_RX_L6N
37	BI	PCIE_SDA_A	E1	IO_DIFFIO_RX_L6P
37	OUT	PCIE_ALERTn_A	G3	IO_DIFFIO_RX_L7N
43	BI	FMC_A_SDA_3V3	F2	IO_DIFFIO_RX_L7P
43	BI	FMC_A_SCL_3V3	F1	IO_DIFFIO_RX_L8N
55	OUT	PTP_CLK_RST_N	G1	IO_DIFFIO_RX_L8P
70 55	OUT	PTP_CLK_LOL	G7	IO_DIFFIO_RX_L9N
			H7	IO_JTAGEN_DIFFIO_RX_L9P
67	OUT	OPT_FAN_RPM	H2	IO_DIFFIO_RX_L10N
67	OUT	FAN_RPM	H1	IO_DIFFIO_RX_L10P
48	IN	SYS_M10_TMS	J7	IO_TMS_DIFFIO_RX_L11N
48	IN	SYS_M10_TCK	J8	IO_TCK_DIFFIO_RX_L11P
48	IN	SYS_M10_TDI	H3	IO_TDI_DIFFIO_RX_L12N
48	IN	SYS_M10_TDO	H4	IO_TDO_DIFFIO_RX_L12P
48	OUT	MUX_SEL4	J2	IO_DIFFIO_RX_L13N
62	OUT	MUX_SEL5	J1	IO_DIFFIO_RX_L13P
62	OUT	MUX_SEL7	J6	IO_DIFFIO_RX_L14N
61	OUT	MUX_SEL0	K7	IO_DIFFIO_RX_L14P
61	OUT	MUX_SEL1	K2	IO_DIFFIO_RX_L15N
61	OUT	MUX_SEL2	K1	IO_DIFFIO_RX_L15P
61	OUT	MUX_SEL3	K4	IO_DIFFIO_RX_L16N
62	OUT	MUX_SEL6	K3	IO_DIFFIO_RX_L16P
47	IN	SYS_PWR_OK	L1	IO_DIFFIO_RX_L17N
47	IN	HPS_HW_RSTN	L2	IO_DIFFIO_RX_L17P
60	OUT	PWM_VCXO	J3	IO_VREFB1N0
60	OUT	VCXO_FS_3V3	J4	IO1B

1 OF 5

LEFT BANKS

978988

3.3V_STB

BANK 2

			L3	IO_CLK0N_DIFFIO_RX_L20N
48	IN	SYS_M10_RESETn	M3	IO_CLK0P_DIFFIO_RX_L20P
47	IN	SYS_M10_50M	M1	IO_DIFFIO_RX_L21N
68	IN	SYS_SW0	M2	IO_DIFFIO_RX_L21P
68	IN	SYS_SW1	K8	IO_CLK1N_DIFFIO_RX_L22N
68	IN	SYS_SW2	L8	IO_CLK1P_DIFFIO_RX_L22P
68	IN	SYS_SW3	N1	IO_DIFFIO_RX_L23N
68	IN	SYS_SW4	P1	IO_DIFFIO_RX_L23P
68	IN	SYS_SW5	M4	IO_DPCLK0_DIFFIO_RX_L24N
68	IN	SYS_SW6	N3	IO_DPCLK1_DIFFIO_RX_L24P
68	IN	SYS_SW7	R3	IO_DIFFIO_RX_L25N
68	IN	SYS_PB0	T3	IO_DIFFIO_RX_L25P
68	IN	SYS_PB1	P2	IO_DIFFIO_RX_L26N
68	IN	SYS_PB2	P3	IO_DIFFIO_RX_L26P
68	IN	SYS_PB3	L7	IO_DIFFIO_RX_L27N
68	IN	SYS_PB4	M7	IO_DIFFIO_RX_L27P
68	IN	SYS_PB5	T1	IO_DIFFIO_RX_L28N
68	OUT	SYS_LED0	T2	IO_DIFFIO_RX_L28P
68	OUT	SYS_LED1	N4	IO_PLL_L_CLKOUTN_DIFFIO_RX_L31N
68	OUT	SYS_LED2	P4	IO_PLL_L_CLKOUTP_DIFFIO_RX_L31P
68	OUT	SYS_LED3	U1	IO_DIFFIO_RX_L32N
68	OUT	SYS_LED4	U2	IO_DIFFIO_RX_L32P
68	OUT	SYS_LED5	R1	IO_VREFB2N0
68	OUT	SYS_LED6	R2	IO2
68	OUT	SYS_LED7		

OD 68
OD 68

3.3V_STB

R113
A93548-034
10K
1%
0.0625W
RES
0402

19 OUT FM_GXF_12A_PERST_N — R931 0 5% 0.0625W RES 0402 — R_12A_PERST_IO_N

18 OUT FM_GXF_12C_PERST_N — R1114 0 5% 0.0625W RES 0402 — R_12C_PERST_IO_N

BGA
U3
IC

HC_10M16SC

P1V8_G3

BANK 3

67	OUT	MUX_DIP_SW10	R4	IO_DIFFIO_TX_RX_B1N
18	OUT	CLK_GXF_IO_PLL	T4	IO_DIFFIO_RX_B1P
67	OUT	MUX_DIP_SW11	U3	IO_DIFFIO_RX_B2N
45	OUT	USB2_RESETn	V2	IO_DIFFIO_RX_B2P
69	OUT	VCCL_I2C_EN	P6	IO_DIFFIO_TX_RX_B3N
8	BI	F_SDM_IO11	P5	IO_DIFFIO_TX_RX_B3P
68 8	BI	F_SDM_IO9	V3	IO_DIFFIO_RX_B4N
68 8	BI	F_SDM_IO5	V4	IO_DIFFIO_RX_B4P
8	BI	F_SDM_IO6	R5	IO_DIFFIO_TX_RX_B5N
8	BI	F_SDM_IO4	R6	IO_DIFFIO_TX_RX_B5P
8	BI	F_SDM_IO10	U5	IO_DIFFIO_RX_B6N
8	BI	F_SDM_IO13	V5	IO_DIFFIO_RX_B6P
8	BI	F_SDM_IO1	T5	IO_DIFFIO_TX_RX_B13N
			T6	IO_DIFFIO_TX_RX_B13P
8	BI	F_SDM_IO16	T7	IO_DIFFIO_RX_B14N
69 8	BI	F_SDM_IO8	T8	IO_DIFFIO_RX_B14P
8	BI	F_NSTATUS	N7	IO_DIFFIO_TX_RX_B15N
8	BI	F_SDM_IO14	N8	IO_DIFFIO_TX_RX_B15P
69 8	OUT	F_SDM_IO12	R8	IO_DIFFIO_TX_RX_B16N
68 8	BI	F_SDM_IO7	R9	IO_DIFFIO_TX_RX_B16P
8	BI	F_SDM_IO0	U7	IO_DIFFIO_RX_B17N
8	BI	F_SDM_IO15	V7	IO_DIFFIO_RX_B17P
8	BI	F_NCONFIG	V9	IO_CLK6N_DIFFIO_TX_RX_B18N
8	OUT	FC_RSTN	U9	IO_CLK6P_DIFFIO_TX_RX_B18P
8	BI	F_SDM_IO2	U8	IO_DIFFIO_RX_B19N
8	BI	F_SDM_IO3	V8	IO_DIFFIO_RX_B19P
67	IN	MUX_DIP_SW9	M8	IO_CLK7N_DIFFIO_TX_RX_B20N
			M9	IO_CLK7P_DIFFIO_TX_RX_B20P
		R_12A_PERST_IO_N	V10	IO_DIFFIO_RX_B21N
		R_12C_PERST_IO_N	V11	IO_DIFFIO_RX_B21P
69	OUT	CLK_I2C_EN1	T9	IO_DIFFIO_TX_RX_B22N
67	IN	MUX_DIP_SW8	T10	IO_DIFFIO_TX_RX_B22P
67	IN	USER_PB0	V12	IO_DIFFIO_RX_B23N
67	IN	USER_PB1	U11	IO_DIFFIO_RX_B23P
			U6	IO_VREFB3N0
8	IN	FC_PRSNTn	V6	IO3

BANK 4

			P10	IO_DIFFIO_TX_RX_B30N
		CLK_I2C_SCL	N11	IO_DIFFIO_TX_RX_B30P
57 58 59 69	BI	CLK_I2C_SCL	U12	IO_DIFFIO_RX_B31N
57 58 59 69	BI	SI5391_A_OEN	U13	IO_DIFFIO_RX_B31P
57	OUT	SI5391_A_RSTN	M10	IO_DIFFIO_TX_RX_B32N
67	IN	MUX_DIP_SW0	L10	IO_DIFFIO_TX_RX_B32P
67	IN	MUX_DIP_SW1	R10	IO_DIFFIO_TX_RX_B33N
67	IN	MUX_DIP_SW2	R11	IO_DIFFIO_TX_RX_B33P
67	IN	MUX_DIP_SW3	V13	IO_DIFFIO_RX_B34N
58	OUT	SI5391_B_RSTN	V14	IO_DIFFIO_RX_B34P
59	OUT	SI5332_1_IN3	R12	IO_DIFFIO_TX_RX_B35N
59	OUT	SI5332_1_IN1	R13	IO_DIFFIO_TX_RX_B35P
59	OUT	SI5332_1_IN2	T13	IO_DIFFIO_RX_B36N
67	IN	MUX_DIP_SW4	T14	IO_DIFFIO_RX_B36P
58	OUT	SI5391_B_OEN	R14	IO_DIFFIO_TX_RX_B37N
49	BI	QSPI_D0	T15	IO_DIFFIO_TX_RX_B37P
49	BI	QSPI_D1	V15	IO_DIFFIO_RX_B38N
49	BI	QSPI_D2	U15	IO_DIFFIO_RX_B38P
49	BI	QSPI_D3	U16	IO_DIFFIO_TX_RX_B41N
49	OUT	QSPI_SCLK	U17	IO_DIFFIO_TX_RX_B41P
49	OUT	QSPI_CSN	V16	IO_DIFFIO_RX_B42N
49	OUT	QSPI_RSTN	V17	IO_DIFFIO_RX_B42P
67	IN	MUX_DIP_SW5	T11	IO_VREFB4N0
67	IN	MUX_DIP_SW6	T12	IO4
67	IN	MUX_DIP_SW7		

2 OF 5

BOTTOM BANKS

978988

BGA
U85
IC

HC_MT25QU02GCBB

1.8V_STB

		VCC	B4		
49	IN	QSPI_SCLK	B2	C	DQ0 D3 QSPI_D0 BI 49
49	IN	QSPI_CSN	C2	S_N	DQ1 D2 QSPI_D1 BI 49
				W_N/DQ2	C4 QSPI_D2 BI 49
			B3	DQ3/HOLD_N	D4 QSPI_D3 BI 49
		VSS			
		NC	A2		RFU C5
		RFU	A3		RFU D1
49	IN	QSPI_RSTN	A4	RESET_N	RFU D5
		RFU	A5		RFU E1
		RFU	B1		RFU E2
		RFU	B5		RFU E3
		RFU	C1		RFU E4
		RFU	C3		RFU E5

K14016-001

1.8V_STB

C2711
K19409-001
4.7UF
20%
6.3V
X6S
0402

C2712
A36096-030
0.1UF
10%
16V
X7R
0402

1.8V_STB

R979 2 1 RES 4.7K 1% 0.0625W 0402 QSPI_D0 BI 49
R980 2 1 RES 4.7K 1% 0.0625W 0402 QSPI_D1 BI 49
R981 2 1 RES 4.7K 1% 0.0625W 0402 QSPI_D2 BI 49
R982 2 1 RES 4.7K 1% 0.0625W 0402 QSPI_D3 BI 49
R983 2 1 RES 4.7K 1% 0.0625W 0402 QSPI_CSN OUT 49
R984 2 1 RES 1K 1% 0.063W 0402 QSPI_SCLK OUT 49

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DEPARTMENT	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	SIZE: C+	CODE: 34649	DOCUMENT NUMBER: 150-0330690-D1 / REV: D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 49 OF 97

ALTERA CONFIDENTIAL

SYS MAX-10- II

BGA
U3
IC

HC_10M16SC

3.3V_STB

BANK 7

OUT	MUX_SEL_ZL	D14	IO_DIFFIO_RX_T1N
OUT	SI5394_LOL	E14	IO_DIFFIO_RX_T1P
OUT	FMC_B_PCIE_WAKEN_3V3	B16	IO_DIFFIO_RX_T2N
OUT	SI5394_OE_N	B17	IO_DIFFIO_RX_T2P
IN	PMB_ALERTn	D13	IO_DIFFIO_RX_T9N
IN	SI5394_INT_N	D12	IO_DIFFIO_RX_T9P
IN	OVERTEMP_N	C15	IO_DIFFIO_RX_T10N
OUT	USB_MUX_RESET	C16	IO_DIFFIO_RX_T10P
BI	SYS_MAX10_I2C_SDA	A14	IO_DIFFIO_RX_T11N
OUT	SYS_MAX10_I2C_SCL	A15	IO_DIFFIO_RX_T11P
OUT	MUX_SEL15	C13	IO_DIFFIO_RX_T12N
	GPIO0_AC0_ZL_INTN_SYS	C14	IO_DIFFIO_RX_T12P
OUT	MUX_SEL8	B13	IO_DIFFIO_RX_T13N
OUT	SI5394_RSTN	B14	IO_DIFFIO_RX_T13P
OUT	MUX_SEL9	F12	IO_DIFFIO_RX_T20N
OUT	TUSB1146_SCL	F11	IO_DIFFIO_RX_T20P
BI	TUSB1146_SDA	B12	IO_DIFFIO_RX_T21N
OUT	MUX_SEL10	C12	IO_DIFFIO_RX_T21P
OUT	FMC_A_PCIE_WAKEN_3V3	B11	IO_DIFFIO_RX_T22N
OUT	FMC_A_PCIE_PERSTN_3V3	C11	IO_DIFFIO_RX_T22P
OUT	MUX_SEL11	A12	IO_DIFFIO_RX_T23N
OUT	MUX_SEL12	A13	IO_DIFFIO_RX_T23P
OUT	MUX_SEL13	A16	IO_VREFB7N0
OUT	MUX_SEL14	A17	IO7

IN GPIO0_AC0_ZL_INTn — R584 1 2 RES 0

TUSB1146_SCL — R258 1 2 RES 0

TUSB1146_SDA — R259 1 2 RES 0

BANK 8

	IO_CLK4N_DIFFIO_RX_T24N	D9	USER_SW0
	IO_CLK4P_DIFFIO_RX_T24P	D10	USER_SW1
	IO_DIFFIO_RX_T25N	A10	USER_SW2
	IO_DIFFIO_RX_T25P	A11	USER_SW3
	IO_CLK5N_DIFFIO_RX_T26N	G10	USER_SW4
	IO_CLK5P_DIFFIO_RX_T26P	F10	USER_SW5
	IO_DIFFIO_RX_T27N	A8	USER_SW6
	IO_DIFFIO_RX_T27P	A9	USER_SW7
	IO_DEV_CLRN_DIFFIO_RX_T28N	B8	MCIO_CLK_SEL_EPn
	IO_DIFFIO_RX_T28P	B9	VCCL_SDA
	IO_DIFFIO_RX_T29N	C9	VCCL_SCL
	IO_DIFFIO_RX_T29P	C10	PMB_SDA
	IO_DIFFIO_RX_T30N	D7	PMB_SCL
	IO_DEV_OE_DIFFIO_RX_T30P	D8	MCIO_CLK_ENN
	IO_DIFFIO_RX_T31N	A5	IVSNS_I2C_SDA
	IO_DIFFIO_RX_T31P	A6	IVSNS_I2C_SCL
	IO_DIFFIO_RX_T32N	B5	FMC_B_PCIE_PERSTN_3V3
	IO_DIFFIO_RX_T32P	C6	TPS65987D_HPD
	IO_DIFFIO_RX_T33N	C7	FMC_B_SDA_3V3
	IO_DIFFIO_RX_T33P	C8	FMC_B_SCL_3V3
	IO_CRC_ERROR_DIFFIO_RX_T34N	C4	SYS_PWR_RSV3
	IO_DIFFIO_RX_T34P	C5	SYS_PWR_RSV2
	IO_DIFFIO_RX_T35N	B4	SYS_PWR_RSV1
	IO_DIFFIO_RX_T35P	A4	SYS_PWR_RSV0
	IO_CONF_DONE_DIFFIO_RX_T36N	H8	SYS_M10_CONF_DONE
	IO_NSTATUS_DIFFIO_RX_T36P	G8	R117 1 2 RES 4.7K 1% 0.0625W 0402
	IO_DIFFIO_RX_T37N	B3	USER_LED3
	IO_DIFFIO_RX_T37P	A3	USER_LED4
	IO_DIFFIO_RX_T38N	D5	USER_LED5
	IO_DIFFIO_RX_T38P	D6	USER_LED6
	IO_DIFFIO_RX_T39N	B2	USER_LED7
	IO_DIFFIO_RX_T39P	A2	USER_LED0
	IO_CONFIG_SEL	G9	R118 1 2 RES 4.7K 1% 0.0625W 0402
	INPUT_ONLY_NCONFIG	H9	R119 1 2 RES 4.7K 1% 0.0625W 0402
	IO_VREFB8N0	B7	USER_LED1
	IO8	A7	USER_LED2

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TOP BANKS

978988

3.3V_STB

BGA
U3
IC

P1V2_IO — HC_10M16SC — P1V2_IO

BANK 5

OUT	FPGA_SGPI	P14	IO_RDN_DIFFIO_RX_R1N
OUT	DIMM_IO_EN	N14	IO_RUP_DIFFIO_RX_R1P
BI	F_GPIO_03	R16	IO_DIFFIO_RX_R2N
BI	F_GPIO_01	T16	IO_DIFFIO_RX_R2P
OUT	FM_GXF_13C_SPARE_IO_N — R654 1 2 RES 0	M11	R_13C_SPARE_IO_N
OUT	CLK_I2C_EN0	M12	IO_DIFFIO_RX_R7N
BI	F_GPIO_06	T18	IO_DIFFIO_RX_R7P
OUT	FM_GXF_13A_PERST_IO_N — R745 1 2 RES 0	U18	R_13A_PERST_IO_N
BI	FPGA_SGPIO_CLK	M15	IO_DIFFIO_RX_R8N
BI	VCXO_INC	N15	IO_DIFFIO_RX_R8P
BI	F_GPIO_04	M16	IO_DIFFIO_RX_R9N
		N16	IO_DIFFIO_RX_R9P
BI	F_GPIO_02	P15	IO_DIFFIO_RX_R10N
BI	VCXO_DEC	R15	IO_DIFFIO_RX_R10P
		L11	IO_DIFFIO_RX_R11N
OUT	ZL_I2C_EN	L12	IO_DIFFIO_RX_R11P
BI	F_GPIO_07	R17	IO_DIFFIO_RX_R12N
		T17	IO_DIFFIO_RX_R12P
BI	FPGA_SGPIO_SYNC	K15	IO_DIFFIO_RX_R13N
BI	FPGA_SGPO	L15	IO_DIFFIO_RX_R13P
BI	F_GPIO_05	K16	IO_DIFFIO_RX_R14N
BI	F_GPIO_08	L16	IO_DIFFIO_RX_R14P
BI	F_GPIO_00	P18	IO_DIFFIO_RX_R15N
OUT	FPGA_RESETn	R18	IO_DIFFIO_RX_R15P
BI	F_GPIO_09	M18	IO_DIFFIO_RX_R16N
BI	F_GPIO_10	N18	IO_DIFFIO_RX_R16P
BI	F_GPIO_12	P17	IO_DIFFIO_RX_R17N
BI	F_GPIO_11	P16	IO_DIFFIO_RX_R17P
			IO_VREFB5N0
			IO5

BANK 6

	IO_CLK2N_DIFFIO_RX_R18N	K11	AVST_D20
	IO_CLK2P_DIFFIO_RX_R18P	K12	
	IO_DIFFIO_RX_R19N	L18	AVST_D26
	IO_DIFFIO_RX_R19P	M17	AVST_D31
	IO_CLK3N_DIFFIO_RX_R20N	K17	AVST_D17
	IO_CLK3P_DIFFIO_RX_R20P	L17	AVST_D30
	IO_DIFFIO_RX_R21N	J18	AVST_D28
	IO_DIFFIO_RX_R21P	K18	AVST_D27
	IO_DIFFIO_RX_R22N	G18	AVST_D9
	IO_DIFFIO_RX_R22P	H18	AVST_D16
	IO_DIFFIO_RX_R23N	G17	AVST_D7
	IO_DIFFIO_RX_R23P	H17	AVST_READY
	IO_DIFFIO_RX_R24N	J12	AVST_D25
	IO_DIFFIO_RX_R24P	J11	AVST_D24
	IO_DIFFIO_RX_R26N	H15	AVST_D8
	IO_DIFFIO_RX_R26P	J15	AVST_D21
	IO_DIFFIO_RX_R27N	H16	AVST_D29
	IO_DIFFIO_RX_R27P	J16	VCXO_FS_1V2
	IO_DPCLK2_DIFFIO_RX_R30N	H12	AVST_D22
	IO_DPCLK3_DIFFIO_RX_R30P	H11	AVST_CLK
	IO_DIFFIO_RX_R31N	G15	AVST_D13
	IO_DIFFIO_RX_R31P	F15	AVST_VALID
	IO_DIFFIO_RX_R32N	F16	AVST_D11
	IO_DIFFIO_RX_R32P	G16	AVST_D10
	IO_DIFFIO_RX_R33N	D16	AVST_D12
	IO_DIFFIO_RX_R33P	E16	AVST_D15
	IO_DIFFIO_RX_R34N	E17	AVST_D2
	IO_DIFFIO_RX_R34P	D18	AVST_D3
	IO_DIFFIO_RX_R35N	G12	AVST_D23
	IO_DIFFIO_RX_R35P	G11	AVST_D19
	IO_DIFFIO_RX_R36N	B18	AVST_D6
	IO_DIFFIO_RX_R36P	C18	AVST_D1
	IO_DIFFIO_RX_R37N	D15	AVST_D18
	IO_DIFFIO_RX_R37P	E15	AVST_D14
	IO_DIFFIO_RX_R38N	C17	AVST_D5
	IO_DIFFIO_RX_R38P	D17	AVST_D4
	IO_VREFB6N0	F18	CA_13C_GXF_IO_DFX_1
	IO6	E18	AVST_D0

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RIGHT BANKS

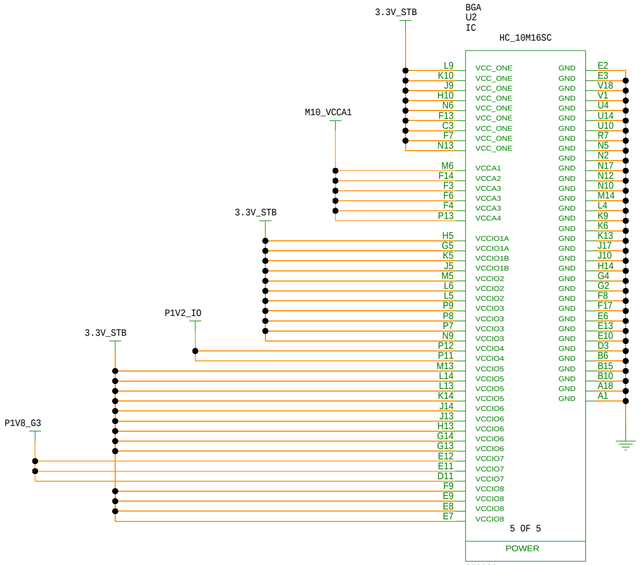
978988

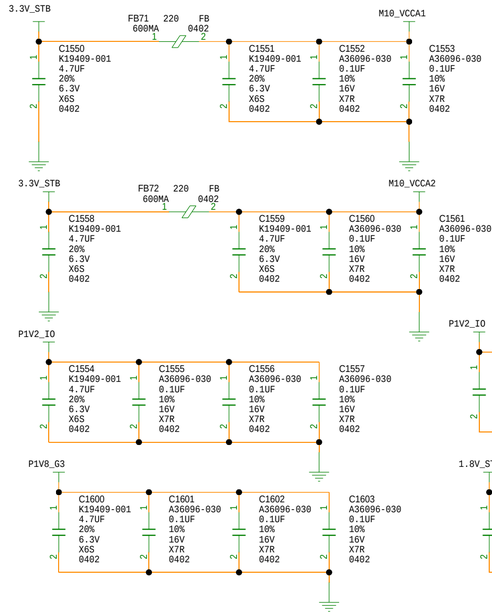
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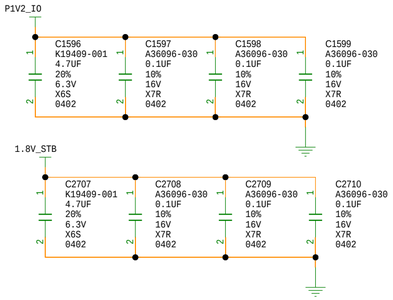
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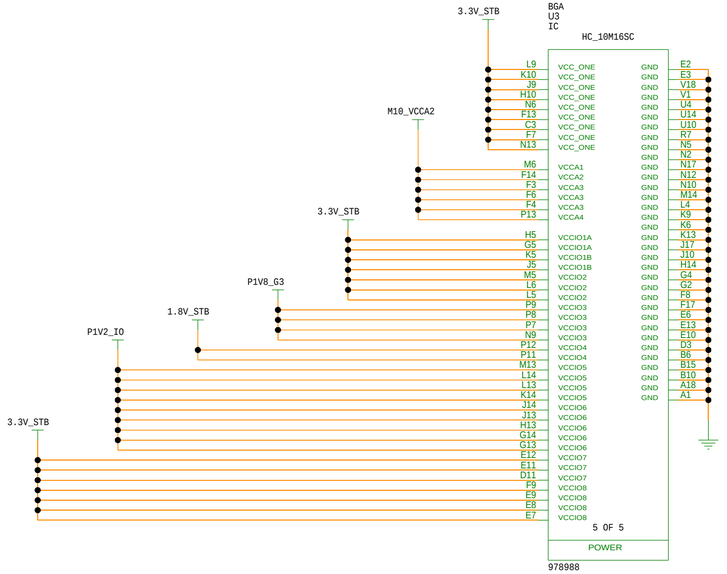
ALTERA CONFIDENTIAL

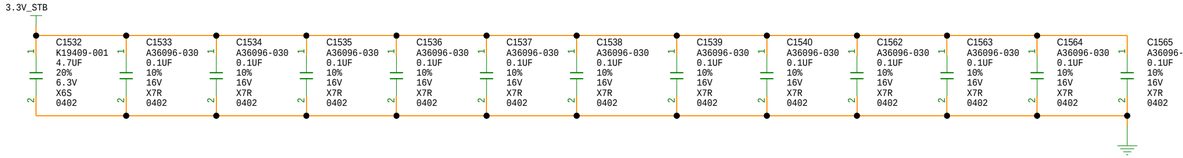
MAX-10 POWER

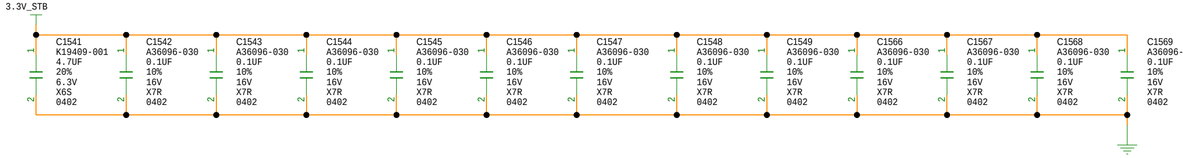


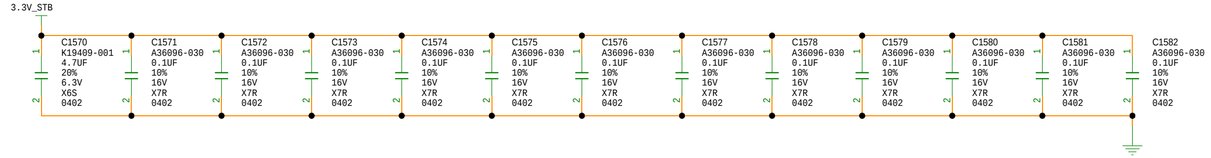


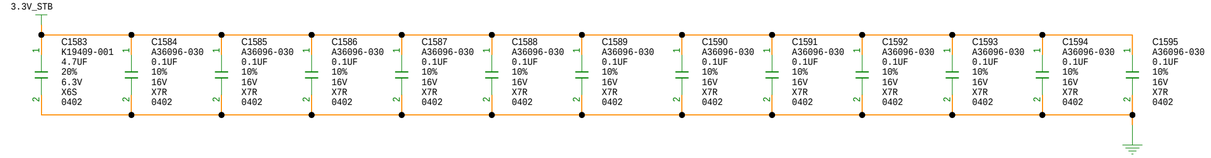












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		SCALE:	DO NOT SCALE DRAWING	SHEET	51 OF 97

ALTERA CONFIDENTIAL

USB BLASTER

3.3V_STB
R78 A36092-023 1K 1% 0.063W EMPTY 0402
R79 RES 10K 1% 0.0625W 0402
R80 K32862-001 20K 0.1% 0.063W RES 0402
C159 A36096-030 0.1UF 10% 16V X7R 0402
FX2_WAKEUP 52
TH J10 CON
D83384-001
GND_EARTH
CONN_USB_DM
CONN_USB_DP
U15 HC_DIODE_TVS_ARRAY_2CH
K69553-001
R275 EMPTY 0402 0 FTDI_USB_DM
R273 EMPTY 0402 0 FTDI_USB_DP
R275&R409, R273&R412 PLACED AS TRIPAD
R409 RES 0402 0 FX2_USB_DM
R672 RES 0402 0 FX2_USB_DP
FX_IFCLK 48
R85 RES 33.2
Y1 XTAL 10PPM 24MHZ HC_XTAL_4P
K15540-001
C160 K13748-001 12PF 5% 25V C0G 0402
C161 K13748-001 12PF 5% 25V C0G 0402
U16 IC HC_CY7C68013A
K33816-001
FX2_PA0 FX2_PA1 FX2_PA2 FX2_PA3 FX2_PA4 FX2_PA5 FX2_PA6 FX2_PA7
RESERVED
AGND AGND
GND GND GND GND GND GND
AVCC AVCC
VCC VCC VCC VCC VCC VCC
DMINUS DPLUS
IFCLK XTALIN XTALOUT
CLKOUT
RESET_N SCL SDA
WAKEUP
CTL0/FLAGA CTL1/FLAGB CTL2/FLAGC
RDY0/SLRD RDY1/SLWR
THPAD
FX2_PB0 FX2_PB1 FX2_PB2 FX2_PB3 FX2_PB4 FX2_PB5 FX2_PB6 FX2_PB7
FX2_PD0 FX2_PD1 FX2_PD2 FX2_PD3 FX2_PD4 FX2_PD5 FX2_PD6 FX2_PD7
FX2_RESETn FX2_SCL FX2_SDA
FX2_WAKEUP
FX2_FLAGA FX2_FLAGB FX2_FLAGC
FX2_SLRDn FX2_SLWRN
R81 A93548-034 10K 1% 0.0625W EMPTY 0402
R82 RES 5% 0 0402 0.0625W
R83 A93548-596 100K 1% 0.0625W RES 0402
FX2_FTDI_RESETN
U14 IC HC_MAX81X
D16587-001
MAX811SEUS+T 1
C158 K19409-001 4.7UF 20% 6.3V X6S 0402
PLACED CLOSE TO CY7C68013
C162 A36096-030 0.1UF 10% 16V X7R 0402
C163 A36096-030 0.1UF 10% 16V X7R 0402
C164 A36096-030 0.1UF 10% 16V X7R 0402
C165 A36096-030 0.1UF 10% 16V X7R 0402
C166 A36096-030 0.1UF 10% 16V X7R 0402
C167 A36096-030 0.1UF 10% 16V X7R 0402
C168 A36096-030 0.1UF 10% 16V X7R 0402
C169 H80885-001 10UF 20% 6.3V X6S 0402
C170 A36096-030 0.1UF 10% 16V X7R 0402
R86 RES 4.7K 1% 0.0625W 0402
R87 RES 4.7K 1% 0.0625W 0402
FX2_SCL
FX2_SDA

FOLLOW TRIPAD PLACEMENT FOR BELOW RESISTORS

R276 EMPTY 0 FTDI_JTAG_NPROC_RST
R1486 RES 240 EXT_PROC_RSTN_R
EXT_PROC_RSTN
R279 EMPTY0 FTDI_JTAG_NTRST
R1487 RES 240 EXT_RSV0_R
EXT_RSV0
R285 EMPTY 0 FTDI_JTAG_TCK
R88 RES 0 FX2_PD0
PWR_M10_TCK
R281 EMPTY 0 FTDI_JTAG_TDO
R91 RES 0 FX2_PD3
PWR_M10_TDO
R280 EMPTY 0 FTDI_JTAG_TMS
R97 RES 0 FX2_PD1
PWR_M10_TMS
R282 EMPTY 0 FTDI_JTAG_TDI
R102 RES 0 FX2_PD2
PWR_M10_TDI

R1726 RES 1M 1% 0.25W 1206
C3045 X7R 1000V 1000PF 10% 1206
GND_EARTH

R92 RES 10K 1% 0402 0.0625W
R96 RES 10K 1% 0402 0.0625W
R98 RES 10K 1% 0402 0.0625W
R101 RES 1K 1% 0402 0.063W
JTAG_EXT_TDO
JTAG_EXT_TMS
JTAG_EXT_TDI
JTAG_EXT_TCK

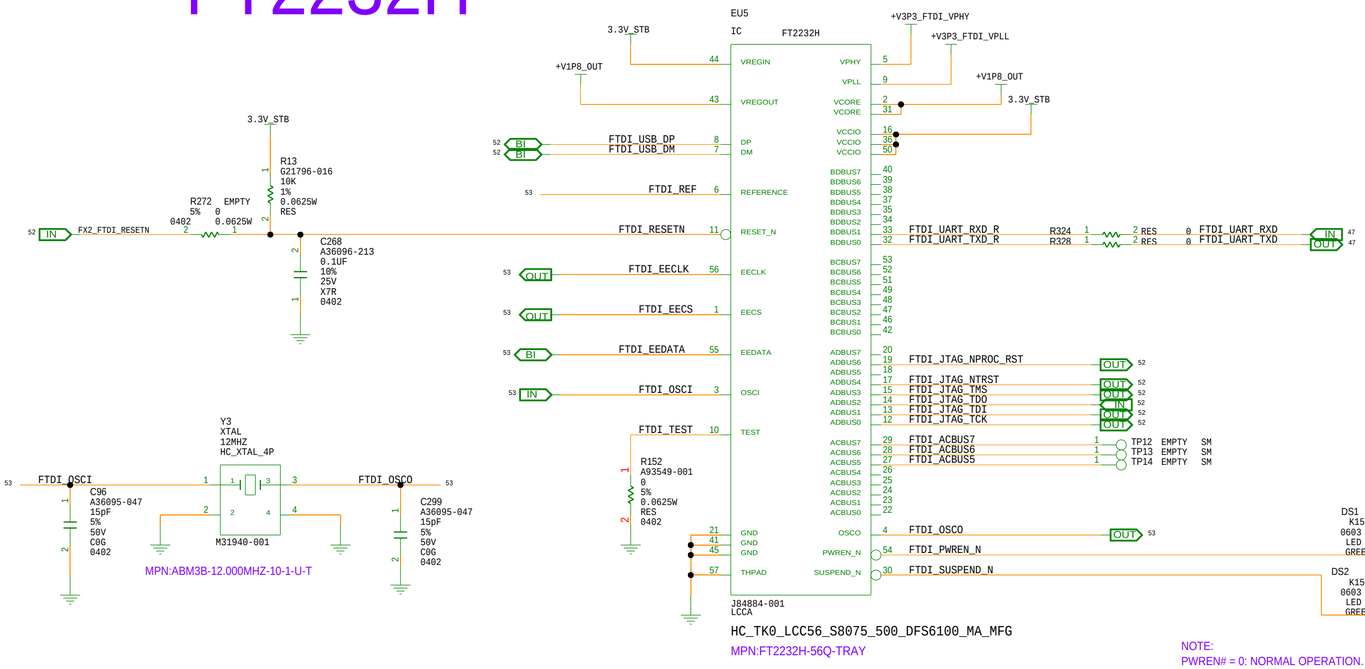
R1727 RES 10K 1% 0402 0.0625W
R1728 RES 10K 1% 0402 0.0625W
R1729 RES 10K 1% 0402 0.0625W
R1730 RES 1K 1% 0402 0.063W
PWR_M10_TDO
PWR_M10_TMS
PWR_M10_TDI
PWR_M10_TCK

R1664 A93548-517 4.7K 1% 0.0625W RES 0402
R95 A93548-517 4.7K 1% 0.0625W RES 0402
TH J11 CON
K10905-001
R90 RES 0
R94 RES 0
R100 RES 0
R104 RES 0
JTAG_EXT_TCK
JTAG_EXT_TDO
JTAG_EXT_TMS
JTAG_EXT_TDI
R1485 RES 240 JTAG_EXT_PRSN
EXT_PROC_RSTN_R
EXT_RSV0_R
R1488 RES 240 EXT_RSV1
C171 A36096-030 0.1UF 10% 16V X7R 0402
C3024 A36095-047 15PF 5% 50V C0G 0402
C3025 A36095-047 15PF 5% 50V C0G 0402
C3026 A36095-047 15PF 5% 50V C0G 0402
C172 A36095-047 15PF 5% 50V C0G 0402
C173 A36095-047 15PF 5% 50V C0G 0402
C174 A36095-047 15PF 5% 50V C0G 0402
C175 A36095-047 15PF 5% 50V C0G 0402
C3027 A36095-047 15PF 5% 50V C0G 0402
R1662 RES 4.7K 1% 0.0625W 0402
R1663 RES 4.7K 1% 0.0625W 0402
EXT_RSV0
EXT_PROC_RSTN

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DEPARTMENT	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	SIZE	CODE	DOCUMENT NUMBER	REV
		C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET	52 OF 97

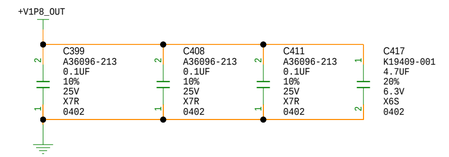
FT2232H

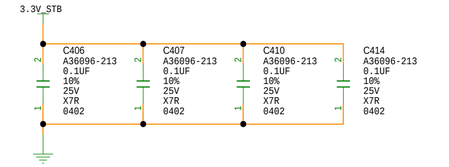


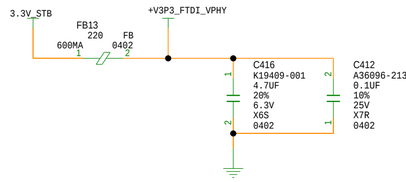
NOTE:
PWREN# = 0: NORMAL OPERATION.
PWREN# = 1: USB SUSPEND MODE OR DEVICE HAS NOT BEEN CONFIGURED.

NOTE:
SUSPEND# = 0: USB IS IN SUSPEND MODE

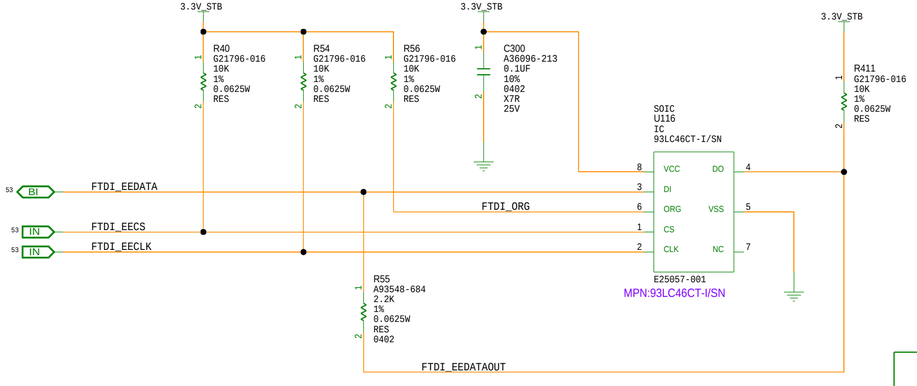
DECAP

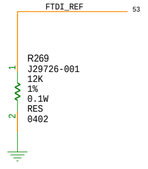


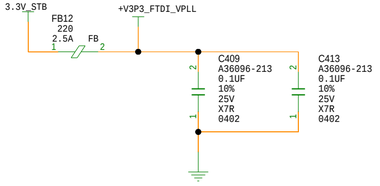




EEPROM





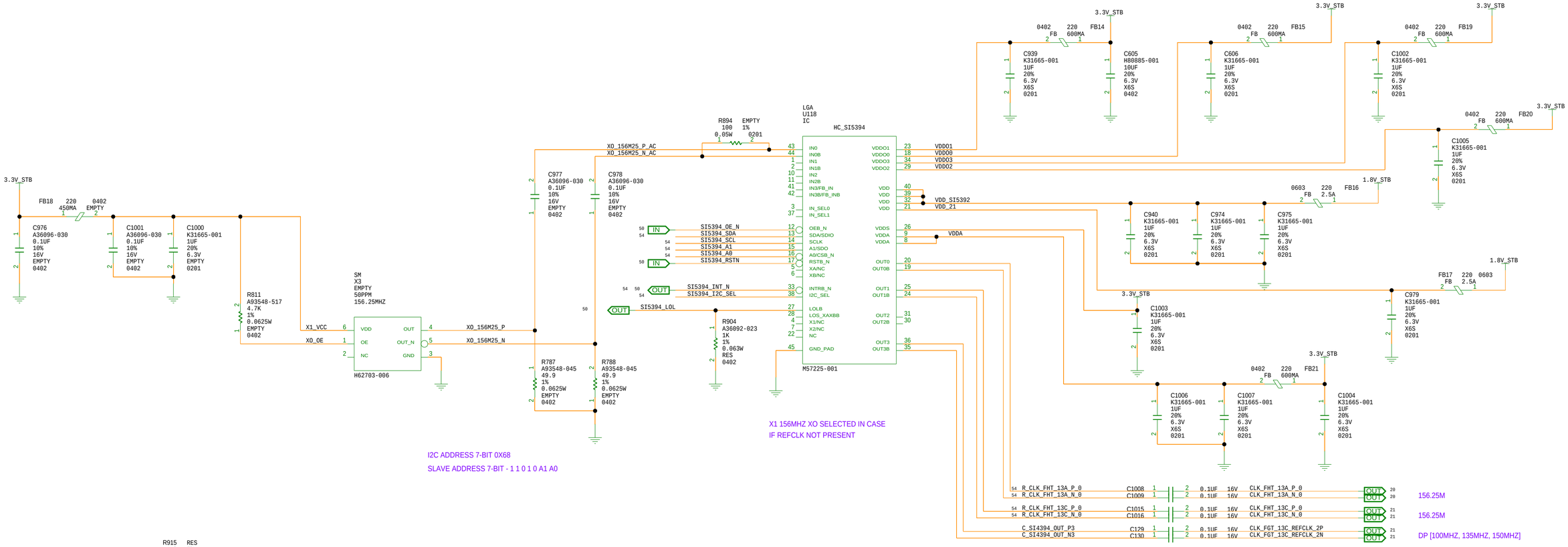


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DEPARTMENT	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	SIZE: C+ / CODE: 34649	DOCUMENT NUMBER: 150-0330690-D1	REV: D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 53 OF 97

ALTERA CONFIDENTIAL

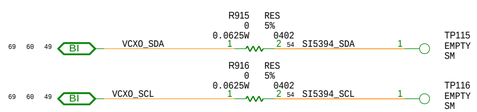
SI5394 CLOCK GENERATOR



I2C ADDRESS 7-BIT 0X68

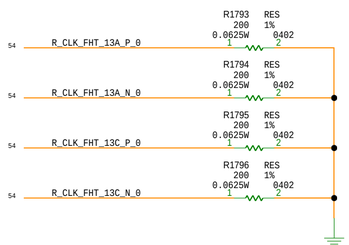
SLAVE ADDRESS 7-BIT - 1 1 0 1 0 A1 A0

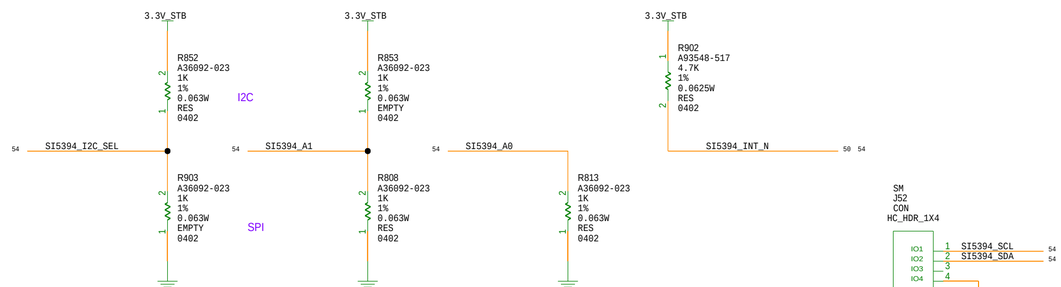
X1 156MHZ XO SELECTED IN CASE
IF REFCLK NOT PRESENT



IO_VDD_SEL = 0 (DEFAULT)

CHANGE IO_VDD_SEL = 1 FOR 3.3V OPERATION





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DEPARTMENT		SIZE	CODE	DOCUMENT NUMBER	REV
	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 54 OF 97	

ALTERA CONFIDENTIAL

ZL30733 PLL CLOCK GENERATOR-1

U123
IC
HC_ZL30733

OUT0P	52	C_TOD_CLK_156.25M_P	C924	0.1UF	16V	TOD_CLK_156.25M_P	OUT	12
OUT0N	51	C_TOD_CLK_156.25M_N	C927	0.1UF	16V	TOD_CLK_156.25M_N	OUT	12
OUT1P	55	R_1PPS_FPGA_CLK	R444	RES 36		1PPS_FPGA_CLK	OUT	70
OUT1N	54	R_1PPS_SMA_OUT	R777	RES 0		1PPS_SMA_OUT	OUT	46
OUT2P	58	R_10MHz_SMA_OUT	R151	RES 0		10MHz_SMA_OUT	OUT	46
OUT2N	57	R_ZL_SPARE_CLK_100M	R933	EMPTY 0		ZL_SPARE_CLK	OUT	21
OUT3P	61	C_ETH_REFCLK_156.25M_P	C289	0.1UF	16V	ETH_REFCLK_156.25M_P	OUT	21
OUT3N	60	C_ETH_REFCLK_156.25M_N	C290	0.1UF	16V	ETH_REFCLK_156.25M_N	OUT	21
OUT4P	64	C_CPRI_REFCLK_153.6M_P	C291	0.1UF	16V	CPRI_REFCLK_153.6M_P	OUT	21
OUT4N	63	C_CPRI_REFCLK_153.6M_N	C292	0.1UF	16V	CPRI_REFCLK_153.6M_N	OUT	21
OUT5P	17	C_CPRI_REFCLK_184.32M_P	C301	0.1UF	16V	CPRI_REFCLK_184.32M_P	OUT	21
OUT5N	18	C_CPRI_REFCLK_184.32M_N	C302	0.1UF	16V	CPRI_REFCLK_184.32M_N	OUT	21
OUT6P	20	C_FMCA_REFCLK_122.88M_P	C943	0.1UF	16V	FMCA_REFCLK_122.88M_P	OUT	61
OUT6N	21	C_FMCA_REFCLK_122.88M_N	C952	0.1UF	16V	FMCA_REFCLK_122.88M_N	OUT	61
OUT7P	23	C_FMCA_REFCLK_156.25M_P	C993	0.1UF	16V	FMCA_REFCLK_156.25M_P	OUT	62
OUT7N	24	C_FMCA_REFCLK_156.25M_N	C994	0.1UF	16V	FMCA_REFCLK_156.25M_N	OUT	62
OUT8P	26	C_SI5391_A_156.25M_REFIN_P	C303	0.1UF	16V	SI5391_A_156.25M_REFIN_P	OUT	57
OUT8N	27	C_SI5391_A_156.25M_REFIN_N	C496	0.1UF	16V	SI5391_A_156.25M_REFIN_N	OUT	57
OUT9P	29	C_SI5391_B_156.25M_REFIN_P	C1013	0.1UF	16V	SI5391_B_156.25M_REFIN_P	OUT	58
OUT9N	30	C_SI5391_B_156.25M_REFIN_N	C1018	0.1UF	16V	SI5391_B_156.25M_REFIN_N	OUT	58

M41980-001

TP84 EMPTY SM — ZL_SCK_SCL
R810 EMPTY 1% 0 0201 0.05W — ZL_SPI_SCK — IN 70
R812 RES 1% 0 0201 0.05W — ZL_I2C_SCL — IN 69
R820 RES 1% 0 0201 0.05W — ZL_I2C_SDA — BI 69
TP86 EMPTY SM — ZL_SI_SDA
R822 EMPTY 1% 0 0201 0.05W — ZL_SPI_SI — IN 70
R823 EMPTY 1% 0 0201 0.05W — ZL_CS_IF0 — ZL_SPI_CS — IN 70
R824 EMPTY 1% 0 0201 0.05W — ZL_SO_ADDR0 — ZL_SPI_SO — OUT 70

CONNECT TO DEBUG CONNECTOR

P2V5_G2

FB110 220 FB 600MA 0402 — VDD00_ZL30733 55
C941 K31665-001 1UF 20% 6.3V X6S 0201
C950 H80885-001 10UF 20% 6.3V X6S 0402
C951 K31665-001 1UF 20% 6.3V X6S 0201

FB112 220 FB 600MA 0402 — VDD01_ZL30733 55
C926 K31665-001 1UF 20% 6.3V X6S 0201
C922 H80885-001 10UF 20% 6.3V X6S 0402
C925 K31665-001 1UF 20% 6.3V X6S 0201

FB113 220 FB 600MA 0402 — VDD02_ZL30733 55
C948 K31665-001 1UF 20% 6.3V X6S 0201
C938 H80885-001 10UF 20% 6.3V X6S 0402
C942 K31665-001 1UF 20% 6.3V X6S 0201

FB22 220 FB 600MA 0402 — VDD03_ZL30733 55
C955 K31665-001 1UF 20% 6.3V X6S 0201
C953 H80885-001 10UF 20% 6.3V X6S 0402
C954 K31665-001 1UF 20% 6.3V X6S 0201

FB23 220 FB 600MA 0402 — VDD04_ZL30733 55
C958 K31665-001 1UF 20% 6.3V X6S 0201
C956 H80885-001 10UF 20% 6.3V X6S 0402
C957 K31665-001 1UF 20% 6.3V X6S 0201

FB25 220 FB 600MA 0402 — VDD06_ZL30733 55
C964 K31665-001 1UF 20% 6.3V X6S 0201
C962 H80885-001 10UF 20% 6.3V X6S 0402
C963 K31665-001 1UF 20% 6.3V X6S 0201

FB26 220 FB 600MA 0402 — VDD07_ZL30733 55
C1019 K31665-001 1UF 20% 6.3V X6S 0201
C965 H80885-001 10UF 20% 6.3V X6S 0402
C966 K31665-001 1UF 20% 6.3V X6S 0201

FB24 220 FB 600MA 0402 — VDD05_ZL30733 55
C961 K31665-001 1UF 20% 6.3V X6S 0201
C959 H80885-001 10UF 20% 6.3V X6S 0402
C960 K31665-001 1UF 20% 6.3V X6S 0201

FB27 220 FB 600MA 0402 — VDD08_ZL30733 55
C1022 K31665-001 1UF 20% 6.3V X6S 0201
C1020 H80885-001 10UF 20% 6.3V X6S 0402
C1021 K31665-001 1UF 20% 6.3V X6S 0201

FB28 220 FB 600MA 0402 — VDD09_ZL30733 55
C1025 K31665-001 1UF 20% 6.3V X6S 0201
C1023 H80885-001 10UF 20% 6.3V X6S 0402
C1024 K31665-001 1UF 20% 6.3V X6S 0201

3.3V_STB

FB104 220 FB 600MA 0402
C921 K31665-001 1UF 20% 6.3V X6S 0201
C920 H80885-001 10UF 20% 6.3V X6S 0402

3.3V_STB

FB106 220 FB 2.5A 0603
C923 H80885-001 10UF 20% 6.3V X6S 0402
C928 H80885-001 10UF 20% 6.3V X6S 0402
C929 K31665-001 1UF 20% 6.3V X6S 0201
C931 K31665-001 1UF 20% 6.3V X6S 0201
C930 K31665-001 1UF 20% 6.3V X6S 0201
C932 K31665-001 1UF 20% 6.3V X6S 0201

1.8V_STB

FB108 220 FB 2.5A 0603
C936 K31665-001 1UF 20% 6.3V X6S 0201
C949 H80885-001 10UF 20% 6.3V X6S 0402
C944 K31665-001 1UF 20% 6.3V X6S 0201
C945 K31665-001 1UF 20% 6.3V X6S 0201
C946 K31665-001 1UF 20% 6.3V X6S 0201
C947 K31665-001 1UF 20% 6.3V X6S 0201

U123
IC
HC_ZL30733

VDDIO 44
VDD33 3, 14, 38, 43
VDD18 1, 2, 37, 15

55	VDD00_ZL30733	50
55	VDD01_ZL30733	53
55	VDD02_ZL30733	56
55	VDD03_ZL30733	59
55	VDD04_ZL30733	62
55	VDD05_ZL30733	19
55	VDD06_ZL30733	22
55	VDD07_ZL30733	25
55	VDD08_ZL30733	28
55	VDD09_ZL30733	31

EPAD_VSS 65

1 OF 4
M41980-001

U123
IC
HC_ZL30733

RST_B	16	PTP_CLK_RST_n	IN	49 55
SCK_SCL	36	ZL_SCK_SCL		55
SI_SDA	35	ZL_SI_SDA		55
SO_IF1	33	ZL_SO_ADDR0		55
CS_B_IF0	34	ZL_CS_IF0		55
GPIO0_AC0	49	R_GPIO0_AC0	R782 RES 0	
GPIO1_AC1	48	ZRL_LOCK	R783 RES 0	
GPIO2_AC2	47	R_FPGA_HPS_ZL_HO	R794 RES 0	
GPIO3_AC3	46	GPIO3_AC3_R	R331 RES 0	
GPIO4_AC4	45	GPIO4_AC4_R	R555 RES 0	
IC0	32	1 — TP8 EMPTY SM		

4 OF 4
M41980-001

ZL_CS_IF0="0" : I2C (DEFAULT)
ZL_CS_IF0="1": SPI

3.3V_STB

R938 A93548-034 10K 1% 0.0625W EMPTY 0402
R867 A36092-023 1K 1% 0.063W EMPTY 0402
R778 A36092-023 1K 1% 0.063W RES 0402
R892 A36092-023 1K 1% 0.063W RES 0402
R779 A36092-023 1K 1% 0.063W EMPTY 0402
R780 A36092-023 1K 1% 0.063W EMPTY 0402

GPIO0_AC0_ZL_INTn — OUT 58 47
PTP_CLK_LOL — OUT 49 70
FPGA_HPS_ZL_HO — OUT 70
GPIO3_AC3 — OUT 47
GPIO4_AC4 — OUT 47

R936 A36092-023 1K 1% 0.063W RES 0402
R781 A36092-023 1K 1% 0.063W RES 0402
R789 A36092-023 1K 1% 0.063W RES 0402
R790 A36092-023 1K 1% 0.063W EMPTY 0402
R880 A36092-023 1K 1% 0.063W EMPTY 0402
R791 A36092-023 1K 1% 0.063W RES 0402
R792 A36092-023 1K 1% 0.063W RES 0402

ZL_SO_ADDR0='0' FOR 550MS SETS I2C INTERFACE

I2C ADDRESS = 70H

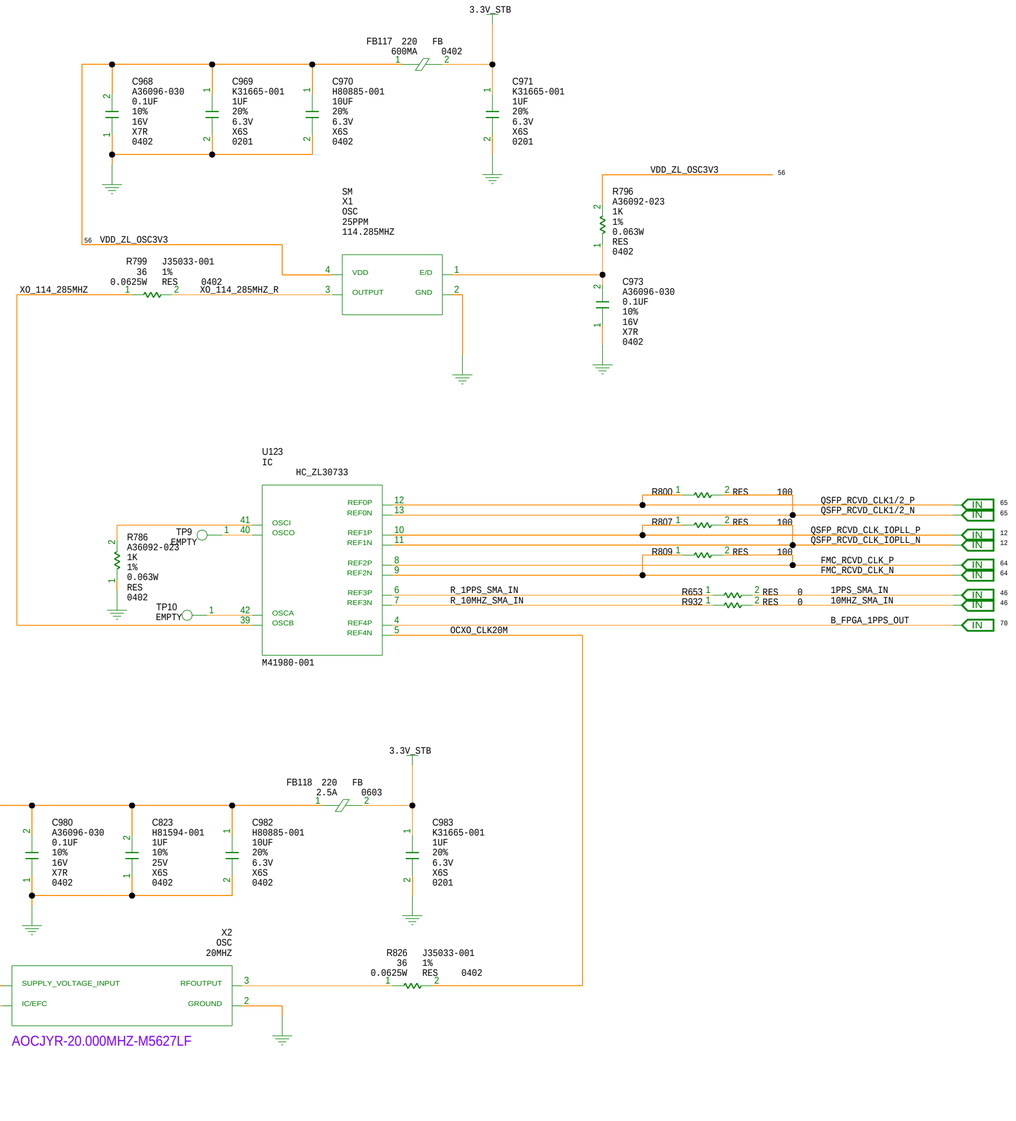
PTP_CLK_RST_n — IN 49 55
R935 C48657-004 10K 1% 0.05W RES 0201

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DEPARTMENT	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	SIZE: C+	CODE: 34649	DOCUMENT NUMBER: 150-0330690-D1
		SCALE:	DO NOT SCALE DRAWING	REV: D1
				SHEET: 55 OF 97

ALTERA CONFIDENTIAL

ZL30733 PLL CLOCK GENERATOR-2

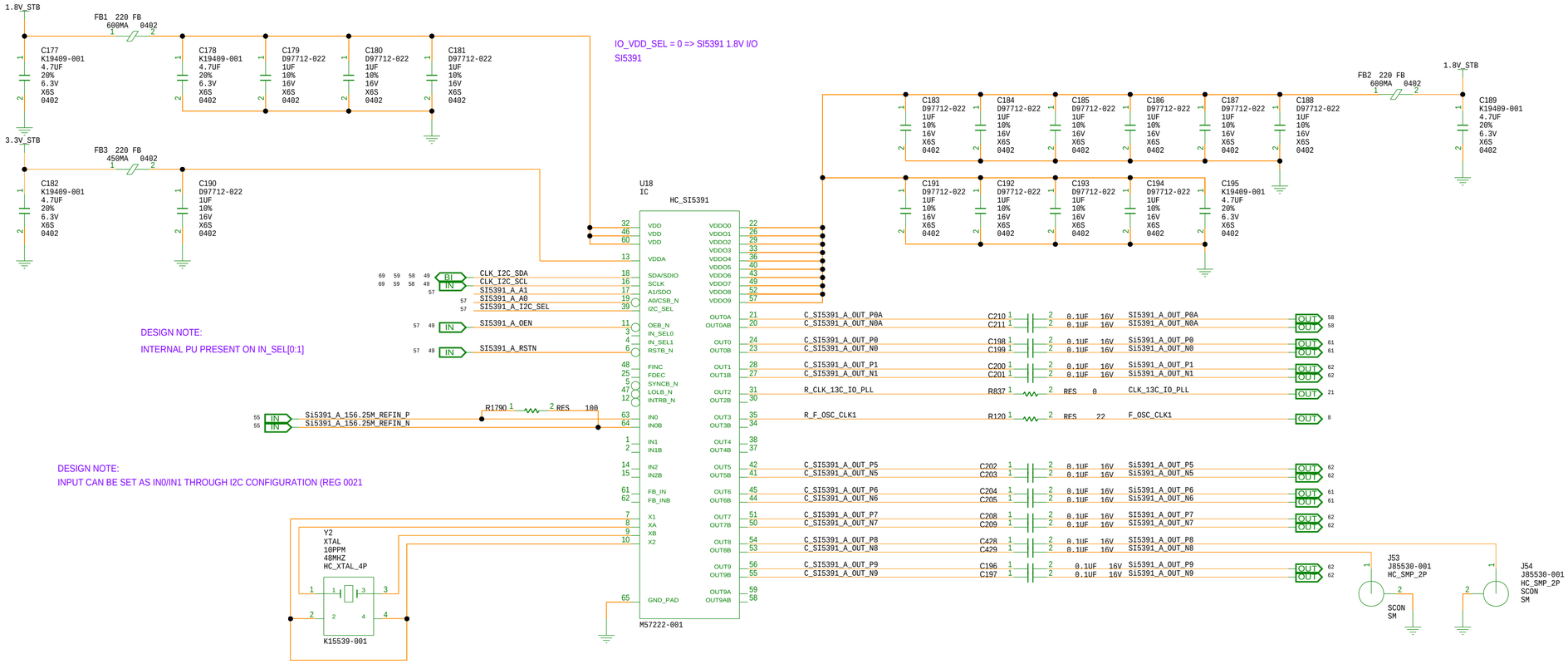


Mon Sep 15 10:27:18 2025

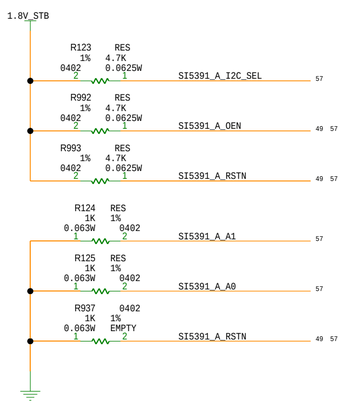
DEPARTMENT		SIZE	CODE	DOCUMENT NUMBER	REV
	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 56 OF 97	

ALTERA CONFIDENTIAL

SI5391 CLOCK GENERATOR-I



PULL-UP/DOWN RESISTORS



I2C_SEL = 1B'1, I2C INTERFACE

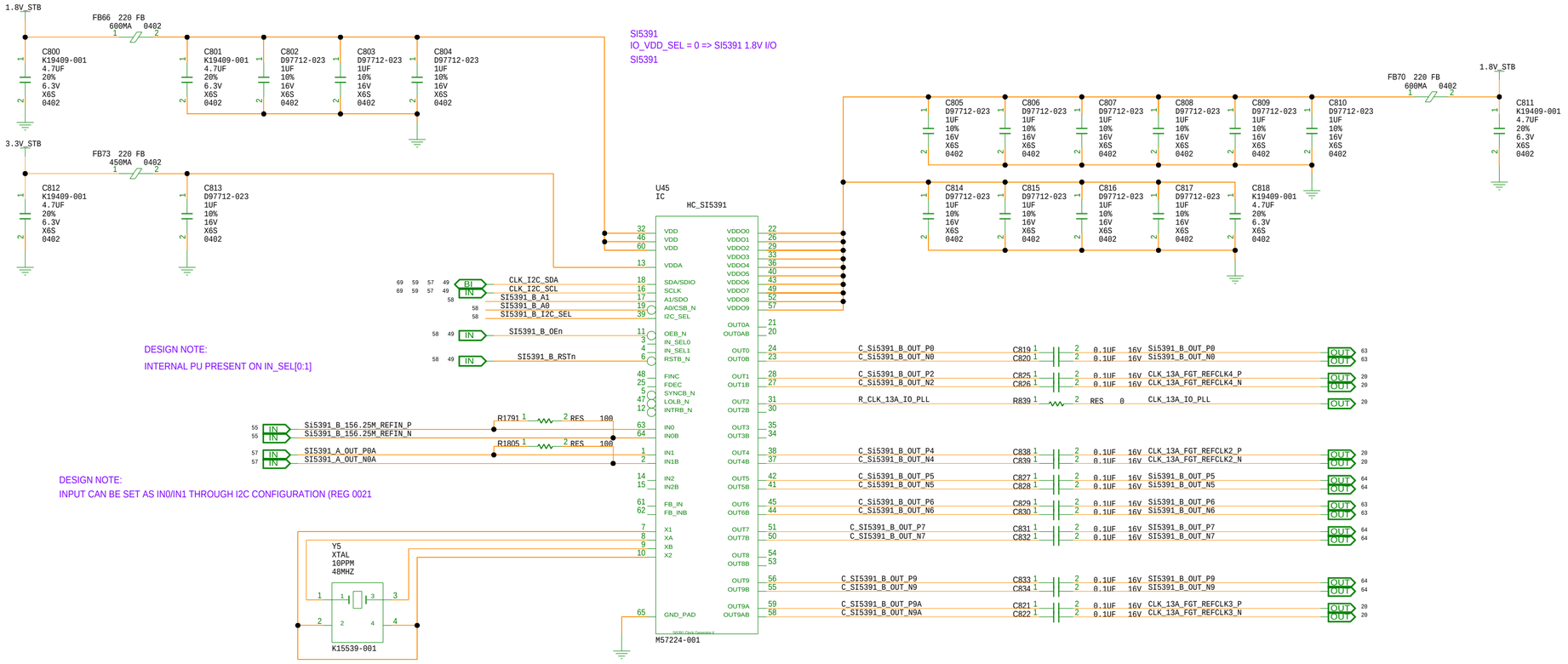
I2C ADDRESS 7B'111 0100

Mon Sep 15 10:27:18 2025

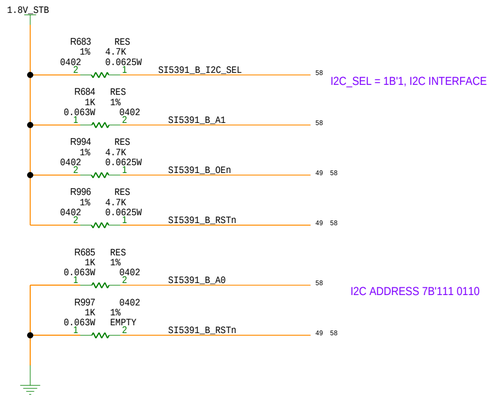
DEPARTMENT	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	SIZE	CODE	DOCUMENT NUMBER	REV
		C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET	57 OF 97

ALTERA CONFIDENTIAL

SI5391 CLOCK GENERATOR-II



PULL-UP/DOWN RESISTORS

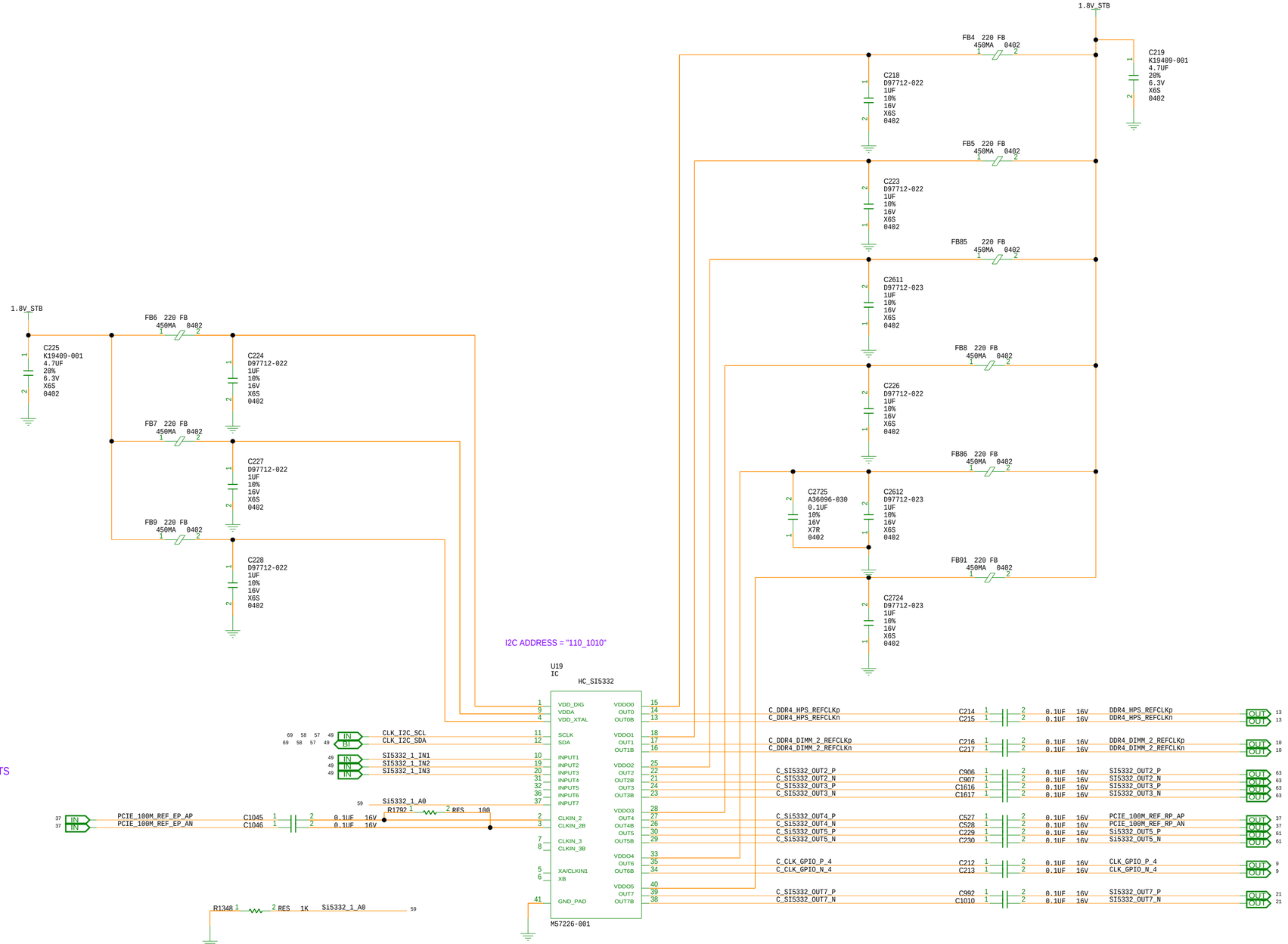


Mon Sep 15 10:27:18 2025

DEPARTMENT	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	SIZE	CODE	DOCUMENT NUMBER	REV
		C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 58 OF 97	

ALTERA CONFIDENTIAL

SI5332 CLOCK GENERATOR



INPUT1: OUTPUT ENABLE FOR ALL CLOCK OUTPUTS
INPUT2: SPREAD SPECTRUM ENABLE

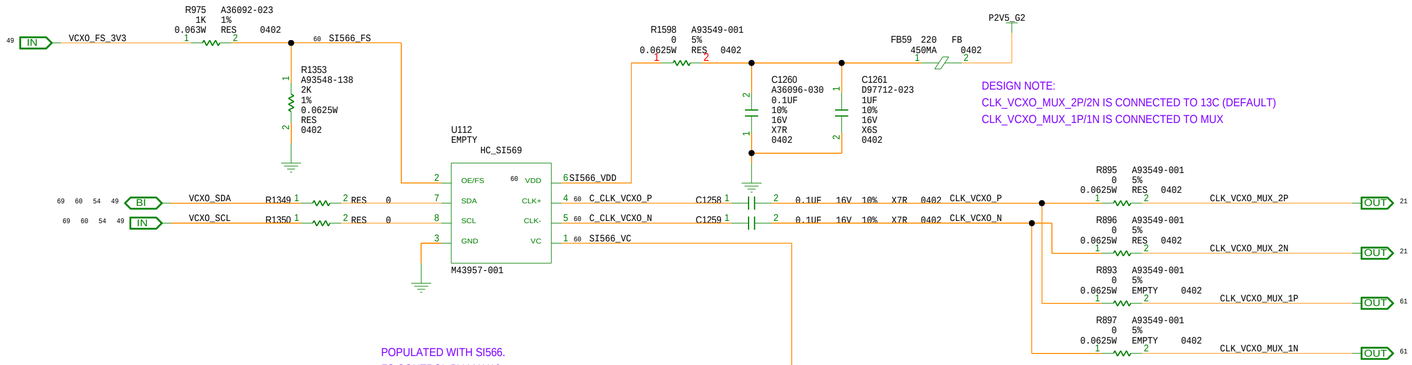
Mon Sep 15 10:27:18 2025

DEPARTMENT		SIZE	CODE	DOCUMENT NUMBER	REV
	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 59 OF 97	

VCXO

CUSTOM PART NUMBER SPECIFICATION:
PART NUMBER566BCA000132ABG
MODEL NUMBERSI566
FREQUENCY (MHZ):
FS=0;148.350000
FS=1:148.500000
OUTPUT FORMAT AND SUPPLY VOLTAGELVDS: 1.8 V, 2.5 V, 3.3 V
TEMPERATURE STABILITY+/- 20 PPM
TUNING SLOPE105 PPM/V
OE PIN LOCATION AND POLARITYVC PIN1, FS PIN2, OE ALWAYS ON
PACKAGE OPTION5X7 MM
OPERATING TEMP RANGE (°C): -40 TO +85

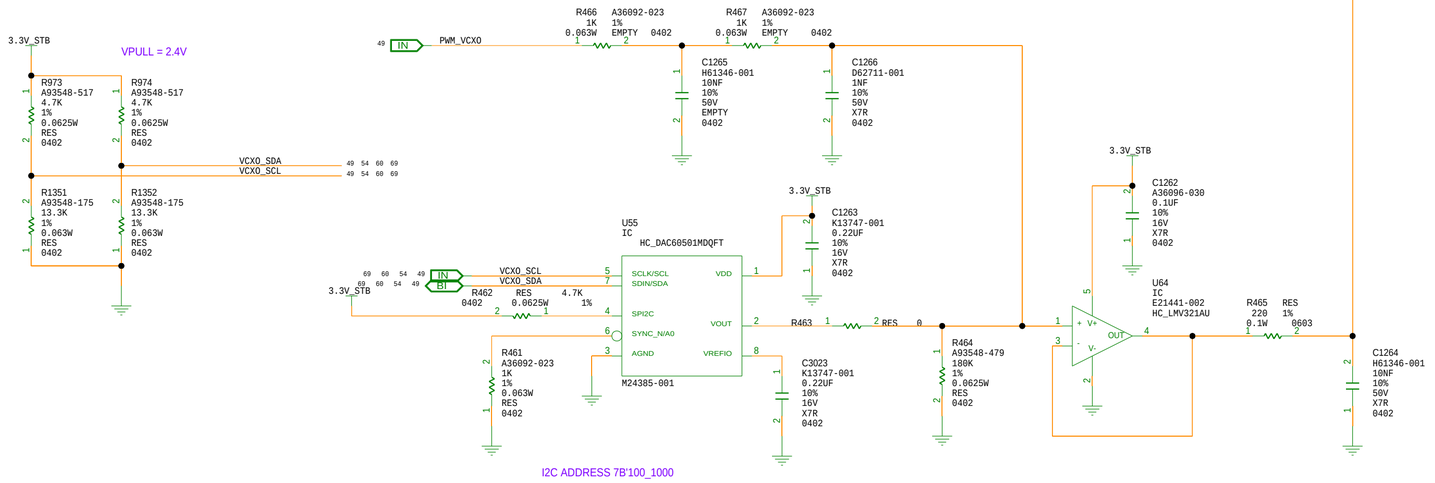




DESIGN NOTE:
CLK_VCXO_MUX_2P/2N IS CONNECTED TO 13C (DEFAULT)
CLK_VCXO_MUX_1P/1N IS CONNECTED TO MUX

POPULATED WITH SI566.
FS CONTROL BY MAX10
FS=0: 148.35MHZ
FS=1: 148.50MHZ
I2C ONLY AVAILABLE IF SI569 IS POPULATED.

DESIGN NOTE:
CLK_VCXO_MUX_2P/2N IS CONNECTED TO 13C (DEFAULT)
CLK_VCXO_MUX_1P/1N IS CONNECTED TO MUX



I2C ADDRESS 7B'100_1000

DEFAULT MIDRANG, 2.5V/2, NO OFFSET FROM THE VDD/2 OF THE VCXO INPUT

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		C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 60 OF 97	

ALTERA CONFIDENTIAL

CLOCK MUX-I

J19 K10975-001 CON TH 1

J20 K10975-001 CON TH 1

S15332_OUT5_P / S15332_OUT5_N

C305 / C307 0.1UF 16V SI53307_SMA1_CLKIN1_P / SI53307_SMA1_CLKIN1_N

R155 RES 100 1% 0.0625W 0402

R156 RES 100

MUX_SEL0 — P2V5_G2

R1521 RES 1K / R1522 RES 2K

R157 RES 1K

R158 / R159 RES 0

U22 IC HC_SI53307_B_GM

CLK0 / CLK0B / CLK1 / CLK1B / CLK_SEL / OE / SFOUT0 / SFOUT1 / Q0 / Q0B / Q1 / Q1B / VDD / VDDO / GND / GND / GND_PAD

K13822-001

C_CLK_A_12C_FGT_P_0 / C_CLK_A_12C_FGT_N_0

C_CLK_3A_GPIO_P_0 / C_CLK_3A_GPIO_N_0

C304 / C306 0.1UF 16V CLK_A_12C_FGT_P_0 / CLK_A_12C_FGT_N_0

C2616 / C2617 0.1UF 16V CLK_3A_GPIO_P_0 / CLK_3A_GPIO_N_0

OUT

FB32 FB 450MA 220 0402 — P2V5_G2

FB33 FB 450MA 220 0402

C308 A36096-030 0.1UF 10% 16V X7R 0402

C309 D97712-022 1UF 10% 16V X6S 0402

C310 A36096-030 0.1UF 10% 16V X7R 0402

C311 D97712-022 1UF 10% 16V X6S 0402

CAD NOTE:
PLACE AS FOUR-PAD RESISTOR, TO AVOID ANY STUB

CAD NOTE:
PLACE AS TRI-PAD WITH AC_CAP TO AVOID ANY STUB

148.5M

J49 K10975-001 CON TH 1

J50 K10975-001 CON TH 1

CLK_VCXO_MUX_1P / CLK_VCXO_MUX_1N

C281 / C598 0.1UF 16V

R1060 / R1075 EMPTY 0

R899 / R1237 RES 0

R1273 / R1277 EMPTY 0

FMC_A_CLK_M2C_P0 / FMC_A_CLK_M2C_N0

R898 / R1242 RES 0

MUX_IN0_P / MUX_IN0_N

R1262 / R1263 RES 0

R1264 / R1265 EMPTY 0 — STUB_CLK_A_12C_FGT_P_1 / STUB_CLK_A_12C_FGT_N_1

R1043 / R1044 EMPTY 0

R1372 RES 100 1% 0.0625W 0402

R_MUX_IN0_P / R_MUX_IN0_N

R1373 RES 100

R_FMC_A_CLK_M2C_P0 / R_FMC_A_CLK_M2C_N0

U108 IC HC_SI53307_B_GM

MUX_SEL1 — P2V5_G2

R1527 / R1528 RES 1K 2K

R1374 RES 1K

R1375 / R1376 RES 0

K13822-001

C_CLK_A_12C_FGT_P_1 / C_CLK_A_12C_FGT_N_1

C_CLK_B_12A_FGT_P_1 / C_CLK_B_12A_FGT_N_1

C2618 / C2619 0.1UF 16V CLK_A_12C_FGT_P_1 / CLK_A_12C_FGT_N_1

C999 / C999 0.1UF 16V CLK_B_12A_FGT_P_1 / CLK_B_12A_FGT_N_1

FB87 FB 450MA 220 0402 — P2V5_G2

FB88 FB 450MA 220 0402

C2622 A36096-030 0.1UF 10% 16V X7R 0402

C2623 D97712-023 1UF 10% 16V X6S 0402

C2624 A36096-030 0.1UF 10% 16V X7R 0402

C2625 D97712-023 1UF 10% 16V X6S 0402

CAD NOTE:
PLACE THESE SMAS CLOSE TO FMC-A

CAD NOTE:
PLACE AS TRI-PAD RESISTOR, TO AVOID ANY STUB

U23 IC HC_SI53307_B_GM

SI5391_A_OUT_P0 / SI5391_A_OUT_N0

C312 / C315 0.1UF 16V C_FMC_A_CLK_M2C_P1 / C_FMC_A_CLK_M2C_N1

FMC_A_CLK_M2C_P1 / FMC_A_CLK_M2C_N1

R160 RES 100 1% 0.0625W 0402

R161 RES 100

MUX_SEL2 — P2V5_G2

R1525 / R1526 RES 1K 2K

R162 RES 1K

R163 / R164 RES 0

K13822-001

C_CLK_A_12C_FGT_P_2 / C_CLK_A_12C_FGT_N_2

C_CLK_3A_GPIO_P_1 / C_CLK_3A_GPIO_N_1

C313 / C314 0.1UF 16V CLK_A_12C_FGT_P_2 / CLK_A_12C_FGT_N_2

C2718 / C2719 0.1UF 16V CLK_3A_GPIO_P_1 / CLK_3A_GPIO_N_1

FB34 FB 450MA 220 0402 — P2V5_G2

FB35 FB 450MA 220 0402

C316 A36096-030 0.1UF 10% 16V X7R 0402

C317 D97712-022 1UF 10% 16V X6S 0402

C318 A36096-030 0.1UF 10% 16V X7R 0402

C319 D97712-022 1UF 10% 16V X6S 0402

STUB_CLK_A_12C_FGT_P_3 / STUB_CLK_A_12C_FGT_N_3

R1266 / R1272 EMPTY 0

U24 IC HC_SI53307_B_GM

SI5391_A_OUT_P6 / SI5391_A_OUT_N6

FMCA_REFCLK_122.88M_P / FMCA_REFCLK_122.88M_N

R1065 / R1074 / R1261 EMPTY 0 / RES 0 / EMPTY 0

R_FMCA_REFCLK_122.88M_P / R_FMCA_REFCLK_122.88M_N

R166 RES 100

MUX_SEL3 — P2V5_G2

R1523 / R1524 RES 1K 2K

R167 RES 1K

R168 / R169 RES 0

K13822-001

C_CLK_A_12C_FGT_P_3 / C_CLK_A_12C_FGT_N_3

C_CLK_3A_GPIO_P_2 / C_CLK_3A_GPIO_N_2

C321 / C322 0.1UF 16V CLK_A_12C_FGT_P_3 / CLK_A_12C_FGT_N_3

C266 / C267 0.1UF 16V CLK_3A_GPIO_P_2 / CLK_3A_GPIO_N_2

FB36 FB 450MA 220 0402 — P2V5_G2

FB37 FB 450MA 220 0402

C324 A36096-030 0.1UF 10% 16V X7R 0402

C325 D97712-022 1UF 10% 16V X6S 0402

C326 A36096-030 0.1UF 10% 16V X7R 0402

C327 D97712-023 1UF 10% 16V X6S 0402

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	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 61 OF 97	

ALTERA CONFIDENTIAL

CLOCK MUX-II

156.25M

U26 IC HC_SI53307_B_GM K13822-001

R175 RES 100 1% 0.0625W 0402

Si5391_A_OUT_P9 / Si5391_A_OUT_N9

FMC_A_CLK_M2C_P2 / FMC_A_CLK_M2C_N2

C338 / C339 0.1UF 16V C_FMC_A_CLK_M2C_P2 / C_FMC_A_CLK_M2C_N2

R176 RES 100

MUX_SEL4

R1529 / R1530 RES 1K / RES 2K

P2V5_G2

R177 RES 1K

R178 / R179 RES 0

C_CLK_A_12C_FGT_P_4 / C_CLK_A_12C_FGT_N_4

C336 / C337 0.1UF 16V CLK_A_12C_FGT_P_4 / CLK_A_12C_FGT_N_4

FB40 FB 450MA 220 0402

FB41 FB 450MA 220 0402

C340 A36096-030 0.1UF 10% 16V X7R 0402

C341 D97712-023 1UF 10% 16V X6S 0402

C342 A36096-030 0.1UF 10% 16V X7R 0402

C343 D97712-023 1UF 10% 16V X6S 0402

368.63M

J39 K10975-001 CON TH 1

J40 K10975-001 CON TH 1

U109 IC HC_SI53307_B_GM K13822-001

R1377 RES 100 1% 0.0625W 0402

SI5391_A_OUT_P7 / SI5391_A_OUT_N7

C2628 / C2629 0.1UF 16V SI53307_SMA2_CLKIN1_P / SI53307_SMA2_CLKIN1_N

R1378 RES 100

MUX_SEL6

R1531 / R1532 RES 1K / RES 2K

P2V5_G2

R1379 RES 1K

R1380 / R1381 RES 0

C_CLK_A_12C_FGT_P_6 / C_CLK_A_12C_FGT_N_6

C2626 / C2627 0.1UF 16V CLK_A_12C_FGT_P_6 / CLK_A_12C_FGT_N_6

FB89 FB 450MA 220 0402

FB90 FB 450MA 220 0402

C2630 A36096-030 0.1UF 10% 16V X7R 0402

C2631 D97712-023 1UF 10% 16V X6S 0402

C2632 A36096-030 0.1UF 10% 16V X7R 0402

C2633 D97712-023 1UF 10% 16V X6S 0402

CLK_SEL: LOW - CLKIN0 (BY DEFAULT), HIGH - CLKIN1

184.32M

U27 IC HC_SI53307_B_GM K13822-001

R180 RES 100 1% 0.0625W 0402

SI5391_A_OUT_P5 / SI5391_A_OUT_N5

FMC_A_CLK_M2C_P3 / FMC_A_CLK_M2C_N3

C346 / C347 0.1UF 16V C_FMC_A_CLK_M2C_P3 / C_FMC_A_CLK_M2C_N3

R181 RES 100

MUX_SEL5

R1533 / R1534 RES 1K / RES 2K

P2V5_G2

R182 RES 1K

R183 / R184 RES 0

C_CLK_A_12C_FGT_P_5 / C_CLK_A_12C_FGT_N_5

C344 / C345 0.1UF 16V CLK_A_12C_FGT_P_5 / CLK_A_12C_FGT_N_5

FB42 FB 450MA 220 0402

FB43 FB 450MA 220 0402

C348 A36096-030 0.1UF 10% 16V X7R 0402

C349 D97712-023 1UF 10% 16V X6S 0402

C350 A36096-030 0.1UF 10% 16V X7R 0402

C351 D97712-023 1UF 10% 16V X6S 0402

CAD NOTE:
PLACE AS TRI-PAD RESISTOR, TO AVOID ANY STUB

156.25

FMCA_REFCLK_156.25M_P / FMCA_REFCLK_156.25M_N

R1096 / R1111 / R1112 / R1113 EMPTY 0 / RES 0 / RES 0 / EMPTY 0

STUB_CLK_A_12A_FGT_P_7 / STUB_CLK_A_12A_FGT_N_7

R1083 / R1085 EMPTY 0 / EMPTY 0

CAD NOTE:
PLACE AS TRI-PAD WITH AC_CAP TO AVOID ANY STUB

U25 IC HC_SI53307_B_GM K13822-001

R170 RES 100 1% 0.0625W 0402

SI5391_A_OUT_P1 / SI5391_A_OUT_N1

FMCA_REFCLK_156.25M_P / FMCA_REFCLK_156.25M_N

R171 RES 100

MUX_SEL7

R1535 / R1536 RES 1K / RES 2K

P2V5_G2

R172 RES 1K

R173 / R174 RES 0

C_CLK_A_12C_FGT_P_7 / C_CLK_A_12C_FGT_N_7

C_CLK_B_12A_FGT_P_7 / C_CLK_B_12A_FGT_N_7

C328 / C329 0.1UF 16V CLK_A_12C_FGT_P_7 / CLK_A_12C_FGT_N_7

C845 / C846 0.1UF 16V CLK_B_12A_FGT_P_7 / CLK_B_12A_FGT_N_7

FB38 FB 450MA 220 0402

FB39 FB 450MA 220 0402

C332 A36096-030 0.1UF 10% 16V X7R 0402

C333 D97712-023 1UF 10% 16V X6S 0402

C334 A36096-030 0.1UF 10% 16V X7R 0402

C335 D97712-023 1UF 10% 16V X6S 0402

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	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 62 OF 97	

ALTERA CONFIDENTIAL

CLOCK MUX-III

368.63M

J6 K10975-001 CON TH 1
J33 K10975-001 CON TH 1
C837 1 2 0.1UF 16V
C840 1 2 0.1UF 16V
SI5332_OUT3_P
SI5332_OUT3_N
SI53307_SMA3_CLKIN1_P
SI53307_SMA3_CLKIN1_N
R686 RES 100 0.0625W 0402
R687 1 2 RES 100
MUX_SEL8
P2V5_G2
R688 1 2 RES 1K
R689 1 2 RES 2K
R690 1 2 RES 1K
R691 1 2 RES 0
R692 1 2 RES 0
U46 IC HC_SI53307_B_GM
CLK0 CLK0B CLK1 CLK1B CLK_SEL OE SFOUT0 SFOUT1
Q0 Q0B Q1 Q1B VDD VDDO GND GND GND_PAD
K13822-001
C_CLK_B_12A_FGT_P_0
C_CLK_B_12A_FGT_N_0
C835 1 2 0.1UF 16V CLK_B_12A_FGT_P_0
C836 1 2 0.1UF 16V CLK_B_12A_FGT_N_0
FB74 FB 220 450MA 0402
FB75 FB 220 450MA 0402
C841 A36096-030 0.1UF 10% 16V X7R 0402
C842 D97712-023 1UF 10% 16V X6S 0402
C843 A36096-030 0.1UF 10% 16V X7R 0402
C844 D97712-023 1UF 10% 16V X6S 0402

CLK_SEL: LOW - CLKIN0 (BY DEFAULT), HIGH - CLKIN1

J46 K10975-001 CON TH 1
J47 K10975-001 CON TH 1
C855 1 2 0.1UF 16V
C856 1 2 0.1UF 16V
SI5332_OUT2_P
SI5332_OUT2_N
SI53307_SMA4_CLKIN1_P
SI53307_SMA4_CLKIN1_N
R693 RES 100 0.0625W 0402
R694 1 2 RES 100
MUX_SEL9
P2V5_G2
R695 1 2 RES 1K
R696 1 2 RES 2K
R697 1 2 RES 1K
R698 1 2 RES 0
R700 1 2 RES 0
U48 IC HC_SI53307_B_GM
K13822-001
C_CLK_13A_FGT_REFCLK0_P
C_CLK_13A_FGT_REFCLK0_N
C847 1 2 0.1UF 16V CLK_13A_FGT_REFCLK0_P
C848 1 2 0.1UF 16V CLK_13A_FGT_REFCLK0_N
FB76 FB 220 450MA 0402
FB77 FB 220 450MA 0402
C902 A36096-030 0.1UF 10% 16V X7R 0402
C903 D97712-023 1UF 10% 16V X6S 0402
C904 A36096-030 0.1UF 10% 16V X7R 0402
C905 D97712-023 1UF 10% 16V X6S 0402

J46 K10975-001 CON TH 1
J47 K10975-001 CON TH 1
C849 1 2 0.1UF 16V
C852 1 2 0.1UF 16V
Si5391_B_OUT_P0
Si5391_B_OUT_N0
FMC_B_CLK_M2C_P0
FMC_B_CLK_M2C_N0
C_FMC_B_CLK_M2C_P0
C_FMC_B_CLK_M2C_N0
R701 RES 100 0.0625W 0402
R699 1 2 RES 100
MUX_SEL10
P2V5_G2
R702 1 2 RES 1K
R703 1 2 RES 2K
R704 1 2 RES 1K
R705 1 2 RES 0
R706 1 2 RES 0
U53 IC HC_SI53307_B_GM
K13822-001
C_CLK_B_12A_FGT_P_2
C_CLK_B_12A_FGT_N_2
C_CLK_3C_GPIO_P_1
C_CLK_3C_GPIO_N_1
C850 1 2 0.1UF 16V CLK_B_12A_FGT_P_2
C851 1 2 0.1UF 16V CLK_B_12A_FGT_N_2
C272 1 2 0.1UF 16V CLK_3C_GPIO_P_1
C273 1 2 0.1UF 16V CLK_3C_GPIO_N_1
FB78 FB 220 450MA 0402
FB80 FB 220 450MA 0402
C857 A36096-030 0.1UF 10% 16V X7R 0402
C858 D97712-023 1UF 10% 16V X6S 0402
C859 A36096-030 0.1UF 10% 16V X7R 0402
C860 D97712-023 1UF 10% 16V X6S 0402

J46 K10975-001 CON TH 1
J47 K10975-001 CON TH 1
C863 1 2 0.1UF 16V
C866 1 2 0.1UF 16V
Si5391_B_OUT_P6
Si5391_B_OUT_N6
FMC_B_CLK_M2C_P1
FMC_B_CLK_M2C_N1
C_FMC_B_CLK_M2C_P1
C_FMC_B_CLK_M2C_N1
R707 RES 100 0.0625W 0402
R708 1 2 RES 100
MUX_SEL11
P2V5_G2
R709 1 2 RES 1K
R710 1 2 RES 2K
R711 1 2 RES 1K
R712 1 2 RES 0
R713 1 2 RES 0
U56 IC HC_SI53307_B_GM
K13822-001
C_CLK_B_12A_FGT_P_3
C_CLK_B_12A_FGT_N_3
C_CLK_3C_GPIO_P_0
C_CLK_3C_GPIO_N_0
C861 1 2 0.1UF 16V CLK_B_12A_FGT_P_3
C862 1 2 0.1UF 16V CLK_B_12A_FGT_N_3
C270 1 2 0.1UF 16V CLK_3C_GPIO_P_0
C271 1 2 0.1UF 16V CLK_3C_GPIO_N_0
FB81 FB 220 450MA 0402
FB84 FB 220 450MA 0402
C867 A36096-030 0.1UF 10% 16V X7R 0402
C868 D97712-023 1UF 10% 16V X6S 0402
C869 A36096-030 0.1UF 10% 16V X7R 0402
C870 D97712-023 1UF 10% 16V X6S 0402

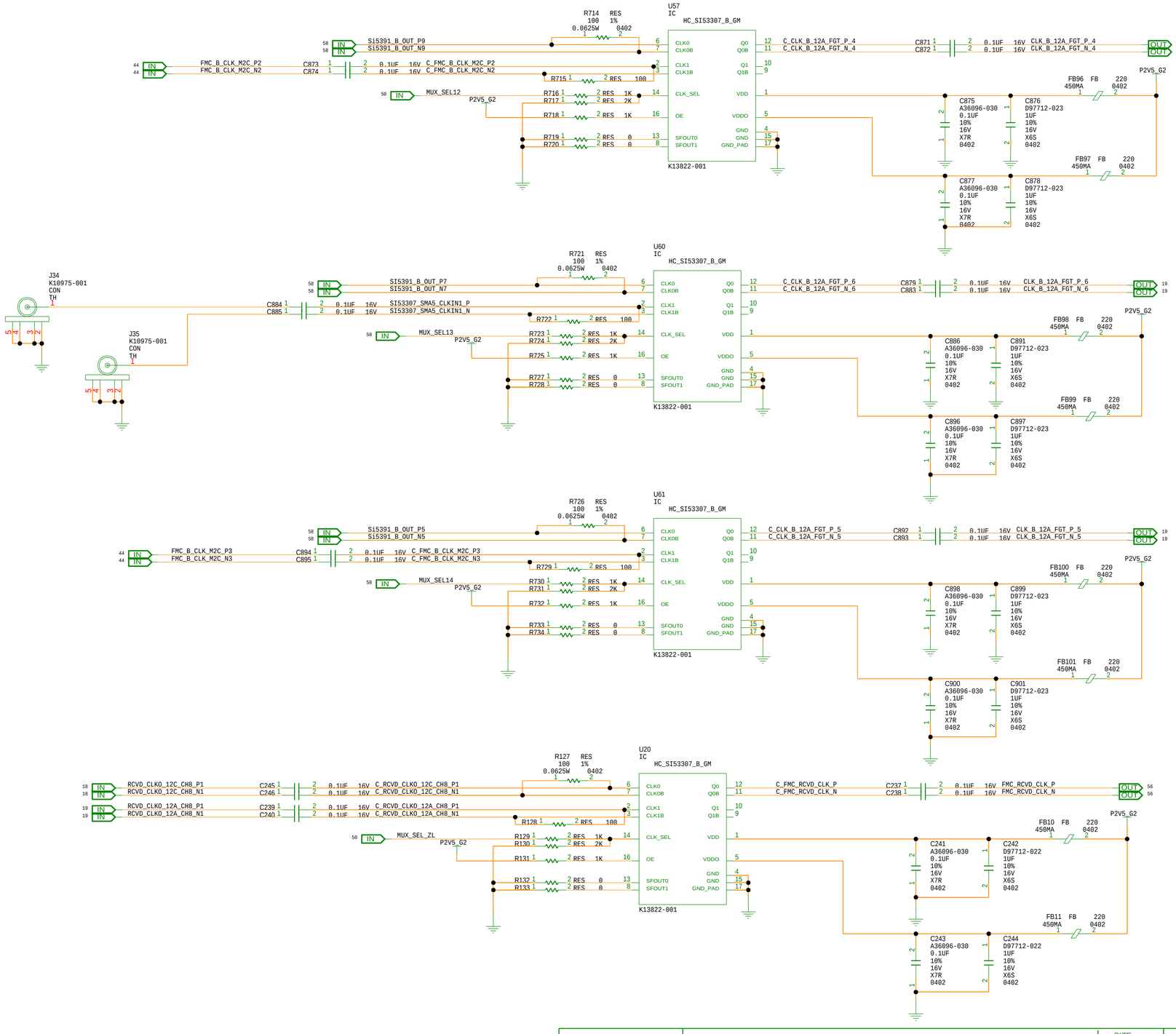
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DEPARTMENT		SIZE	CODE	DOCUMENT NUMBER	REV
	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 63 OF 97	

ALTERA CONFIDENTIAL

CLOCK MUX-IV

CLK_SEL: LOW - CLKIN0 (BY DEFAULT), HIGH - CLKIN1



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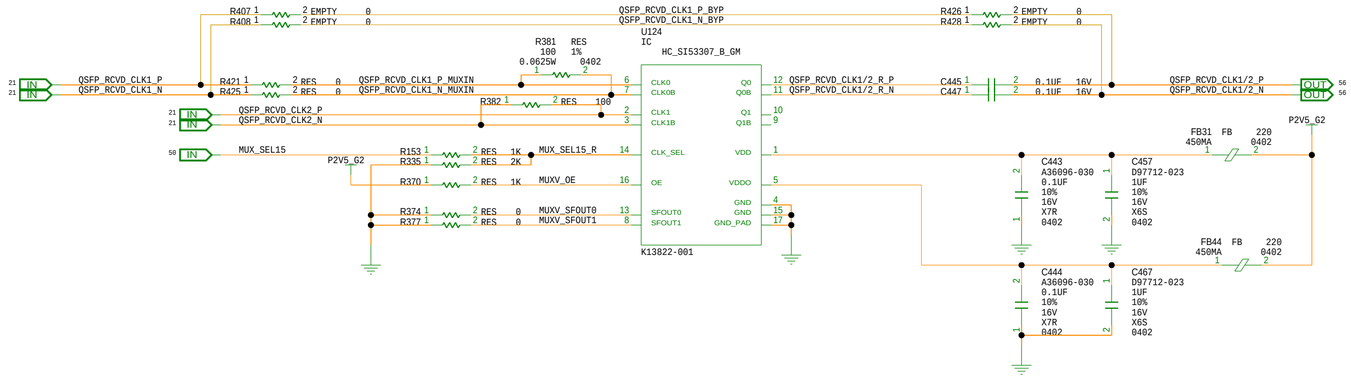
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	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 64 OF 97	

ALTERA CONFIDENTIAL

CLOCK MUX-V

R407 & R421 SHOULD BE TRI-PAD
R408 & R425 SHOULD BE TRI-PAD

R426 & C445 SHOULD BE TRI-PAD
R428 & C447 SHOULD BE TRI-PAD



CLK_SEL: LOW - CLKIN0 (BY DEFAULT), HIGH - CLKIN1

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	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 65 OF 97	

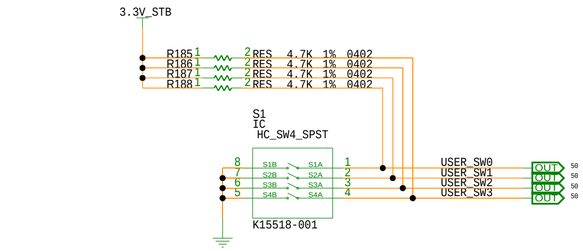
ALTERA CONFIDENTIAL

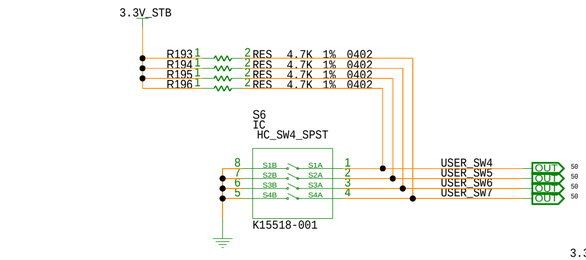
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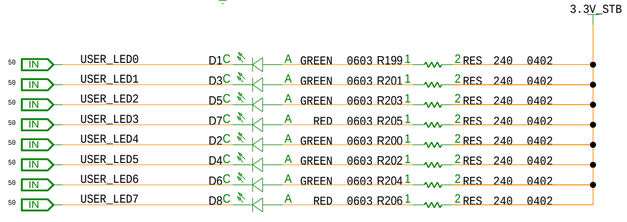
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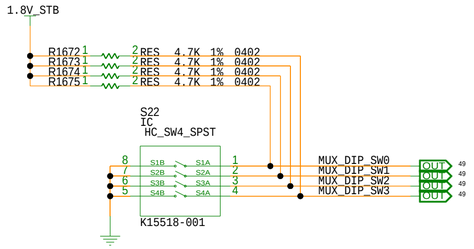
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	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 66 OF 97	

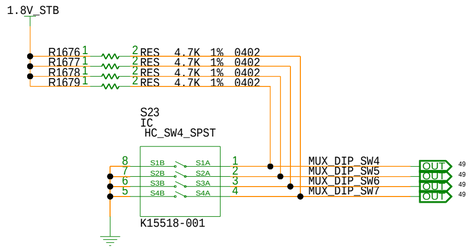
DIP SWITCH, LEDS

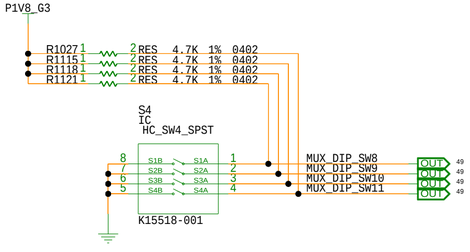


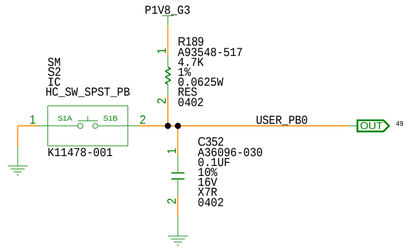


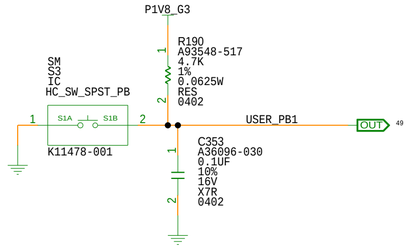


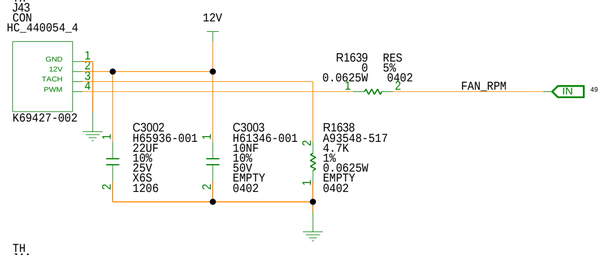


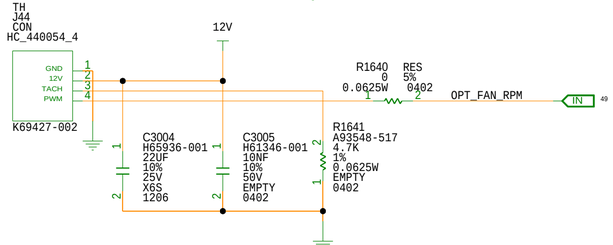








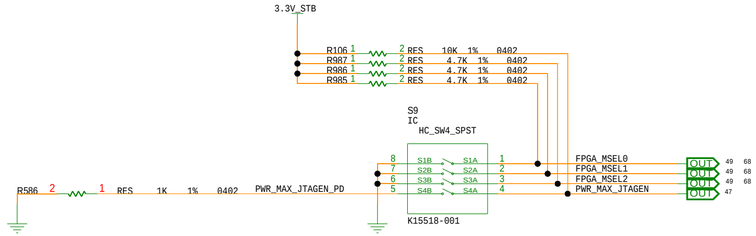




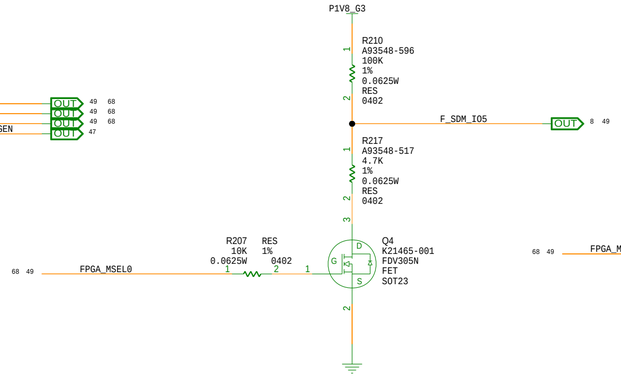
Mon Sep 15 10:27:20 2025

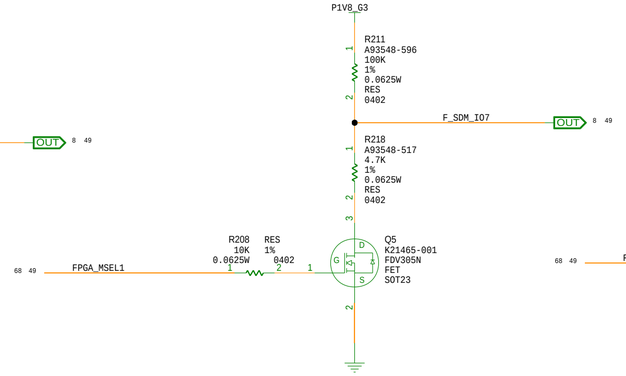
DEPARTMENT		SIZE	CODE	DOCUMENT NUMBER	REV
	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 67 OF 97	

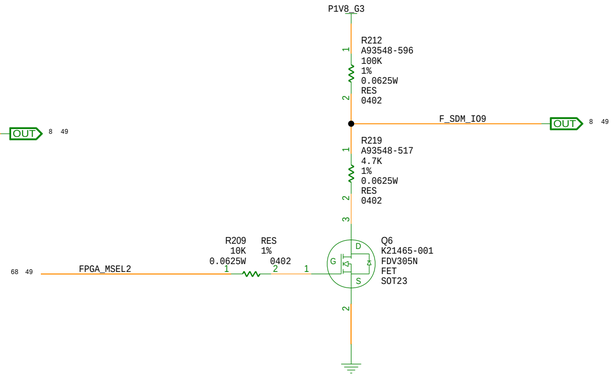
DIP SWITCH, LEDS

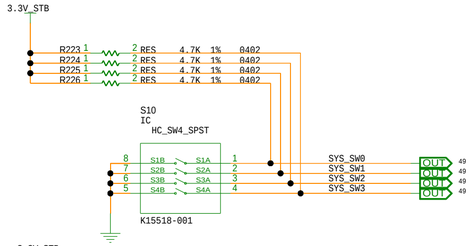


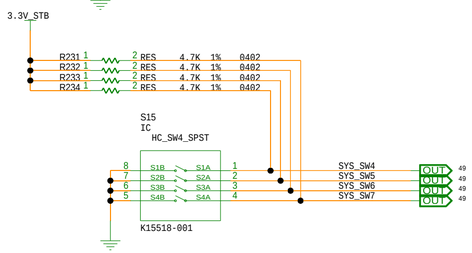
TO REMOTE CONTROL MSEL THROUGH MAX10, ALL S9 BITS MUST BE OFF

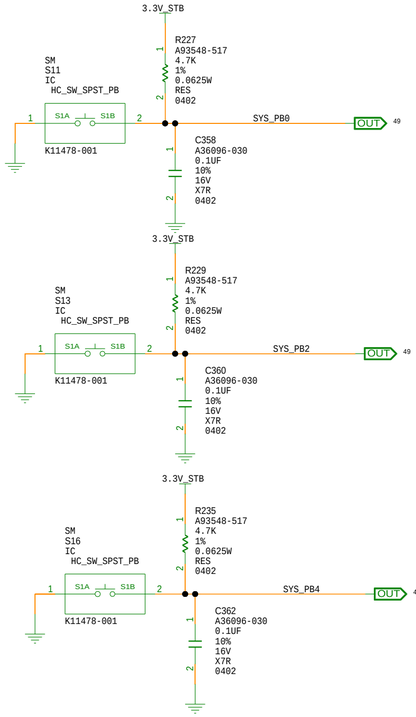


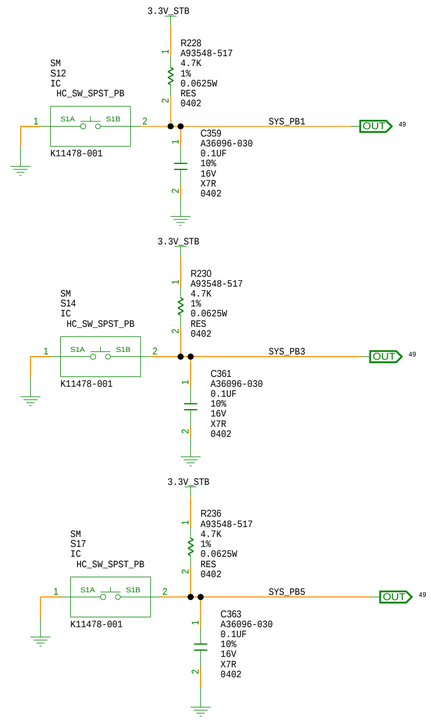


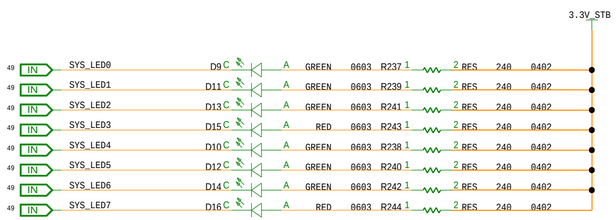










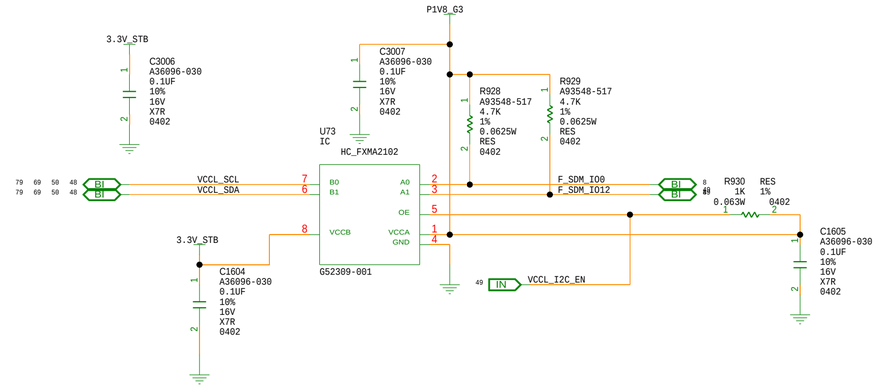


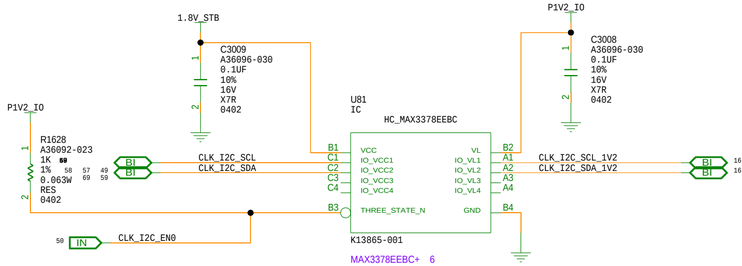
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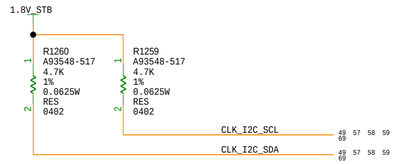
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	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 68 OF 97	

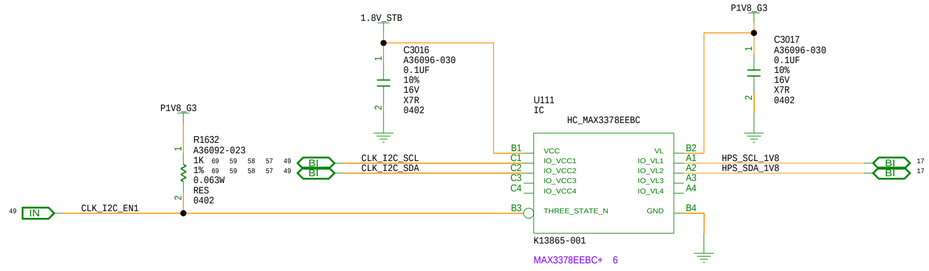
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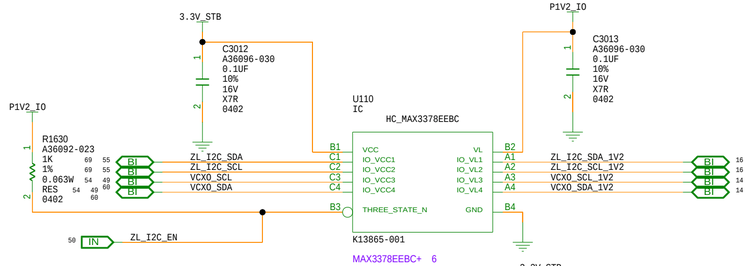
LVL SHIFTER-I

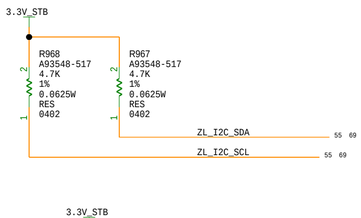


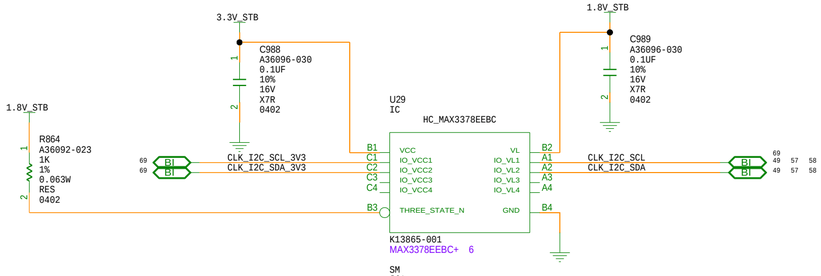


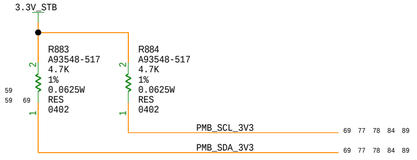


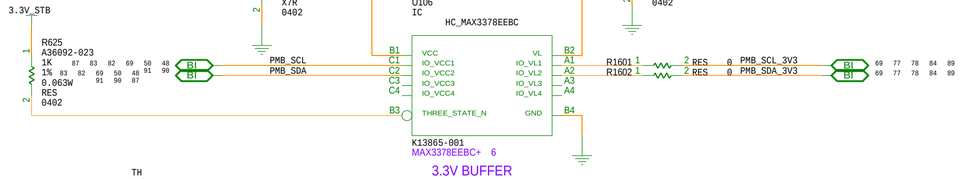


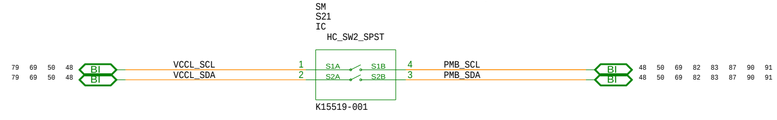


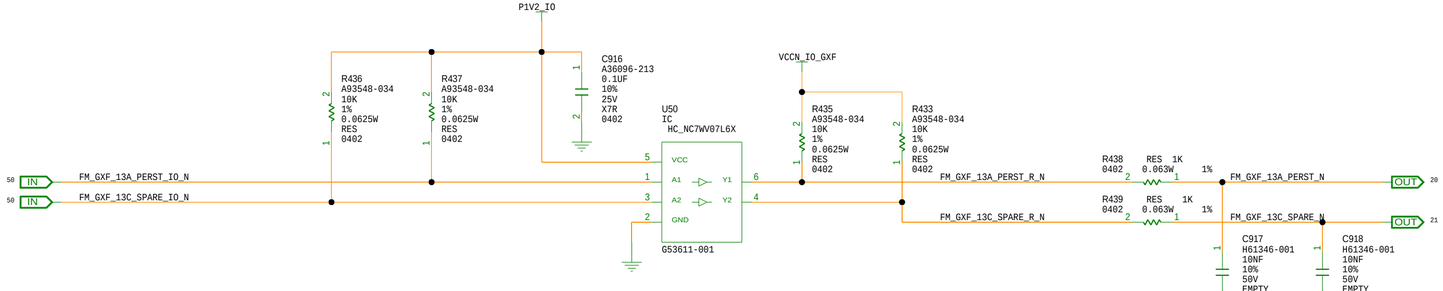


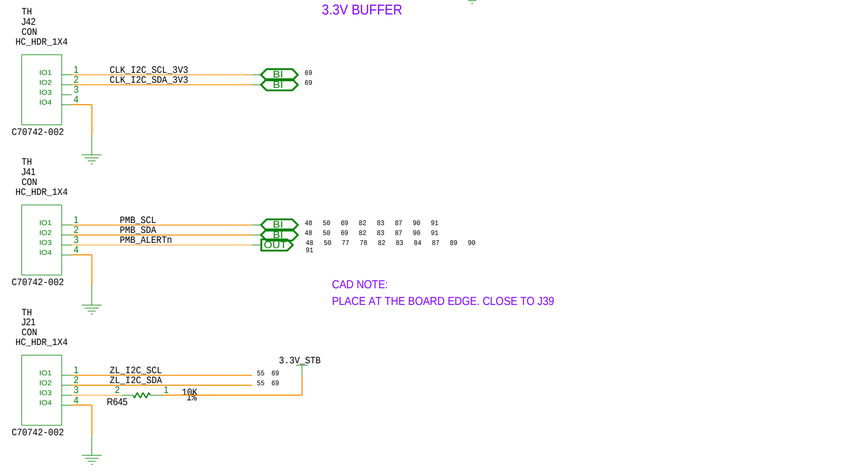












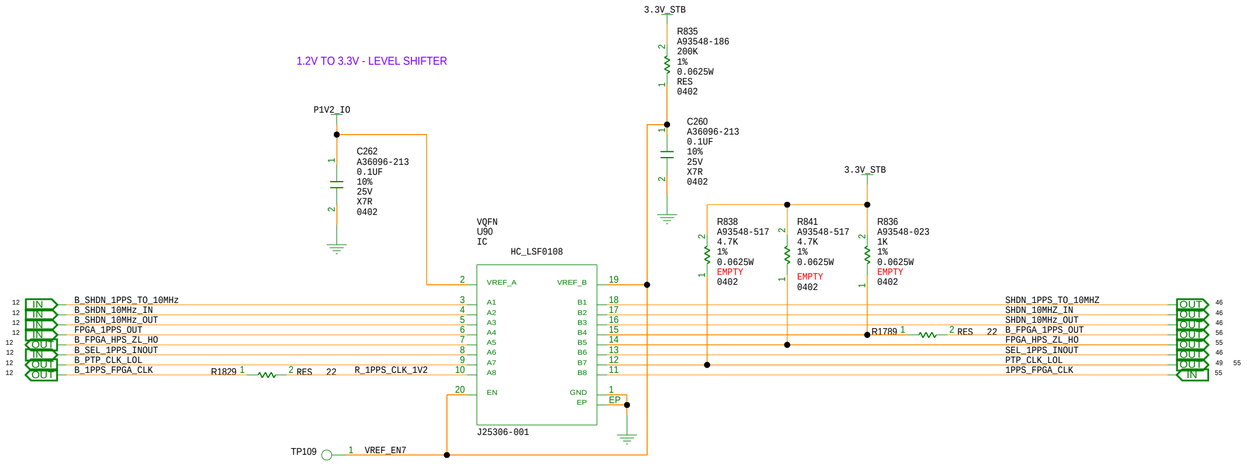
Mon Sep 15 10:27:20 2025

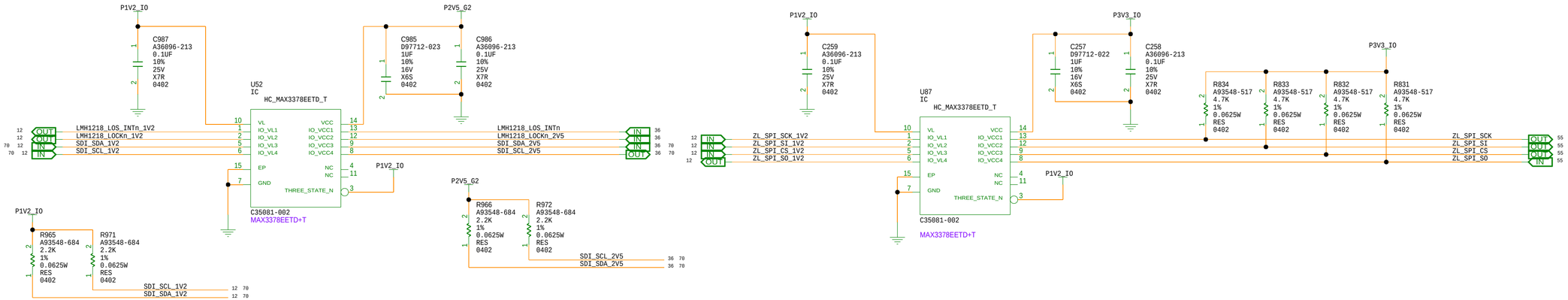
DEPARTMENT	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	SIZE	CODE	DOCUMENT NUMBER	REV
		C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET	69 OF 97

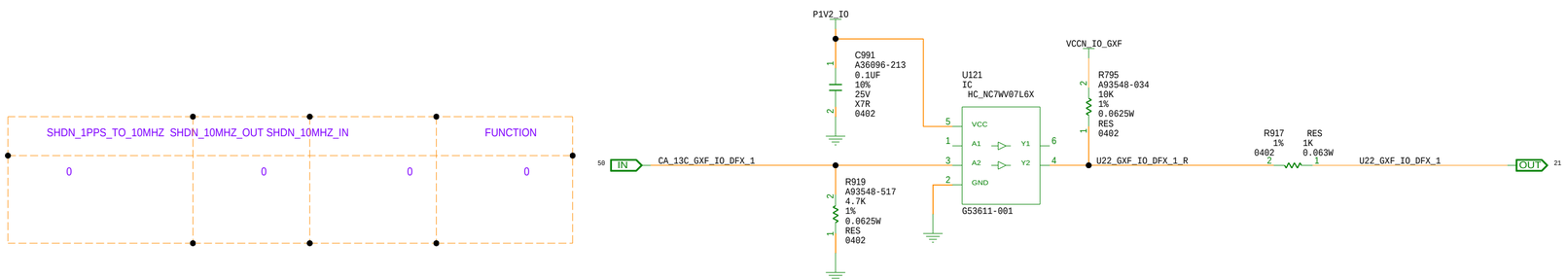
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LVL SHIFTER-II

1.2V TO 3.3V - LEVEL SHIFTER







SHDN_1PPS_TO_10MHZ	SHDN_10MHZ_OUT	SHDN_10MHZ_IN	FUNCTION
0	0	0	0

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DEPARTMENT		SIZE	CODE	DOCUMENT NUMBER	REV
	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 70 OF 97	

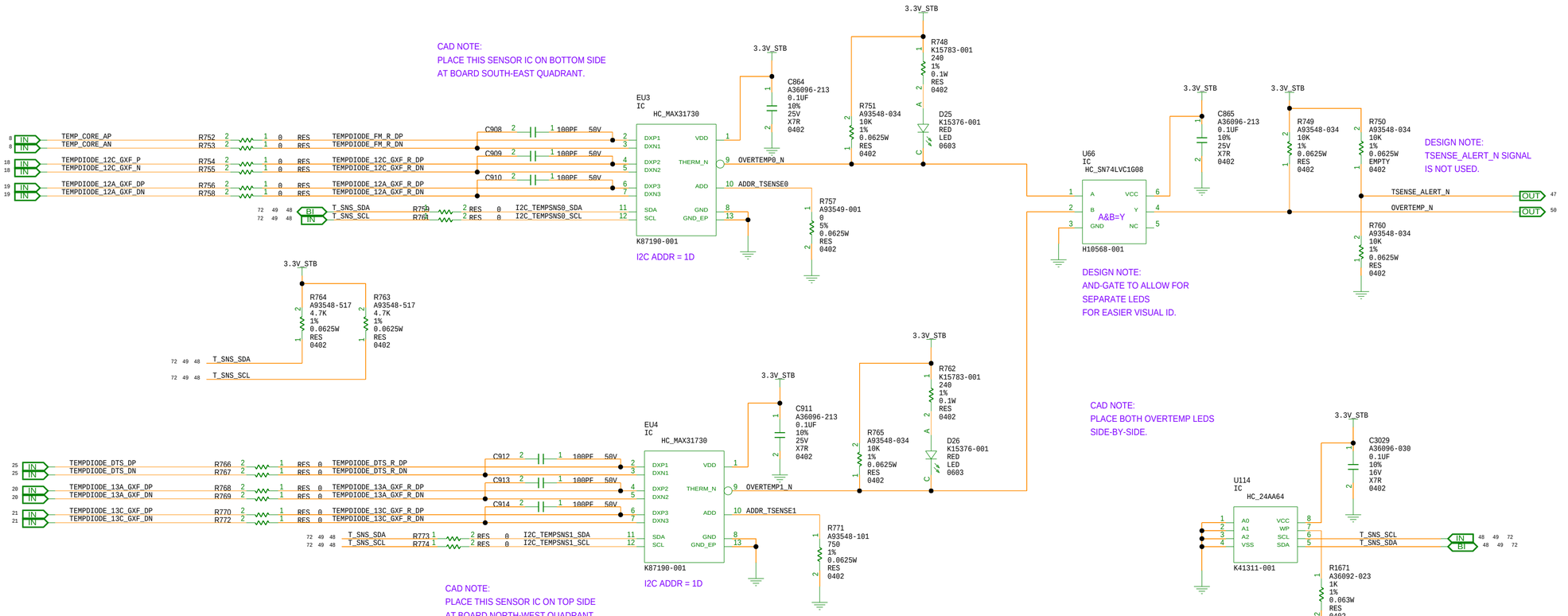
RESERVED

Mon Sep 15 10:26:44 2025

DEPARTMENT	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	SIZE C+	CODE 34649	DOCUMENT NUMBER 150-0330690-D1	REV D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 71 OF 97	

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BOARD TEMP SENSORS



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DEPARTMENT		SIZE	CODE	DOCUMENT NUMBER	REV
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		SCALE:	DO NOT SCALE DRAWING	SHEET 72 OF 97	

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CURRENT/VOLTAGE MONITOR

1.8V_STB
C2615 D97712-023 1UF 10% 16V X6S 0402

RES VALUE NEED TO CHECK

SOT23 U67 IC HC_ADR1581
M79415-001
R644 K15783-001 240 1% 0.1W RES 0402
R1326 11.5K 0.0625W RES 1% 0402
VREF = 0.1V
R478 A36092-023 1K 1% 0.063W RES 0402
C1476 H61346-001 10NF 10% 50V X7R 0402
C1108 A36096-003 470PF 10% 50V X7R 0402

U65 IC HC_LTC2497
K13870-001

IVSNS_I2C_SDA
IVSNS_I2C_SCL

3.3V_STB
FB64 220 450MA FB 0402
C1109 K19409-001 4.7UF 20% 6.3V X6S 0402
C1110 K19409-001 4.7UF 20% 6.3V X6S 0402
C1111 A36096-030 0.1UF 10% 16V X7R 0402

C2634 A36096-030 0.1UF 10% 16V X7R 0402
C2635 A36096-030 0.1UF 10% 16V X7R 0402
C2636 A36096-030 0.1UF 10% 16V X7R 0402
C2637 A36096-030 0.1UF 10% 16V X7R 0402
C2638 A36096-030 0.1UF 10% 16V X7R 0402
C2639 A36096-030 0.1UF 10% 16V X7R 0402
C2640 A36096-030 0.1UF 10% 16V X7R 0402
C2641 A36096-030 0.1UF 10% 16V X7R 0402
C2642 A36096-030 0.1UF 10% 16V X7R 0402
C2643 A36096-030 0.1UF 10% 16V X7R 0402
C2644 A36096-030 0.1UF 10% 16V X7R 0402
C2645 A36096-030 0.1UF 10% 16V X7R 0402
C2646 A36096-030 0.1UF 10% 16V X7R 0402
C2647 A36096-030 0.1UF 10% 16V X7R 0402

R1390 R1391 RES 100 ISNS_VCCERT1_FHT_13C_GXFN ISNS_VCCERT1_FHT_13C_GXFP
R1392 R1393 RES 100 ISNS_P1V0_G2_P ISNS_P1V0_G2_N
R1394 R1395 RES 100 ISNS_VCCERT1_FHT_13A_GXFN ISNS_VCCERT1_FHT_13A_GXFP
R1400 R1401 RES 100 ISNS_VCCERT2_FHT_U2x_GXFN ISNS_VCCERT2_FHT_U2x_GXFP

I2C ADDRESS 7B'001 0100

CURRENT/VOLTAGE MONITOR

3.3V_STB
R479 A93548-517 4.7K 1% 0.0625W RES 0402
R480 A93548-517 4.7K 1% 0.0625W RES 0402
IVSNS_I2C_SDA
IVSNS_I2C_SCL

C1937 A36096-003 470PF 10% 50V X7R 0402
U99 IC HC_LTC2497
K13870-001

PVIN_IMON
12V_VMON
BIAS3V3_VMON
EM5V_VMON
VCCERT1_FHT_GXF_13A
VCCERT2_FHT_GXF_U2X
VCCERT1_FHT_GXF_13C
P1V8_G2_VSNS
P1V8_G2
P1V5_G2
P2V5_G2_VSNS
P1V8_G3_VSNS
VCCFUSE_VSNS
P1V8_STB_VSNS

1.5V/30 1/10
1/3 47/197

3.3V_STB
FB83 220 450MA FB 0402
C1939 K19409-001 4.7UF 20% 6.3V X6S 0402
C1940 K19409-001 4.7UF 20% 6.3V X6S 0402
C1941 A36096-030 0.1UF 10% 16V X7R 0402

I2C ADDRESS 7B'001 0110

P2V5_G2
R1756 R1757 RES 1K 1% 0.063W 0402
P2V5_G2_VSNS

VCCFUSEWR_SDM
R1760 R1761 RES 1K 1% 0.063W 0402
VCCFUSE_VSNS

1.8V_STB
R1758 R1759 RES 1K 1% 0.063W 0402
P1V8_STB_VSNS

VCCERT2_FHT_GXF_U2X
P1V0_G2
R1748 R1749 R1750 R1751 R1752 RES 1K 1% 0.063W 0402
VCCERT1_FHT_GXF_13A
VCCERT1_FHT_GXF_13C
P1V5_G2

P1V8_G2
R1741 R1753 RES 1K 1% 0.063W 0402
P1V8_G2_VSNS

P1V8_G3
R1754 R1755 RES 1K 1% 0.063W 0402
P1V8_G3_VSNS

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		SCALE:	DO NOT SCALE DRAWING	SHEET 73 OF 97	

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POWER INPUT

TH
J18
EMPTY
HC_01_0004_26

12V_DCIN

GND 12V
SENSE0 12V
GND 12V
GND 12V

M44881-001

OVERLAP 2X3 CONNECTOR
WITH 2X4 CONNECTOR
FOR FUTURE USE IF ANY

12V DCIN
2X3 ATX

SM
J30
SCON
HC_01_0004_36

12V_DCIN

GND0 12V0
GND1 12V1
GND2 12V2

K12408-001

30A CURRENT LIMIT
INRUSH: ~5A/5MS

30A 2MOHM = 60MW PDISS

CAD NOTE:
PLACE THE TP AT EASILY ACCESSIBLE SPOT ON THE PCB.
AVOID UNDER HEATSINK AND ADD-ON CARD SHADOW AREA

12V_DCIN

C881 H81594-001 1UF 10% 25V X6S 0402
C882 A36096-015 680PF 10% 50V X7R 0402
D17 K12744-001 SMCJ12A 75.4A DIO SM

R1267 A93548-246 249K 1% 0.0625W RES 0402
R1269 K21612-001 1.74M 1% 0.063W RES 0402

OV = 13.5V
UV = 6.7V

74 PVIN_UV
PVIN_OV

R1268 A93548-181 20K 1% 0.0625W EMPTY 0402
C2732 D62711-001 1NF 10% 50V X7R 0402
R1270 A93548-181 20K 1% 0.0625W EMPTY 0402
C2496 A36096-030 0.1UF 10% 16V X7R 0402
C1157 H81594-001 1UF 10% 25V X6S 0402

U103
IC
HC_LTC4218

NC SENSE+
VDD SENSE-
UV ISET
OV IMON
TIMER FB
INTVCC FLT_N
GND PG
SOURCE GATE
THPAD

K13978-001

SNS_P_12V
SNS_N_12V
PVIN_IMON
PVIN_PG

R1271 0 5% 0.0625W 0402 EMPTY

R471 0.001 1% 1W 1206 RES
R470 0.001 1% 1W 1206 RES

V12 SNS RSNS
V11 SNS RSNS

Q7
G62364-001
FDMC8010
FET
QFN

R1274 K48915-001 150K 0.1% 0.0625W RES 0402
C2733 D62711-001 1NF 10% 50V X7R 0402
R1335 A93548-181 20K 1% 0.0625W EMPTY 0402

R472 A93548-031 10 1% 0.0625W RES 0402
R475 A36092-023 1K 1% 0.063W RES 0402
R476 A36092-023 1K 1% 0.063W EMPTY 0402
C887 G21127-001 0.1UF 10% 50V X7R 0402

R1477 H89932-001 18K 1% 0.0625W RES 0402
R1478 A93548-138 2K 1% 0.0625W RES 0402

12V_VMON OUT 73

C880 M24380-001 330UF 20% 16V ALUM SM
C1155 G46681-001 150UF 20% 16V TANT 7343
C1156 G46681-001 150UF 20% 16V TANT 7343
C1429 G46681-001 150UF 20% 16V TANT 7343
C1430 G46681-001 150UF 20% 16V TANT 7343
C1431 G46681-001 150UF 20% 16V TANT 7343
C1432 G46681-001 150UF 20% 16V TANT 7343

12V
TP6

3.3V_STB

R1275 A93548-034 10K 1% 0.0625W RES 0402

74 47 PVIN_PG
74 73 PVIN_IMON

R1537 A93548-132 15K 1% 0.0625W RES 0402

TH
S18
IC
K15516-001

R1538 A36092-043 100 1% 0.0625W RES 0402
PVIN_UV 74
C2497 D62711-001 1NF 10% 50V X7R 0402

REVIEW NOTES:
1. ADDED 2X4 ATX CONNECTOR OPTION FOR FUTURE USE. (FP OVERLAP)

OV = 13.772V
UV = 10.463V

12V

R600 RES 200K 0.0625W 1% 0402
R601 A93548-543 1.91K 1% 0.0625W RES 0402
C888 H61346-001 10NF 10% 50V EMPTY 0402
C889 H61346-001 10NF 10% 50V EMPTY 0402
R603 A93548-197 6.04K 1% 0.063W RES 0402

U59
IC
HC_TPS3700

INA+ OUTA
INB- OUTB
GND VDD

J4211-001

3.3V_STB

R598 A93548-517 4.7K 1% 0.0625W RES 0402
R599 A93548-517 4.7K 1% 0.0625W RES 0402

12V_UV OUT 47
12V_OV OUT 47

12V
R602 RES 0 5% 0.0625W 0402

C890 H81594-001 1UF 10% 25V X6S 0402

VCC_12V	STATUS	OUTA	OUTB
10.46~13.77V 1	NORMAL	1	1
<10.46V	UV	0	1
>13.77V	OV	1	0

12V

R1358 G21796-299 3.3K 1% 0.0625W RES 0402

D18 GREEN LED 0603

Q17
K21465-001
FDV305N
FET
SOT23

74 47 PVIN_PG

CAD NOTE:
PLACE THE TP AT EASILY ACCESSIBLE SPOT ON THE PCB.
AVOID UNDER HEATSINK AND ADD-ON CARD SHADOW AREA
SPREAD THE GND TEST POINTS ACROSS THE PCB AREA. PROVIDE GND SYMBOL SILK

TP132
TP133
TP134
TP135
TP136
TP137

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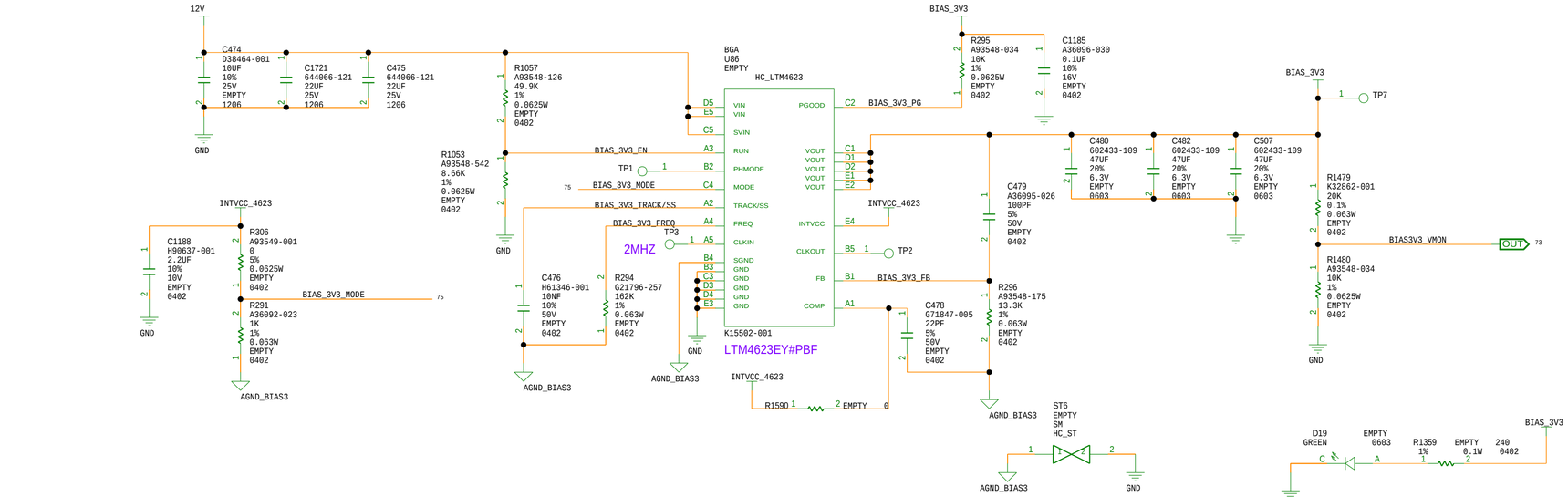
DEPARTMENT		SIZE	CODE	DOCUMENT NUMBER	REV
	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 74 OF 97	

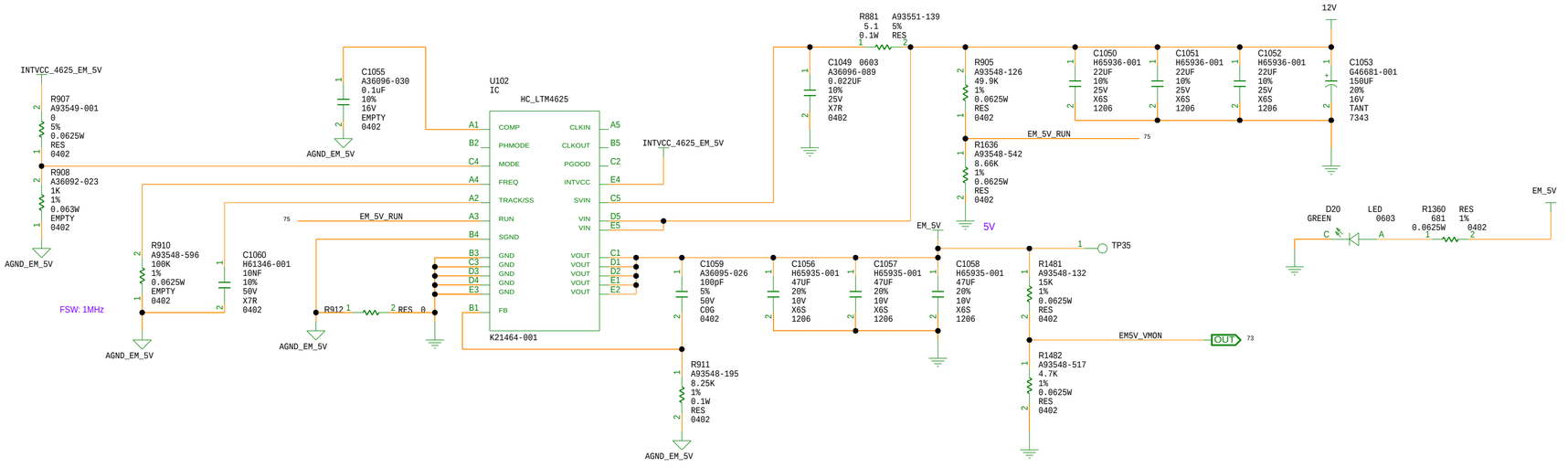
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POWER BIAS

MAX CURRENT = 3A
LOAD CURRENT = 0.5A
VO = 3.3V
SWITCHING FREQ= 2 MHZ
RIPPLE = 5MV
TSS = 3MSEC

CAD NOTE:
PLACE THE TP AT EASILY ACCESSIBLE SPOT ON THE PCB.
AVOID UNDER HEATSINK AND ADD-ON CARD SHADOW AREA





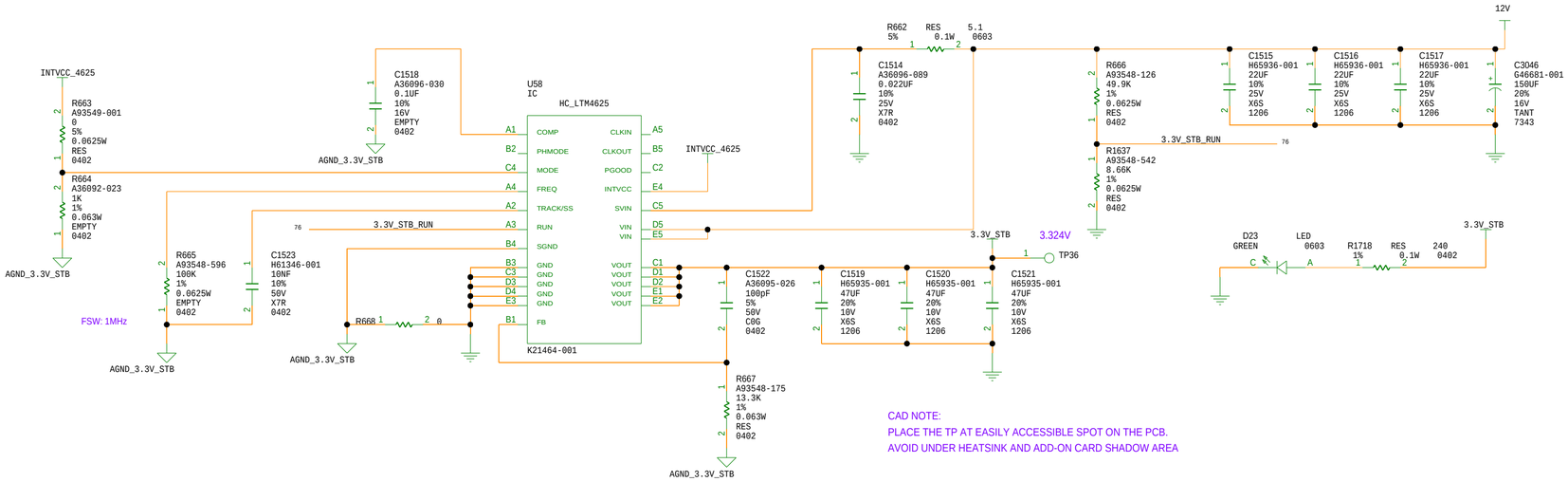
CAD NOTE:
PLACE THE TP AT EASILY ACCESSIBLE SPOT ON THE PCB.
AVOID UNDER HEATSINK AND ADD-ON CARD SHADOW AREA

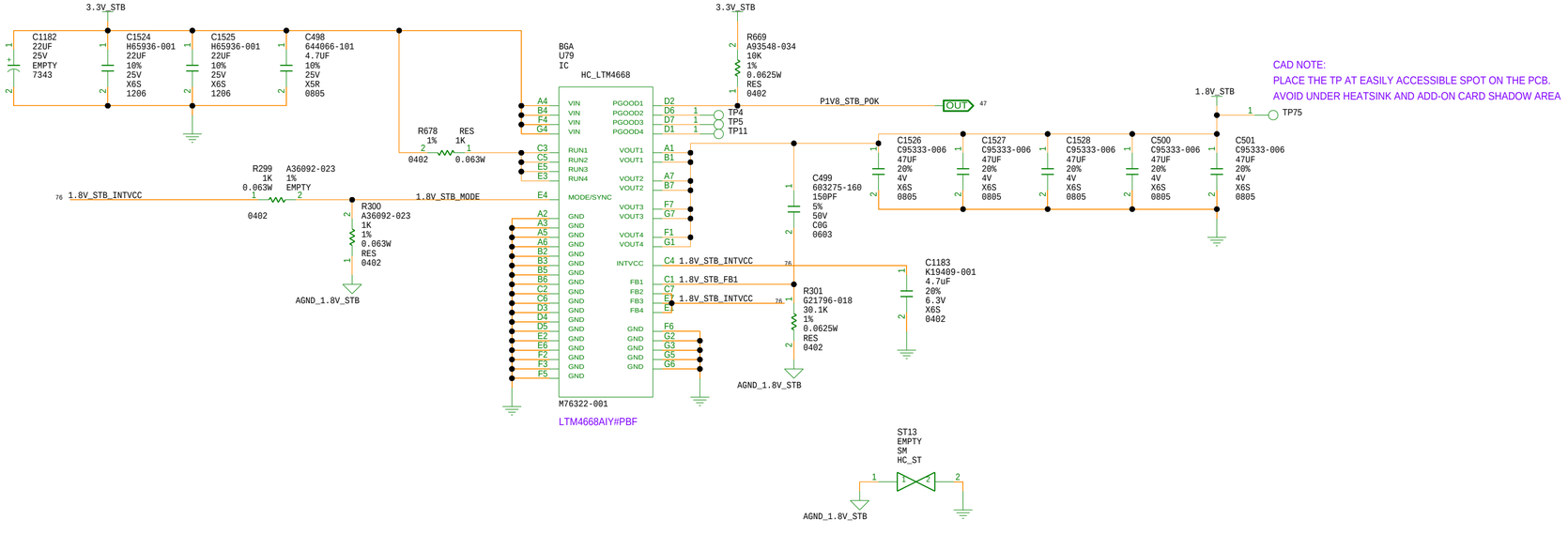
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DEPARTMENT		SIZE	CODE	DOCUMENT NUMBER	REV
	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 75 OF 97	

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3.3V_STB & 1.8V_STB





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DEPARTMENT	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	SIZE	CODE	DOCUMENT NUMBER	REV
		C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 76 OF 97	

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P3V3_BUS

3.3V_STB

77 47 P3V3_BUS_POK — R261 1 — 2 RES 4.7K 1% 0.0625W 0402

R1387 A93548-034 10K 1% 0.0625W EMPTY 0402

C2659 H61346-001 10NF 10% 50V EMPTY 0402

AGND_P3V3_BUS

77 47 P3V3_BUS_EN

R262 A36092-023 1K 1% 0.063W RES 0402

AGND_P3V3_BUS

12V

C1114 G46681-001 150UF 20% 16V TANT 7343

R264 E58715-001 0.005 1% 1W RES 0805

BGA U30 IC

HC_LTM4678

P3V3_BUS_VIN

C400 H65936-001 22UF 10% 25V X6S 1206

C401 H65936-001 22UF 10% 25V X6S 1206

C402 H65936-001 22UF 10% 25V X6S 1206

C403 H65936-001 22UF 10% 25V X6S 1206

C404 H65936-001 22UF 10% 25V X6S 1206

C405 D38464-001 10UF 10% 25V X6S 1206

P3V3_BUS_ASEL

R1281 H89932-001 18K 1% 0.0625W RES 0402

AGND_P3V3_BUS

J1 IIN_DP
K1 IIN_DN
F12 ASEL
D1 SVIN
G1 G2 G3 G4 H1 H2 H3 H4 VIN0
E1 E2 E3 E4 F1 F2 F3 F4 VIN1

VDD25 D12 1 — TP63 EMPTY SM
VDD33 E8 1 — TP65 EMPTY SM
ALERT_N H11 — PMB_ALERTN — OUT 48 50 69 78 82 83 84 87 89 90 91

K7 K8 K9 K10 K11 L7 L8 L9 L10 L11 L12 M7 M8 M9 M10 VOUT0
A7 A8 A9 A10 B7 B8 B9 B10 B11 B12 C7 C8 C9 D7 VOUT1

C390 724613-088 470UF 6.3V TANT 7343

C391 724613-088 470UF 6.3V TANT 7343

C392 K13620-001 100UF 20% 10V X5R 1206

C393 K13620-001 100UF 20% 10V X5R 1206

C388 K13620-001 100UF 20% 10V X5R 1206

C394 K13620-001 100UF 20% 10V X5R 1206

C395 K13620-001 100UF 20% 10V X5R 1206

C389 K13620-001 100UF 20% 10V X5R 1206

C397 K13620-001 100UF 20% 10V X5R 1206

C2510 K13620-001 100UF 20% 10V X5R 1206

P3V3_BUS — 1 — TP76 EMPTY SM

CAD NOTE:
PLACE THE TP AT EASILY ACCESSIBLE SPOT ON THE PCB.
AVOID UNDER HEATSINK AND ADD-ON CARD SHADOW AREA

VOUT SET TO 3.3V
PMBUS ADDRESS = "7BIT 0X4D"

C396 D97712-022 1UF 10% 16V X6S 0402

P3V3_BUS

R266 2 — 1 RES 0 — P3V3_BUS_R_VOSNS0_DN — M12 VOSNS0_DN
R267 2 — 1 RES 0 — P3V3_BUS_R_VOSNS0_DP — M11 VOSNS0_DP

R1453 1 — 2 RES 0 — P3V3_BUS_R_VOSNS1_DN — A12 VOSNS1_DN
R1454 1 — 2 RES 0 — P3V3_BUS_R_VOSNS1_DP — A11 VOSNS1_DP

J11 TSNS0A
J8 TSNS0B
J10 TSNS1A
D8 TSNS1B

L2 M1 M2 SW0
A1 A2 B2 SW1

VOUT0_CFG E11
VOUT1_CFG E12 — P3V3_BUS_VOUTCFG — R1396 1 — 2 RES 11.3K

VTRIM0_CFG C12
VTRIM1_CFG E10 — P3V3_BUS_VTRIMCFG — R1458 1 — 2 EMPTY 0

PGOOD0 J7
PGOOD1 D9 — P3V3_BUS_POK — OUT 47 77 — AGND_P3V3_BUS

INTVCC E7 — P3V3_BUS_INTVCC — 2.2UF C1116 1 — 2 16V

EXTVCC F8 — P3V3_BUS_EXTVCC — R305 1 — 2 RES 0

FSWPH_CFG E9 — P3V3_BUS_FSWPH_CFG

FAULT1_N G11 — P3V3_BUS_FAULT_N — 77
FAULT0_N H12

SHARE_CLK D11 — P3V3_BUS_SHARE_CLK — 77

SYNC K12 — P3V3_BUS_SYNC — 77

77 47 IN P3V3_BUS_EN — G12 RUN0, F11 RUN1

89 84 78 69 IN BI PMB_SCL_3V3 PMB_SDA_3V3 — J12 SCL, H10 SDA

P3V3_BUS_COMP0A — J9 COMP0A
P3V3_BUS_COMP0B — H9 COMP0B
D10 COMP1A
C10 COMP1B

C1107 D62711-001 1nF 10% 50V X7R 0402

C1112 K13768-001 100pF 5% 50V C0G 0402

AGND_P3V3_BUS

NEED CONFIG
Rht0 20k
gm0: 3.69m

P3V3_BUS_WP — C11 WP

R1388 A36092-023 1K 1% 0.063W RES 0402

AGND_P3V3_BUS

F9 F10 SGND
A3 A4 A5 A6 B1 B3 B4 B5 B6 C1 C2 C3 C4 C5 C6 D2 D3 D4 D5 D6 E5 E6 F5 F6 F7 G5 GND
G9 G10 SGND
M6 M5 M4 M3 L6 L5 L4 L3 L1 K6 K5 K4 K3 K2 J6 J5 J4 J3 J2 H8 H7 H6 H5 G8 G7 G6 GND

AGND_P3V3_BUS

J99380-001

LTM4678EY#PBF

MAX CURRENT = 50A(SINGLE)
LOAD CURRENT = 28.52A
VO = 3.3V
SWITCHING FREQ= 750 KHZ
RIPPLE = 10MV

CAD NOTE:CONNECT NEAR FPGA BGA REGION. NEED TO ROUTE AS KELVIN SENSE

R1397 G21796-315 10.7k 1% 0.0625W RES 0402

AGND_P3V3_BUS

3.3V_STB

R1399 A93548-034 10K 1% 0.0625W RES 0402

R1402 A93548-034 10K 1% 0.0625W RES 0402

R1403 A93548-202 4.99K 1% 0.0625W RES 0402

77 P3V3_BUS_FAULT_N
77 P3V3_BUS_SHARE_CLK
77 P3V3_BUS_SYNC

R274 5% 1 — 2 RES 0 0.0625W 0402

AGND_P3V3_BUS

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		SCALE:	DO NOT SCALE DRAWING	SHEET 77 OF 97	

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P1V2_BUS

3.3V_STB

78 47 P1V2_BUS_POK — R320 1 — 2 RES 4.7K 1% 0.0625W 0402

C2658
H61346-001
10NF
10%
50V
EMPTY
0402

R1311
A93548-034
10K
1%
0.0625W
EMPTY
0402

78 47 P1V2_BUS_EN

R321
A36092-023
1K
1%
0.063W
RES
0402

AGND_P1V2_BUS

R327 G21796-315
10.7K 1%
0.0625W RES
P1V2_BUS_ASEL
0402

AGND_P1V2_BUS

BGA
U37
IC

HC_LTM4677

12V

C448
150UF
16V
TANT
7343

C566 H65936-001 22UF 10% 25V X6S 1206
C567 H65936-001 22UF 10% 25V X6S 1206
C568 H65936-001 22UF 10% 25V X6S 1206
C569 H65936-001 22UF 10% 25V X6S 1206
C570 H65936-001 22UF 10% 25V X6S 1206
C571 D38464-001 10UF 10% 25V X6S 1206

VOUT SET TO 1.2V

PMBUS ADDRESS = "7BIT 0X4A"

P1V2_BUS

C564
D97712-022
1UF
10%
16V
X6S
0402

R333 A93549-001 0 5% 0.0625W RES 0402

R336 A93549-001 0 5% 0.0625W RES 0402

P1V2_BUS_VOSNS0_DP
P1V2_BUS_VOSNS0_DN
P1V2_BUS_VOSNS1

R1564 1 — 2 RES 0

TP27
TP28
TP29

CAD NOTE:CONNECT NEAR FPGA BGA REGION. NEED TO ROUTE AS KELVIN SENSE

MAX CURRENT = 36A(SINGLE)
LOAD CURRENT = 9.99A
VO = 1.2V
SWITCHING FREQ= 500 KHZ
RIPPLE = 10MV

78 47 IN P1V2_BUS_EN

89 84 77 69 IN PMB_SCL_3V3
89 84 77 69 BI PMB_SDA_3V3

P1V2_BUS_COMPA

C572
A36096-022
2200PF
10%
50V
X7R
0402

R1605 1 — 2 EMPTY 0
R1606 1 — 2 EMPTY 0

C1197
A36095-026
100pF
5%
50V
C0G
0402

R332
A93548-202
4.99K
1%
0.0625W
RES
0402

AGND_P1V2_BUS

P1V2_BUS_WP

R1063
A36092-023
1K
1%
0.063W
RES
0402

AGND_P1V2_BUS

R334 5% 1 — 2 RES 0 0.0625W 0402

AGND_P1V2_BUS

K15467-001

LTM4677EY#PBF

VDD25 J6 — TP30 EMPTY SM
VDD33 J7 — TP31 EMPTY SM

ALERT_N E5 — PMB_ALERTN — OUT 48 50 69 77 82 83 84 87 89 90

VOUT0 A1 A2 A3 B1 B2 B3 C1 C2 C3 D1 D2 D3

C2506
699013-049
470UF
20%
2.5V
ALUM
SM

C2508 H65271-001 100UF 20% 4V X6S 1206
C2509 H65271-001 100UF 20% 4V X6S 1206
C559 H65271-001 100UF 20% 4V X6S 1206
C560 H65271-001 100UF 20% 4V X6S 1206
C561 H65271-001 100UF 20% 4V X6S 1206
C563 H65271-001 100UF 20% 4V X6S 1206

P1V2_BUS

TP77
EMPTY
SM

CAD NOTE: PLEASE THIS BULK CAP ON THE SAME SIDE AS IC

C269
699013-049
470UF
20%
2.5V
EMPTY
SM

C320 H65271-001 100UF 20% 4V X6S 1206
C323 H65271-001 100UF 20% 4V X6S 1206
C599 H65271-001 100UF 20% 4V X6S 1206
C600 H65271-001 100UF 20% 4V X6S 1206
C601 H65271-001 100UF 20% 4V X6S 1206
C602 H65271-001 100UF 20% 4V X6S 1206

PLACEHOLDER

VOUT1 J1 J2 J3 K1 K2 K3 L1 L2 L3 M1 M2 M3

VOUT0CFG G5
VOUT1CFG G6

P1V2_BUS_VOUTCFG
P1V2_BUS_VTRIMCFG

VTRIM0CFG H5
VTRIM1CFG H6

R1312
A93549-001
0
5%
0.0625W
RES
0402

R1361
A93548-116
1.62K
1%
0.0625W
RES
0402

AGND_P1V2_BUS

INTVCC F9 G9

P1V2_BUS_INTVCC 1 — TP32 EMPTY SM

FSWPHCFG H4 — R1565 1 — 2 EMPTY 6.34K

SHARE_CLK H7 — P1V2_BUS_SHARE_CLK 78

SYNC E7 — P1V2_BUS_SYNC 78

SNUB0 A5
SNUB1 M5

P1V2_BUS_SNUB0
P1V2_BUS_SNUB1

R1313
A93548-034
10K
1%
0.0625W
EMPTY
0402

R1320
A93548-034
10K
1%
0.0625W
EMPTY
0402

GPIO0_N E4
GPIO1_N F4

P1V2_BUS_POK — OUT 47 78

DNC C10 E11 H11 K10

ISN0A_DP F2
ISN0A_DN E2
ISN0B_DP F1
ISN0B_DN E1
ISN1A_DP H2
ISN1A_DN G2
ISN1B_DP H1
ISN1B_DN G1

SGND G7 G8

AGND_P1V2_BUS

GND G11 G12 H3 H10 J4 J10 K4 K7 K8 K9 L4 L5 L6 L7 L8 L9 M4 M6 M7 M8 M9 M10

3.3V_STB

R1576
A93548-202
4.99K
1%
0.0625W
RES
0402

R1577
A93548-034
10K
1%
0.0625W
RES
0402

78 P1V2_BUS_SYNC
78 P1V2_BUS_SHARE_CLK

CAD NOTE:
PLACE THE TP AT EASILY ACCESSIBLE SPOT ON THE PCB.
AVOID UNDER HEATSINK AND ADD-ON CARD SHADOW AREA

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VCCL CONTROLLER

C1212 150uF 16V TANT 7343
C1213 150uF 16V TANT 7343
C1214 150uF 16V TANT 7343
C1184 150uF 16V TANT 7343
C645 150uF 16V TANT 7343
C669 H65936-001 22UF 10% 25V X6S 1206
C670 H65936-001 22UF 10% 25V X6S 1206
C679 H65936-001 22UF 10% 25V X6S 1206
C680 M13510-001 2.2UF 20% 25V X6S 0402

12V

QFN U31 IC

HC_LTC3888_1

C638 A36096-109 2.2UF 20% 10V X5R 0402
C646 A36096-109 2.2UF 20% 10V X5R 0402
R604 K32884-001 18.7K 1% 0.0625W RES 0402

VIN
VDD25
VDD33
RSET
ISENSE0
ISENSE1
ISENSE2
ISENSE3
ISENSE4
ISENSE5
ISENSE6
ISENSE7
VSENSE0_DP
VSENSE0_DN
VSENSE1_DP
VSENSE1_DN
TSNS0/VDR_MON
TSNS1
ITH0
ITH1
ITHR0
ITHR1
SCL
SDA
VFB0
VFB1
ASEL0
ASEL1
GND_TPD

PGOOD0/CLKOUT
PGOOD1
FAULT0_N
FAULT1_N
ALERT_N
SYNC
RUN0
RUN1
INTVCC
ILIM0_CFG
ILIM1_CFG
DAOUT0
DAOUT1
SHARE_CLK
IREF
PWM0
PWM1
PWM2
PWM3
PWM4
PWM5
PWM6
PWM7
PWM_CFG

M76261-001

LTC3888IUHG-1#PBF

PMBUS ADDRESS = "110_0010"

3.3V_STB

R609 A93548-517 4.7K 1% 0.0625W RES 0402

VCCL_POK
VCCL_FAULT
R1607 0402 EMPTY 0
VCCL_ALERTN
VCCL_SYNC 0402 R611 RES 4.7K
VCCL_EN
VCCL_INTVCC 0402 C682 X5R 4.7UF
VCCL_ILIMCFG 0402 R610 RES 6.49K
VCCL_DAOUT0
VCCL_SH_CLK 0402 R443 RES 4.7K
VCCL_IREF
VCCL_PWM0
VCCL_PWM2
VCCL_PWM3
VCCL_PWM4
C1177 H81594-001 1UF 10% 25V X6S 0402
0402 R608 RES 6.49K

OC_WARN: 44A
OC_FAULT: 55A
PER PHASE

PWM CONFIG:
4+2
550kHz

VCCL_IREF
VCCL_ITH0
VCCL_ITHR0
VCCL_SCL
VCCL_SDA
R605 A93548-034 10K 1% 0.0625W RES 0402
C1205 A36095-026 100PF 5% 50V EMPTY 0402
R606 A93548-034 10K 1% 0.0625W RES 0402
0402 R607 RES 1.74K
0402 R551 RES 4.42K

VCCL_TSNS0

R309 A93548-517 4.7K 1% 0.0625W EMPTY 0402
R289 A93548-517 4.7K 1% 0.0625W RES 0402
R288 A93548-517 4.7K 1% 0.0625W RES 0402
R287 A93548-517 4.7K 1% 0.0625W RES 0402
R283 A93548-517 4.7K 1% 0.0625W RES 0402

VCCL_POK
VCCL_ALERTN
VCCL_SDA
VCCL_SCL
VCCL_EN

R286 A36092-023 1K 1% 0.063W RES 0402

VCCL_SENSE_DN
VCCL_SENSE_DP

C415 K13546-001 10PF 5% 50V C0G 0402
R284 A93549-001 0 5% 0.0625W RES 0402

R613 RES 0 5% 0.0625W 0402 VCCL_VGND
R1303 RES 0 5% 0.0625W 0402 VCCL_VOUT
R1304 RES 0 5% 0.0625W 0402 VCCL_SENSE_N
VCCL_SENSE_P

IN FUTURE IF 3V IS REQUIRED ON VCCL_BIAS,
REMOVE FB95 AND INSTALL FB46 AND OTHERS

-MODIFIED FOR 5V

EM_5V
VCCL_BIAS
VCCL_VDD33
BIAS_3V3
FB95 450MA FB 220 0402
FB46 450MA EMPTY 220 0402
R290 5% EMPTY 0.0625W 0 0402
C420 H90637-001 2.2UF 10% 10V EMPTY 0402
C421 H90637-001 2.2UF 10% 10V X6S 0402
C422 H90637-001 2.2UF 10% 10V EMPTY 0402

R310 0402 RES 499 VCCL_ISENSE0
R594 0402 RES 499 VCCL_ISENSE4
R419 0402 RES 499 VCCL_ISENSE2
R416 0402 RES 499 VCCL_ISENSE3
R626 A93549-001 0 5% 0.0625W RES 0402 VCCL_IREF
C1193 A36096-085 0.47UF 10% 6.3V X5R 0402
C1194 A36096-085 0.47UF 10% 6.3V X5R 0402
C1195 A36096-085 0.47UF 10% 6.3V X5R 0402
C637 A36096-085 0.47UF 10% 6.3V X5R 0402
C1196 A36096-030 0.1uF 10% 16V X7R 0402

C398 A36096-067 330PF 10% 50V X7R 0402
C688 A36095-026 100PF 5% 50V C0G 0402
C690 A36096-022 2200PF 10% 50V X7R 0402

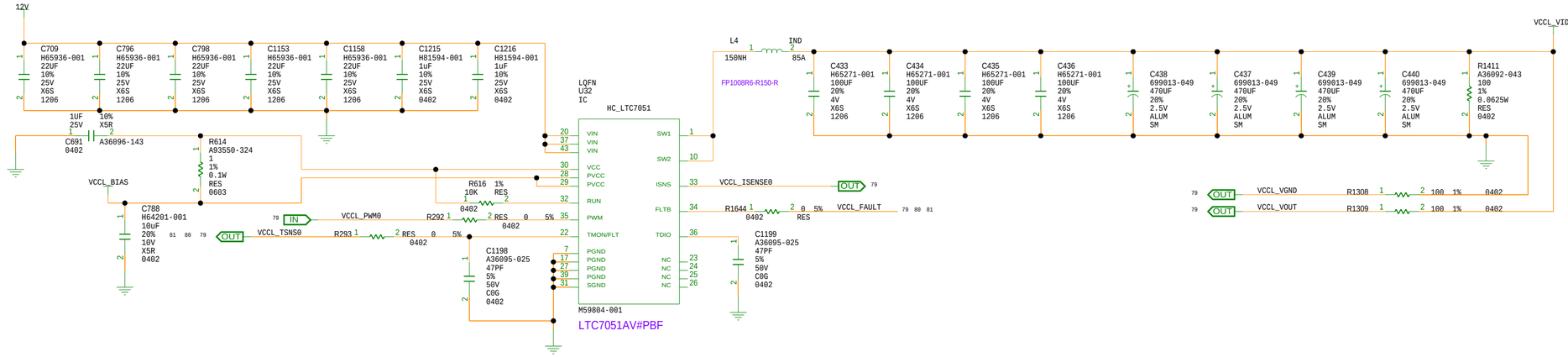
VCCL_TSNS0
VCCL_ITH0
VCCL_ITHR0

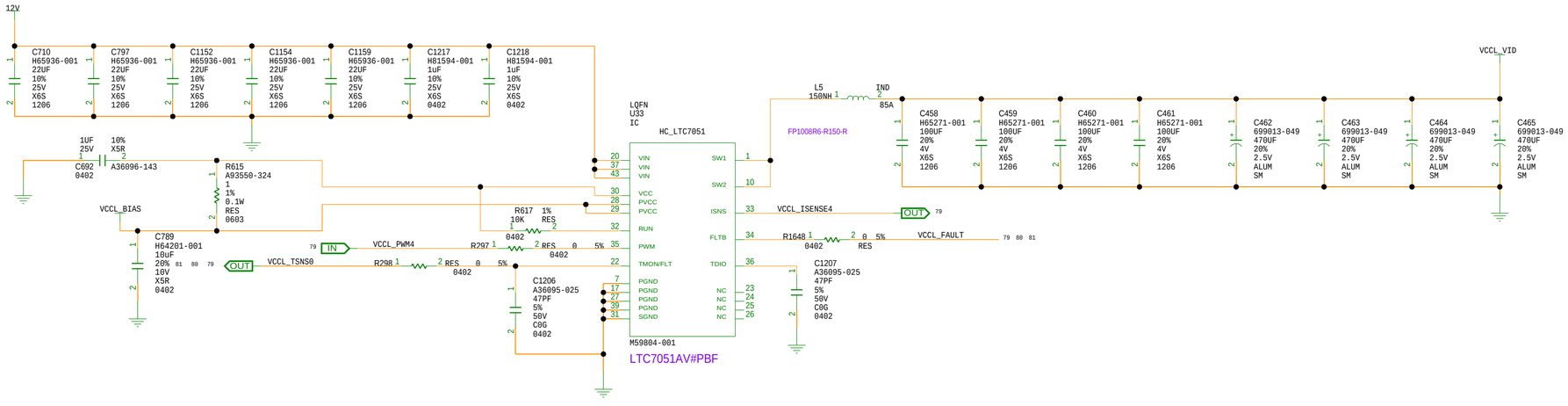
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		SCALE:	DO NOT SCALE DRAWING	SHEET 79 OF 97	

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CORE FETS



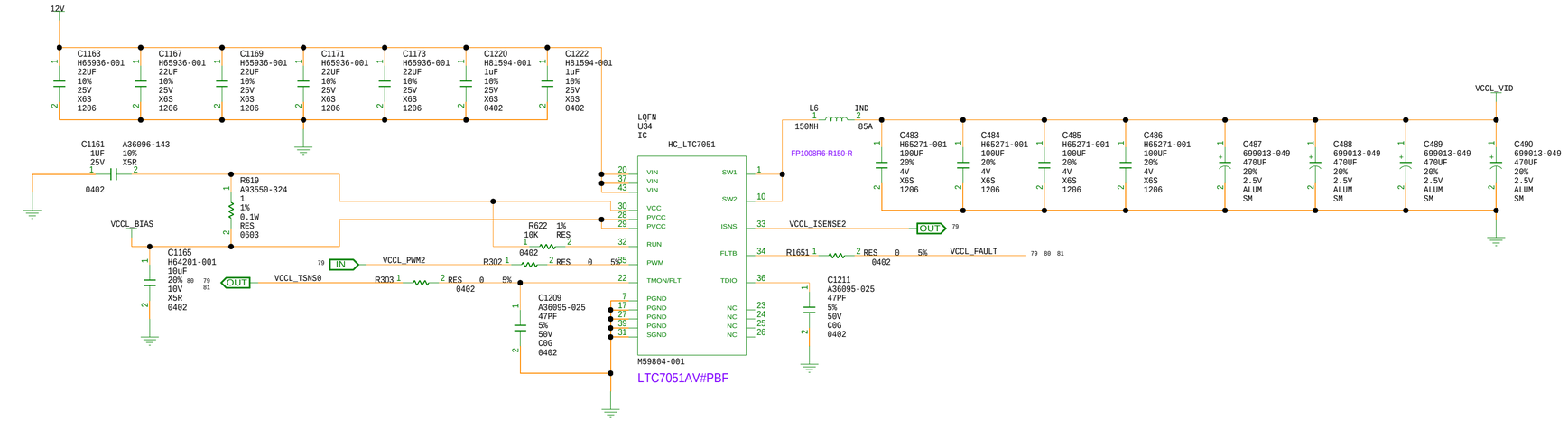


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	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
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CORE FETS



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		SCALE:	DO NOT SCALE DRAWING	SHEET 81 OF 97	

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P0V8_G1

3.3V_STB

PMB_SCL R1298 2.7K 1% 0.0625W 0402
PMB_SDA R1299 2.7K 1% 0.0625W 0402
PMB_ALERTN R1300 2.7K 1% 0.0625W 0402
P0V8_POK R366 4.7K 1% 0.0625W 0402

C2660 H61346-001 10NF 10% 50V EMPTY 0402

AGND_P0V8

12V

C449 150UF 16V TANT 7343
C630 H65936-001 22UF 10% 25V X6S 1206
C631 H65936-001 22UF 10% 25V X6S 1206
C632 H65936-001 22UF 10% 25V X6S 1206
C633 H65936-001 22UF 10% 25V X6S 1206
C634 H65936-001 22UF 10% 25V X6S 1206
C635 D38464-001 10UF 10% 25V X6S 1206

R372 A93548-494 5.23K 1% 0.0625W RES 0402
P0V8_G1_ASEL

BGA U44 IC HC_LTM4677

VOUT SET TO 0.8V
PMBUS ADDRESS = "7BIT 0X46"

P0V8_G1

C624 D97712-022 1UF 10% 16V X6S 0402

R1282 A93549-001 0 5% 0.0625W RES 0402
R1283 A93549-001 0 5% 0.0625W RES 0402

P0V8_G1_VOSNS0_DP
P0V8_G1_VOSNS0_DN
R1566 0 RES 0402 P0V8_G1_VOSNS1

TP33 TP34 TP45

CAD NOTE:CONNECT NEAR FPGA BGA REGION. NEED TO ROUTE AS KELVIN SENSE

MAX CURRENT = 36A(SINGLE)
LOAD CURRENT = 11.6A
VO = 0.8V
SWITCHING FREQ= 500 KHZ
RIPPLE = 10MV

P0V8_EN
P0V8_G1_COMPA
P0V8_G1_WP
P0V8_G1_INTVCC
P0V8_G1_SHARE_CLK
P0V8_G1_SYNC
P0V8_G1_SNUB0
P0V8_G1_SNUB1
P0V8_G1_VOUTCFG
P0V8_G1_VTRIMCFG
P0V8_POK
PMB_ALERTN

TP46 EMPTY SM
TP47 EMPTY SM
TP48 EMPTY SM 6.34K
TP80 EMPTY SM

R1567 EMPTY 0
R1608 EMPTY 0
R1609 EMPTY 0

C1200 A36095-026 100PF 5% 50V C0G 0402
C628 A36096-022 2200PF 10% 50V X7R 0402
R379 A93548-202 4.99K 1% 0.0625W RES 0402
R378 A36092-023 1K 1% 0.063W RES 0402
R380 0 5% 0.0625W RES 0402

R1324 A93548-034 10K 1% 0.0625W EMPTY 0402
R367 A36092-023 1K 1% 0.063W RES 0402

R1325 A93549-001 0 5% 0.0625W RES 0402
R1368 G21796-163 825 1% 0.0625W RES 0402
R1327 A93548-034 10K 1% 0.0625W EMPTY 0402
R1328 A93548-034 10K 1% 0.0625W EMPTY 0402

CAD NOTE:
PLACE THE TP AT EASILY ACCESSIBLE SPOT ON THE PCB.
AVOID UNDER HEATSINK AND ADD-ON CARD SHADOW AREA

CAD NOTE: PLEASE THIS BULK CAP ON THE SAME SIDE AS IC

C2516 699013-049 470UF 20% 2.5V ALUM SM
C2518 H65271-001 100UF 20% 4V X6S 1206
C2519 H65271-001 100UF 20% 4V X6S 1206
C2674 H65271-001 100UF 20% 4V X6S 1206
C2675 H65271-001 100UF 20% 4V X6S 1206
C2676 H65271-001 100UF 20% 4V X6S 1206
C2677 H65271-001 100UF 20% 4V X6S 1206
C2513 699013-049 470UF 20% 2.5V EMPTY SM
C2512 H65271-001 100UF 20% 4V X6S 1206
C2514 H65271-001 100UF 20% 4V X6S 1206
C2515 H65271-001 100UF 20% 4V X6S 1206
C621 H65271-001 100UF 20% 4V X6S 1206
C622 H65271-001 100UF 20% 4V X6S 1206
C623 H65271-001 100UF 20% 4V X6S 1206

PLACEHOLDER

R1568 A93548-202 4.99K 1% 0.0625W RES 0402
R1569 A93548-034 10K 1% 0.0625W RES 0402

LTM4677EY#PBF

K15467-001

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DEPARTMENT		SIZE	CODE	DOCUMENT NUMBER	REV
	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 82 OF 97	

ALTERA CONFIDENTIAL

P0V8_VCC_HSSI_1

VOUT SET TO 0.8V
PMBUS ADDRESS = "7BIT 0X40"

CAD NOTE:CONNECT NEAR FPGA BGA REGION. NEED TO ROUTE AS KELVIN SENSE

MAX CURRENT = 36A(SINGLE)
LOAD CURRENT = 29.2A
VO = 0.8V
SWITCHING FREQ= 500 KHZ
RIPPLE = 10MV

CAD NOTE:
PLACE THE TP AT EASILY ACCESSIBLE SPOT ON THE PCB.
AVOID UNDER HEATSINK AND ADD-ON CARD SHADOW AREA

CAD NOTE: PLEASE THIS BULK CAP ON THE SAME SIDE AS IC

PLACEHOLDER

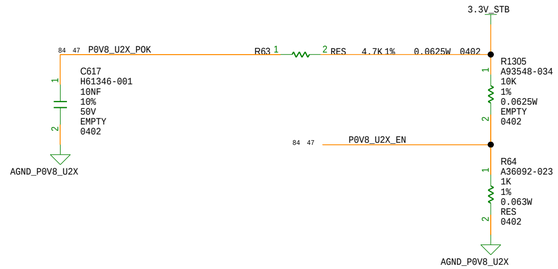
LTM4677EY#PBF

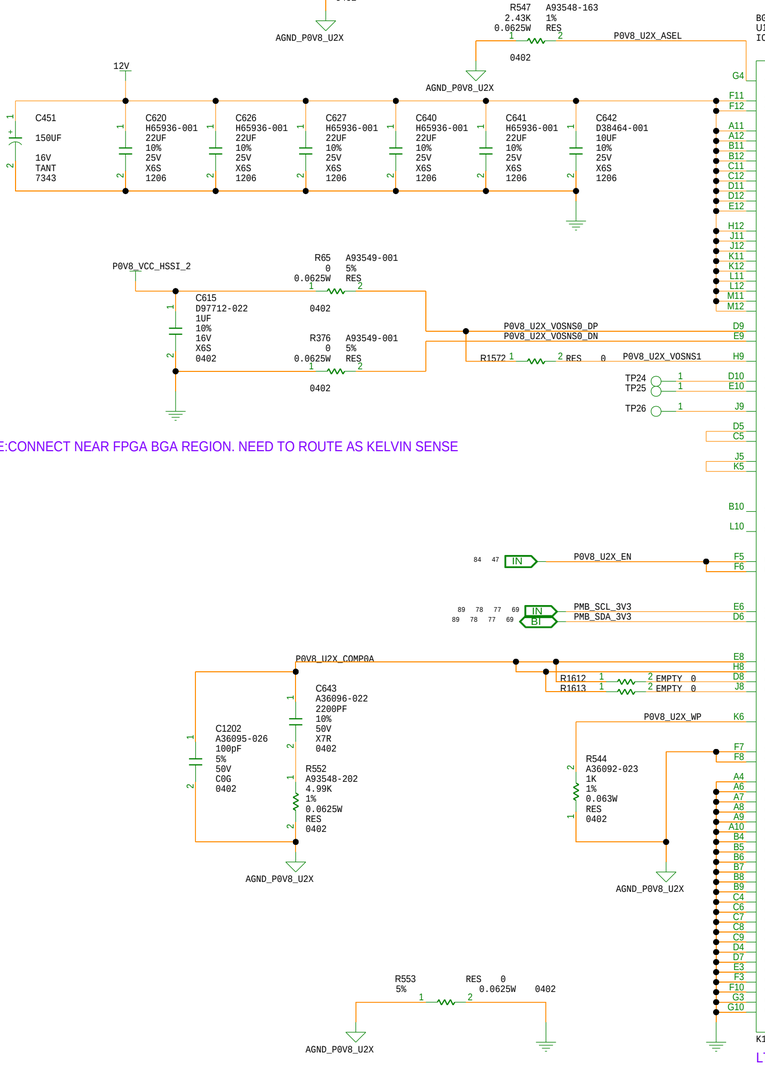
Mon Sep 15 10:27:23 2025

DEPARTMENT	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	SIZE	CODE	DOCUMENT NUMBER	REV
		C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET	83 OF 97

ALTERA CONFIDENTIAL

P0V8_VCC_HSSI_2





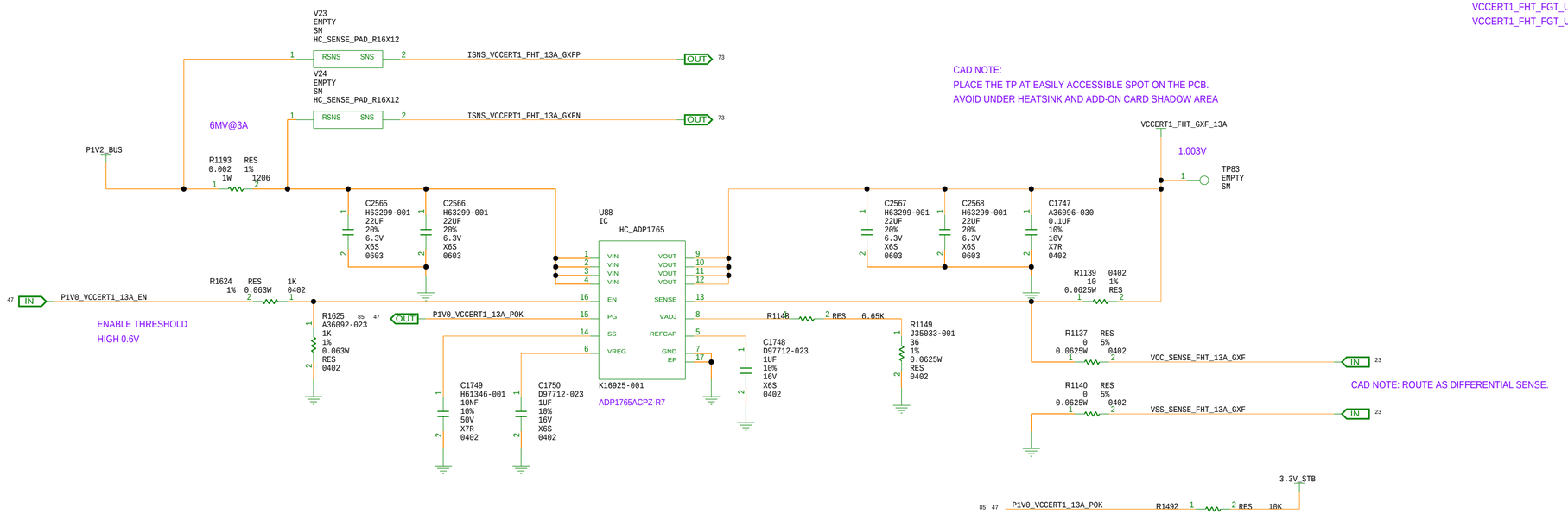
Mon Sep 15 10:27:23 2025

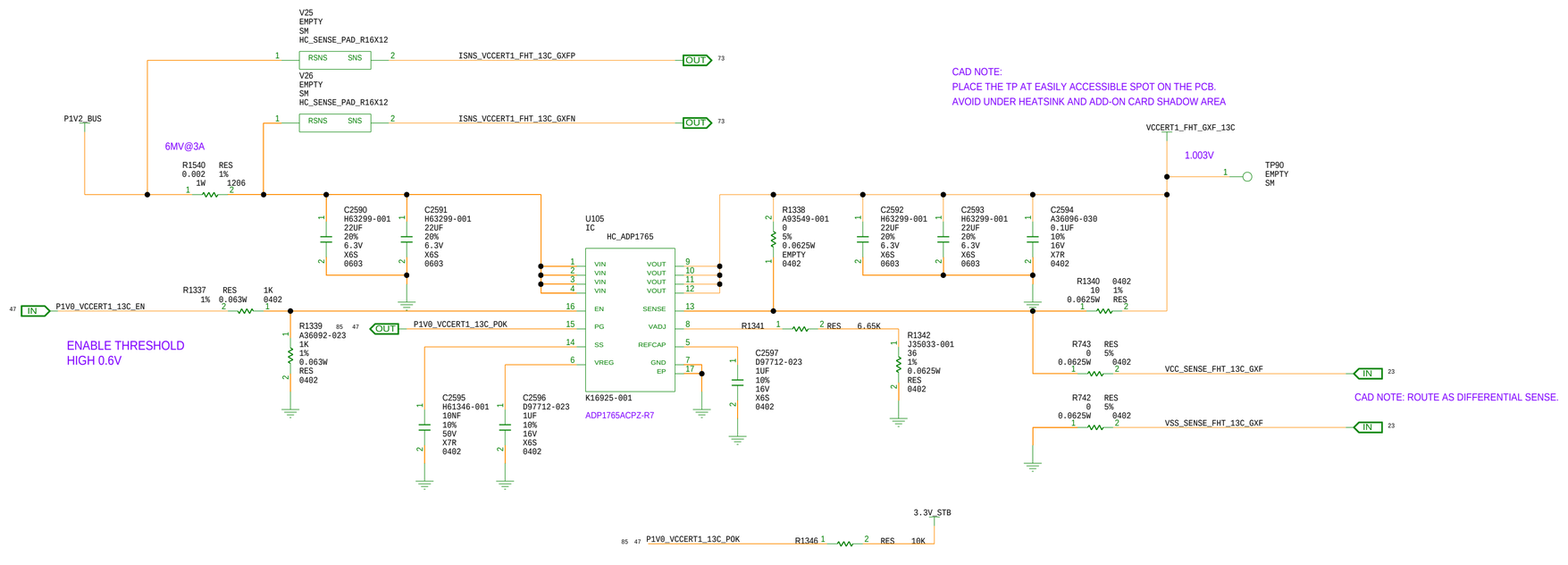
DEPARTMENT		SIZE	CODE	DOCUMENT NUMBER	REV
	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 84 OF 97	

ALTERA CONFIDENTIAL

PWR VCCERT1_13A/13C

VCCERT1_FHT_FGT_U20
VCCERT1_FHT_FGT_U22



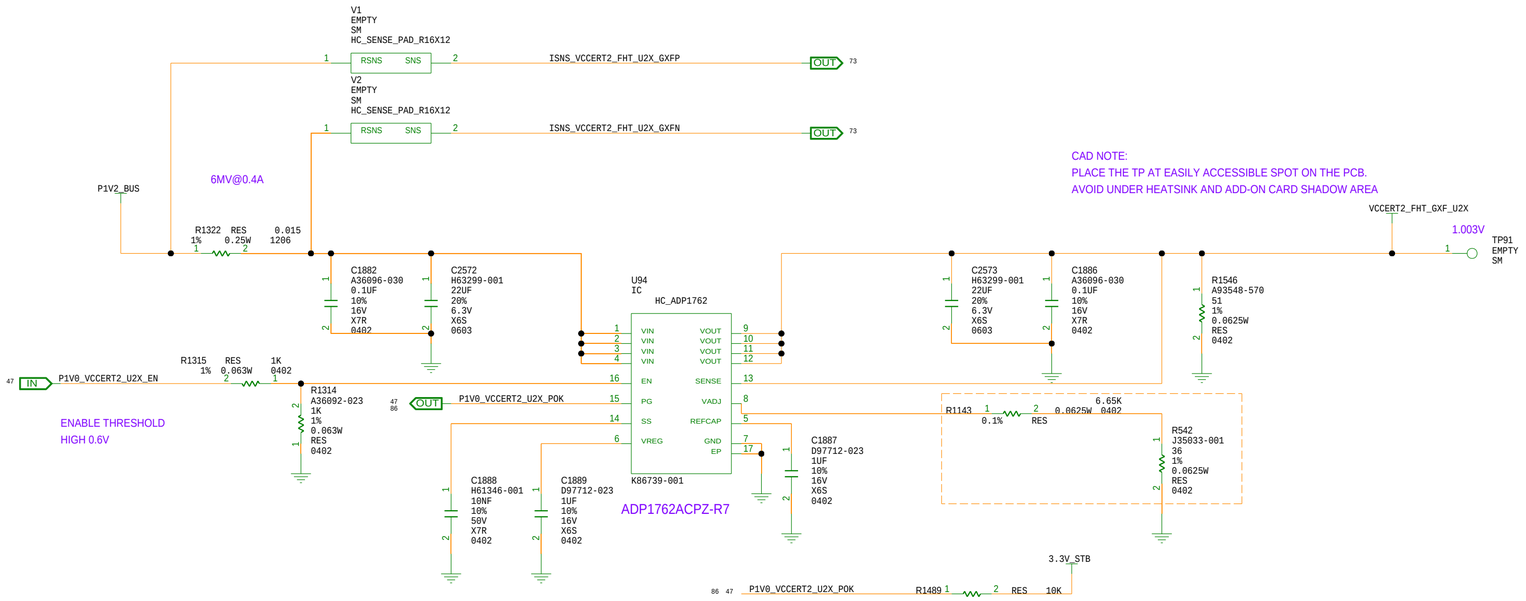


Mon Sep 15 10:27:24 2025

DEPARTMENT		SIZE	CODE	DOCUMENT NUMBER	REV
	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 85 OF 97	

ALTERA CONFIDENTIAL

PWR VCCERT2_U20/U22



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DEPARTMENT		SIZE	CODE	DOCUMENT NUMBER	REV
	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 86 OF 97	

ALTERA CONFIDENTIAL

VCCERT_UX_GXF

3.3V_STB
C2661 H61346-001 10NF 10% 50V EMPTY 0402
AGND_P1V0_G1
87 47 P1V0_G1_POK
R1080 RES 4.7K 1% 0.0625W 0402
R1336 A93548-034 10K 1% 0.0625W EMPTY 0402
87 47 P1V0_G1_EN
R1081 A36092-023 1K 1% 0.063W RES 0402
AGND_P1V0_G1

R1289 15.4K 0.05W C48657-449 1% RES 0201
P1V0_G1_ASEL
AGND_P1V0_G1
12V
C453 150UF 16V TANT 7343
C1779 H65936-001 22UF 10% 25V X6S 1206
C1780 H65936-001 22UF 10% 25V X6S 1206
C1781 H65936-001 22UF 10% 25V X6S 1206
C1782 H65936-001 22UF 10% 25V X6S 1206
C1783 H65936-001 22UF 10% 25V X6S 1206
C1784 D38464-001 10UF 10% 25V X6S 1206

VCCERT_FGT_GXF_R
C1777 D97712-022 1UF 10% 16V X6S 0402
R1087 0 5% 0.0625W RES 0402
R1413 0 5% 0.0625W RES 0402
R1580 RES 0
P1V0_G1_VOSNS0_DP
P1V0_G1_VOSNS0_DN
P1V0_G1_VOSNS1

VOUT SET TO 1V
PMBUS ADDRESS = "7BIT 0X4C"

CAD NOTE:CONNECT NEAR FPGA BGA REGION. NEED TO ROUTE AS KELVIN SENSE

MAX CURRENT = 36A(SINGLE)
LOAD CURRENT = 24.56A
VO = 1V
SWITCHING FREQ= 500 KHZ
RIPPLE = 10MV

TP55 TP56 TP57
R1090 5% RES 0 0.0625W 0402
AGND_P1V0_G1

BGA U104 IC
HC_LTM4677

G4	ASEL
F11 F12	SVIN
A11 A12 B11 B12 C11 C12 D11 D12 E12	VIN0
H12 J11 J12 K11 K12 L11 L12 M11 M12	VIN1
D9 E9	VOSNS0_DP VOSNS0_DN
H9	VOSNS1
D10 E10	VORB0_DP VORB0_DN
J9	VORB1
D5 C5	TSNS0A TSNS0B
J5 K5	TSNS1A TSNS1B
B10	SW0
L10	SW1
F5 F6	RUN0 RUN1
E6 D6	SCL SDA
E8 H8 D8 J8	COMP0A COMP1A COMP0B COMP1B
K6	WP
F7 F8	SGND
A4 A6 A7 A8 A9 A10 B4 B5 B6 B7 B8 B9 C4 C6 C7 C8 C9 D4 D7 E3 F3 F10 G3 G10	GND

87 47 IN P1V0_G1_EN
91 90 83 82 69 50 48 IN PMB_SCL
91 90 83 82 69 50 48 BI PMB_SDA
P1V0_G1_COMPA
R1614 2 EMPTY 0
R1615 2 EMPTY 0
C1788 A36096-022 2200PF 10% 50V X7R 0402
C1203 A36095-026 100pF 5% 50V C0G 0402
R1089 A93548-202 4.99K 1% 0.0625W RES 0402
P1V0_G1_WP
R1088 A36092-023 1K 1% 0.063W RES 0402
AGND_P1V0_G1
AGND_P1V0_G1

K15467-001
LTM4677EY#PBF

J6 J7	VDD25 VDD33
E5	ALERT_N
A1 A2 A3 B1 B2 B3 C1 C2 C3 D1 D2 D3	VOUT0
J1 J2 J3 K1 K2 K3 L1 L2 L3 M1 M2 M3	VOUT1
G5	VOUT0CFG
G6	VOUT1CFG
H5	VTRIM0CFG
H6	VTRIM1CFG
F9	INTVCC
G9	INTVCC
H4	FSWPHCFG
H7	SHARE_CLK
E7	SYNC
A5 M5	SNUB0 SNUB1
E4 F4	GPIO0_N GPIO1_N
C10 E11 H11 K10	DNC
F2 E2 F1 E1	ISNS0A_DP ISNS0A_DN ISNS0B_DP ISNS0B_DN
H2 G2 H1 G1	ISNS1A_DP ISNS1A_DN ISNS1B_DP ISNS1B_DN
G7 G8	SGND
G11 G12 H3 H10 J4 J10 K4 K7 K8 K9 L4 L5 L6 L7 L8 L9 M4 M6 M7 M8 M9 M10	GND

J6 J7 TP58EMPTY SM TP59EMPTY SM
PMB_ALERTN
OUT 48 50 69 77 78 82 83 84 89 90 91
P1V0_G1_VOUTCFG
P1V0_G1_VTRIMCFG
P1V0_G1_INTVCC
TP60 EMPTY SM
R1574 EMPTY 6.34K
P1V0_G1_SHARE_CLK 87
P1V0_G1_SYNC 87
P1V0_G1_SNUB0
P1V0_G1_SNUB1
P1V0_G1_POK
OUT 47 87
AGND_P1V0_G1

CAD NOTE:
PLACE THE TP AT EASILY ACCESSIBLE SPOT ON THE PCB.
AVOID UNDER HEATSINK AND ADD-ON CARD SHADOW AREA

VCCERT_FGT_GXF_R
TP119 EMPTY SM
C2520 699013-049 470UF 20% 2.5V ALUM SM
C2522 H65271-001 100UF 20% 4V X6S 1206
C2524 H65271-001 100UF 20% 4V X6S 1206
C1772 H65271-001 100UF 20% 4V X6S 1206
C1773 H65271-001 100UF 20% 4V X6S 1206
C1774 H65271-001 100UF 20% 4V X6S 1206
C2525 H65271-001 100UF 20% 4V X6S 1206

CAD NOTE: PLEASE THIS BULK CAP ON THE SAME SIDE AS IC
C1101 699013-049 470UF 20% 2.5V EMPTY SM
PLACEHOLDER
C1102 H65271-001 100UF 20% 4V X6S 1206
C1103 H65271-001 100UF 20% 4V X6S 1206
C1104 H65271-001 100UF 20% 4V X6S 1206
C1105 H65271-001 100UF 20% 4V X6S 1206
C1106 H65271-001 100UF 20% 4V X6S 1206
C2528 H65271-001 100UF 20% 4V X6S 1206

PLACED UNDER FPGA

R1343 A93549-001 0 5% 0.0625W RES 0402
R1382 G24503-019 1.2K 1% 0.125W RES 0402
AGND_P1V0_G1
R1344 A93548-034 10K 1% 0.0625W EMPTY 0402
R1345 A93548-034 10K 1% 0.0625W EMPTY 0402

3.3V_STB
R1581 A93548-202 4.99K 1% 0.0625W RES 0402
R1582 A93548-034 10K 1% 0.0625W RES 0402
87 P1V0_G1_SYNC
87 P1V0_G1_SHARE_CLK

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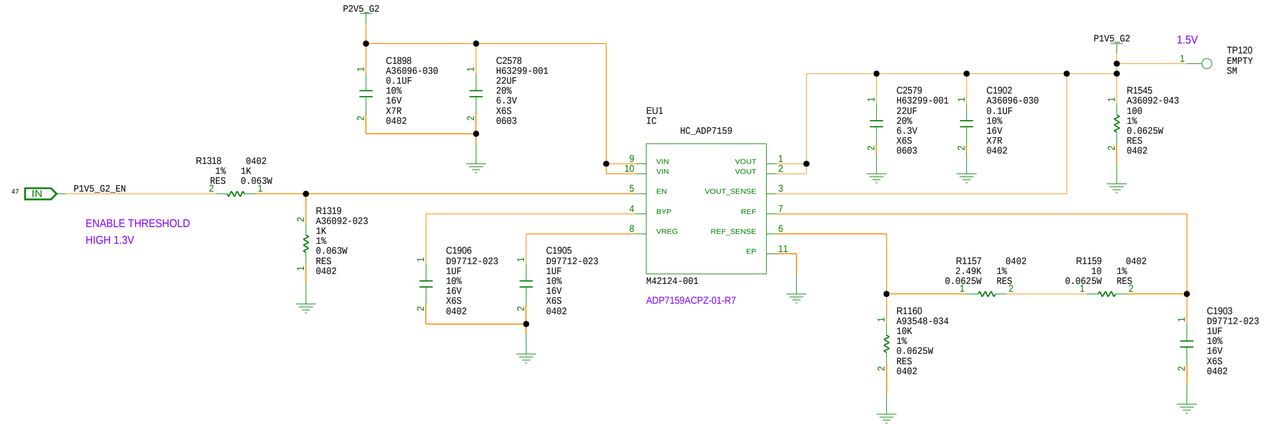
DEPARTMENT		SIZE	CODE	DOCUMENT NUMBER	REV
	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 87 OF 97	

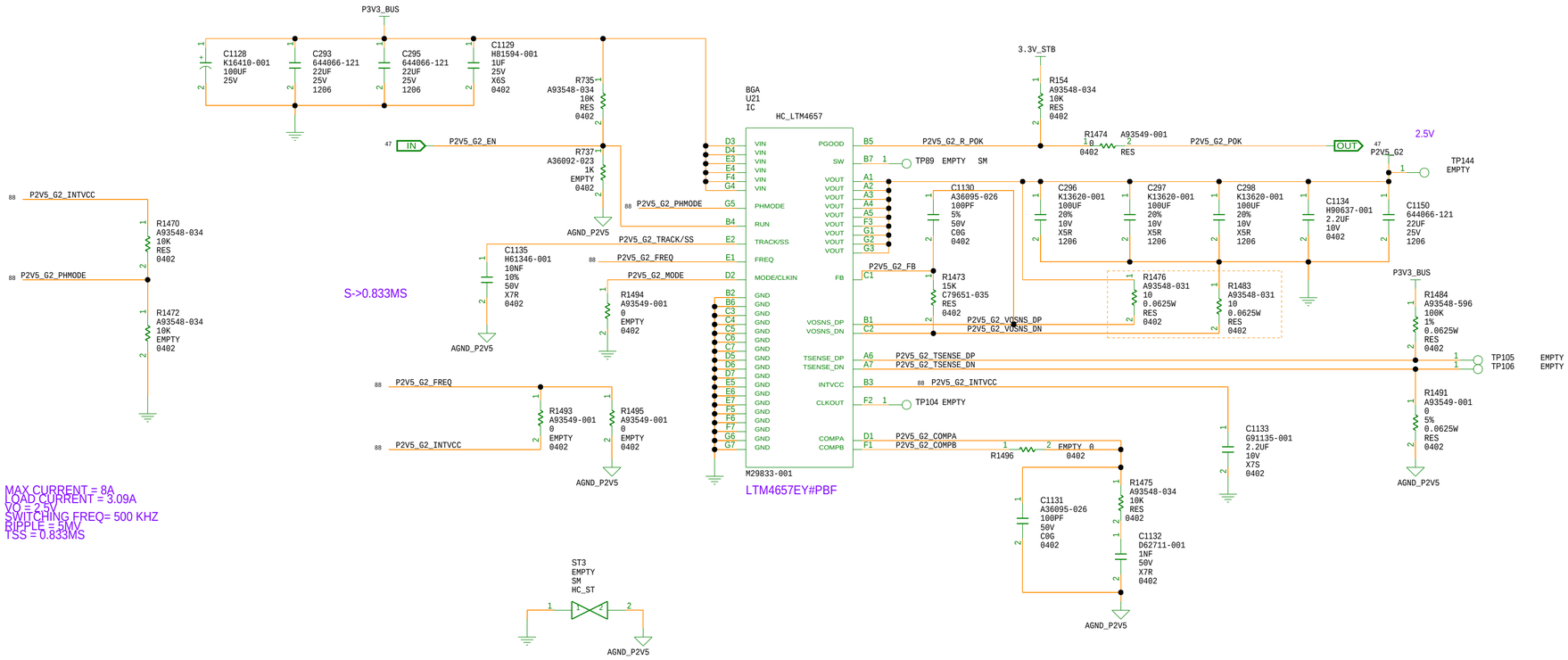
ALTERA CONFIDENTIAL

PWR P1V5_G2/P2V5_G2

CAD NOTE:
PLACE THE TP AT EASILY ACCESSIBLE SPOT ON THE PCB.
AVOID UNDER HEATSINK AND ADD-ON CARD SHADOW AREA

P1V5_G2





P2V5_G2

CAD NOTE:
PLACE THE TP AT EASILY ACCESSIBLE SPOT ON THE PCB.
AVOID UNDER HEATSINK AND ADD-ON CARD SHADOW AREA

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DEPARTMENT	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	SIZE: C+ / CODE: 34649	DOCUMENT NUMBER: 150-0330690-D1	REV: D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 88 OF 97

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P1V8_G2

3.3V_STB

89 47 P1V8_G2_POK — R399 1 — 2 RES 4.7K 1% 0.0625W 0402

C786 H61346-001 10NF 10% 50V EMPTY 0402 — AGND_P1V8_G2

R1407 A93548-034 10K 1% 0.0625W EMPTY 0402

89 47 P1V8_G2_EN

R400 A36092-023 1K 1% 0.063W RES 0402 — AGND_P1V8_G2

12V

C1120 G46681-001 150UF 20% 16V TANT 7343

R643 E58715-001 0.005 1% 1W RES 0805

P1V8_G2_VIN

C790 H65936-001 22UF 10% 25V X6S 1206

C791 H65936-001 22UF 10% 25V X6S 1206

C792 H65936-001 22UF 10% 25V X6S 1206

C793 H65936-001 22UF 10% 25V X6S 1206

C794 H65936-001 22UF 10% 25V X6S 1206

C795 D38464-001 10UF 10% 25V X6S 1206

R650 A93548-212 22.6K 1% 0.0625W RES 0402 — AGND_P1V8_G2

BGA
U43
IC

HC_LTM4678

J1 IIN_DP
K1 IIN_DN
F12 ASEL
D1 SVIN
G1 VIN0
G2 VIN0
G3 VIN0
G4 VIN0
H1 VIN0
H2 VIN0
H3 VIN0
H4 VIN0
E1 VIN1
E2 VIN1
E3 VIN1
E4 VIN1
F1 VIN1
F2 VIN1
F3 VIN1
F4 VIN1

VDD25 D12 — TP78 EMPTY SM
VDD33 E8 — TP79 EMPTY SM
ALERT_N H11 — PMB_ALERTN — OUT 48 50 69 77 78 82 83 84 87 90 91

VOUT0 K7
VOUT0 K8
VOUT0 K9
VOUT0 K10
VOUT0 K11
VOUT0 L7
VOUT0 L8
VOUT0 L9
VOUT0 L10
VOUT0 L11
VOUT0 L12
VOUT0 M7
VOUT0 M8
VOUT0 M9
VOUT0 M10
VOUT1 A7
VOUT1 A8
VOUT1 A9
VOUT1 A10
VOUT1 B7
VOUT1 B8
VOUT1 B9
VOUT1 B10
VOUT1 B11
VOUT1 B12
VOUT1 C7
VOUT1 C8
VOUT1 C9
VOUT1 D7

C455 724613-088 470UF 6.3V TANT 7343
C1174 724613-088 470UF 6.3V TANT 7343
C524 K13620-001 100UF 20% 10V X5R 1206
C525 K13620-001 100UF 20% 10V X5R 1206
C654 K13620-001 100UF 20% 10V X5R 1206
C658 K13620-001 100UF 20% 10V X5R 1206
C661 K13620-001 100UF 20% 10V X5R 1206
C783 K13620-001 100UF 20% 10V X5R 1206
C784 K13620-001 100UF 20% 10V X5R 1206
C785 K13620-001 100UF 20% 10V X5R 1206

P1V8_G2 — TP122 EMPTY SM

CAD NOTE:
PLACE THE TP AT EASILY ACCESSIBLE SPOT ON THE PCB.
AVOID UNDER HEATSINK AND ADD-ON CARD SHADOW AREA

VOUT SET TO 1.8V
PMBUS ADDRESS = "7BIT 0X4E"

C787 D97712-022 1UF 10% 16V X6S 0402 — P1V8_G2

R670 2 — 1 RES 0
R680 2 — 1 RES 0
P1V8_G2_R_VOSNS0_DN — M12 VOSNS0_DN
P1V8_G2_R_VOSNS0_DP — M11 VOSNS0_DP

R1461 1 — 2 RES 0 P1V8_G2_R_VOSNS1_DN — A12 VOSNS1_DN
R1462 1 — 2 RES 0 P1V8_G2_R_VOSNS1_DP — A11 VOSNS1_DP

J11 TSNS0A
J8 TSNS0B
J10 TSNS1A
D8 TSNS1B

L2 SW0
M1 SW0
M2 SW0
A1 SW1
A2 SW1
B2 SW1

VOUT0_CFG E11
VOUT1_CFG E12
P1V8_G2_VOUTCFG — R1420 1 — 2 RES 2.61K

VTRIM0_CFG C12
VTRIM1_CFG E10
P1V8_G2_VTRIMCFG — R1463 1 — 2 RES 16.2K

PGOOD0 J7
PGOOD1 D9
P1V8_G2_POK — OUT 47 89 — AGND_P1V8_G2

INTVCC E7
P1V8_G2_INTVCC — 2.2UF C1121 1 — 2 10V

EXTVCC F8
P1V8_G2_EXTVCC — R304 1 — 2 RES 0

FSWPH_CFG E9
P1V8_G2_FSWPH_CFG

FAULT1_N G11
FAULT0_N H12
P1V8_G2_FAULT_N — 89

SHARE_CLK D11
P1V8_G2_SHARE_CLK — 89

SYNC K12
P1V8_G2_SYNC — 89

R1427 H89932-001 18K 1% 0.0625W RES 0402 — AGND_P1V8_G2

SGND G9
SGND G10 — AGND_P1V8_G2

GND M6
GND M5
GND M4
GND M3
GND L6
GND L5
GND L4
GND L3
GND L1
GND K6
GND K5
GND K4
GND K3
GND K2
GND J6
GND J5
GND J4
GND J3
GND J2
GND H8
GND H7
GND H6
GND H5
GND G8
GND G7
GND G6

J99380-001
LTM4678EY#PBF

CAD NOTE:CONNECT NEAR FPGA BGA REGION. NEED TO ROUTE AS KELVIN SENSE

MAX CURRENT = 50A(SINGLE)
LOAD CURRENT = 16.05A
VO = 1.8V
SWITCHING FREQ= 425 KHZ
RIPPLE = 10MV

89 47 IN P1V8_G2_EN — G12 RUN0, F11 RUN1

84 78 77 69 IN PMB_SCL_3V3 — J12 SCL
84 78 77 69 BI PMB_SDA_3V3 — H10 SDA

P1V8_G2_COMP0A — J9 COMP0A
P1V8_G2_COMP0B — H9 COMP0B
D10 COMP1A
C10 COMP1B

C799 D62711-001 1nF 10% 50V X7R 0402
C1119 G71847-005 22pF 5% 50V C0G 0402 — AGND_P1V8_G2

NEED CONFIG
Rht0 20k
gm0: 3.69m

P1V8_G2_WP — C11 WP

R1408 A36092-023 1K 1% 0.063W RES 0402 — AGND_P1V8_G2

F9 SGND
P10 SGND

R1406 5% — 1 — 2 RES 0 0.0625W 0402 — AGND_P1V8_G2

3.3V_STB

R1428 A93548-034 10K 1% 0.0625W RES 0402
R1449 A93548-202 4.99K 1% 0.0625W RES 0402
R1464 A93548-034 10K 1% 0.0625W RES 0402

89 P1V8_G2_SHARE_CLK
89 P1V8_G2_SYNC
89 P1V8_G2_FAULT_N

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	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 89 OF 97	

ALTERA CONFIDENTIAL

P1V2_G3

3.3V_STB

90 47 P1V2_G3_POK — R413 1 — 2 RES 4.7K 1% 0.0625W 0402

R1550
A93548-034
10K
1%
0.0625W
EMPTY
0402

C2662
H61346-001
10NF
10%
50V
EMPTY
0402

AGND_P1V2_G3

90 47 P1V2_G3_EN

R414
A36092-023
1K
1%
0.063W
RES
0402

AGND_P1V2_G3

NOTE: U95 Default configured for LTM4686EV-1 uModule
If Changing to LTM4686EV
UnMount R1561,R1551 AND mount R1410

CAD NOTE: PLACE THE TP AT EASILY ACCESSIBLE SPOT ON THE PCB.
AVOID UNDER HEATSINK AND ADD-ON CARD SHADOW AREA

P1V2_G3_ASEL

R682
G21796-309
32.4K
1%
0.0625W
RES
0402

AGND_P1V2_G3

LGA
U95
IC

HC_LTM4686

12V

C430 G46681-001 150UF 16V TANT 7343
C1144 G46681-001 150UF 16V TANT 7343
C683 H65936-001 22UF 10% 25V X6S 1206
C684 H65936-001 22UF 10% 25V X6S 1206
C685 H65936-001 22UF 10% 25V X6S 1206
C686 H65936-001 22UF 10% 25V X6S 1206
C687 H65936-001 22UF 10% 25V X6S 1206

EM_5V

C1178
A36096-143
1UF
10%
25V
X5R
0402

R1561 0 5% 0.0625W RES 0402

R1410 1 — 2 EMPTY 0 0402

P1V2_G3

C681
D97712-022
1UF
10%
16V
X6S
0402

R1414 1 — 2 RES 0

R424 1 — 2 RES 0

VOUT SET TO 1.2V
PMBUS ADDRESS = "7BIT 0X4F"

CAD NOTE: CONNECT NEAR FPGA BGA REGION. NEED TO ROUTE AS KELVIN SENSE

MAX CURRENT = 20A(SINGLE)
LOAD CURRENT = 16A
VO = 1.2V
SWITCHING FREQ= 575 KHZ
RIPPLE = 10MV

P1V2_G3_VOSNS0_DP
P1V2_G3_VOSNS0_DN

R1587 1 — 2 RES 0 — P1V2_G3_VOSNS1

TP138 TP139 TP140

P1V2_G3_COMPA

C1192
A36096-047
1.5NF
10%
50V
X7R
0402

C1191
A36095-058
56pF
5%
50V
C0G
0402

R1595
K32862-001
20K
0.1%
0.063W
RES
0402

AGND_P1V2_G3

90 47 IN P1V2_G3_EN

91 87 83 82 69 50 48 IN PMB_SCL
91 87 83 82 69 50 48 BI PMB_SDA

R1593 1 — 2 EMPTY 0
R1594 1 — 2 EMPTY 0

R1552 1 — 2 RES 0402 1% 1K — P1V2_G3_WP

AGND_P1V2_G3

M75350-001
LTM4686EV-1#PBF

TP141 EMPTY SM
TP142 EMPTY SM

PMB_ALERTN — OUT

C2529 699013-049 470UF 20% 2.5V ALUM SM
C2530 699013-049 470UF 20% 2.5V ALUM SM
C2531 699013-049 470UF 20% 2.5V EMPTY SM

C1145 H65271-001 100UF 20% 4V X6S 1206
C1146 H65271-001 100UF 20% 4V X6S 1206
C1147 H65271-001 100UF 20% 4V X6S 1206
C1148 H65271-001 100UF 20% 4V X6S 1206
C2532 H65271-001 100UF 20% 4V X6S 1206
C676 H65271-001 100UF 20% 4V X6S 1206
C677 H65271-001 100UF 20% 4V X6S 1206
C678 H65271-001 100UF 20% 4V X6S 1206

P1V2_G3 — TP123 EMPTY SM

P1V2_G3_VOUTCFG
P1V2_G3_VTRIMCFG

R1554 A93549-001 0 5% 0.0625W RES 0402
R1555 A93548-116 1.62K 1% 0.0625W RES 0402

FOR 1.2V VOUT:
RVOUTnCFG :- 3.24K/2 =1.62K WILL GIVE 1.3V VOUT
RVTRIMnCFG :- 0 OHM WILL REDUCE -99mV FROM 1.3V

R1551 0 5% 0.0625W RES 0402 — EM_5V — TP143 EMPTY SM

P1V2_G3_INTVCC

R1553 1 — 2 RES 15.4K 0402 — P1V2_G3_FSWPHCFG

AGND_P1V2_G3

P1V2_G3_SHARECLK
P1V2_G3_SYNC

R1652 1 — 2 RES 0
R1653 1 — 2 RES 0

P1V2_G3_POK — OUT 47 90

3.3V_STB

R1556 A36092-023 1K 1% 0.063W RES 0402
R1558 A93548-202 4.99K 1% 0.0625W RES 0402

90 P1V2_G3_SHARECLK
90 P1V2_G3_SYNC

R427 5% 1 — 2 RES 0 0.0625W 0402

AGND_P1V2_G3

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P1V2_IO

3.3V_STB
91 47 P1V2_IO_POK
R1162 1 2 RES 4.7K 1% 0.0625W 0402
R1330 A93548-034 10K 1% 0.0625W EMPTY 0402
C2663 H61346-001 10NF 10% 50V X7R 0402
AGND_P1V2IO
91 47 P1V2_IO_EN
R1163 A36092-023 1K 1% 0.063W RES 0402
AGND_P1V2IO

R1297 G21796-029 12.7K 1% 0.0625W RES 0402
P1V2_IO_ASEL
AGND_P1V2IO
BGA U97 IC
HC_LTM4677

12V
C454 150UF 16V TANT 7343
C1919 H65936-001 22UF 10% 25V X6S 1206
C1920 H65936-001 22UF 10% 25V X6S 1206
C1921 H65936-001 22UF 10% 25V X6S 1206
C1922 H65936-001 22UF 10% 25V X6S 1206
C1923 H65936-001 22UF 10% 25V X6S 1206
C2541 D38464-001 10UF 10% 25V X6S 1206

VOUT SET TO 1.2V
PMBUS ADDRESS = "7BIT 0X4B"

P1V2_IO
C1917 D97712-022 1UF 10% 16V X6S 0402
R1415 A93549-001 0 5% 0.0625W RES 0402
R1171 A93549-001 0 5% 0.0625W RES 0402
R1583 P1V2_IO_VOSNS0_DP P1V2_IO_VOSNS0_DN
RES 0 P1V2_IO_VOSNS1
TP49 TP50 TP51

CAD NOTE:CONNECT NEAR FPGA BGA REGION. NEED TO ROUTE AS KELVIN SENSE

MAX CURRENT = 36A(SINGLE)
LOAD CURRENT = 29.17A
VO = 1.2V
SWITCHING FREQ= 500 KHZ
RIPPLE = 10MV

91 47 IN P1V2_IO_EN
90 87 83 82 69 50 48 IN PMB_SCL
90 87 83 82 69 50 48 BI PMB_SDA

P1V2_IO_COMPA
C1204 A36095-026 100pF 5% 50V C0G 0402
C1928 A36096-022 2200PF 10% 50V X7R 0402
R1173 A93548-202 4.99K 1% 0.0625W RES 0402
R1616 2 EMPTY 0
R1617 2 EMPTY 0
AGND_P1V2IO

P1V2_IO_WP
R1172 A36092-023 1K 1% 0.063W RES 0402
AGND_P1V2IO

R1174 5% RES 0 0.0625W 0402
AGND_P1V2IO

K15467-001
LTM4677EY#PBF

CAD NOTE:
PLACE THE TP AT EASILY ACCESSIBLE SPOT ON THE PCB.
AVOID UNDER HEATSINK AND ADD-ON CARD SHADOW AREA

VDD25 VDD33 J6 J7 TP52 EMPTY SM TP53 EMPTY SM
ALERT_N E5 PMB_ALERTN OUT 48 50 69 77 78 82 83 84 87 89 90

VOUT0 A1 A2 A3 B1 B2 B3 C1 C2 C3 D1 D2 D3
C2537 699013-049 470UF 20% 2.5V ALUM SM
C1909 H65271-001 100UF 20% 4V X6S 1206
C2539 H65271-001 100UF 20% 4V X6S 1206
C2540 H65271-001 100UF 20% 4V X6S 1206
C1912 H65271-001 100UF 20% 4V X6S 1206
C1913 H65271-001 100UF 20% 4V X6S 1206
C1914 H65271-001 100UF 20% 4V X6S 1206
P1V2_IO TP124 EMPTY SM

CAD NOTE: PLEASE THIS BULK CAP ON THE SAME SIDE AS IC

VOUT1 J1 J2 J3 K1 K2 K3 L1 L2 L3 M1 M2 M3
C1095 699013-049 470uF 20% 2.5V EMPTY SM
PLACEHOLDER
C2678 H65271-001 100UF 20% 4V X6S 1206
C2679 H65271-001 100UF 20% 4V X6S 1206
C1096 H65271-001 100UF 20% 4V X6S 1206
C1097 H65271-001 100UF 20% 4V X6S 1206
C1098 H65271-001 100UF 20% 4V X6S 1206
C1099 H65271-001 100UF 20% 4V X6S 1206

VOUT0CFG G5
VOUT1CFG G6
P1V2_IO_VOUTCFG
VTRIM0CFG H5
VTRIM1CFG H6
P1V2_IO_VTRIMCFG
R1331 A93549-001 0 5% 0.0625W RES 0402
R1383 A93548-116 1.62K 1% 0.0625W RES 0402
AGND_P1V2IO

INTVCC F9 G9
P1V2_IO_INTVCC TP54 EMPTY SM
FSWPHCFG H4
R1575 2 EMPTY 6.34K
SHARE_CLK H7 P1V2_IO_SHARE_CLK 91
SYNC E7 P1V2_IO_SYNC 91

SNUB0 A5 P1V2_IO_SNUB0
SNUB1 M5 P1V2_IO_SNUB1
R1332 A93548-034 10K 1% 0.0625W EMPTY 0402
R1333 A93548-034 10K 1% 0.0625W EMPTY 0402

GPIO0_N E4
GPIO1_N F4
P1V2_IO_POK OUT 47 91

DNC C10 E11 H11 K10
ISNS0A_DP F2
ISNS0A_DN E2
ISNS0B_DP F1
ISNS0B_DN E1
ISNS1A_DP H2
ISNS1A_DN G2
ISNS1B_DP H1
ISNS1B_DN G1
SGND G7 G8
AGND_P1V2IO
GND G11 G12 H3 H10 J4 J10 K4 K7 K8 K9 L4 L5 L6 L7 L8 L9 M4 M6 M7 M8 M9 M10

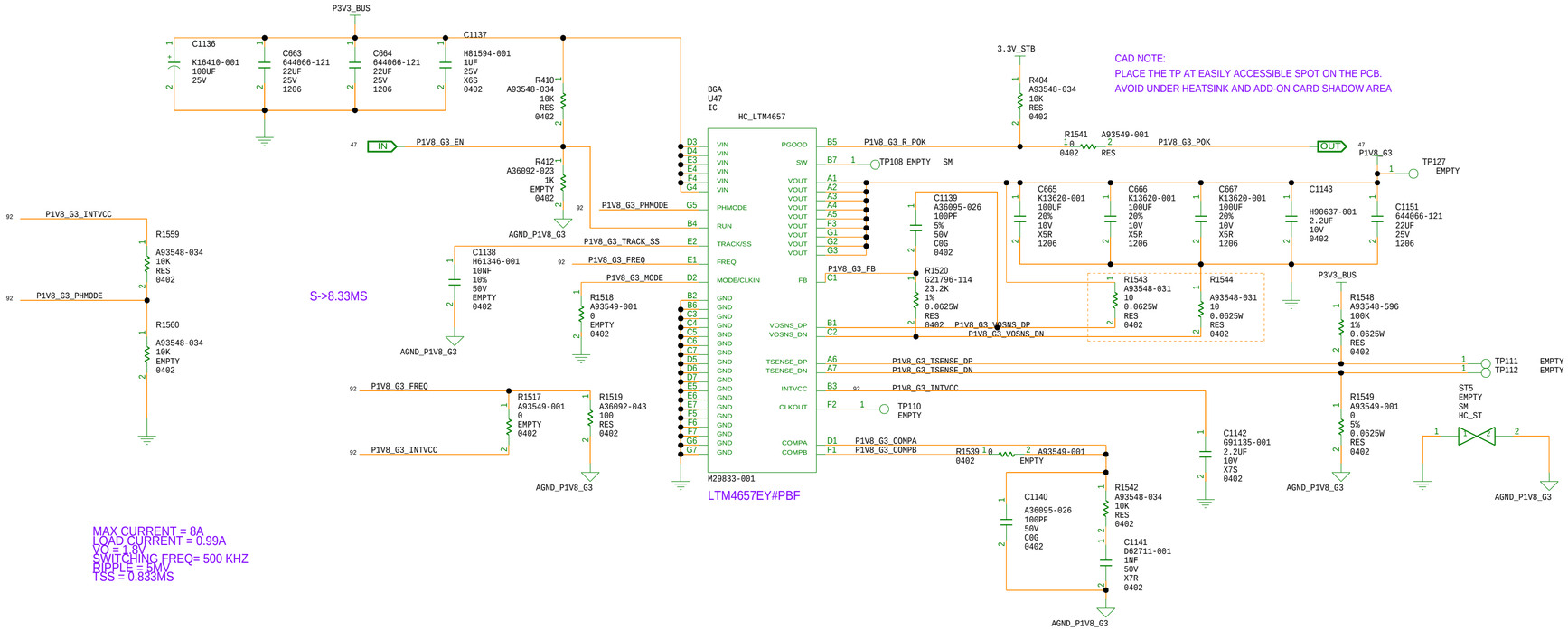
3.3V_STB
R1584 A93548-202 4.99K 1% 0.0625W RES 0402
R1585 A93548-034 10K 1% 0.0625W RES 0402
91 P1V2_IO_SYNC
91 P1V2_IO_SHARE_CLK

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P1V8_G3



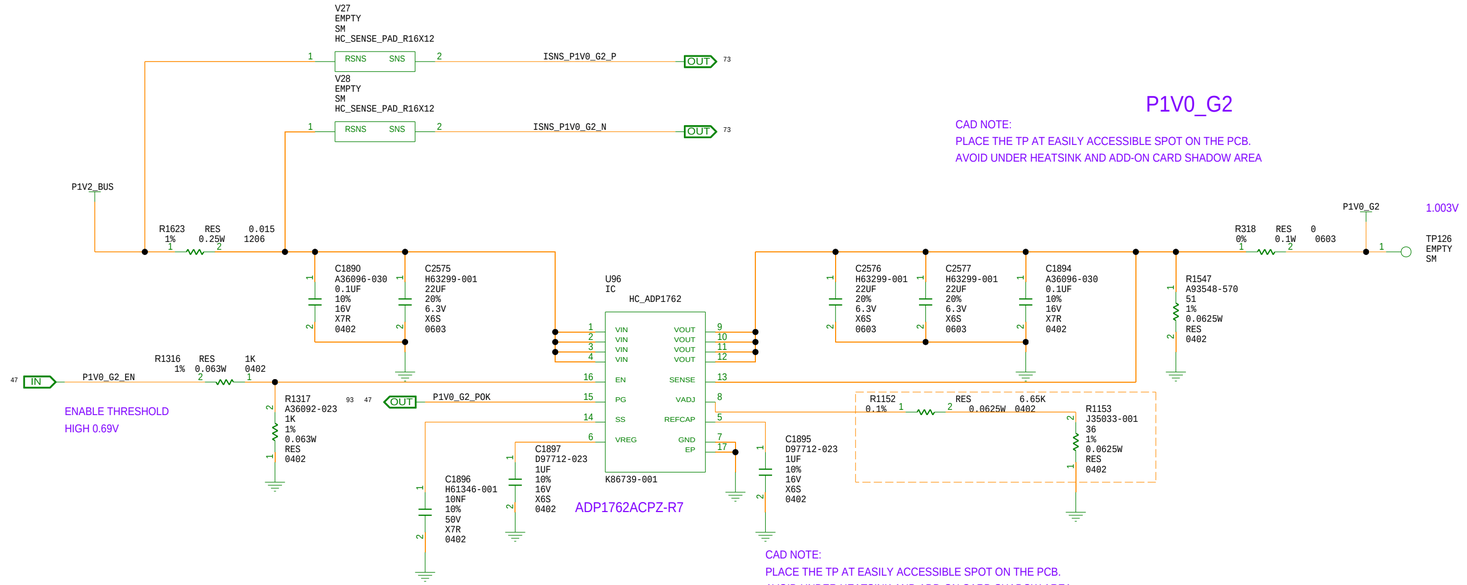
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DEPARTMENT		SIZE	CODE	DOCUMENT NUMBER	REV
	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 92 OF 97	

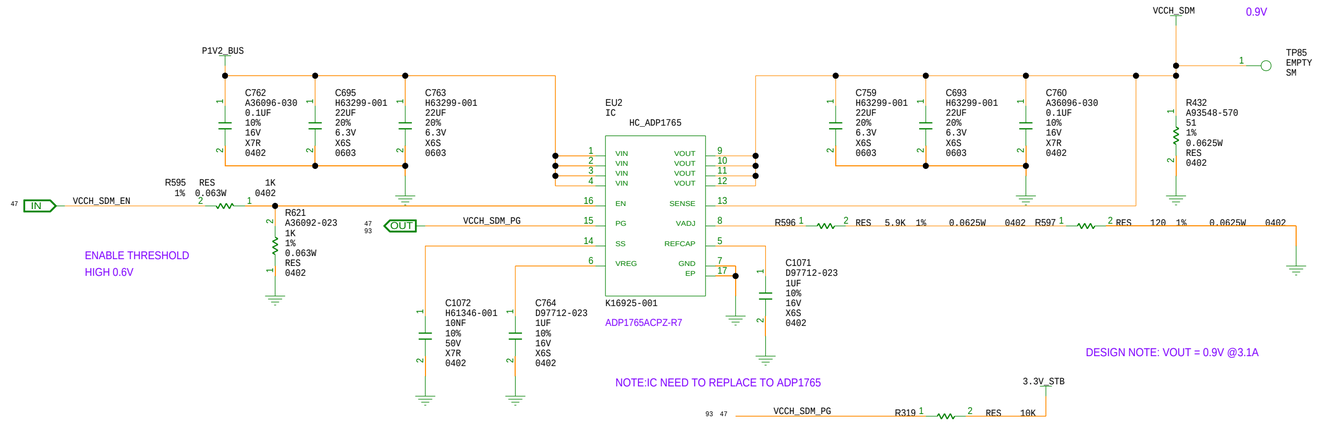
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PWR VCCH_SDM/P1V0_G2

P1V0_G2



VCCH_SDM



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PWR LOAD SWITCHES

7A CURRENT LIMIT
0.34W PDISS

CAD NOTE:
PLACE CLOSE TO THE RESPECTIVE FMC CONNECTOR

CAD NOTE:
PLACE THE TP AT EASILY ACCESSIBLE SPOT ON THE PCB.
AVOID UNDER HEATSINK AND ADD-ON CARD SHADOW AREA

~10V PGOOD THRESHOLD

DESIGN NOTE :ENABLE ON THRESHOLD
1.4V

LTC4211CMS#PBF

CAD NOTE:
PLACE THE TP AT EASILY ACCESSIBLE SPOT ON THE PCB.
AVOID UNDER HEATSINK AND ADD-ON CARD SHADOW AREA

CAD NOTE:
PLACE CLOSE TO THE RESPECTIVE FMC CONNECTOR

DESIGN NOTE :ENABLE ON THRESHOLD
1.4V

LTC4211CMS#PBF

CAD NOTE:
PLACE THE TP AT EASILY ACCESSIBLE SPOT ON THE PCB.
AVOID UNDER HEATSINK AND ADD-ON CARD SHADOW AREA

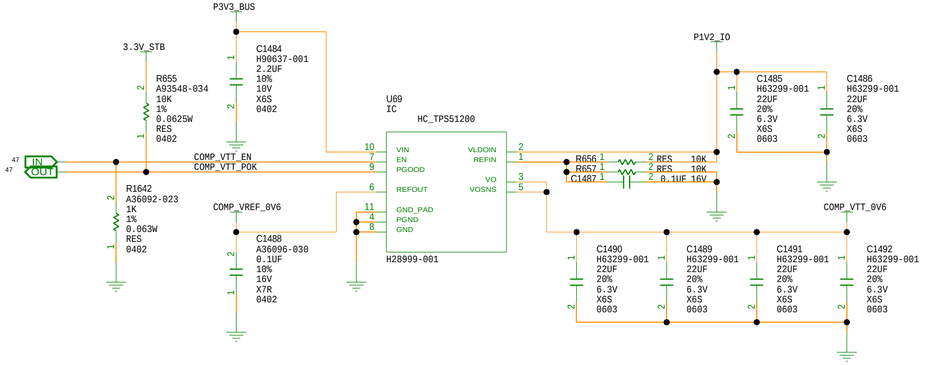
DESIGN NOTE :ENABLE ON THRESHOLD
1.2V

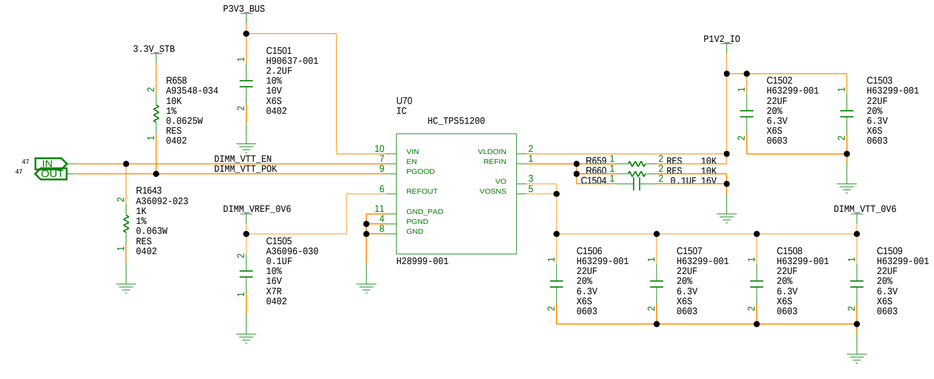
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DEPARTMENT	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	SIZE: C+	CODE: 34649	DOCUMENT NUMBER: 150-0330690-D1 — REV: D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 94 OF 97

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VTT REGULATORS





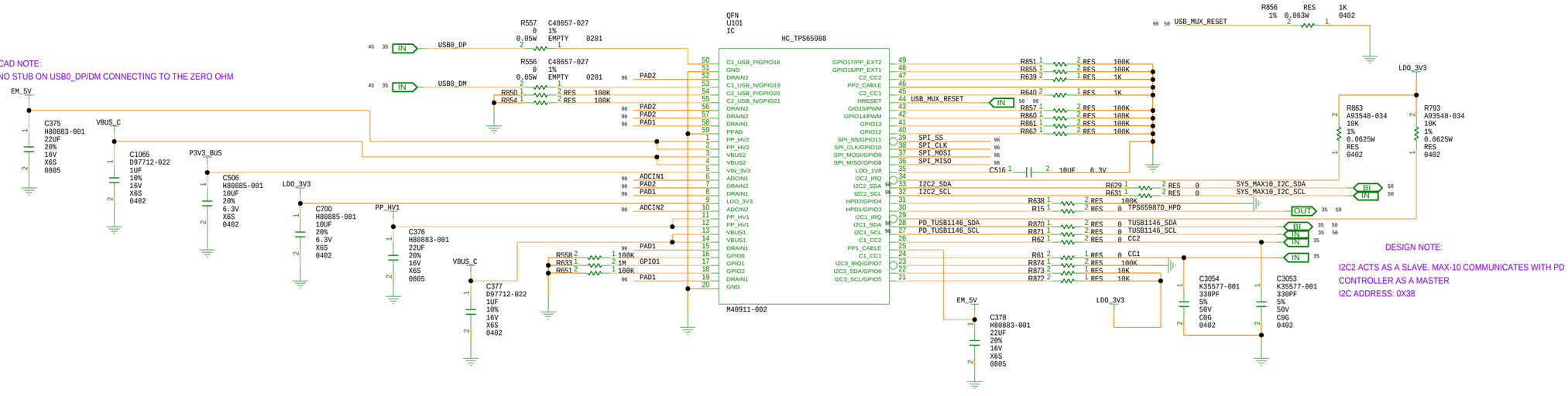
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	ALTERA CORPORATION 101 INNOVATION DR. SAN JOSE, CA 95134	C+	34649	150-0330690-D1	D1
		SCALE:	DO NOT SCALE DRAWING	SHEET 95 OF 97	

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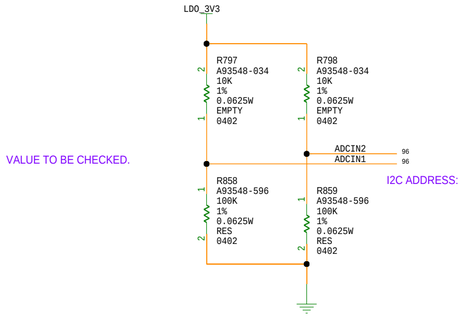
USB PD CONTROLLER

CAD NOTE:
NO STUB ON USB0_DP/DM CONNECTING TO THE ZERO OHM

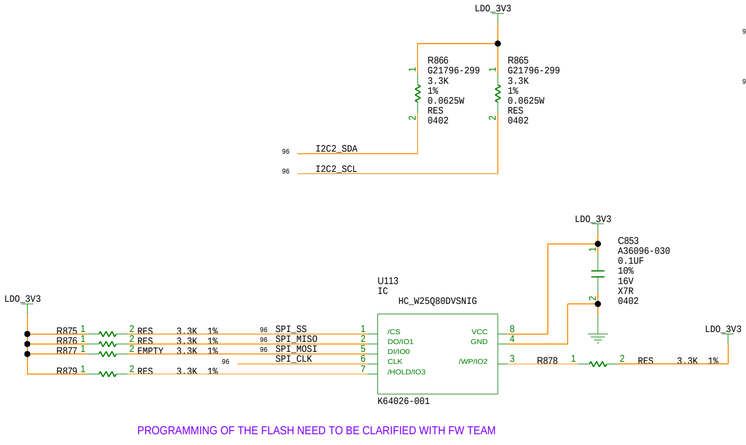


DESIGN NOTE:
I2C2 ACTS AS A SLAVE. MAX-10 COMMUNICATES WITH PD CONTROLLER AS A MASTER
I2C ADDRESS: 0X38

VALUE TO BE CHECKED.



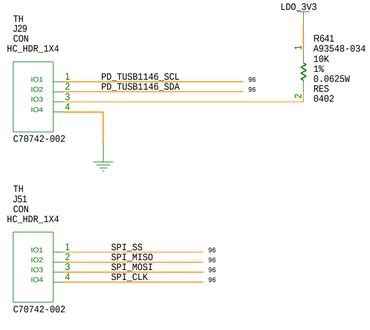
I2C ADDRESS:



PROGRAMMING OF THE FLASH NEED TO BE CLARIFIED WITH FW TEAM



DESIGN NOTE:
I2C1 ACTS AS A MASTER AND COMMUNICATES WTH THE TUSB1146 SS MUX CONTROLLER



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RESERVED

DEPARTMENT		SIZE	CODE	DOCUMENT NUMBER	REV
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		SCALE:	DO NOT SCALE DRAWING	SHEET 97 OF 97	